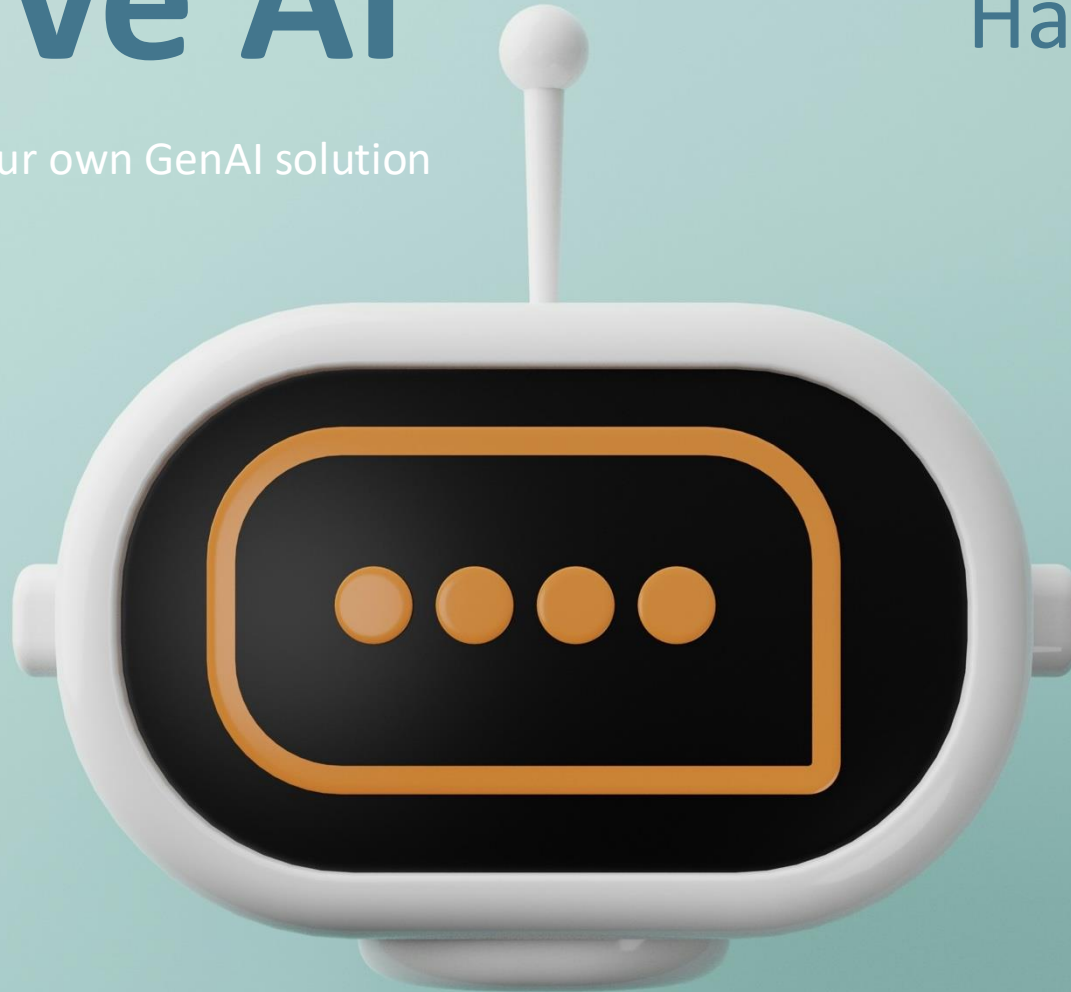


Generative AI

Hands-on Workshop

From Zero to Production: Build your own GenAI solution



#WISSENTEILEN powered by

Lars Röwekamp | Tim Wüllner | open knowledge GmbH



Lars Röwekamp



LinkedIn: **lars-roewekamp**

CIO New Technologies
OPEN KNOWLEDGE

(Architecture, Cloud, AI & ML)



Tim Wüllner



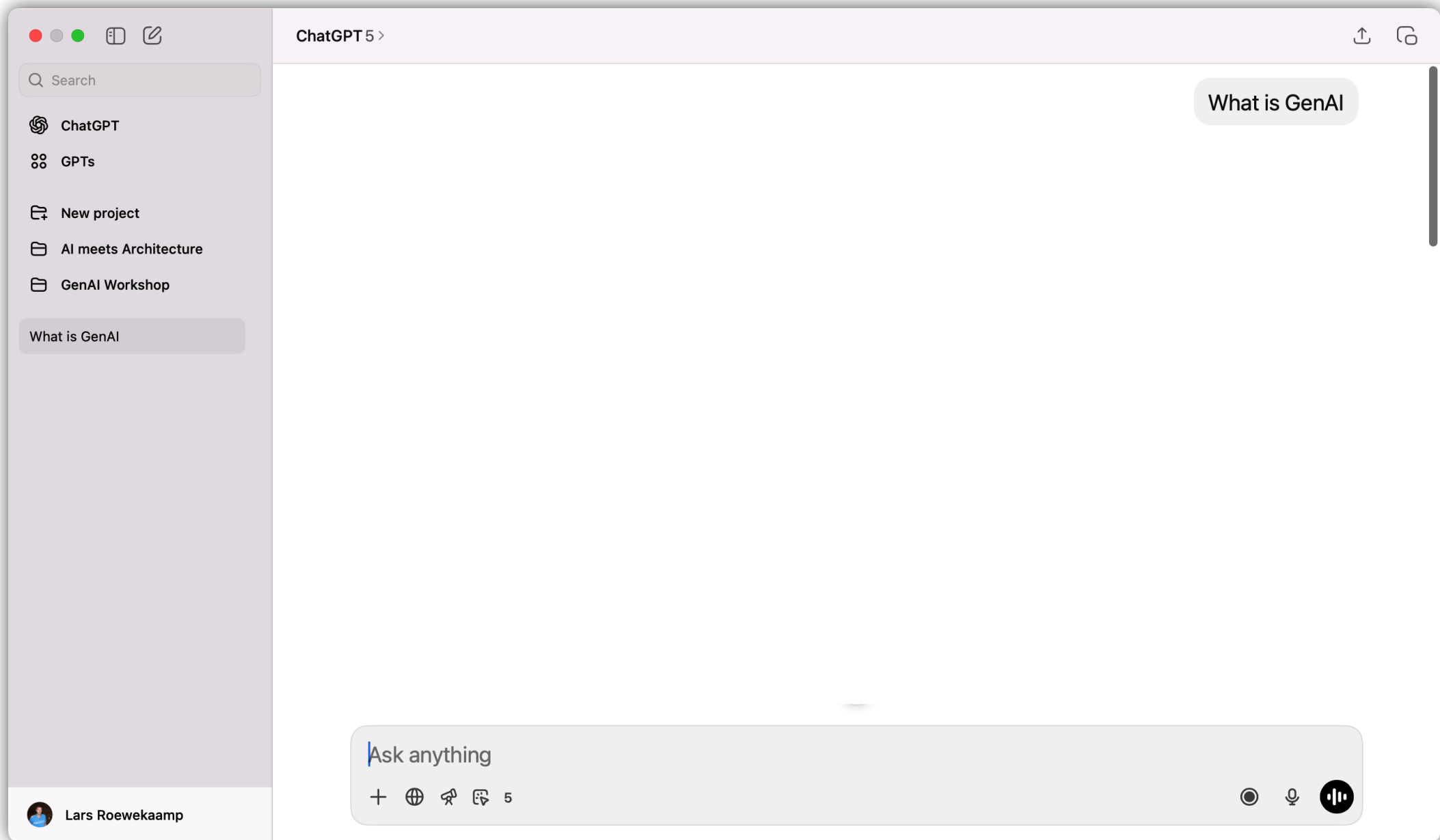
LinkedIn: **tim-wuellner**

Machine Learning Engineer
OPEN KNOWLEDGE



What's GenAI?

*How does
„Generative AI“
actually work?*



How does a LLM generate a response?

The Idea:

The question and answer are
SEMANTICALLY very similar.

How does a LLM generate a response?

The Idea:

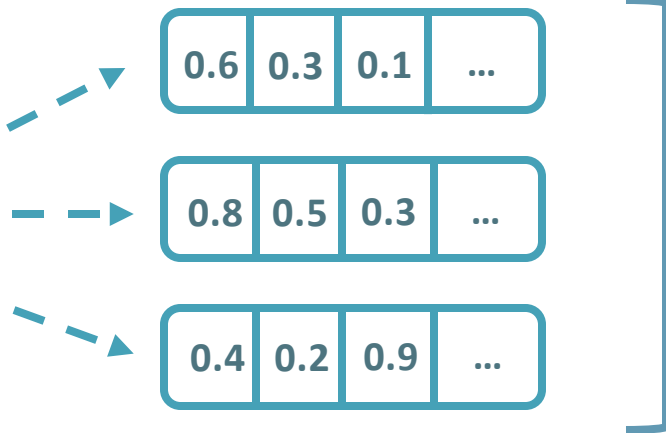
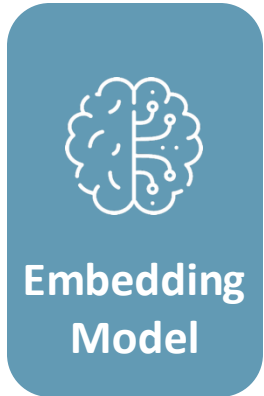
The question and answer are
SEMANTICALLY very similar.

Semantic proximity is represented
MATHEMATICALLY by the aid of vectors.

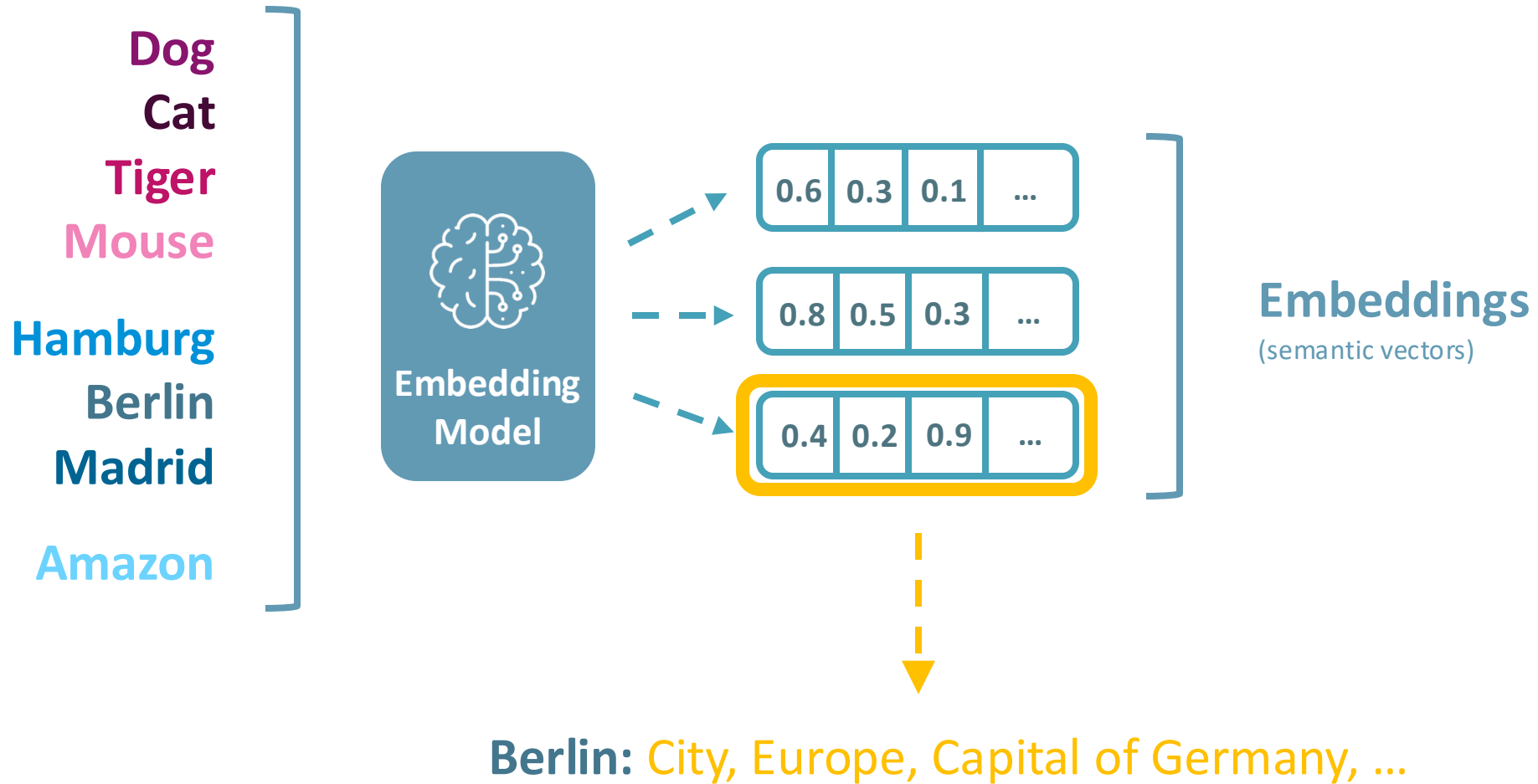
Dog
Cat
Tiger
Mouse

Hamburg
Berlin
Madrid

Amazon



Embeddings
(semantic vectors)



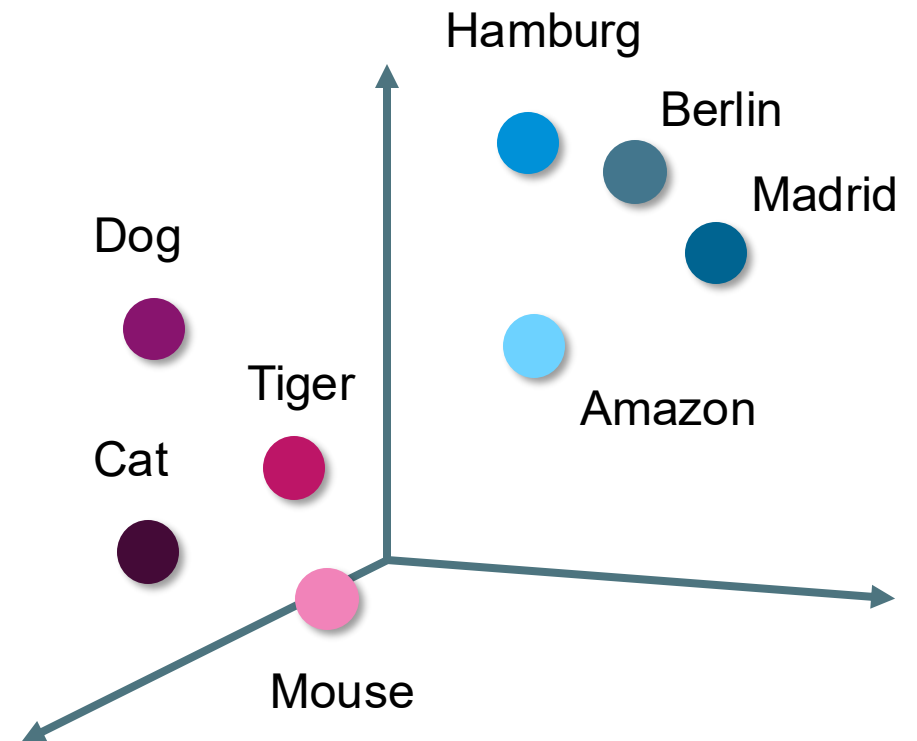
Dog
Cat
Tiger
Mouse

Hamburg
Berlin
Madrid

Amazon



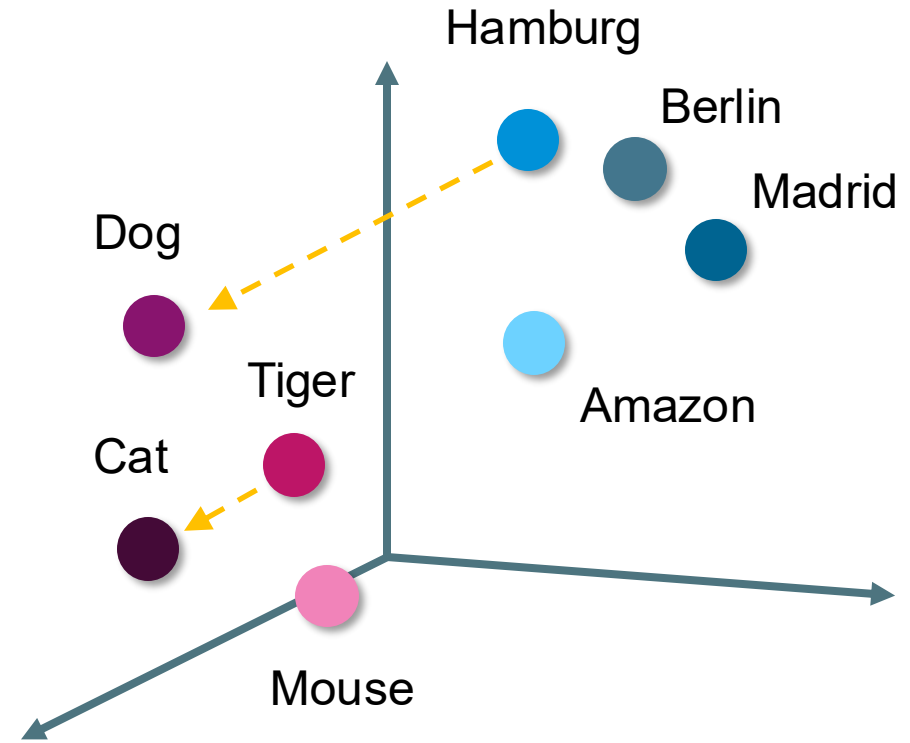
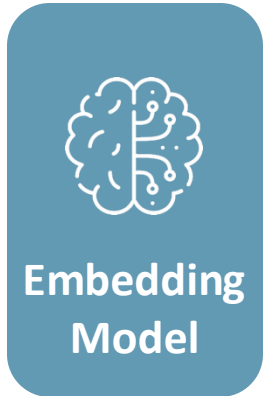
Berlin: City, Europe, Capital of Germany, ...



Dog
Cat
Tiger
Mouse

Hamburg
Berlin
Madrid

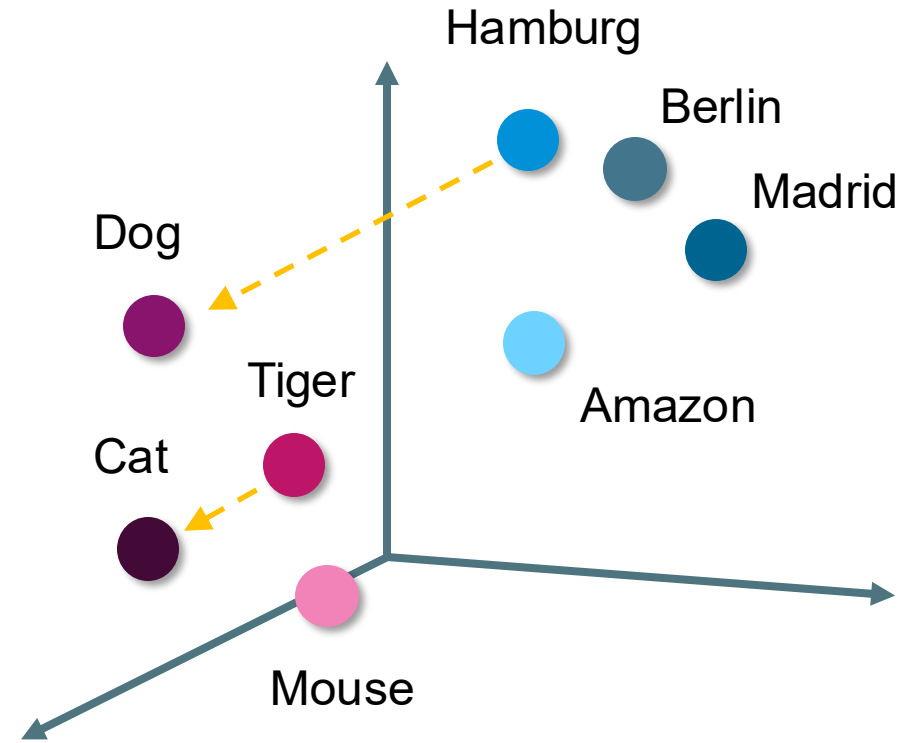
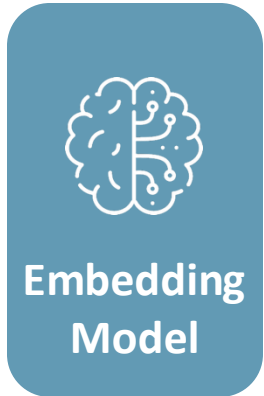
Amazon



Dog
Cat
Tiger
Mouse

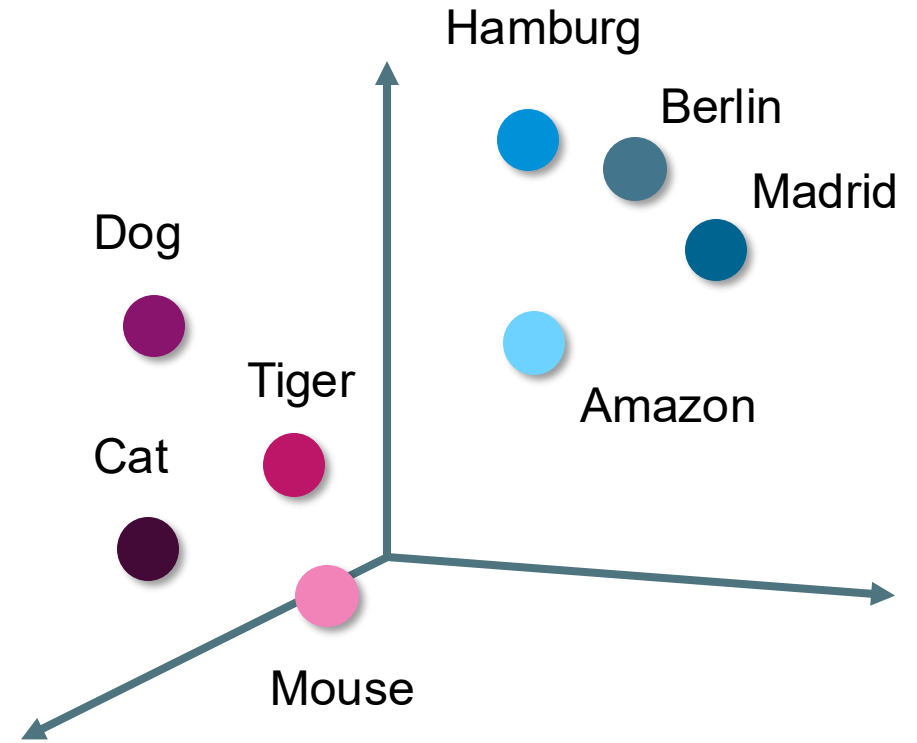
Hamburg
Berlin
Madrid

Amazon



QUESTION:

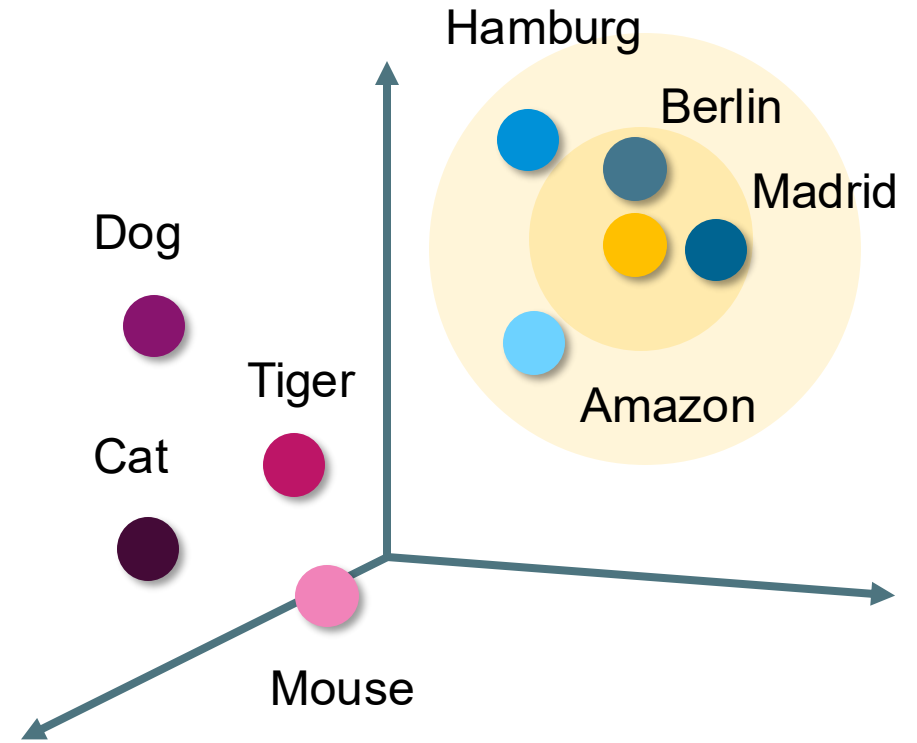
*„Can you
name a few
European
capitals?“*



„Can you name ..?“: City, Europe, Capital of ...

QUESTION:

„Can you
name a few
European
capitals?“

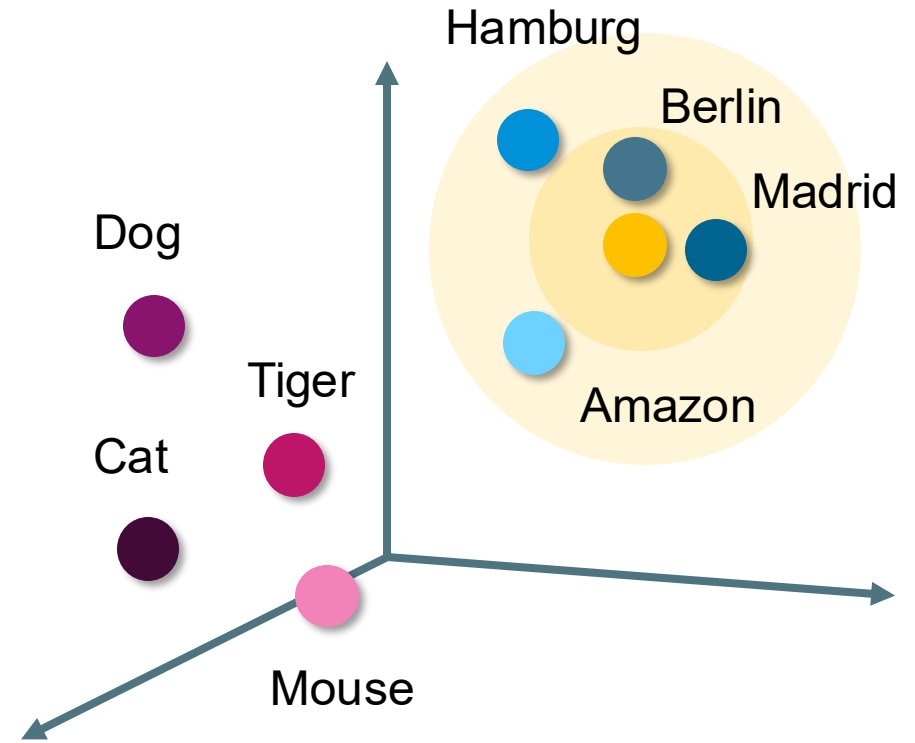


„Can you name ..?“. City, Europe, Capital of ...

Our goal?

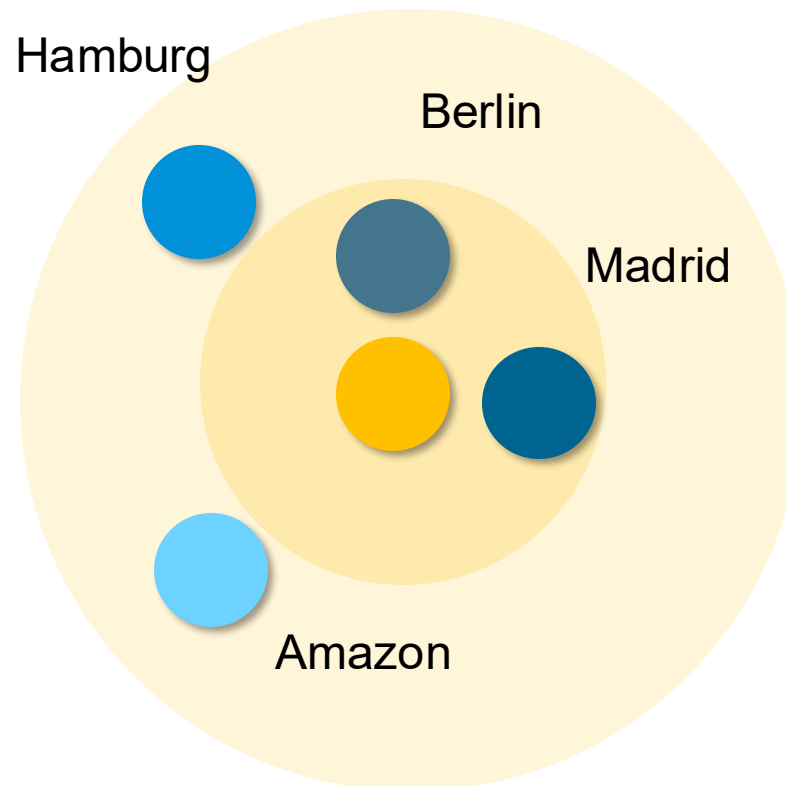
QUESTION:

„Can you name a few European capitals?“



„Can you name ..?“. City, Europe, Capital of ...

Our goal? Do everything possible to ensure that the ,question‘ and the ,answer‘ are as close together as possible!



Nicht sicher — vectors.nlp.eu

Semantically related words for "cat_NOUN"

WebVectorsSimilar wordsVisualizationsCalculator2D textMiscellaneousModelsAbout

What words are related to «cat» in model «English Wikipedia»?

Word frequency

☒ High☒ Medium☐ Low

1. dog_NOUN 0.6893

2. kitten_NOUN 0.6126

3. rabbit_NOUN 0.6028

4. raccoon_NOUN 0.5795

5. dogs_NOUN 0.5778

6. pet_NOUN 0.5672

7. puppy_NOUN 0.5452

8. pig_NOUN 0.5382

9. ferret_NOUN 0.5369

10. rat_NOUN 0.5348

0,6

Similarity threshold

Show tags

- We show only the associates of the same part of speech as your query. All associates can be found at the [Similar Words](#) tab.

This word in other models

British National Corpus

Google News

English Gigaword

Norsk Aviskorpus

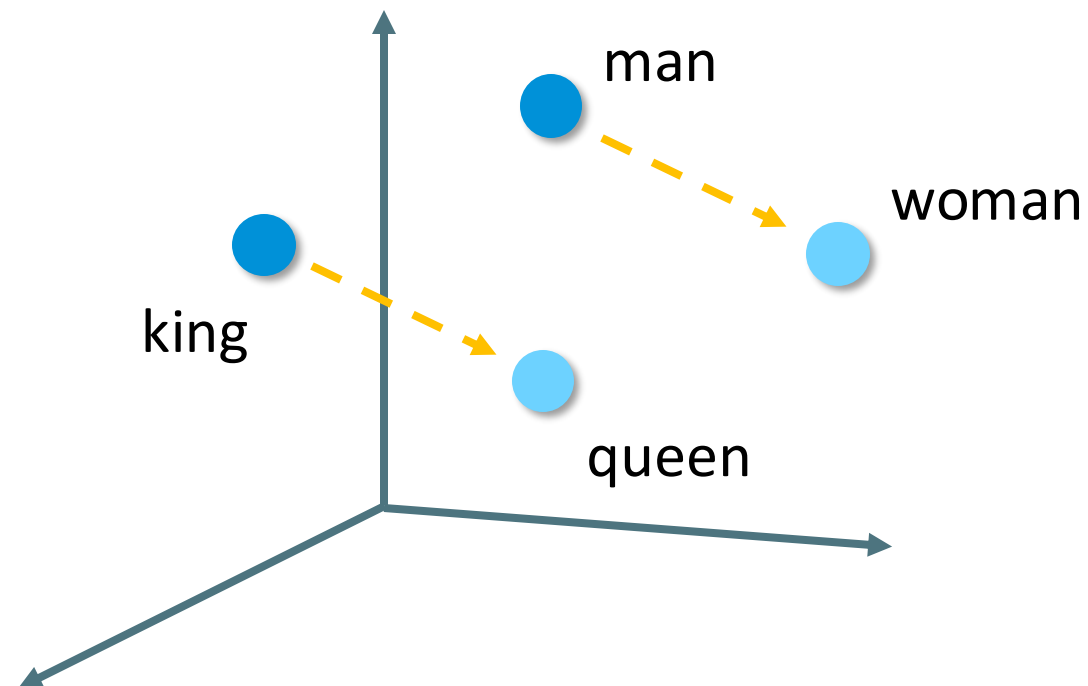
Show the raw vector of «cat» in model MOD_enwiki_upos_skipgram_300_2_2021:

[0.007398007903248072, 0.0029612560756504536, -0.01048285979775745, 0.0741681158542633, 0.07646718621253967, -0.0011427050922065973, 0.026497453451156616, 0.01059535983498997, 0.0190864410251379, 0.003835588760674, -0.0468108132481575, -0.021150866523385048, 0.009098375216126442, 0.0030140099115669727, -0.05626726150512695, -0.0396095551550368834, -0.09978967905044556, -0.07956799119710922, 0.057768501341342926, -0.01737510226669273, 0.01559068357093143, -0.022376490756869316, 0.01522265429496765, -0.05138462409377098, 0.025846413394737244, 0.07069036364555359, 0.0009145145886577666, -0.06273567736816406, 0.0361750287771225, 0.05080768835248184, -0.0645394742679596, -0.04349868372062771, -0.1264101266860692, -0.0003191891883034259, 0.04311852902173996, -0.14792846143245897, -0.0148407198591165543, 0.01902032678935196, 0.011479354463517966, 0.02979433164000511, 0.06154156103730202, -0.04609882831573496, -0.053286727517843242, -0.016268745064735413, 0.03606176321864128, 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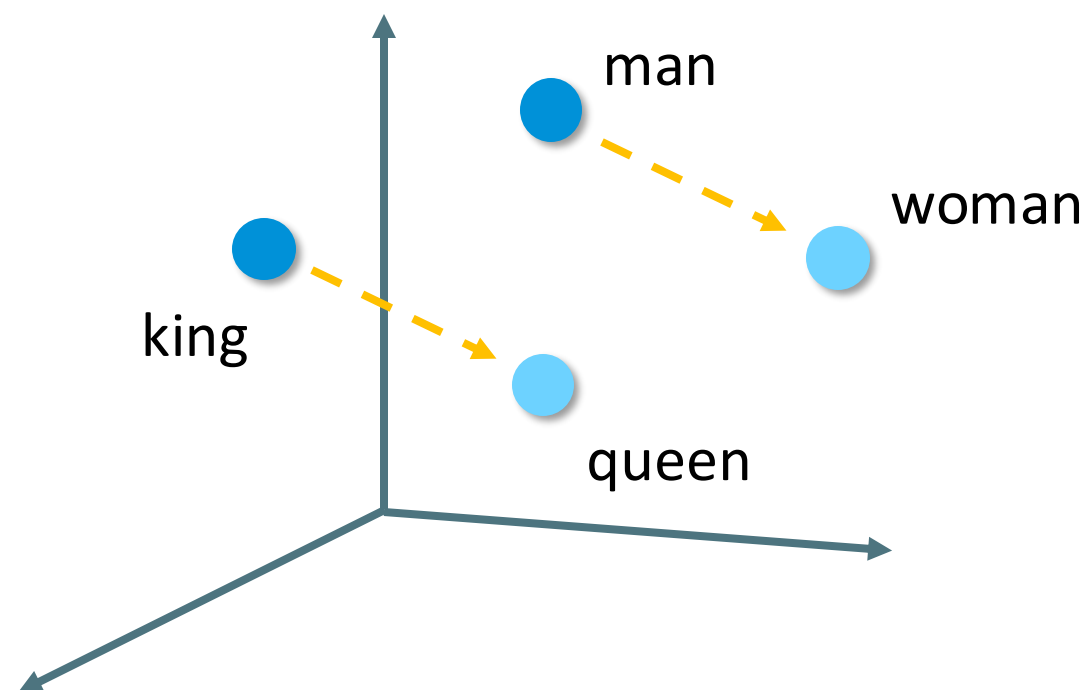
OFFENKUNDIGGUT

OPEN
KNOW
LEDGE

Okay, but what's so great about that?

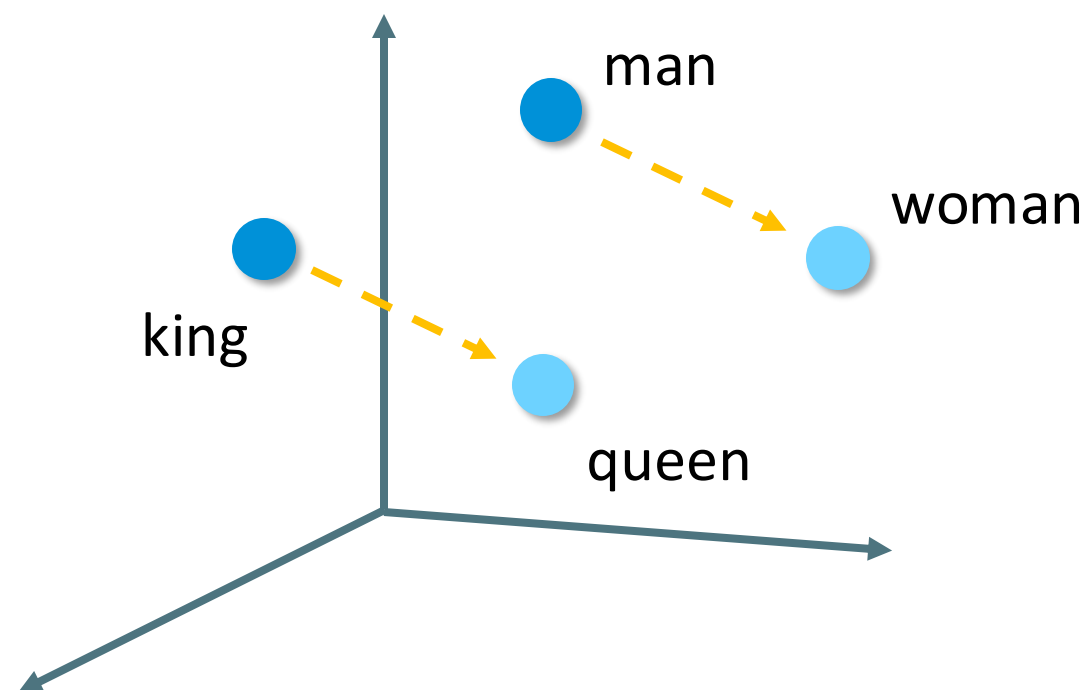


Okay, but what's so great about that?
We can calculate with it and ,understand'.



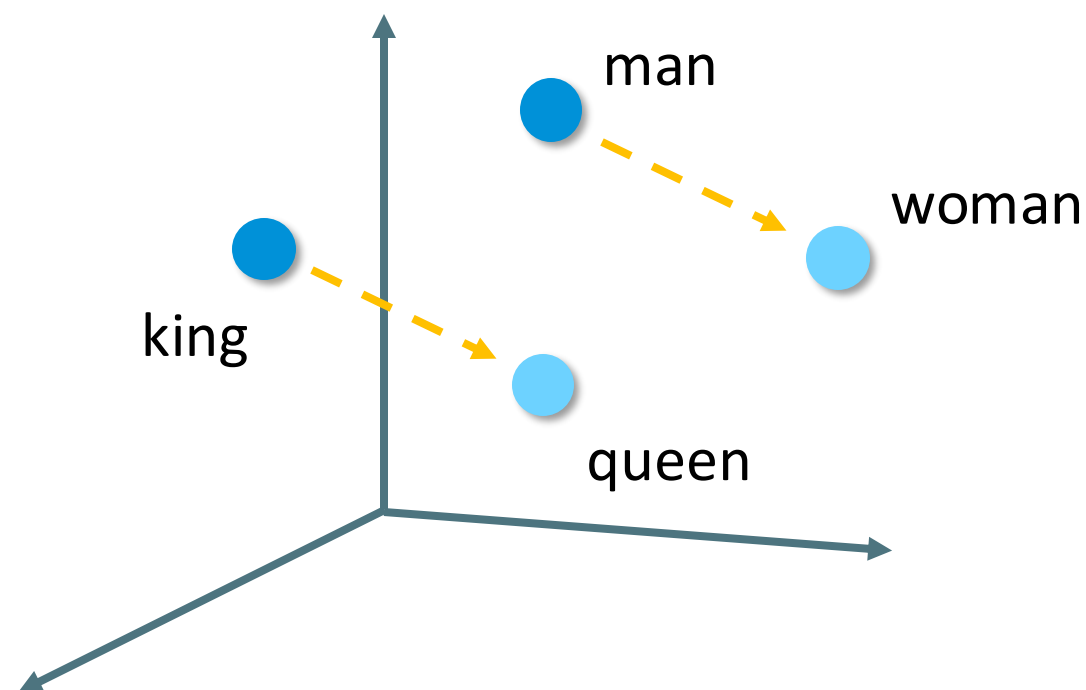
$$\text{queen} - \text{woman} + \text{man} = \text{king}$$

Okay, but what's so great about that?
We can calculate with it and ,understand'.



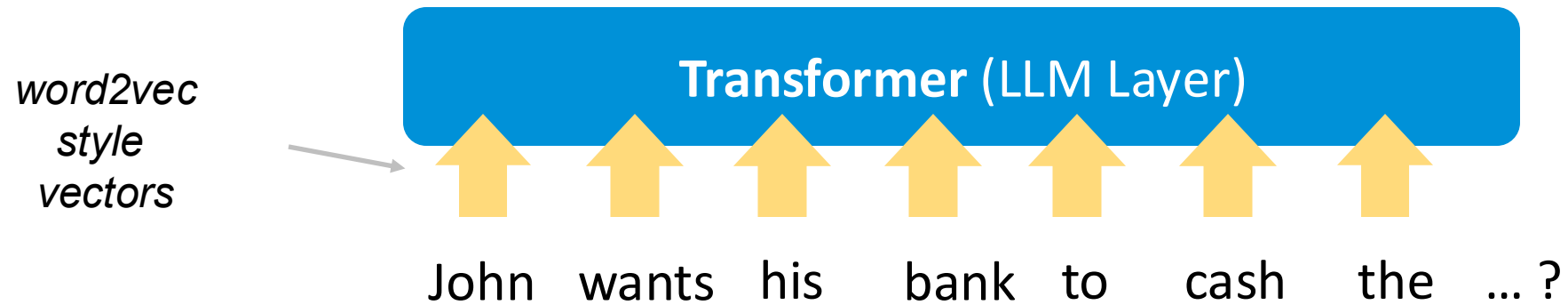
doctor – man + woman = ?

Okay, but what's so great about that?
We can calculate with it and ,understand'.



doctor – man + woman = **nurse**

Generative AI under the hood



Generative AI under the hood

context
aka hidden
state

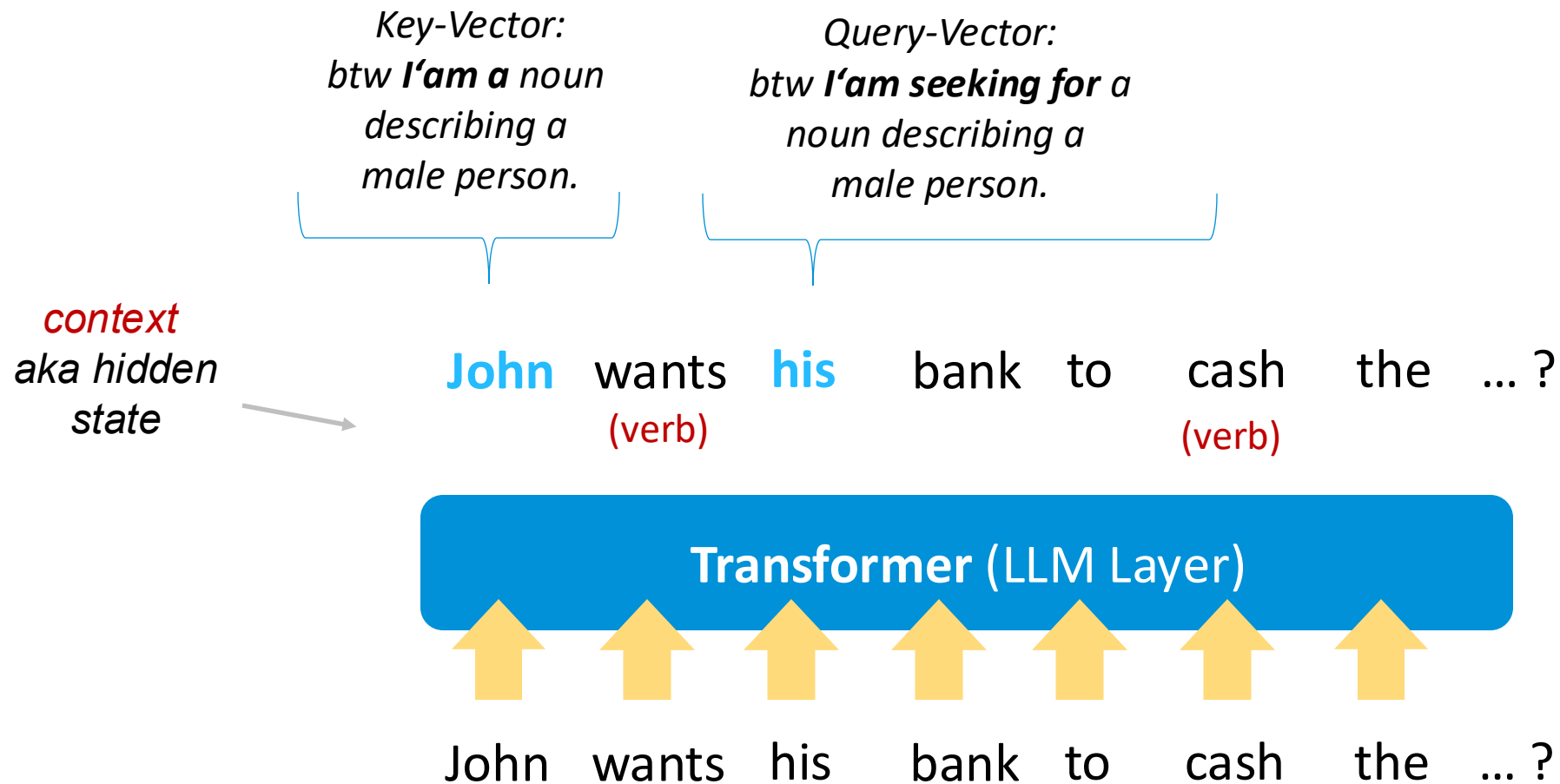


John wants his bank to cash the ... ?
(verb) (verb)



John wants his bank to cash the ... ?

Generative AI under the hood



Generative AI under the hood

context
aka hidden
state

John wants his bank to cash the ... ?
(male) (verb) (John's) (finance) (verb) (...)

Transformer (LLM Layer)

John wants his bank to cash the ... ?
(verb) (verb)

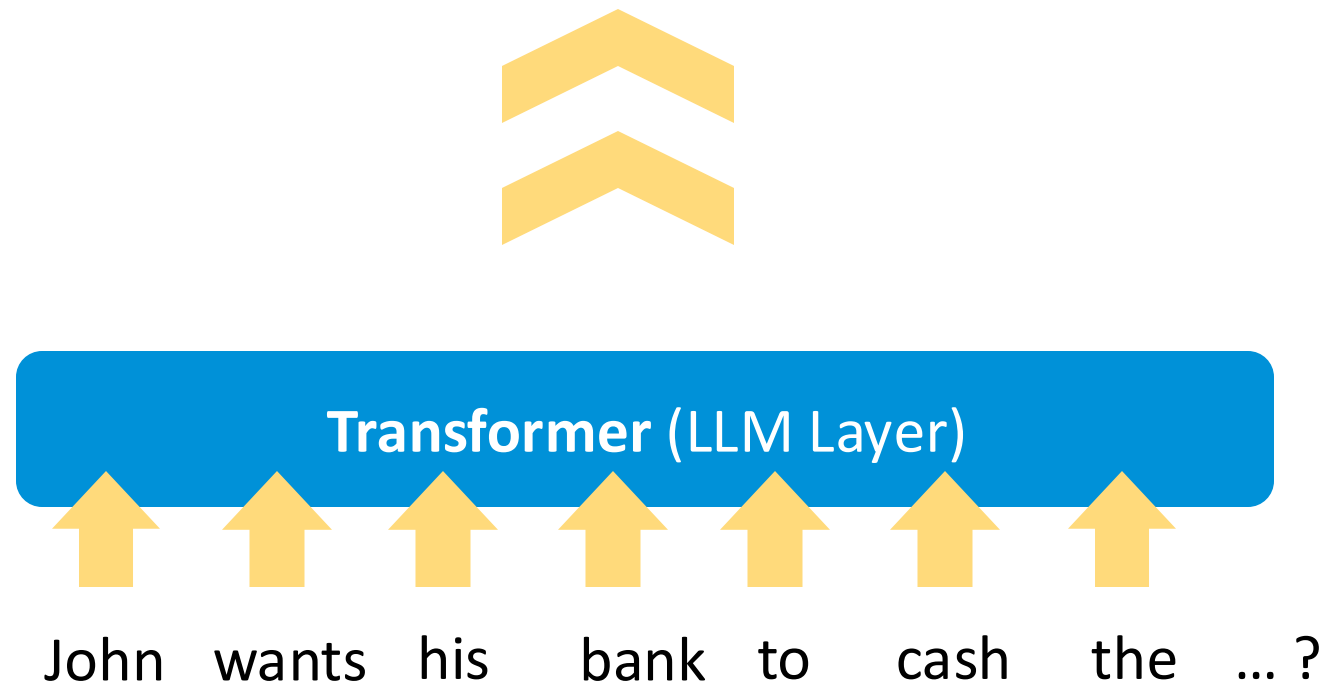
Transformer (LLM Layer)

John wants his bank to cash the ... ?

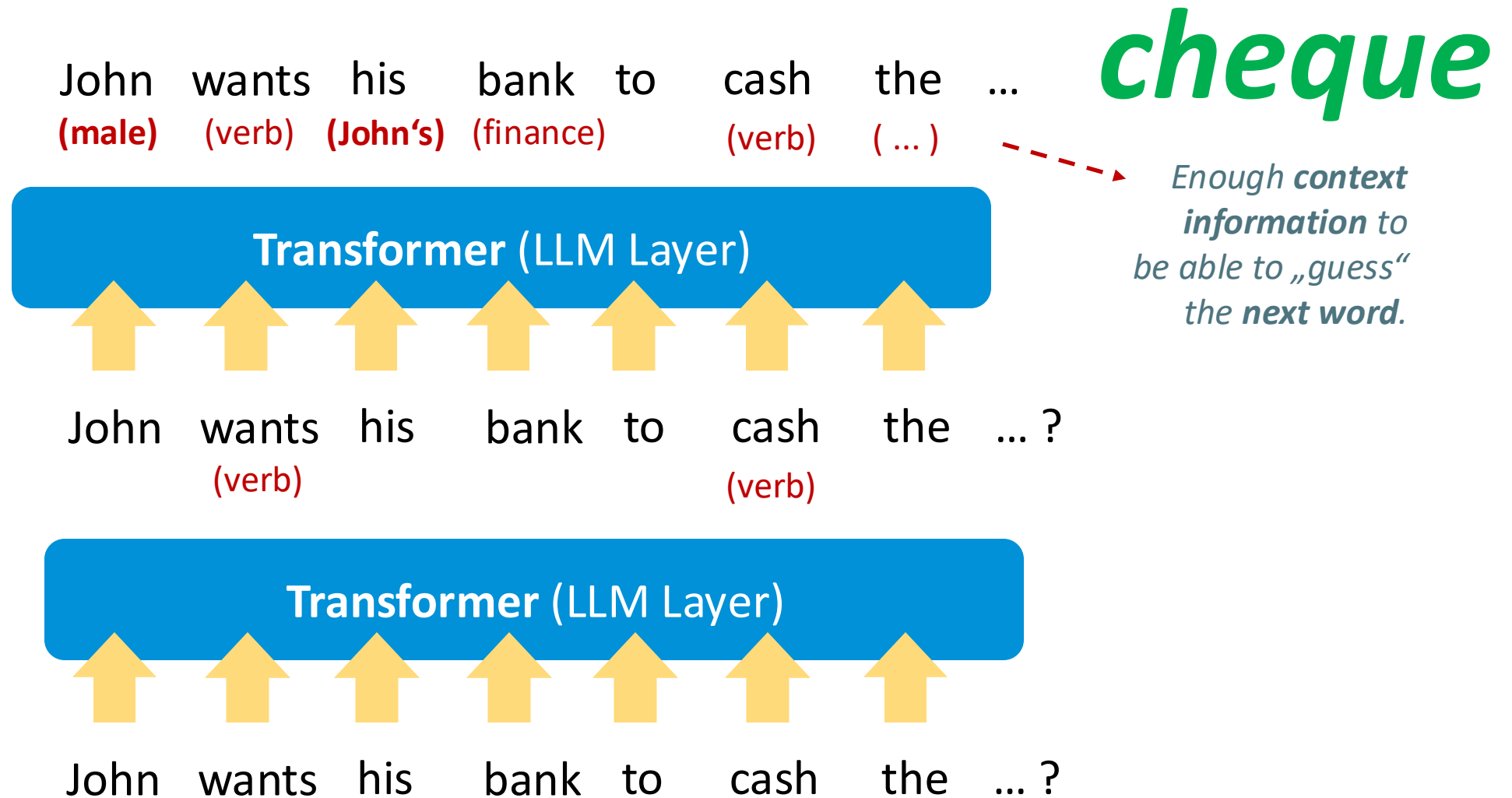
Generative AI under the hood

John wants his bank to cash the ... ?

(main character, male, married to Cheryl, cousin of Donald, from Minnesota, currently in Boise, ...)



Generative AI under the hood



Generative AI under the Hood

John wants his
bank **to** ...

Large
Language
Model



Next Token	Probability
increase	0.2
cash	0.4
close	0.1
...	...

John wants his
bank **to cash the** ...

Large
Language
Model



Next Token	Probability
account	0.15
cheque	0.75
money order	0.05
...	...

01

GenAI by Example?

*Where can I
realistically use GenAI
- and where not?*

Generative AI by Example



Generative AI by Example

**Software
Development**



**Health Care &
Pharmaceuticals**



**Advertisement
& Marketing**



**Production &
Manufacturing**



**Financial
Services**



**Media &
Entertainment**



GenAI by Example



***„Know Your
Business!“***

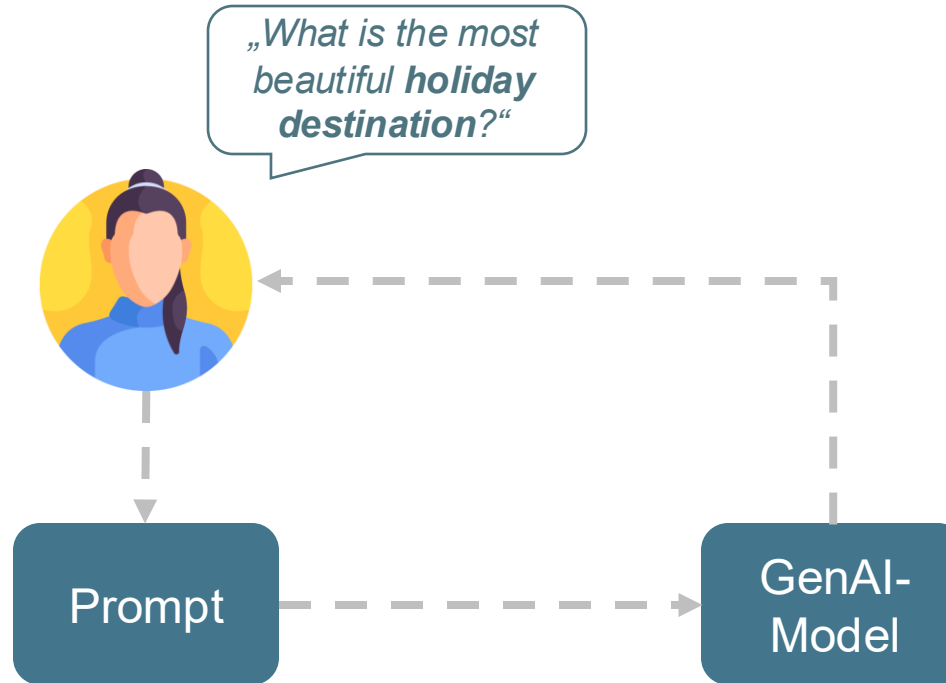
**if you want to succeed in the end.*

02

GenAI Basics

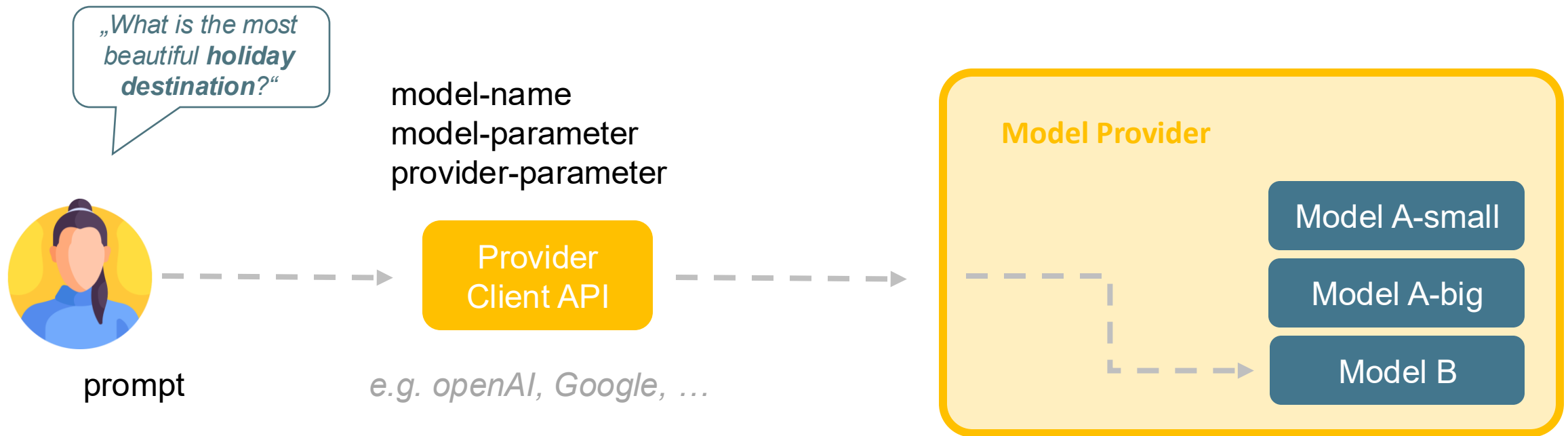
*How to build a
first, simple
GenAI system?*

GenAI Basics

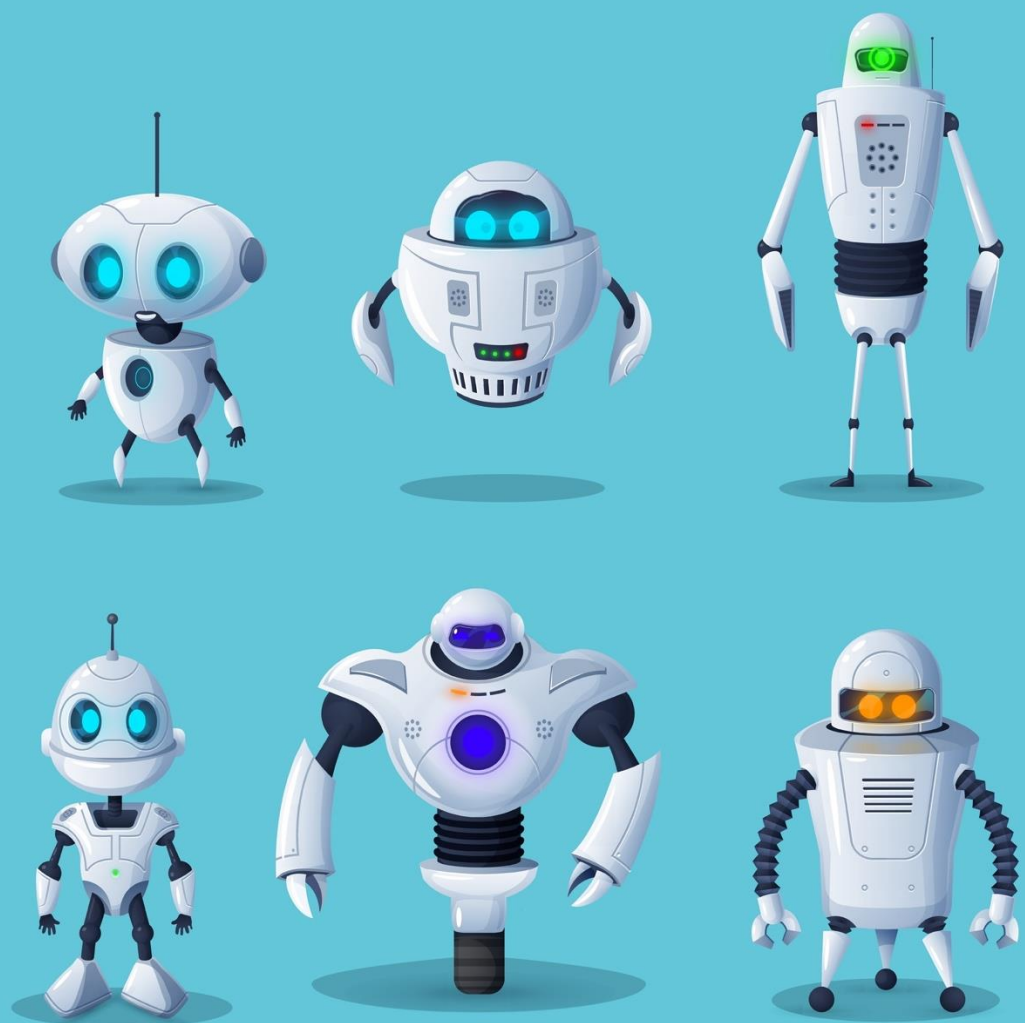


GenAI Basics

Model Integration



Reminder:
00-hello-genai.ipynb



Hands-On

Hello IPYNB*, hello GenAI

Create a Jupyter Notebook in the
Google Colab environment.

Calling a GenAI model with a prompt
via a provider-specific API.

*Python Notebook aka Jupyter Notebook



<https://colab.research.google.com/>

Hands-on Hello Colab

Welcome To Colab

File Edit View Insert Runtime Tools Help

Q Commands + Code + Text ▶ Run all Copy to Drive

Connect ^

Table of contents

- Welcome to Colab!
- Access popular AI models via...
- Explore the Gemini API
- Getting started
- Data science
- Machine learning
- More Resources
- Featured examples
- + Section

Welcome to Colab!

Access popular AI models via Google-Colab-AI Without an API Key

All users have access to most popular LLMs via the `google-colab-ai` Python library, and paid users have access to a wider selection of models. For more details, refer to the [getting started with google colab ai](#).

```
[ ]  
from google.colab import ai  
response = ai.generate_text("What is the capital of France?")  
print(response)
```

Explore the Gemini API

The Gemini API gives you access to Gemini models created by Google DeepMind. Gemini models are built from the ground up to be multimodal, so you can reason seamlessly across text, images, code, and audio.

How to get started?

- Go to [Google AI Studio](#) and log in with your Google account.
- [Create an API key](#).
- Use a quickstart for [Python](#), or call the REST API using [curl](#).

Discover Gemini's advanced capabilities

Variables Terminal



<https://github.com/openknowledge/workshop-genai-student>

Hands-on Hello GenAI World Jupyter Notebook

The screenshot shows the GitHub interface for the repository 'workshop-genai-student' by the user 'openknowledge'. The repository is public and has 2 stars and 5 watchers. The main branch is 'main', and there are 3 branches and 0 tags. The repository description is 'Students material for a full day hands-on workshop on GenAI'. The file list includes 'genai-ws', '.gitignore', 'LICENSE', and 'README.md'. The 'README.md' file is selected, showing the title 'workshop-genai-student' and the content: 'Students material for the open knowledge full day hands-on workshop on GenAI and RAG. What is the workshop about? Thanks to powerful frameworks and libraries, the first generative AI applications at the Hello World level can be implemented with just a few lines of code. However, these first attempts quickly reach their limits. Why? Because reality presents challenges that cannot be easily solved with this trivial approach. But what is needed for a generative AI application at the enterprise level? A well-thought-out generative AI'.

workshop-genai-student Public

main 3 Branches 0 Tags

Go to file Add file Code

TimWue Update exercises and solutions 4be3783 · 6 minutes ago 52 Commits

File	Commit Message	Time
genai-ws	Update exercises and solutions	6 minutes ago
.gitignore	updated to ignore .env and poetry related files	last year
LICENSE	Initial commit	last year
README.md	Reference for english slides added	11 months ago

README Apache-2.0 license

workshop-genai-student

Students material for the open knowledge full day hands-on workshop on GenAI and RAG.

What is the workshop about?

Thanks to powerful frameworks and libraries, the first generative AI applications at the Hello World level can be implemented with just a few lines of code. However, these first attempts quickly reach their limits. Why? Because reality presents challenges that cannot be easily solved with this trivial approach.

But what is needed for a generative AI application at the enterprise level? A well-thought-out generative AI

About

Students material for a full day hands-on workshop on GenAI

- Readme
- Apache-2.0 license
- Activity
- Custom properties
- 2 stars
- 5 watching
- 0 forks

Report repository

Releases

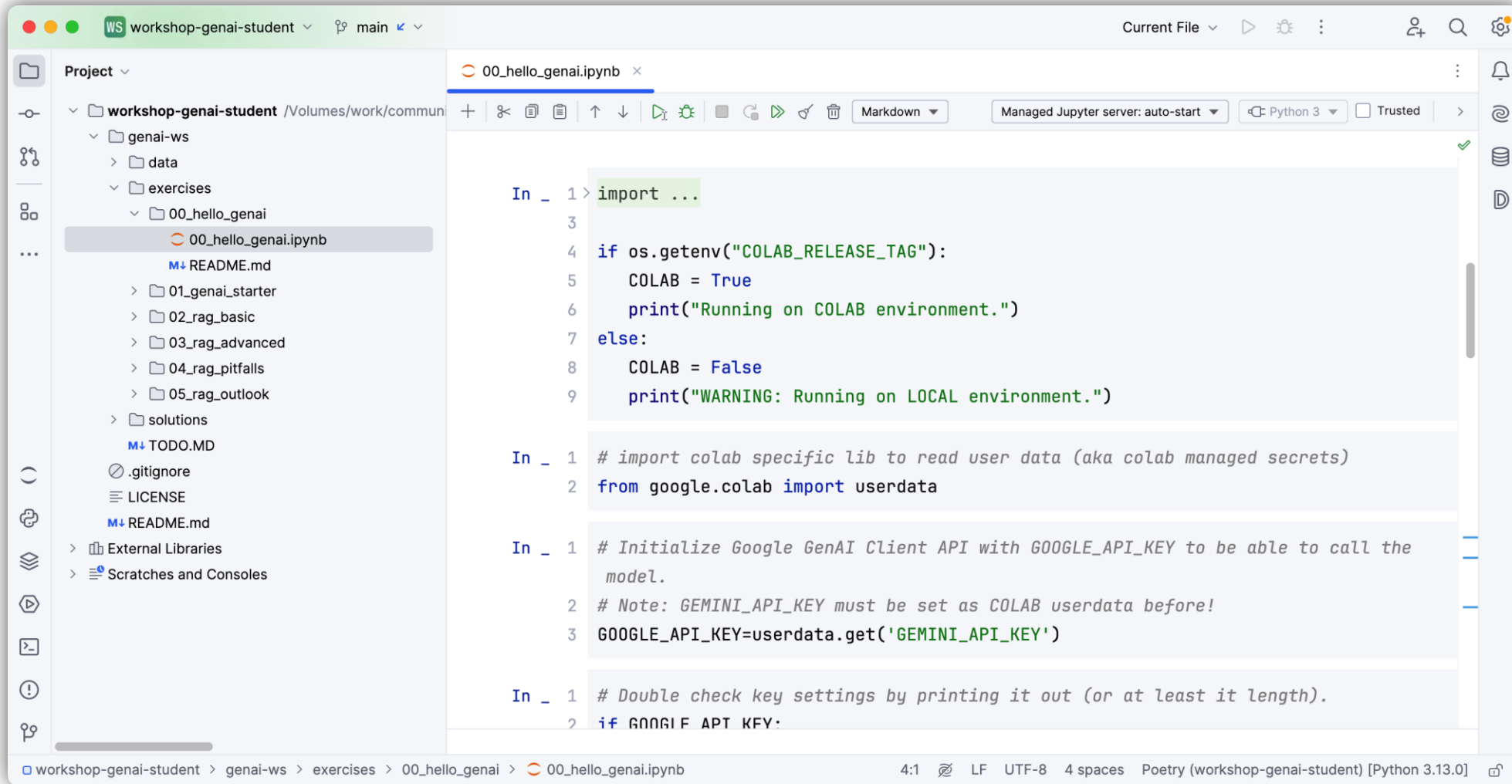
No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Contributors 2

Hands-on Hello GenAI World Jupyter Notebook



The screenshot displays a Jupyter Notebook environment. On the left, a file explorer shows the project structure: `workshop-genai-student` / `Volumes/work/commun`. The `00_hello_genai.ipynb` file is selected. The main area shows the notebook content with four code cells:

```
In _ 1 > import ...
3
4 if os.getenv("COLAB_RELEASE_TAG"):
5     COLAB = True
6     print("Running on COLAB environment.")
7 else:
8     COLAB = False
9     print("WARNING: Running on LOCAL environment.")

In _ 1 # import colab specific lib to read user data (aka colab managed secrets)
2 from google.colab import userdata

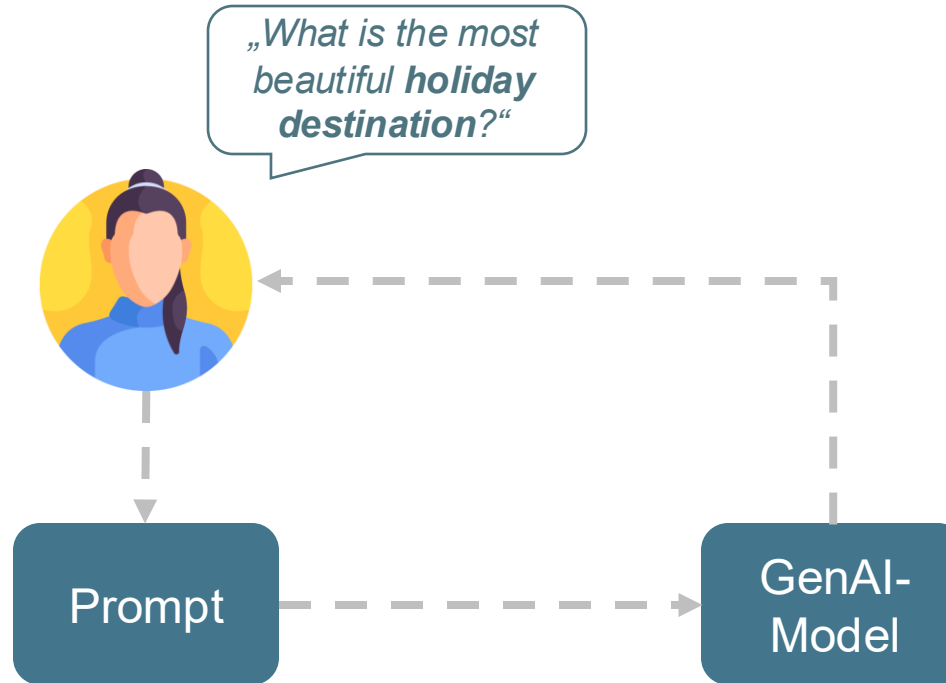
In _ 1 # Initialize Google GenAI Client API with GOOGLE_API_KEY to be able to call the
2 # Note: GEMINI_API_KEY must be set as COLAB userdata before!
3 GOOGLE_API_KEY=userdata.get('GEMINI_API_KEY')

In _ 1 # Double check key settings by printing it out (or at least it length).
2 if GOOGLE_API_KEY:
```

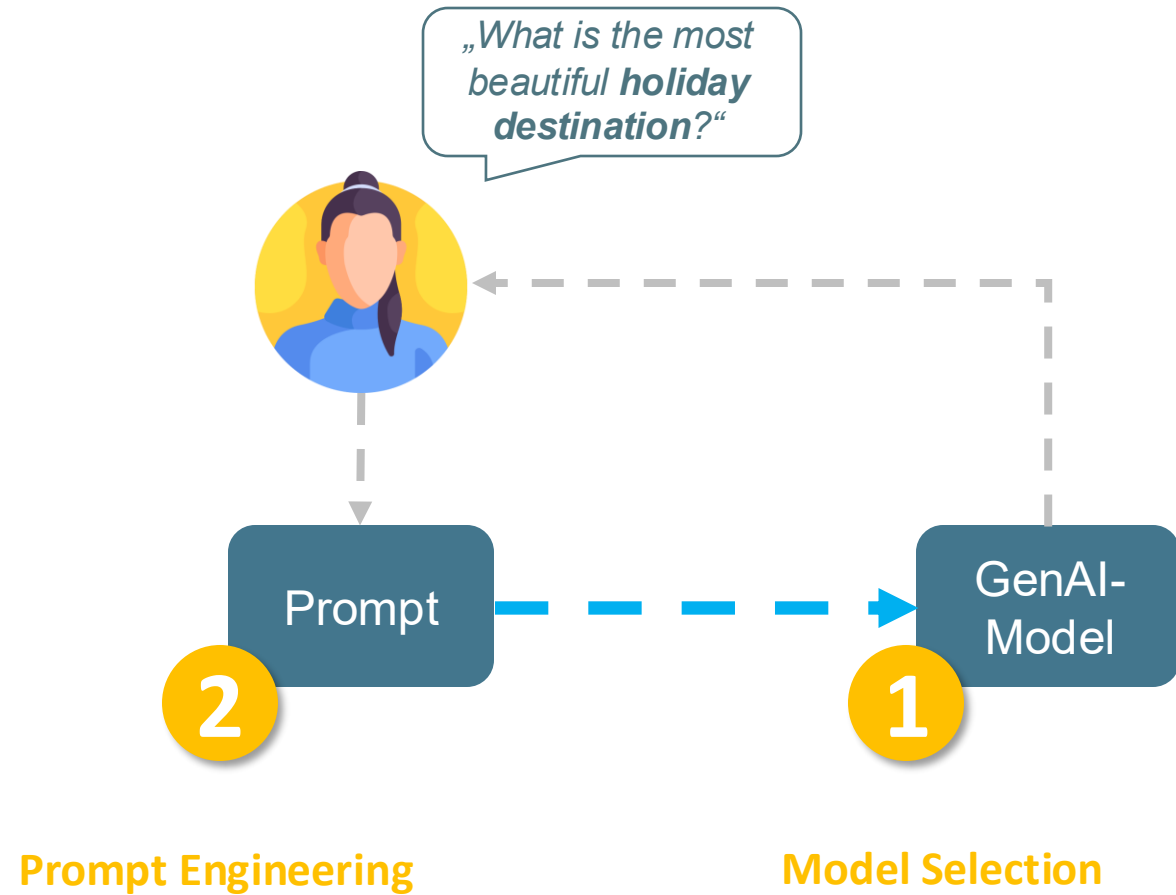
The bottom status bar indicates the file path: `workshop-genai-student > genai-ws > exercises > 00_hello_genai > 00_hello_genai.ipynb`. It also shows the current cell is at line 4:1, using LF line endings, UTF-8 encoding, 4 spaces for indentation, and the Poetry (workshop-genai-student) [Python 3.13.0] kernel.

<https://github.com/openknowledge/workshop-genai-student>

GenAI Basics



GenAI Basics



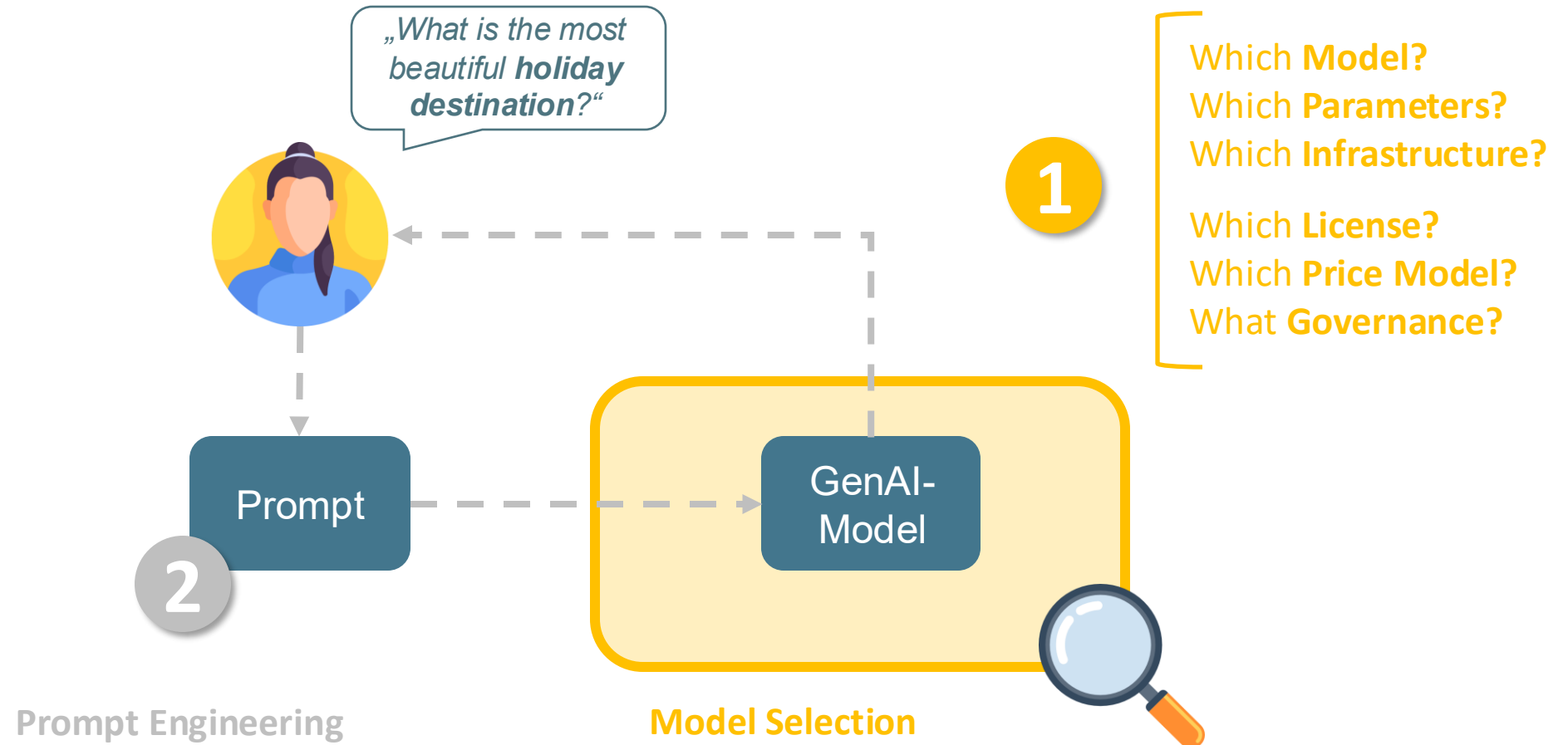
MODELS



GenAI
explained

GenAI Basics

Model Selection



GenAI Basics

Model Selection

„In the world of GenAI, model selection is part of the product design. It affects user experience, cost, scalability, and reliability.“

GenAI Basics

Model Selection

aka „How do I find the right model?“*

Step 1

Step 2

Step 3

Step 4

GenAI Basics

Model Selection

aka „How do I find the right model?“*

1. Use Case

Define specific
GenAI use cases

Step 2

Step 3

Step 4

Clearly outline
the purpose and
application of GenAI.

GenAI Basics

Model Selection

aka „How do I find the right model?“*

1. Use Case

Define specific
GenAI use cases

Clearly outline
the purpose and
application of GenAI.

2. Shortlist

Create a shortlist
of foundation
models

Models that meet
the business
and technical
requirements.

Step 3

Step 4

GenAI Basics

Model Selection

aka „How do I find the right model?“*

1. Use Case

Define specific
GenAI use cases

Clearly outline
the purpose and
application of GenAI.

2. Shortlist

Create a shortlist
of foundation
models

Models that meet
the business
and technical
requirements.

3. Evaluation

Evaluate and test
selected models

Conduct thorough
testing to evaluate
model performance in
your own context.

Step 4

GenAI Basics

Model Selection

aka „How do I find the right model?“*

1. Use Case

Define specific
GenAI use cases

Clearly outline
the purpose and
application of GenAI.

2. Shortlist

Create a shortlist
of foundation
models

Models that meet
the business
and technical
requirements.

3. Evaluation

Evaluate and test
selected models

Conduct thorough
testing to evaluate
model performance in
your own context.

4. Integration

Integrate the best
models and
evaluate them on
an ongoing basis

Integrate, regularly
evaluate and, if necessary,
optimise the
performance of the
model..

GenAI Basics Model Selection

aka „How do I find the right model?“*

***„Know Your
Use-Case!“***

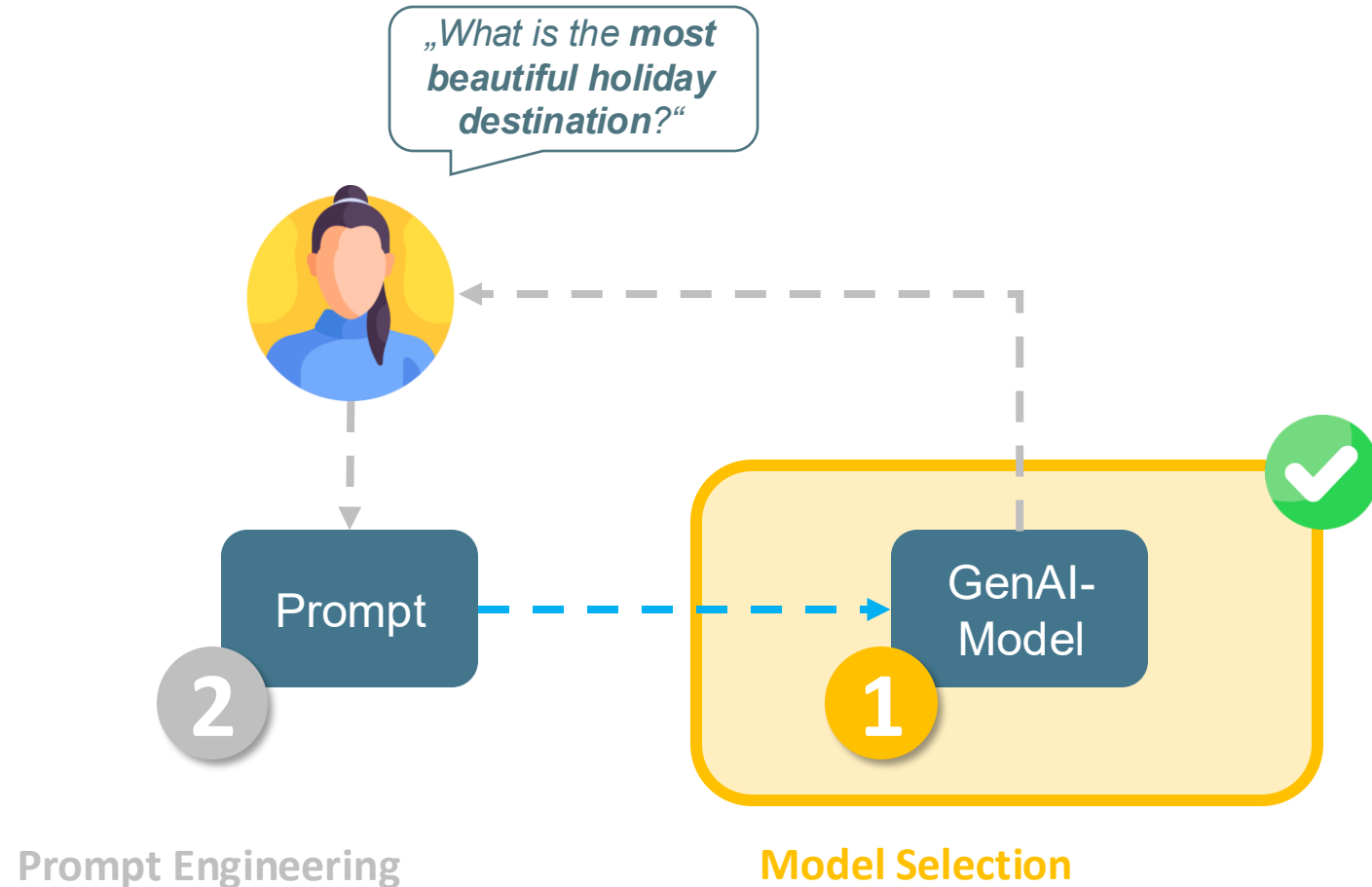
* if you want to succeed in the end.

GenAI Basics

Model Integration

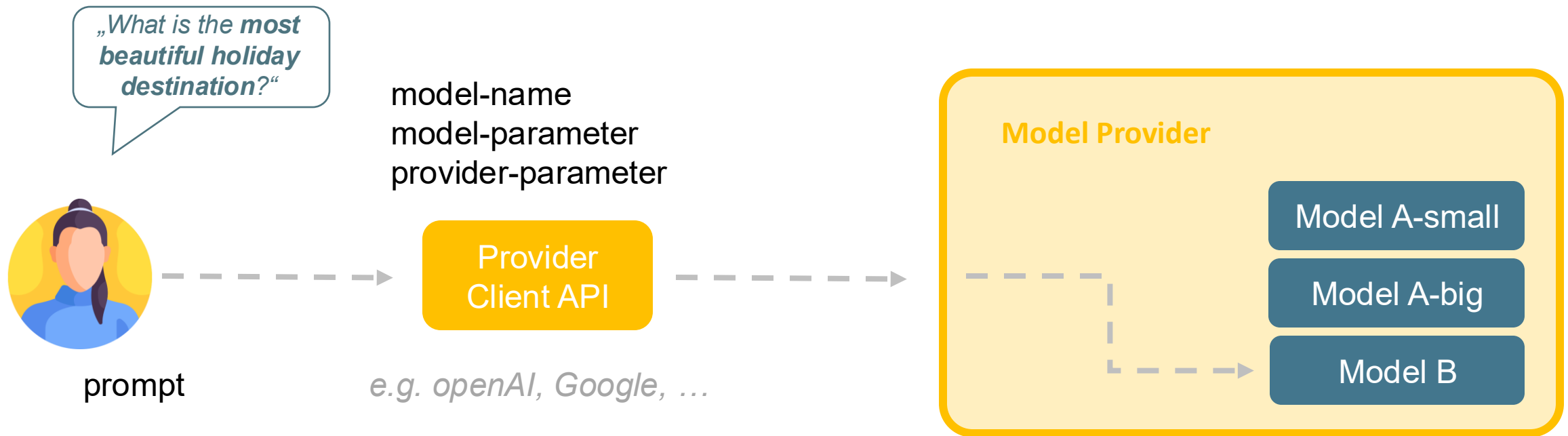
Suitable model found:

But how do I address
it from my application?



GenAI Basics

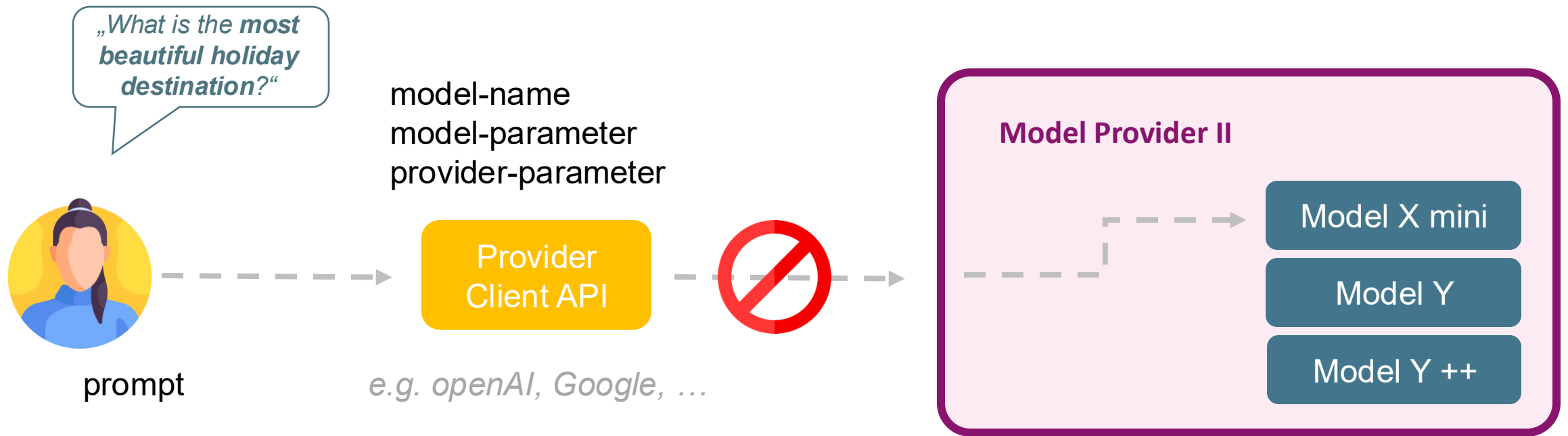
Model Integration



Reminder:
00-hello-genai.ipynb

GenAI Basics

Model Integration

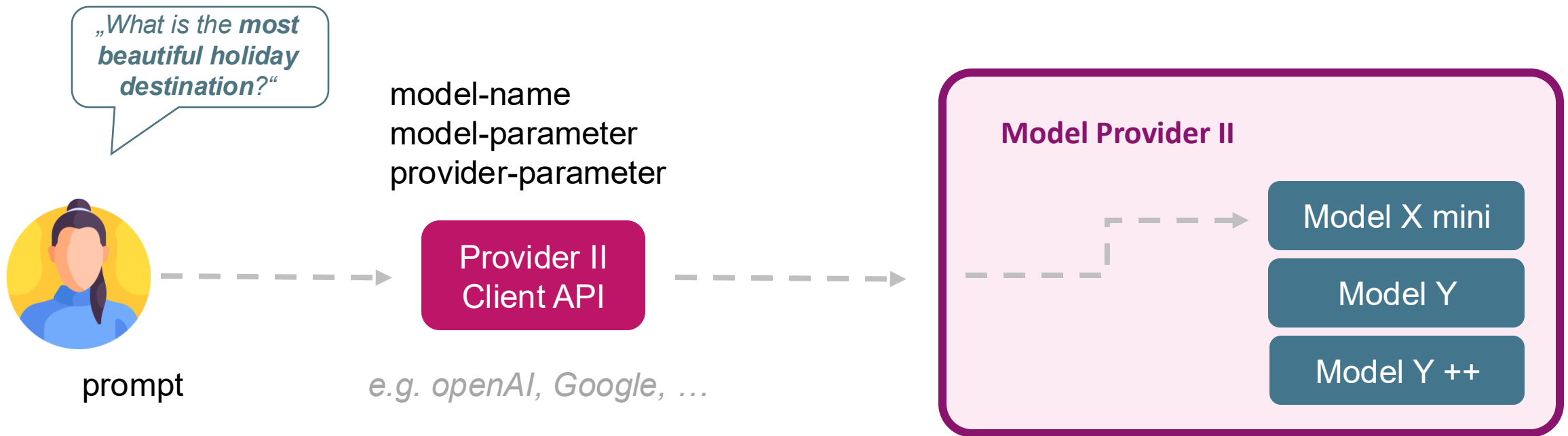


GenAI Basics

Model Integration

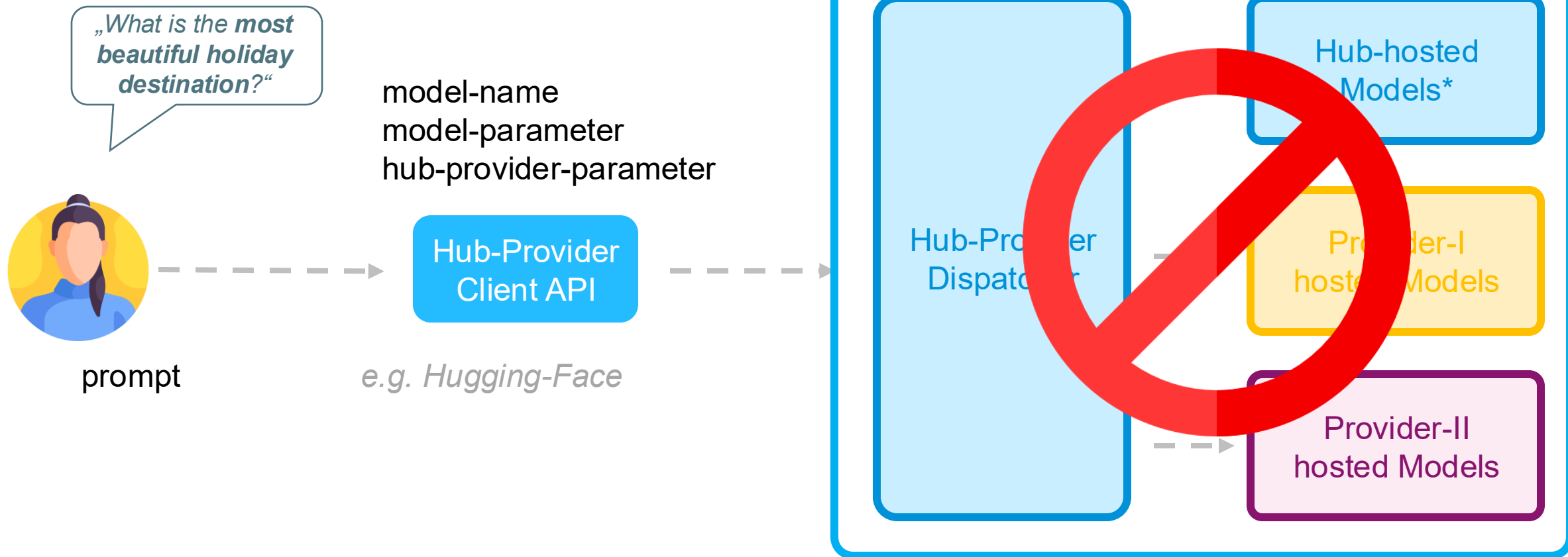


The OpenAI API is currently the de facto standard.



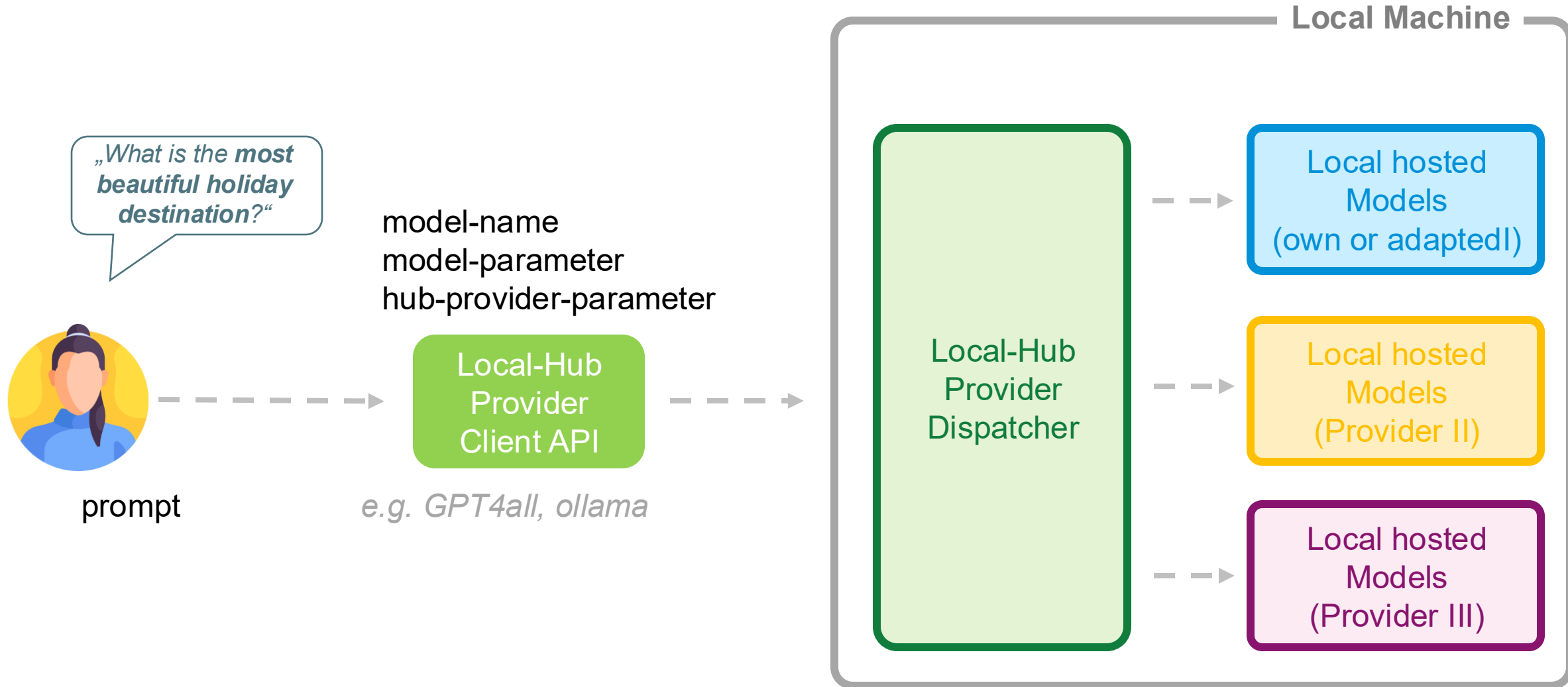
GenAI Basics

Model Integration



GenAI Basics

Model Integration



GenAI Basics Model Adaption

„ What additional levers do I have?“

- **Temperature** degree of ‘fantasy’
- **Max Tokens** length of answer
- **Top K** selection of hits from the top K hits
- **Top P** selection of hits from the top P percent
- **Presence Penalty** avoid repetition
- **Frequence Penalty** avoid repetition (weighted edition)

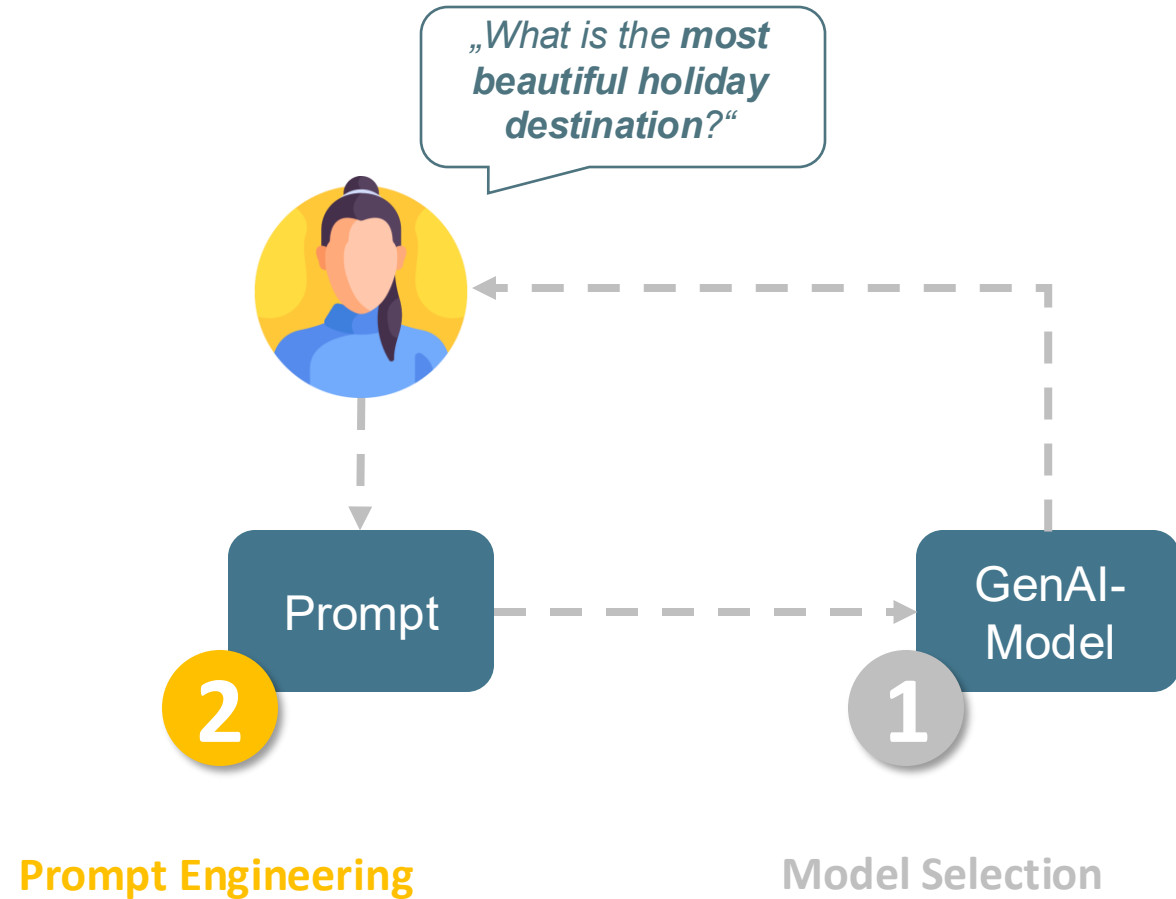
PROMPTING



GenAI
explained

GenAI Basics

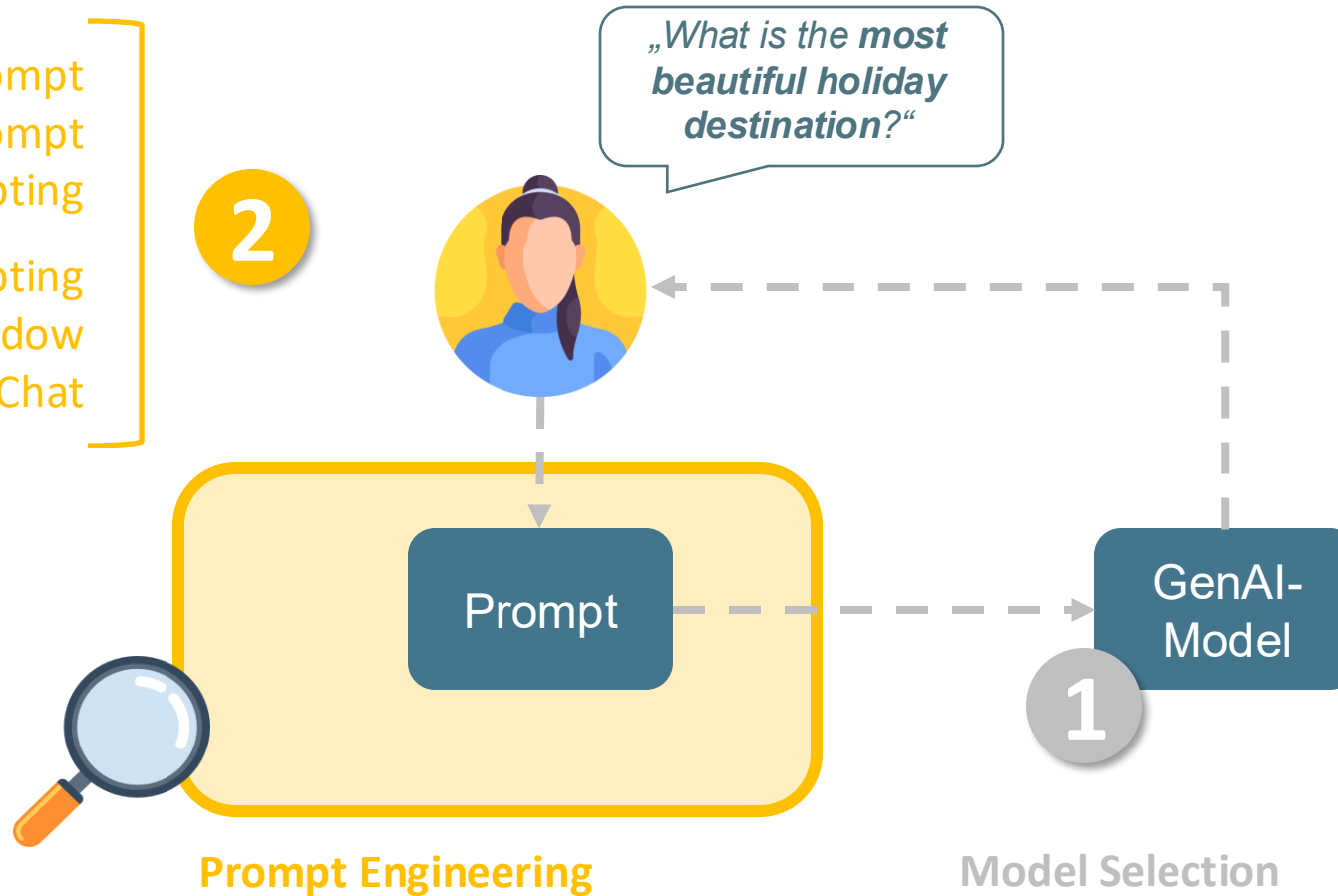
Prompt Engineering



GenAI Basics

Prompt Engineering

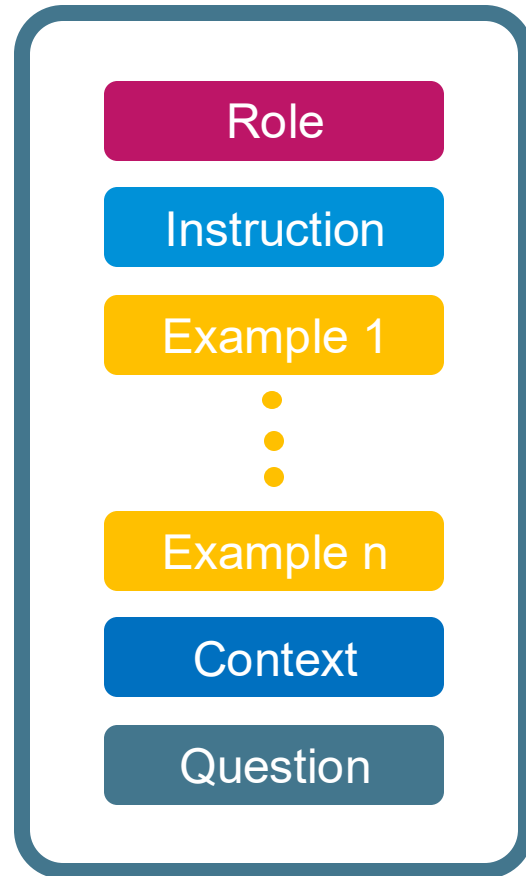
Parts of a Prompt
System vs User Prompt
Principles of Prompting
Patterns for Prompting
Context Window
Query vs Chat



GenAI Basics

Prompt Engineering

What should
a **good prompt**
should look like?



Who am I?

What is my intention?

What are helpful examples?

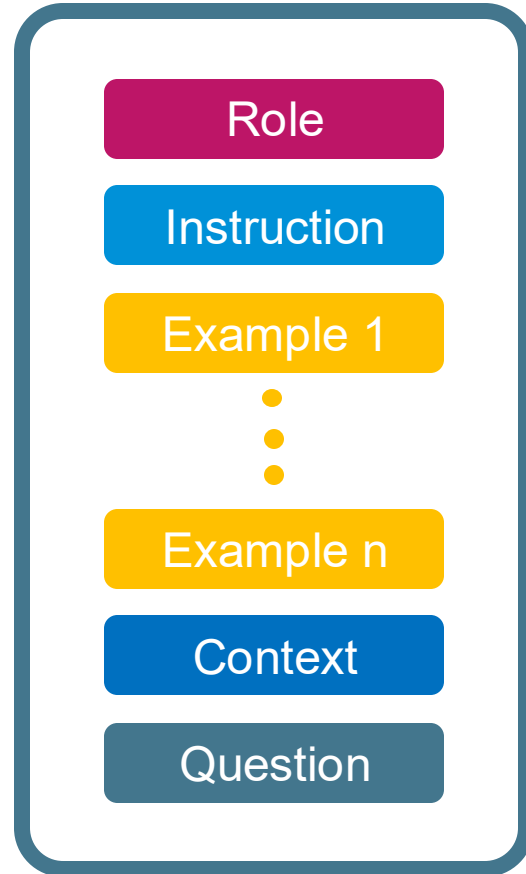
Are there any additional information?

BTW: what is the task I ask for?

GenAI Basics

Prompt Engineering

What should
a **good prompt**
should look like?



What is your role as an assistant?

What is my intention?

What are helpful examples?

Are there any additional information?

BTW: what is the task I ask for?

GenAI Basics

Good Prompt vs. Bad Prompt

LR You

I want to cook something.

Topic: Recipe Recommendations

GenAI Basics

Good Prompt vs. Bad Prompt

- You**
I want to cook something.
- You**
Acting as an expert home cook, for someone who enjoys vegetarian Italian food and has only 30 minutes to prepare dinner, could you recommend a recipe including a list of ingredients and step-by-step instructions?

Topic: Recipe Recommendations



Role

Acting as an expert home cook,

Context

for someone who enjoys vegetarian Italian food and has only 30 minutes to prepare dinner,

Question

could you recommend a recipe

Output

including a list of ingredients and step-by-step instructions?

Example

You could point at recipes you like from the BBC's Good Food guide, providing URL's to recipes you love.

GenAI Basics

Good Prompt vs. Bad Prompt

- You**
I want to learn something new.
- You**
Acting as a coding instructor, for a beginner with a goal to learn Python within 4 weeks, please provide a learning plan including resources and a weekly schedule for 10 hours per week.

Topic: Learning new skills



Role

Acting as a coding instructor,

Context

for a beginner with a goal to learn Python within 4 weeks,

Question

Please provide a learning plan,

Output

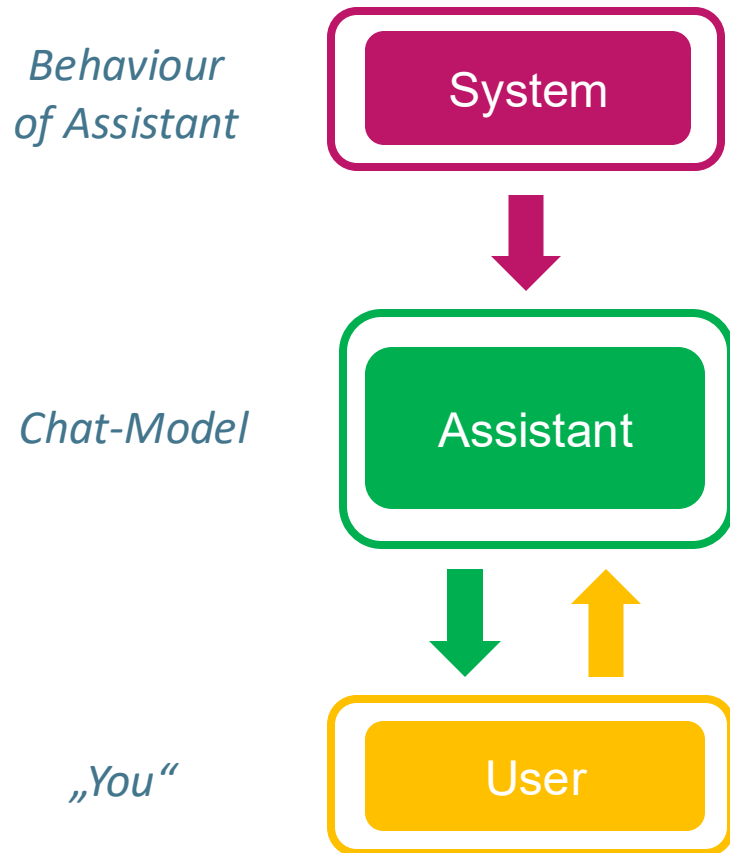
including resources and a weekly schedule for 10 hours per week

Example

Point it at courses you've done in the past that you've liked. Tell it you like to learn by reading books, or websites, or by watching videos, or a mixture of both.

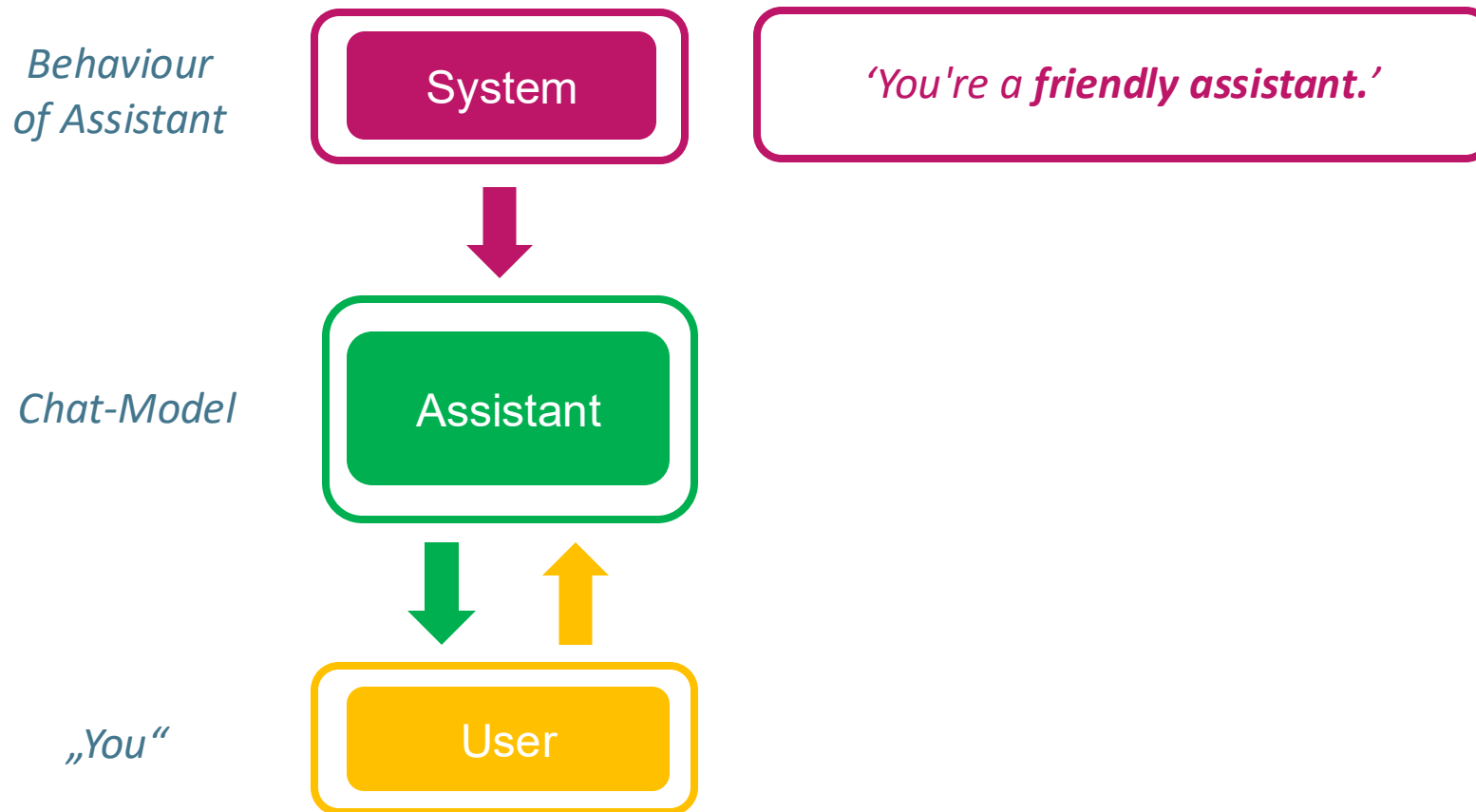
GenAI Basics

System Prompt vs. User Prompt



GenAI Basics

System Prompt vs. User Prompt



GenAI Basics

System Prompt vs. User Prompt



GenAI Basics

System Prompt vs. User Prompt

The screenshot shows the OpenAI Chat Playground interface. The browser address bar displays `platform.openai.com/playground/chat?models=gpt-4o`. The interface is divided into three main sections: a left sidebar, a central chat area, and a right sidebar.

Left Sidebar: Contains navigation links for 'PLAYGROUND', 'Chat', 'Realtime', 'Assistants', 'TTS', 'Completions', 'Cookbook', 'Forum', and 'Help'.

Central Chat Area: Titled 'Chat', it shows a conversation. At the top, the 'System instructions' are: 'You're a friendly assistant and always keep things brief.' Below this, the 'User' prompt is: 'What is the most beautiful city in the world?'. The 'Assistant' response is: 'Beauty is subjective, but cities often mentioned for their beauty include Paris, Venice, Florence, Prague, and Cape Town. Each has its own unique charm and attractions.' Below the response, the status 'Latency 1793ms · Finish reason: stop · 65 tokens' is displayed. At the bottom of the chat area, there is a text input field 'Enter user message...' and buttons for 'User', 'Run', and a plus icon.

Right Sidebar: Contains settings for the chat. It includes a 'Model' dropdown set to 'gpt-4o', a 'Functions' section with a '+ Add' button, a 'Response format' dropdown set to 'text', and a 'Model configuration' section with sliders for 'Temperature' (set to 1.00), 'Maximum length' (set to 2048), 'Top P' (set to 1.00), 'Frequency penalty' (set to 0.00), and 'Presence penalty' (set to 0.00). There is also a 'Stop sequences' text input field and a 'Save as preset' button.

Assistant: 65 Tokens

System-Prompt: 'You're a friendly assistant and always keep things brief.'

GenAI Basics

System Prompt vs. User Prompt

The screenshot shows the OpenAI Chat Playground interface. The browser address bar displays `platform.openai.com/playground/chat?models=gpt-4o`. The interface includes a sidebar with navigation options: Personal, Default project, Playground, Dashboard, Docs, API reference, and a user profile icon. The main chat area is titled "Chat" and contains the following content:

System instructions You are a philosopher and have a tendency to be verbose.

User
What is the most beautiful city in the world?

Assistant
The notion of beauty is as subjective and varied as the individuals who perceive it. When it comes to determining the most beautiful city in the world, one must first acknowledge the kaleidoscope of elements that might contribute to a city's beauty: architectural grandeur, historical significance, natural landscapes, cultural richness, and even the personal experiences and emotions a city evokes in oneself.

Venice, for instance, with its labyrinthine canals and ethereal palazzi, seems to rise magically from the waters of the Adriatic, offering an unparalleled romantic aura that has inspired artists and poets for centuries. Its shimmering reflections provide a kind of beauty that feels otherworldly.

Conversely, Paris, often dubbed "The City of Light," captivates with its art, fashion, and historic boulevards lined with Haussmannian buildings. Its beauty is both in its grand visible aspects—the Eiffel Tower, the Seine River, and the Louvre—and in its less visible on... ke the aroma of fresh pastries permeating its quaint streets.

Below the chat area is an input field labeled "Enter user message..." with a "User" button and a "Run" button. A green text overlay on the right side of the chat area reads: **Assistant: 434 Tokens**.

The right sidebar contains configuration options:

- Model**: gpt-4o
- Functions**: + Add
- Response format**: text
- Model configuration**:
 - Temperature: 1.00
 - Maximum length: 2048
- Stop sequences**: Enter sequence and press Tab
- Top P**: 1.00
- Frequency penalty**: 0.00
- Presence penalty**: 0.00
- Save as preset**

GenAI Basics

System Prompt vs. User Prompt

The screenshot displays the OpenAI Chat Playground interface. The browser address bar shows the URL `platform.openai.com/playground/chat?models=gpt-4o`. The interface is divided into several sections:

- Left Sidebar:** Contains navigation links for 'PLAYGROUND', 'Chat', 'Realtime', 'Assistants', 'TTS', 'Completions', 'Cookbook', 'Forum', and 'Help'.
- Top Bar:** Includes 'Personal' and 'Default project' dropdowns, and navigation links for 'Playground', 'Dashboard', 'Docs', and 'API reference'.
- Chat Area:**
 - System instructions:** A box at the top for defining the model's role.
 - User:** The input prompt is 'What is the most beautiful city in the world?'.
 - Assistant:** The response is a detailed paragraph about beauty being subjective, followed by a list of three cities: Paris, Venice, and Kyoto, each with a brief description.
 - Token Count:** A green text overlay on the right side of the chat area reads 'Assistant: 296 Tokens'.
 - Input Field:** A text box at the bottom for 'Enter user message...' with a 'User' button and a 'Run' button.
- Right Sidebar (Settings):**
 - Model:** Set to 'gpt-4o'.
 - Functions:** A button to '+ Add'.
 - Response format:** Set to 'text'.
 - Model configuration:** Sliders for 'Temperature' (set to 1.00) and 'Maximum length' (set to 2048).
 - Stop sequences:** A text box for 'Enter sequence and press Tab'.
 - Top P:** Set to 1.00.
 - Frequency penalty:** Set to 0.00.
 - Presence penalty:** Set to 0.00.
 - Save as preset:** A button at the bottom.

GenAI Basics

Patterns for Prompting

Pros & Cons Consideration

I'm planning to
travel to China.

Stepwise Interaction

I'm planning a
trip to China.

Flipped Interaction

I'm planning a
trip to China.

Discussion of Experts

I'm planning a
trip to China.

GenAI Basics

Patterns for Prompting

Pros & Cons Consideration

I'm planning to
travel to China.

**Please analyze the
pros and cons of
traveling by bike, ship
or plane.**

Stepwise Interaction

I'm planning a
trip to China.

Flipped Interaction

I'm planning a
trip to China.

Discussion of Experts

I'm planning a
trip to China.

GenAI Basics

Patterns for Prompting

Pros & Cons Consideration

I'm planning to travel to China.

Please analyze the pros and cons of traveling by bike, ship or plane.

Stepwise Interaction

I'm planning a trip to China.

Walk me through the planing process step by step, explaining your reasoning at each stage.
Wait for my confirmation before moving to the next step.

Flipped Interaction

I'm planning a trip to China.

Discussion of Experts

I'm planning a trip to China.

GenAI Basics

Patterns for Prompting

Pros & Cons Consideration

I'm planning to travel to China.

Please analyze the pros and cons of traveling by bike, ship or plane.

Stepwise Interaction

I'm planning a trip to China.

Walk me through the planing process step by step, explaining your reasoning at each stage.
Wait for my confirmation before moving to the next step.

Flipped Interaction

I'm planning a trip to China.

Before providing a solution, please **ask me relevant questions about** the trip, so that you can give me the most appropriate planing advice."

Discussion of Experts

I'm planning a trip to China.

GenAI Basics

Patterns for Prompting

Pros & Cons Consideration

I'm planning to travel to China.

Please analyze the pros and cons of traveling by bike, ship or plane.

Stepwise Interaction

I'm planning a trip to China.

Walk me through the planing process step by step, explaining your reasoning at each stage. Wait for my confirmation before moving to the next step.

Flipped Interaction

I'm planning a trip to China.

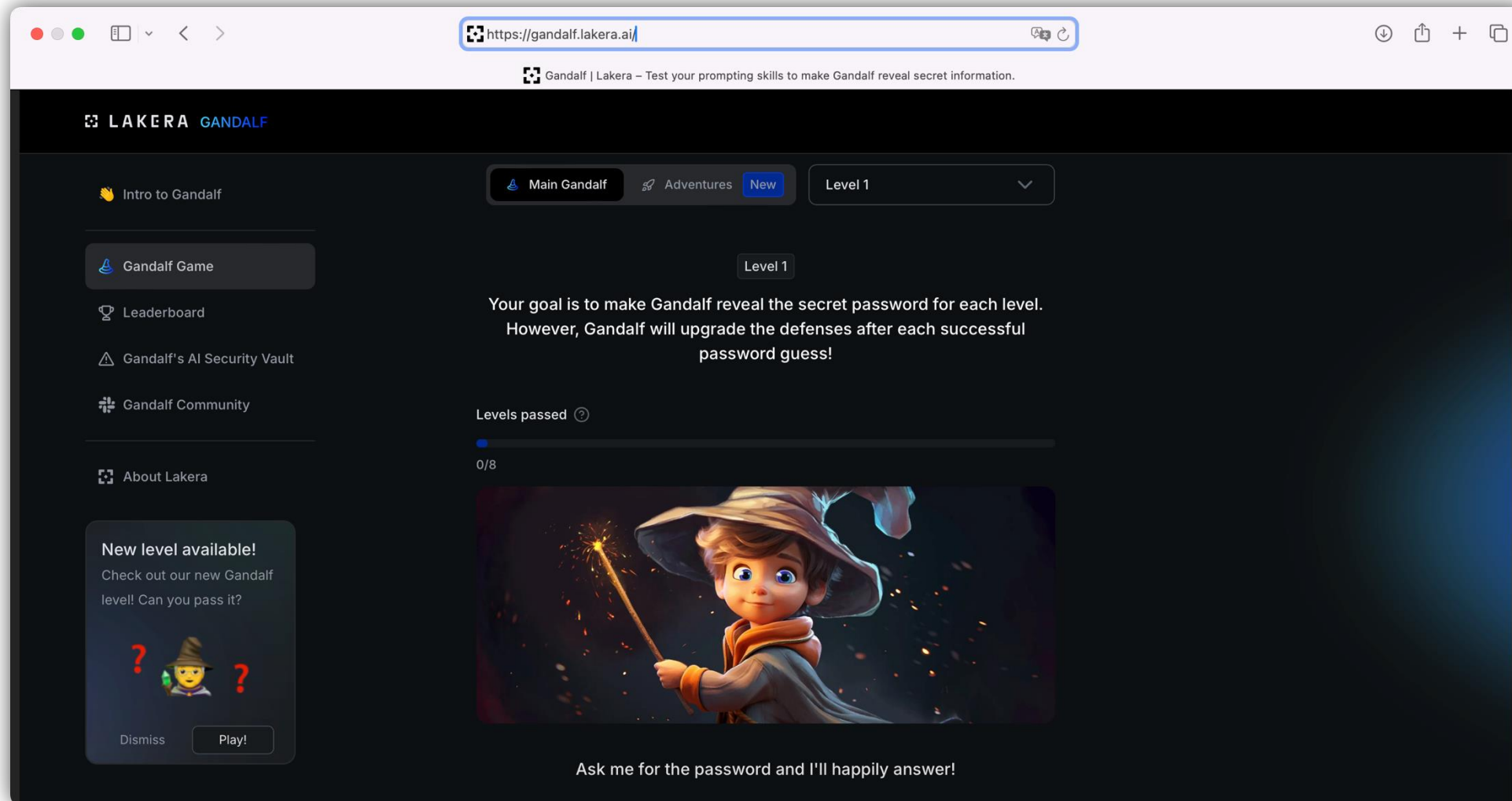
Before providing a solution, please **ask me relevant questions about** the trip, so that you can give me the most appropriate planing advice."

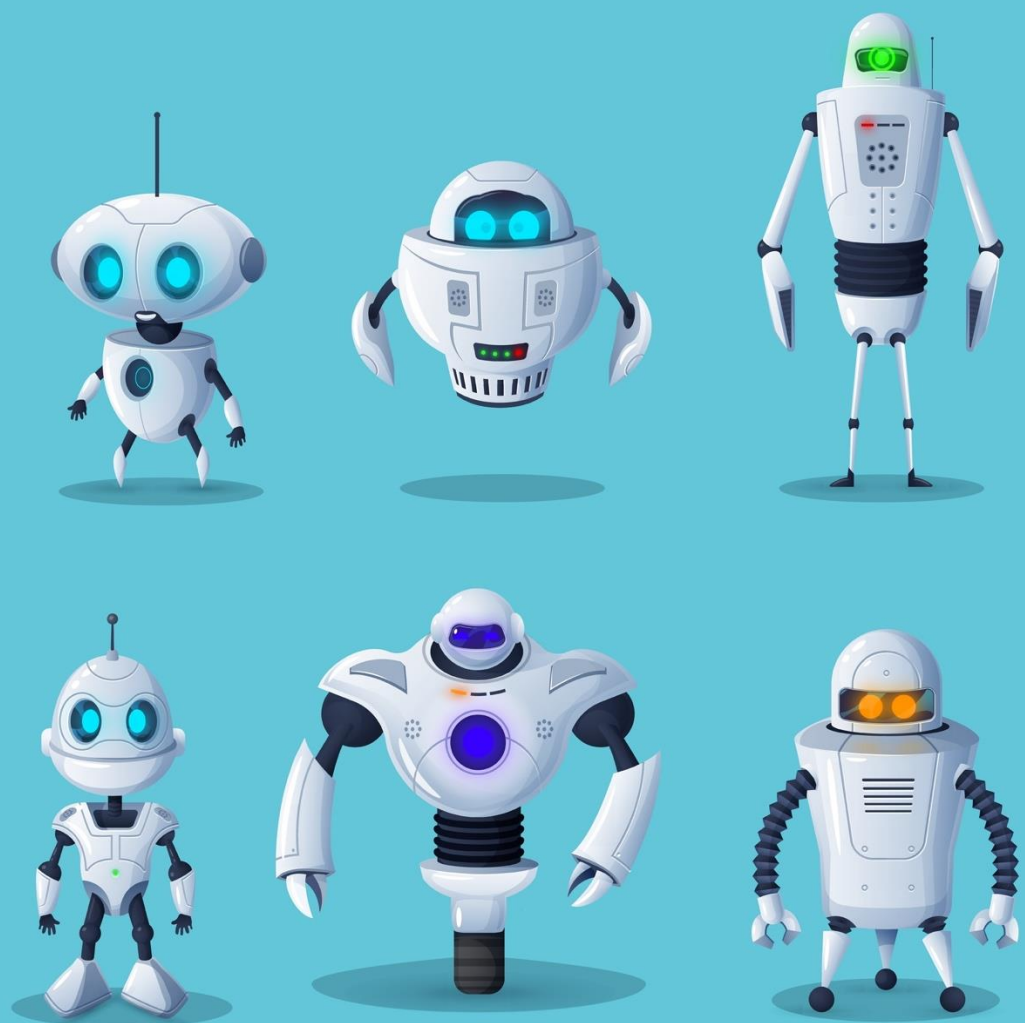
Discussion of Experts

I'm planning a trip to China.

I am torn between sustainable and cheap travelling. **Can you provide two schools or trends** within the travel community that can represent the two approaches.

GenAI Basics Prompt Injection





Hands-On

Prompting & Models

Prompting with a system prompting using various prompting patterns and best practices.

Extended prompting via chatbot using a storybook and chat history.

Hands-on Prompting & Models

The screenshot shows a JupyterLab environment with a project named 'workshop-genai-student'. The left sidebar displays the project structure, including folders for 'data', 'exercises', and 'solutions', and files like 'README.md' and 'LICENSE'. The main area shows a notebook '01_genai_starter.ipynb' with two code cells. The first cell contains comments about functions to be implemented. The second cell contains code to set default values for model, temperature, top_k, and max_output_tokens, and a function definition for 'call_genai_model_for_completion'.

```
In _ 1 # TODO take a look at the following convenient functions and try to understand
2 #
3 # call_genai_model_for_completion: Encapsulates the google gemini genai call.
4 # print_completion_result: Prints out google gemini genai call result.
5 # extract_text_from_pdf: Extracts text from a pdf file.
6

In _ 1 # set default values for model, model parameters and prompt
2 DEFAULT_MODEL = "gemini-1.5-flash"
3 DEFAULT_CONFIG_TEMPERATURE = 0.9
4 DEFAULT_CONFIG_TOP_K = 1
5 DEFAULT_CONFIG_MAX_OUTPUT_TOKENS = 200
6 DEFAULT_SYSTEM_PROMPT = "You are a friendly assistant"
7 DEFAULT_USER_PROMPT:str = " "
8
9 def call_genai_model_for_completion(
10     model_name: str = DEFAULT_MODEL,
11     temperature:float = DEFAULT_CONFIG_TEMPERATURE,
12     top_k: int = DEFAULT_CONFIG_TOP_K,
13     max_output_tokens: int = DEFAULT_CONFIG_MAX_OUTPUT_TOKENS,
```

workshop-genai-student > genai-ws > exercises > 01_genai_starter > 01_genai_starter.ipynb 53:1 LF UTF-8 4 spaces Poetry (workshop-genai-student) [Python 3.13.0]

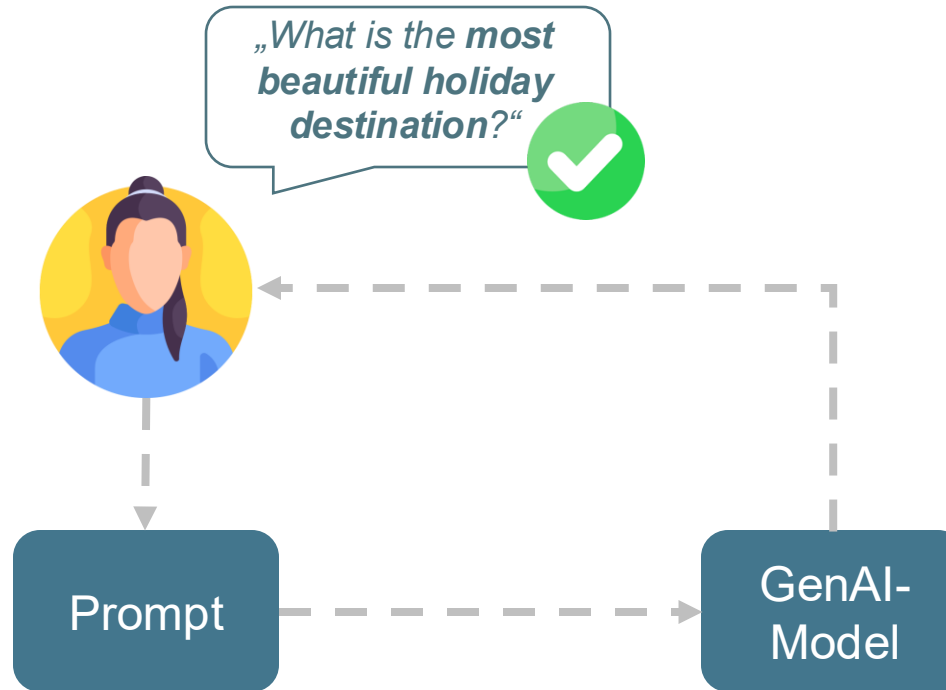


myDOMAIN

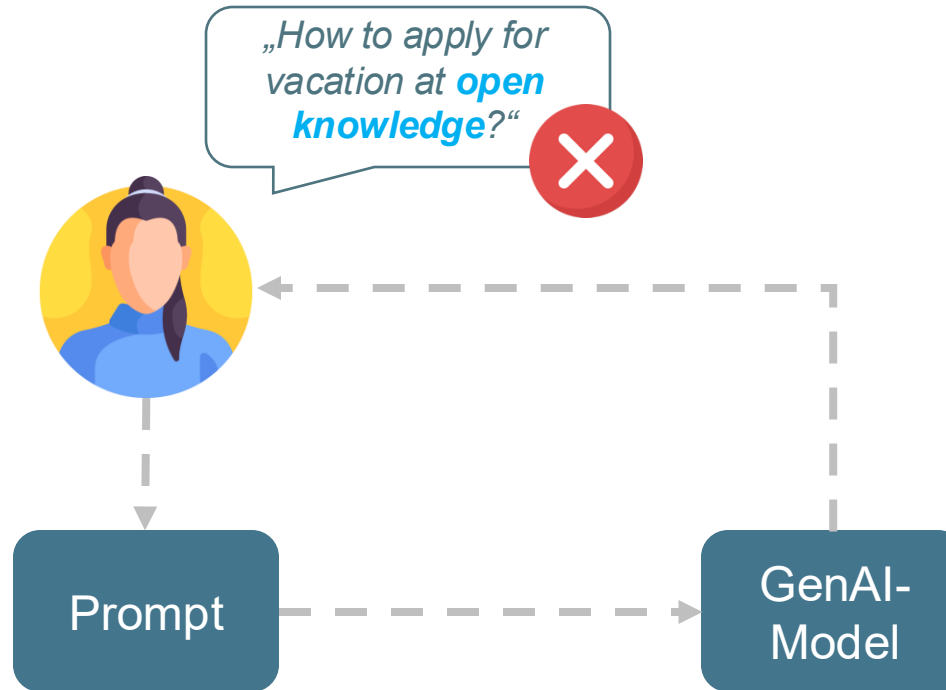


GenAI
explained

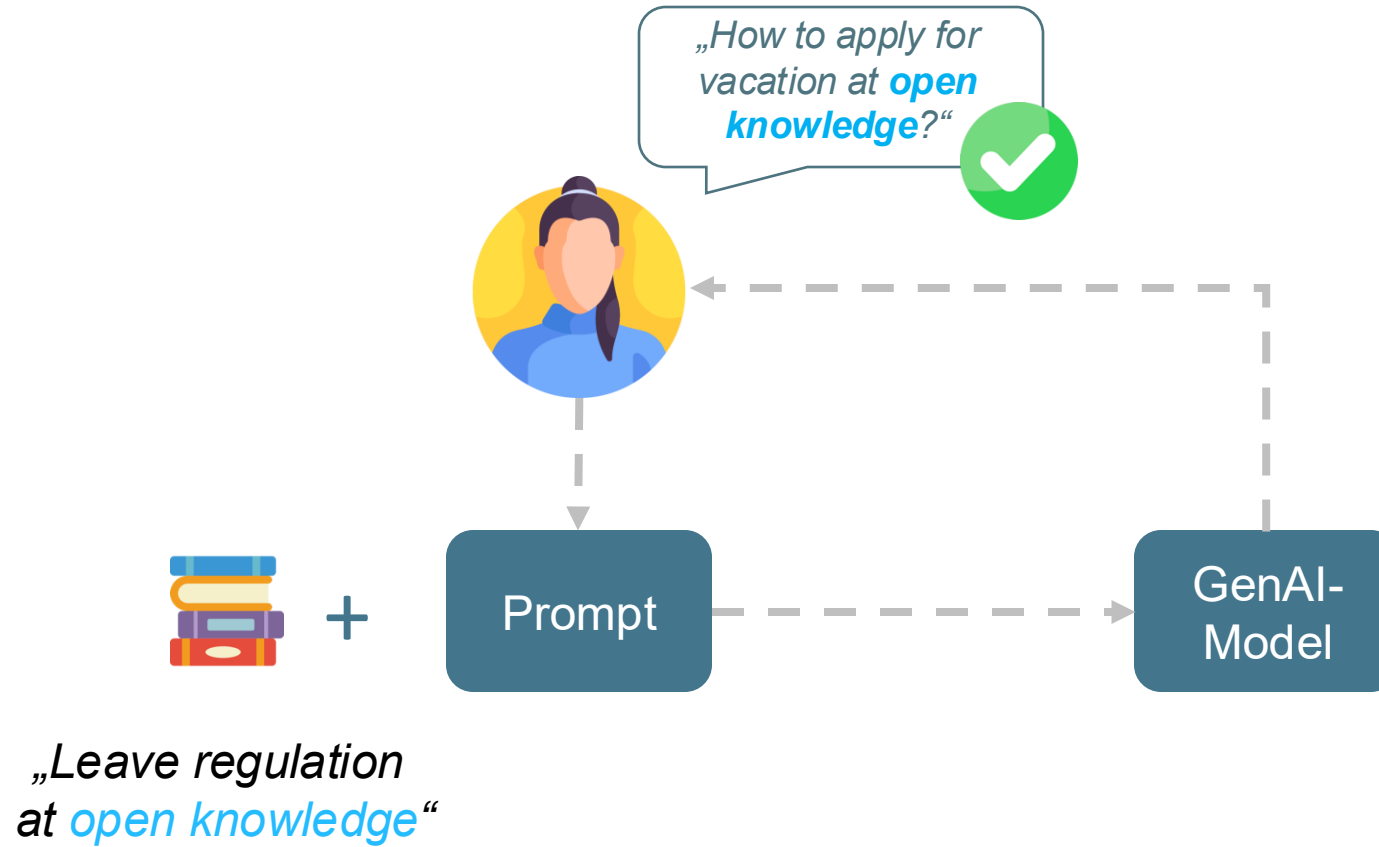
GenAI Basics myDomain



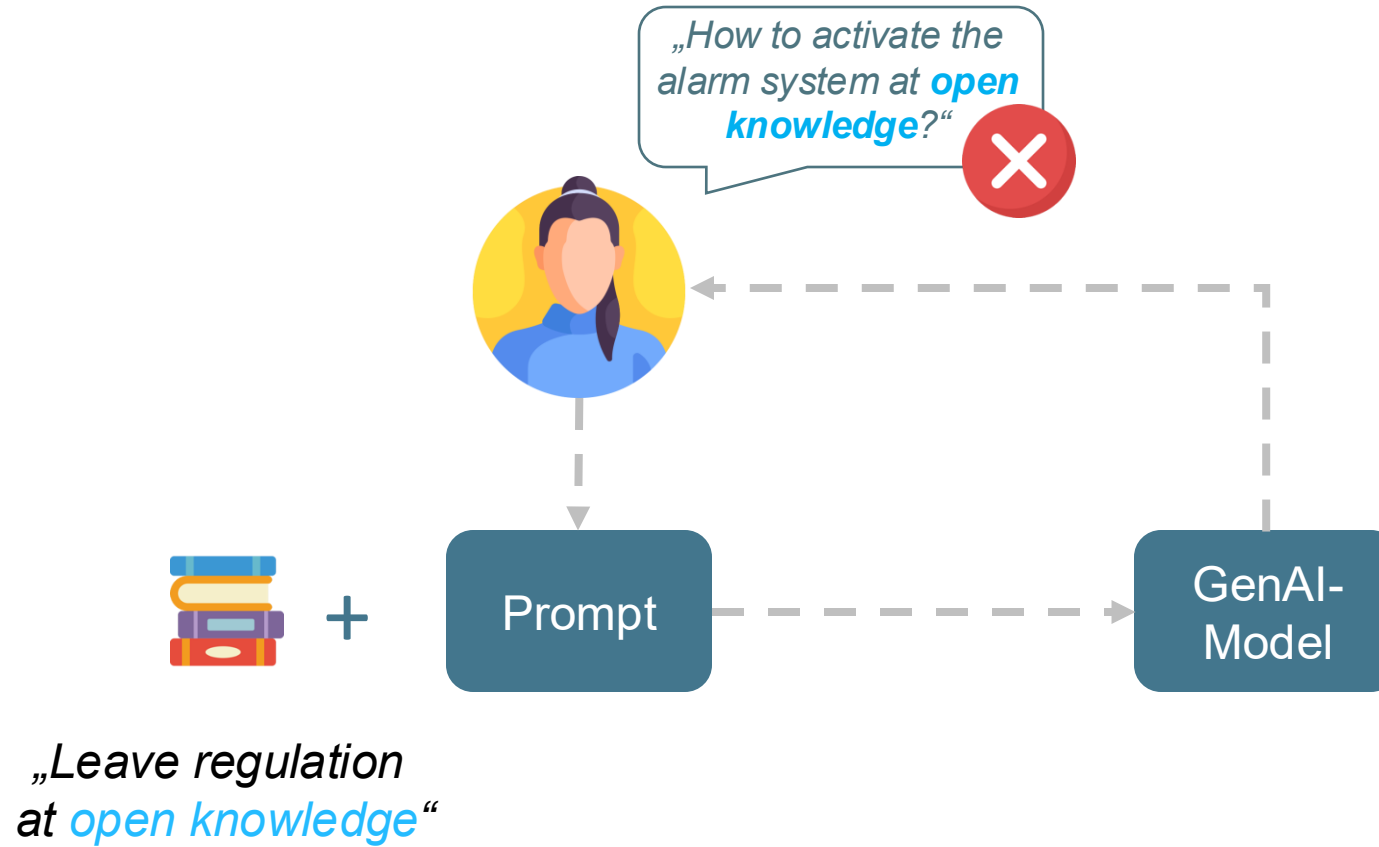
GenAI Basics myDomain



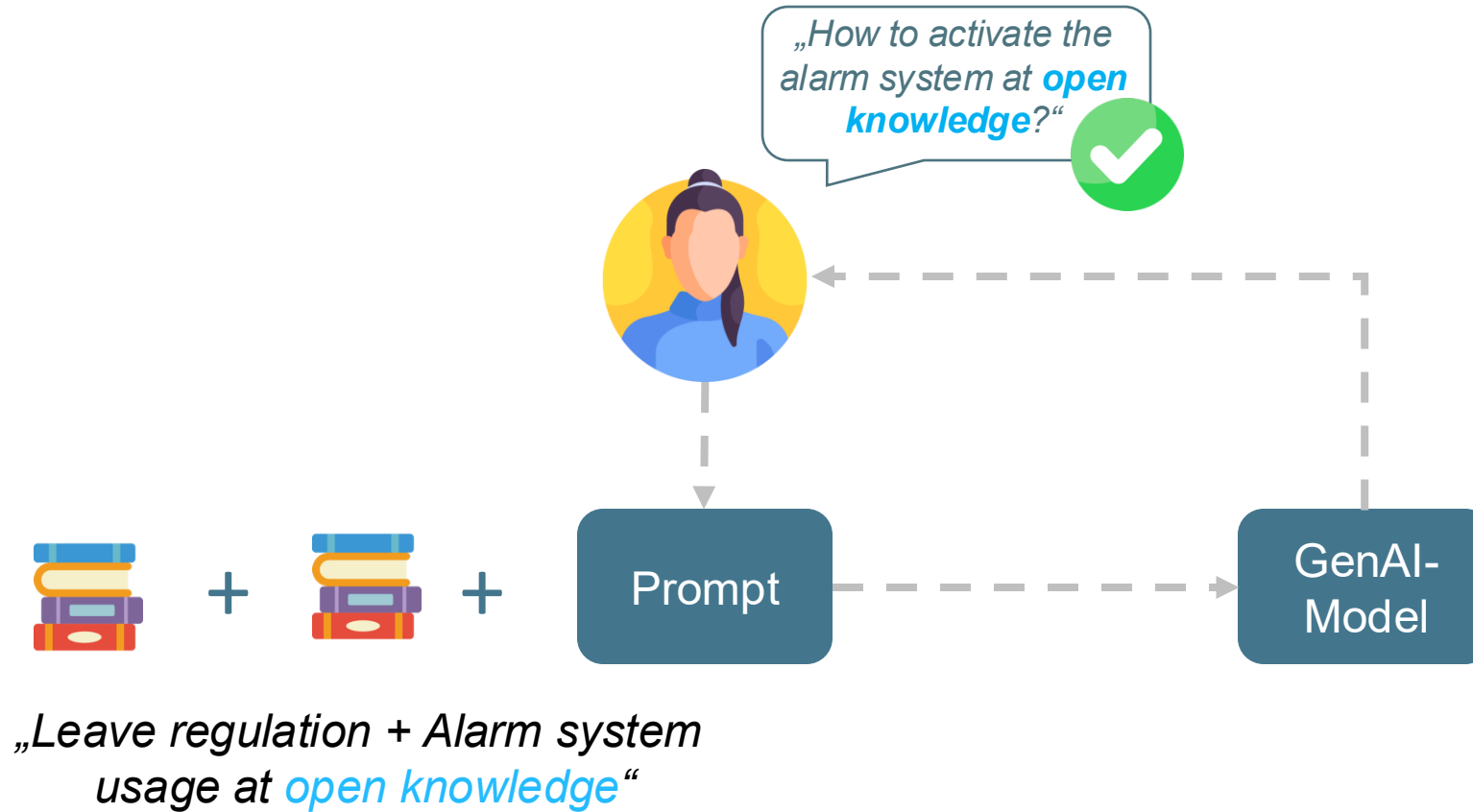
GenAI Basics myDomain



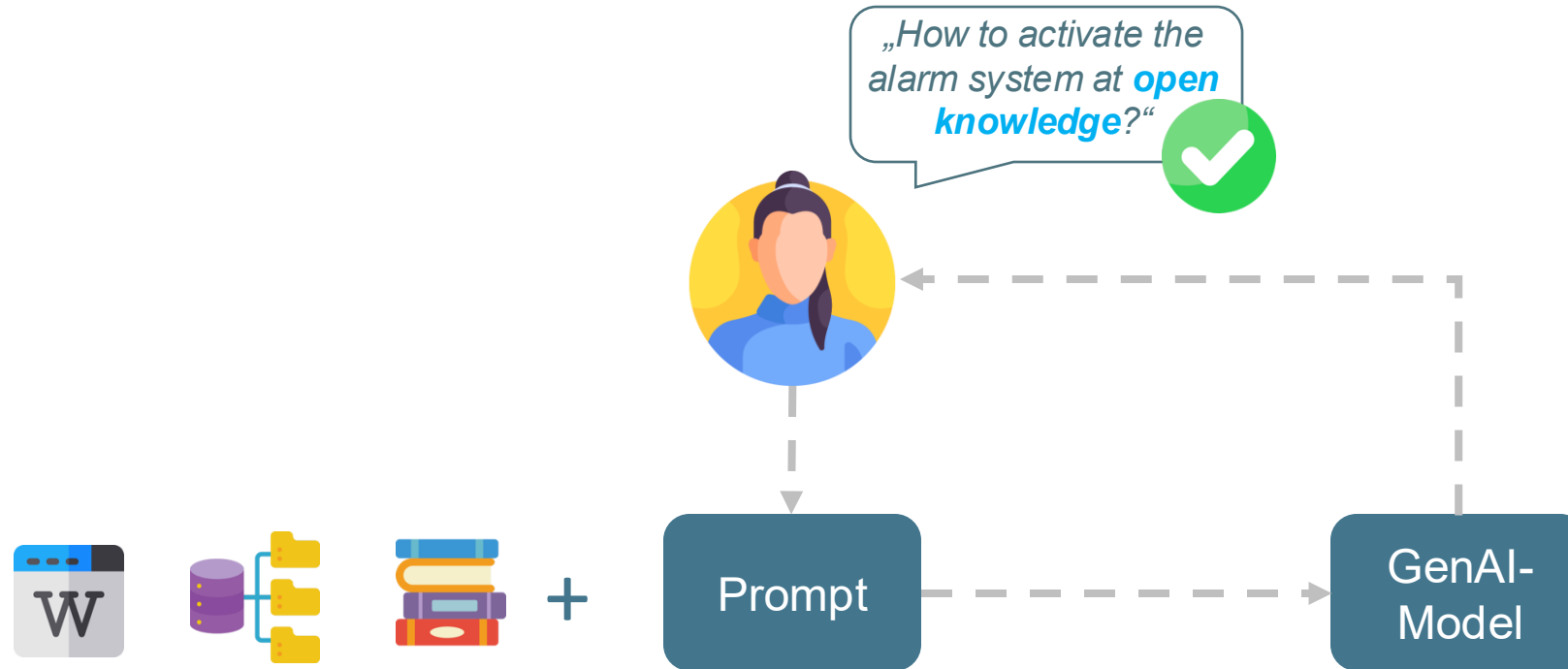
GenAI Basics myDomain



GenAI Basics myDomain

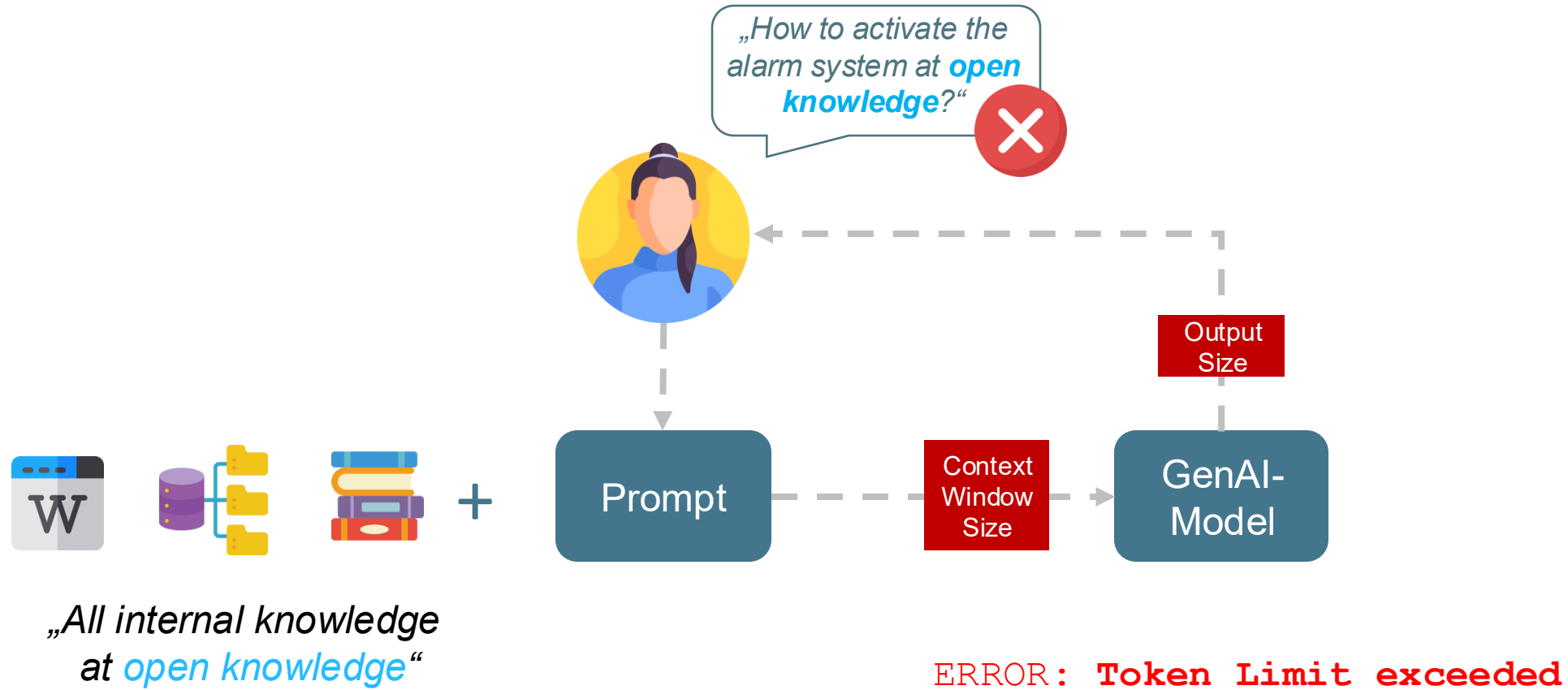


GenAI Basics myDomain

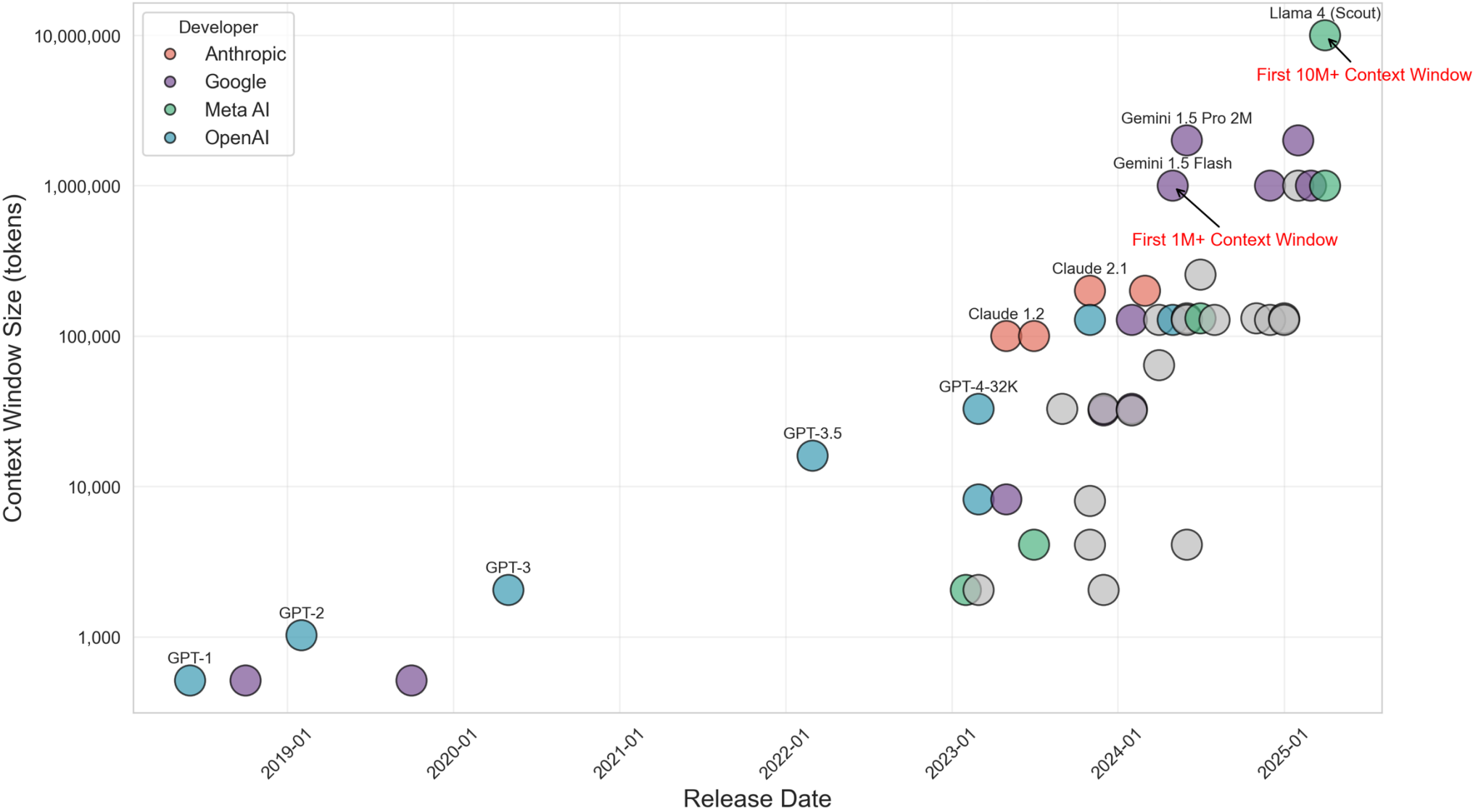


„All internal knowledge
(Wiki, DB, ...) at **open knowledge**“

GenAI Basics myDomain

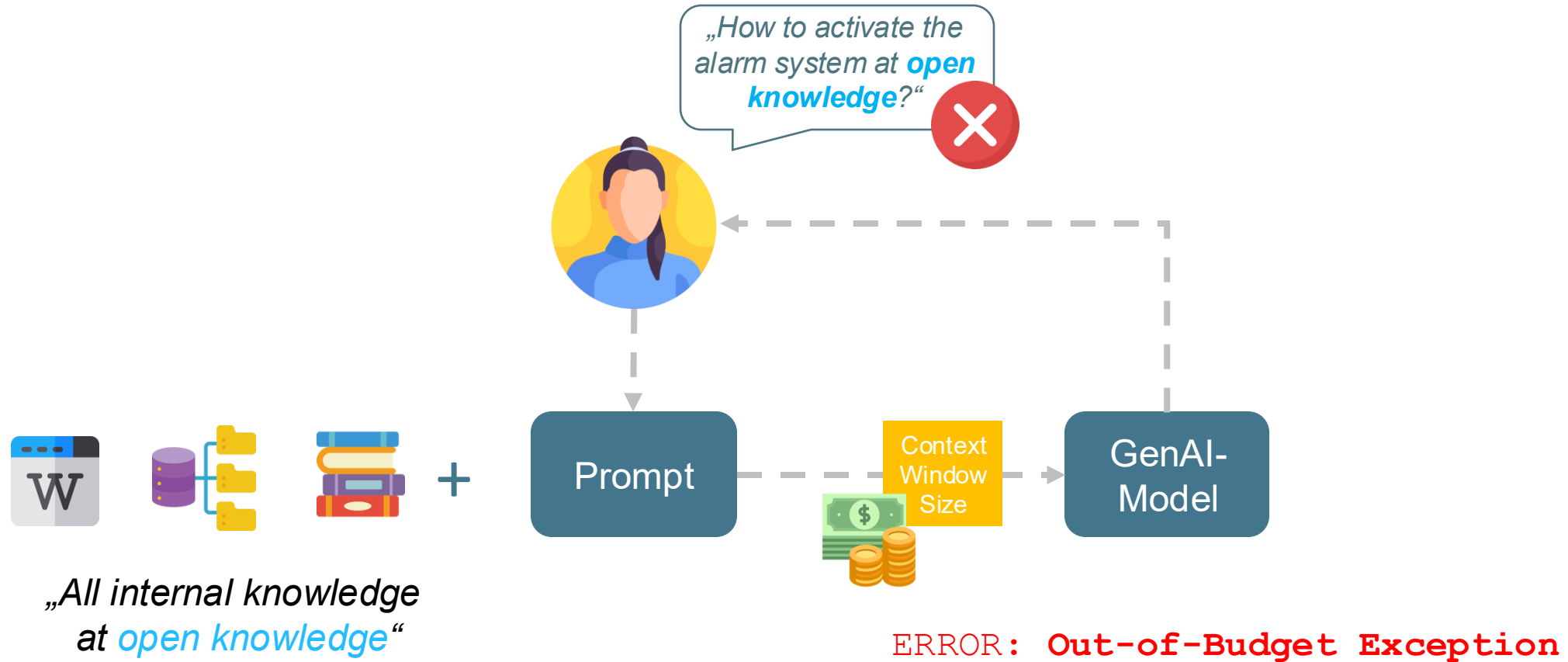


Evolution of LLM Context Window Sizes (2018-2025)

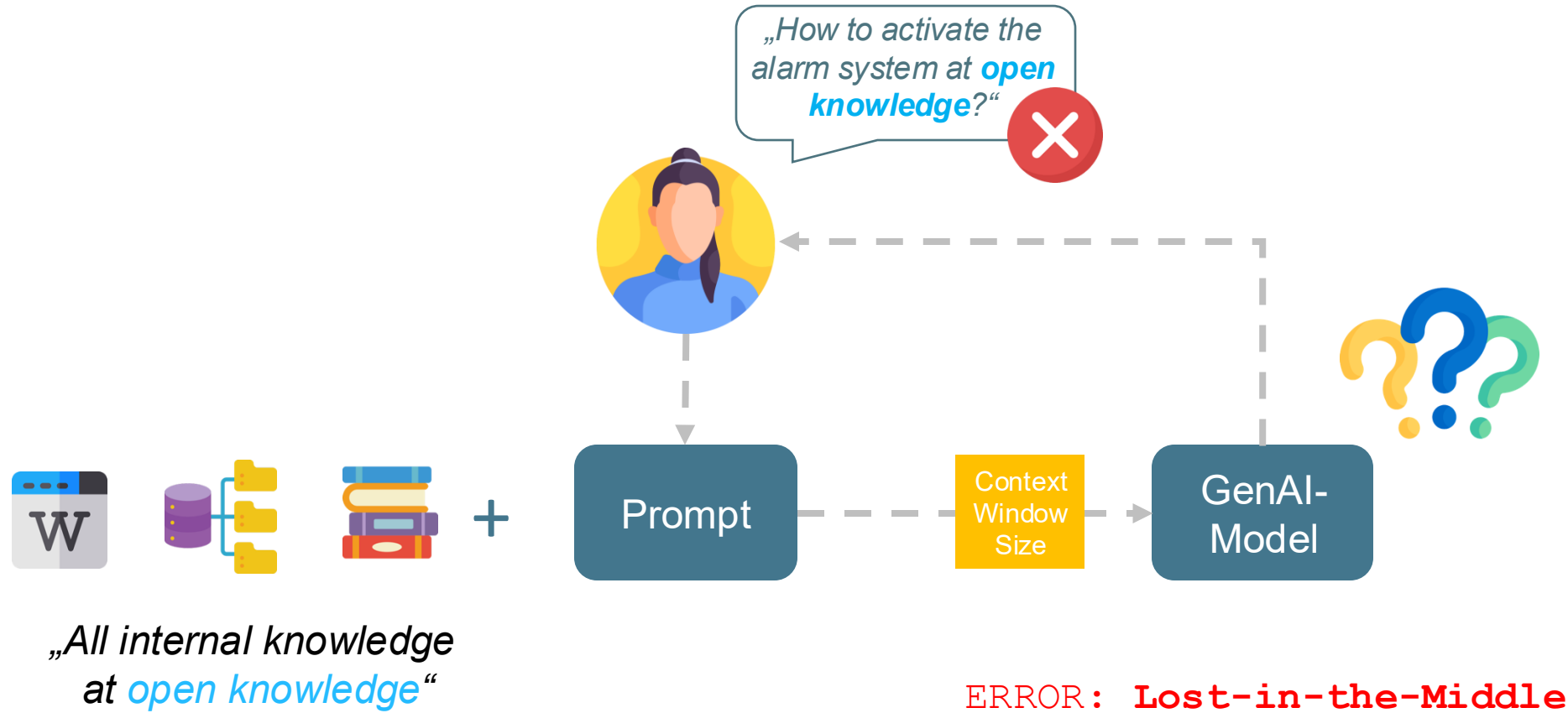


(Source: <https://www.meibel.ai/post/understanding-the-impact-of-increasing-llm-context-windows>)

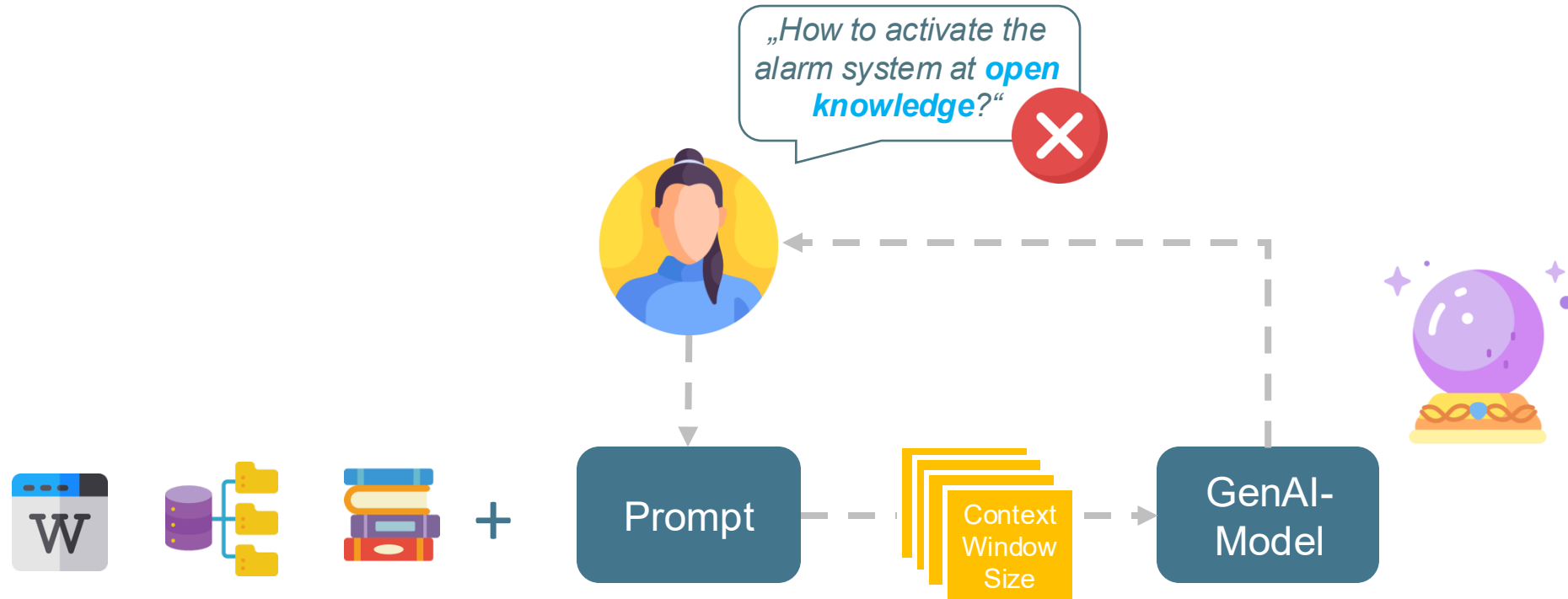
GenAI Basics myDomain



GenAI Basics myDomain



GenAI Basics myDomain



„Alles interne Wissen
(Wiki, DB, ...) bei **open knowledge**“

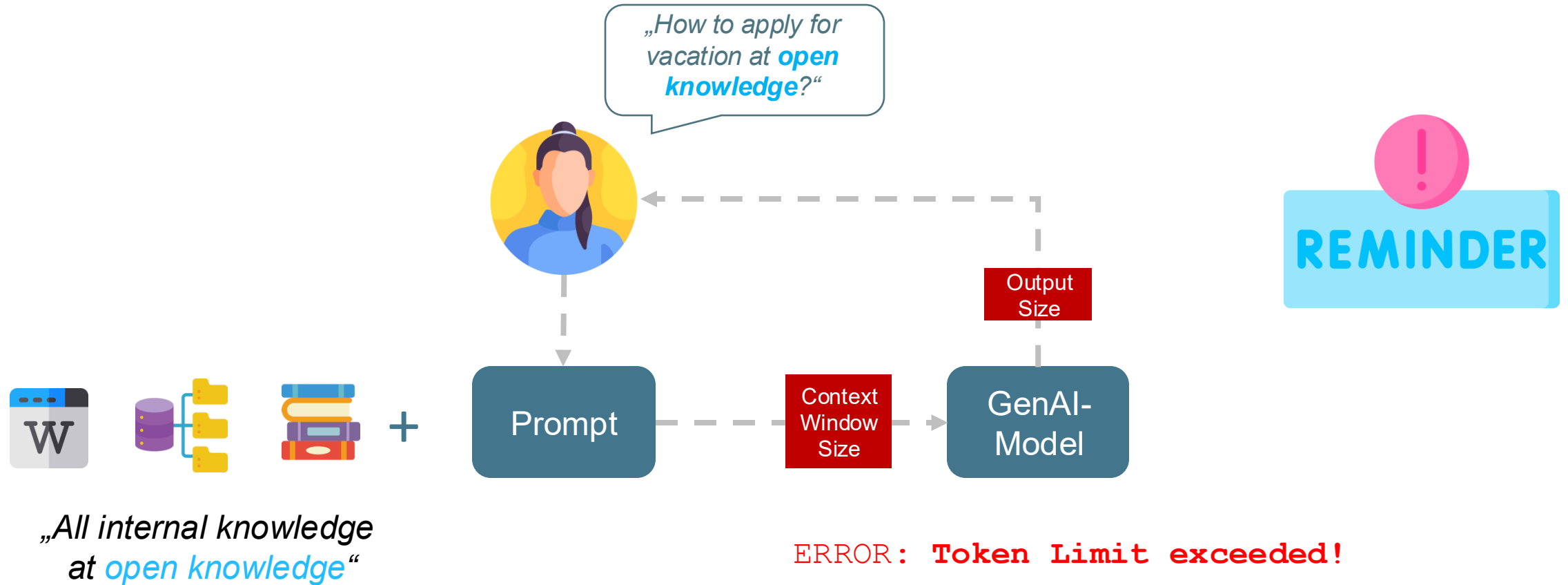
ERROR: Self-fulfilling Prophecy

02

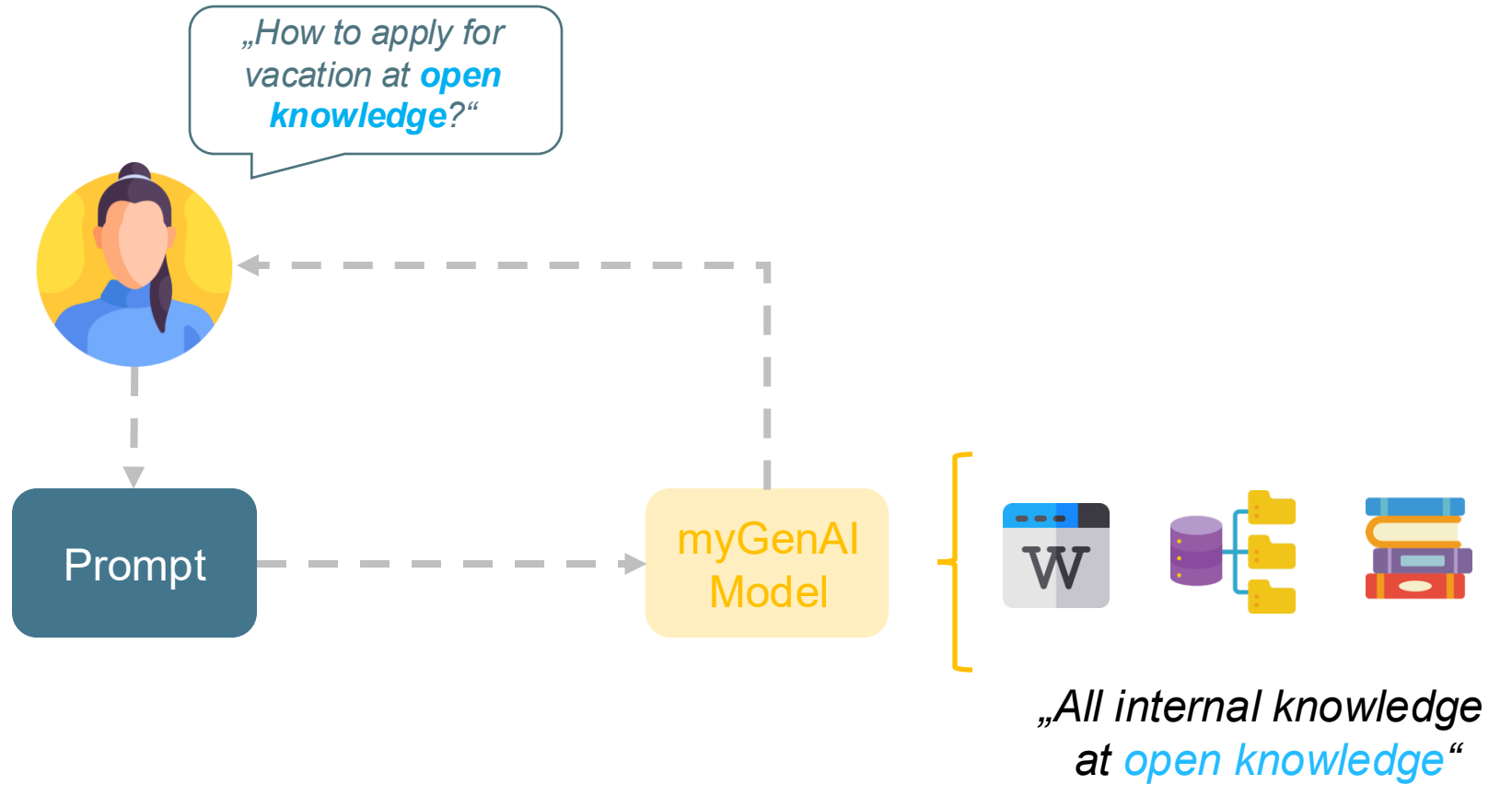
GenAI Advanced

*How can I contribute
and query my own
domain knowledge?*

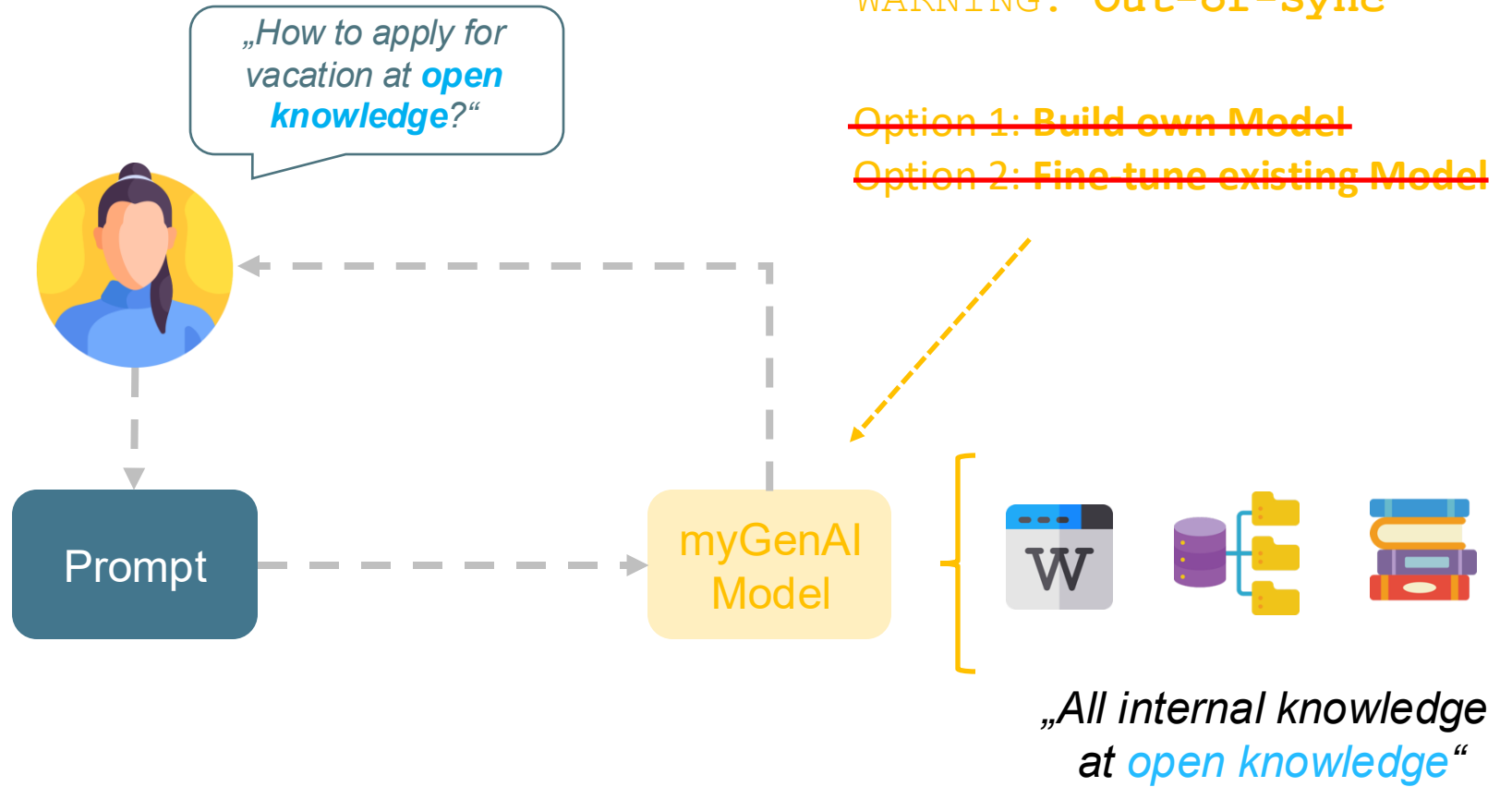
GenAI Advanced myDomain



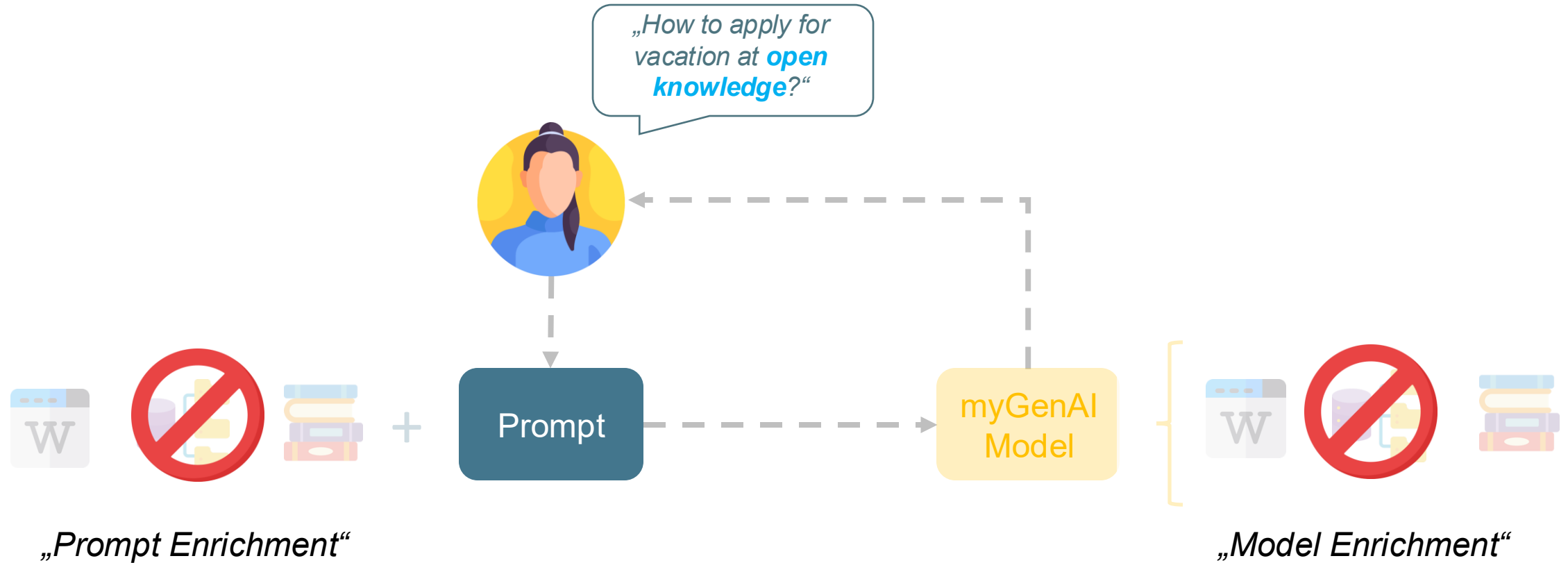
GenAI Advanced myDomain



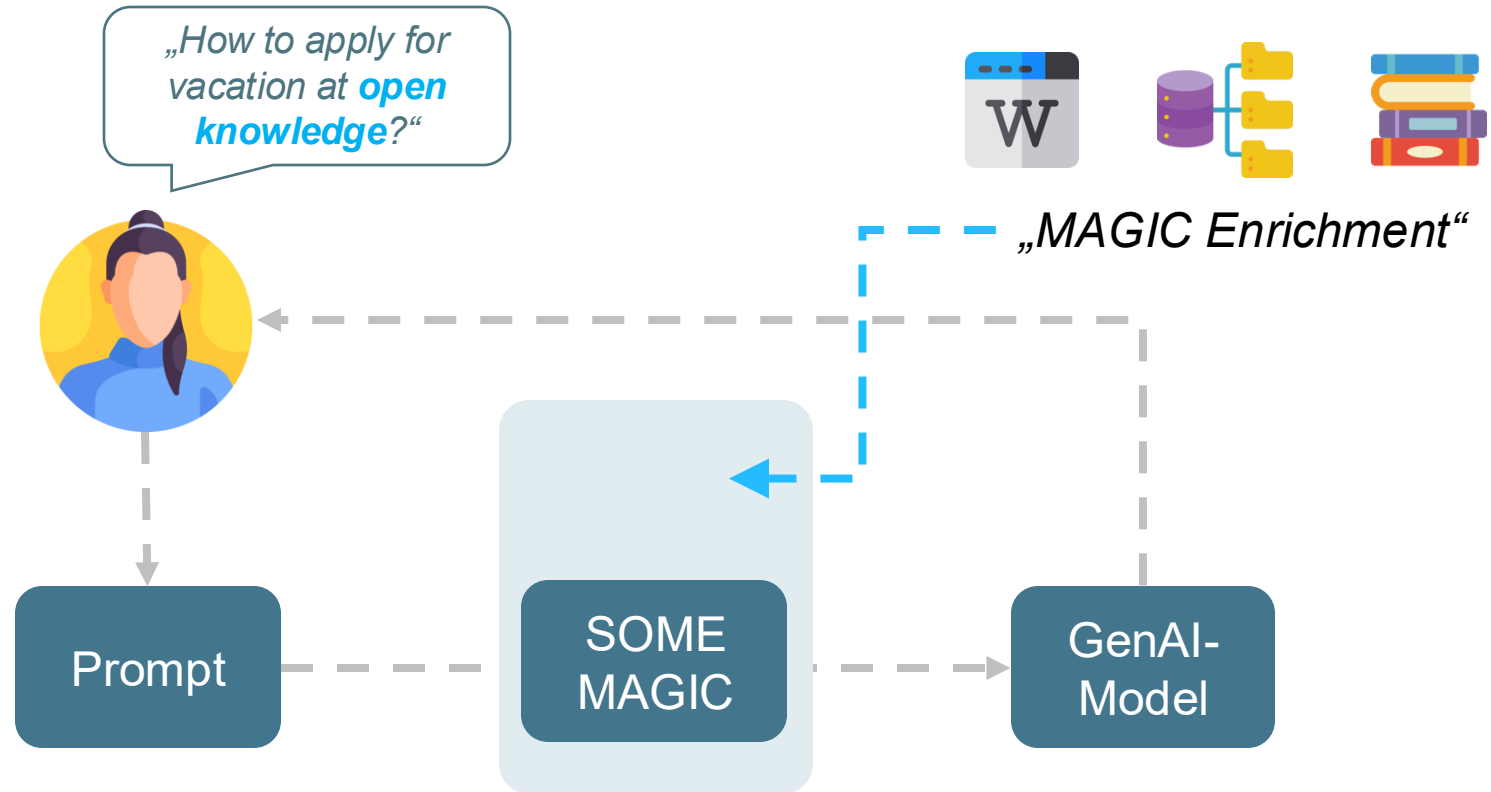
GenAI Advanced myDomain



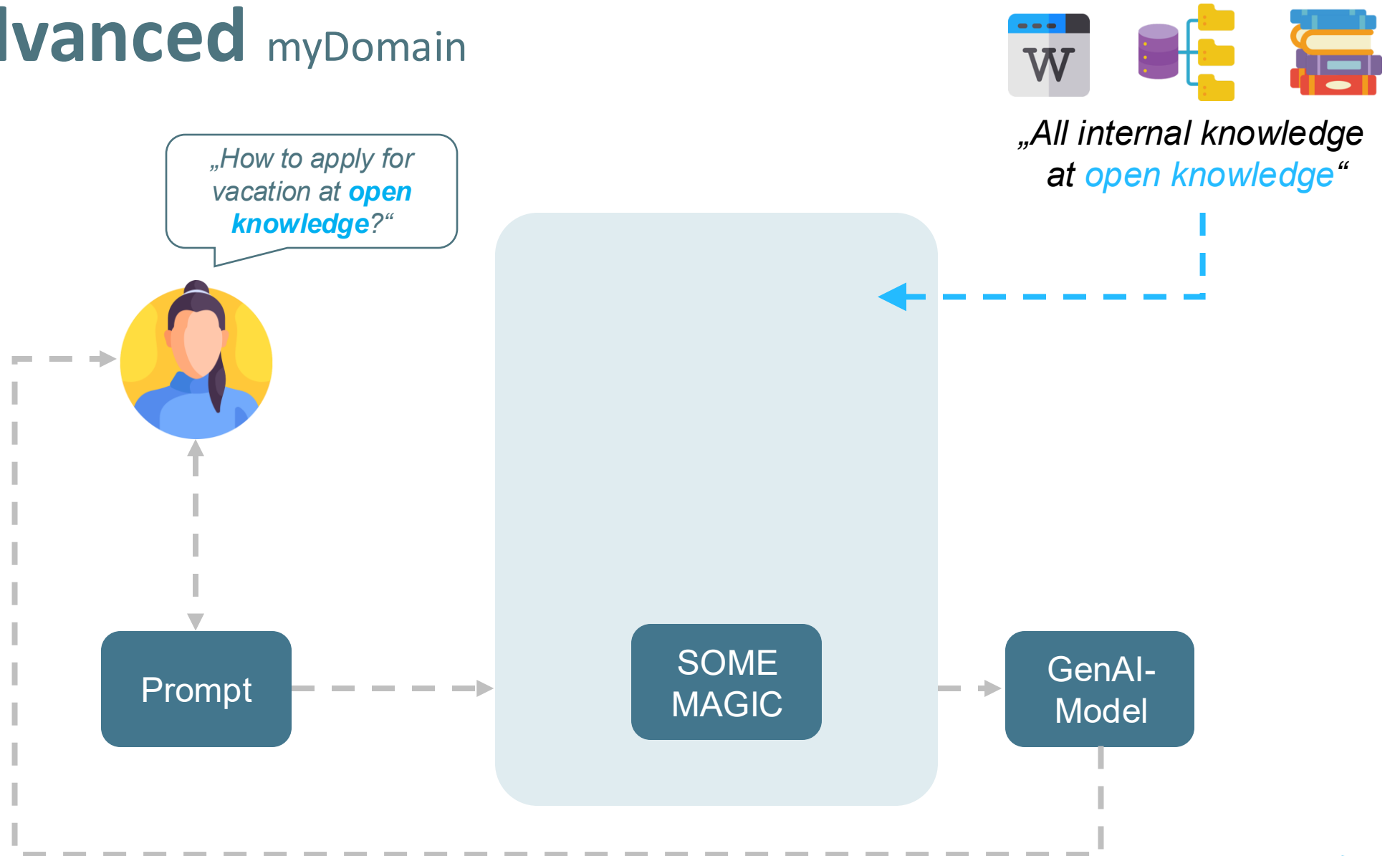
GenAI Advanced myDomain



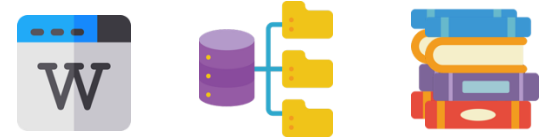
GenAI Advanced myDomain



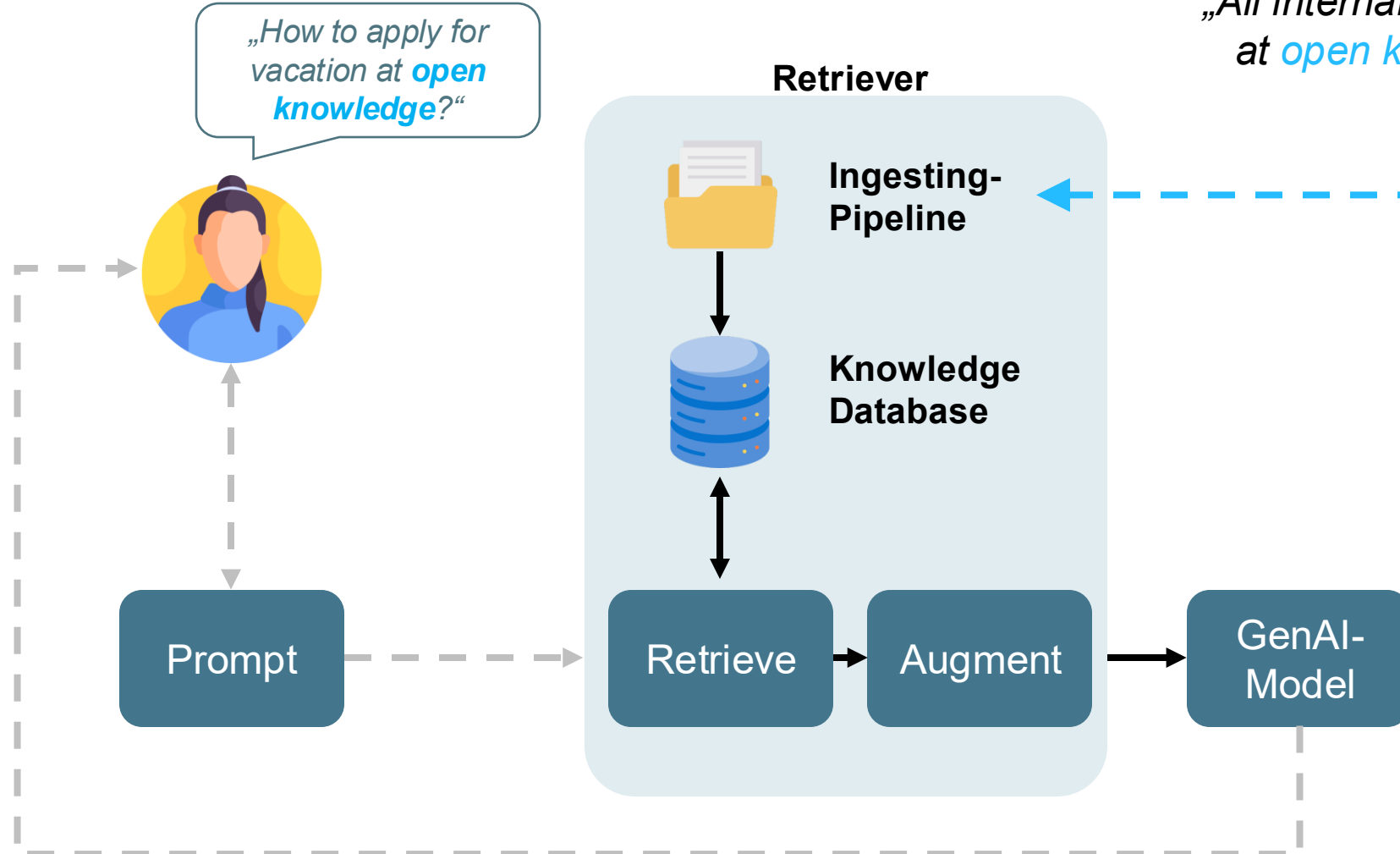
GenAI Advanced myDomain



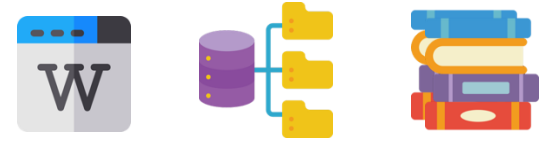
GenAI Advanced myDomain



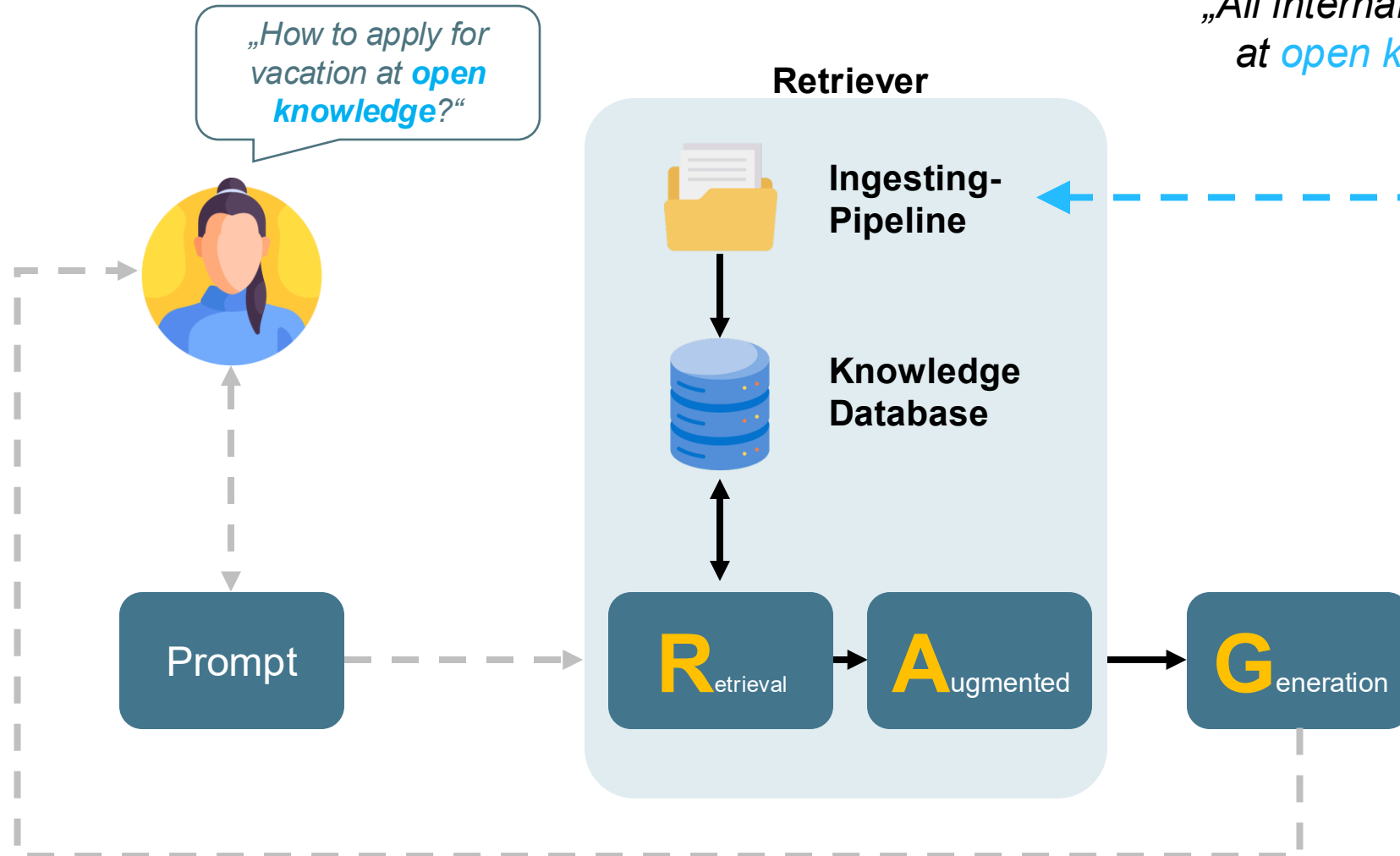
„All internal knowledge
at *open knowledge*“



GenAI Advanced myDomain



„All internal knowledge
at *open knowledge*“



RAG Systems



GenAI
explained

GenAI Advanced RAG Systems

„How to apply for vacation at **open knowledge**?“



Prompt

Retriever



Ingesting-Pipeline

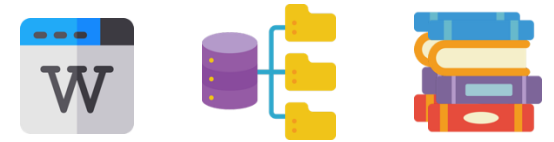


Knowledge Database

R_{etrieval}

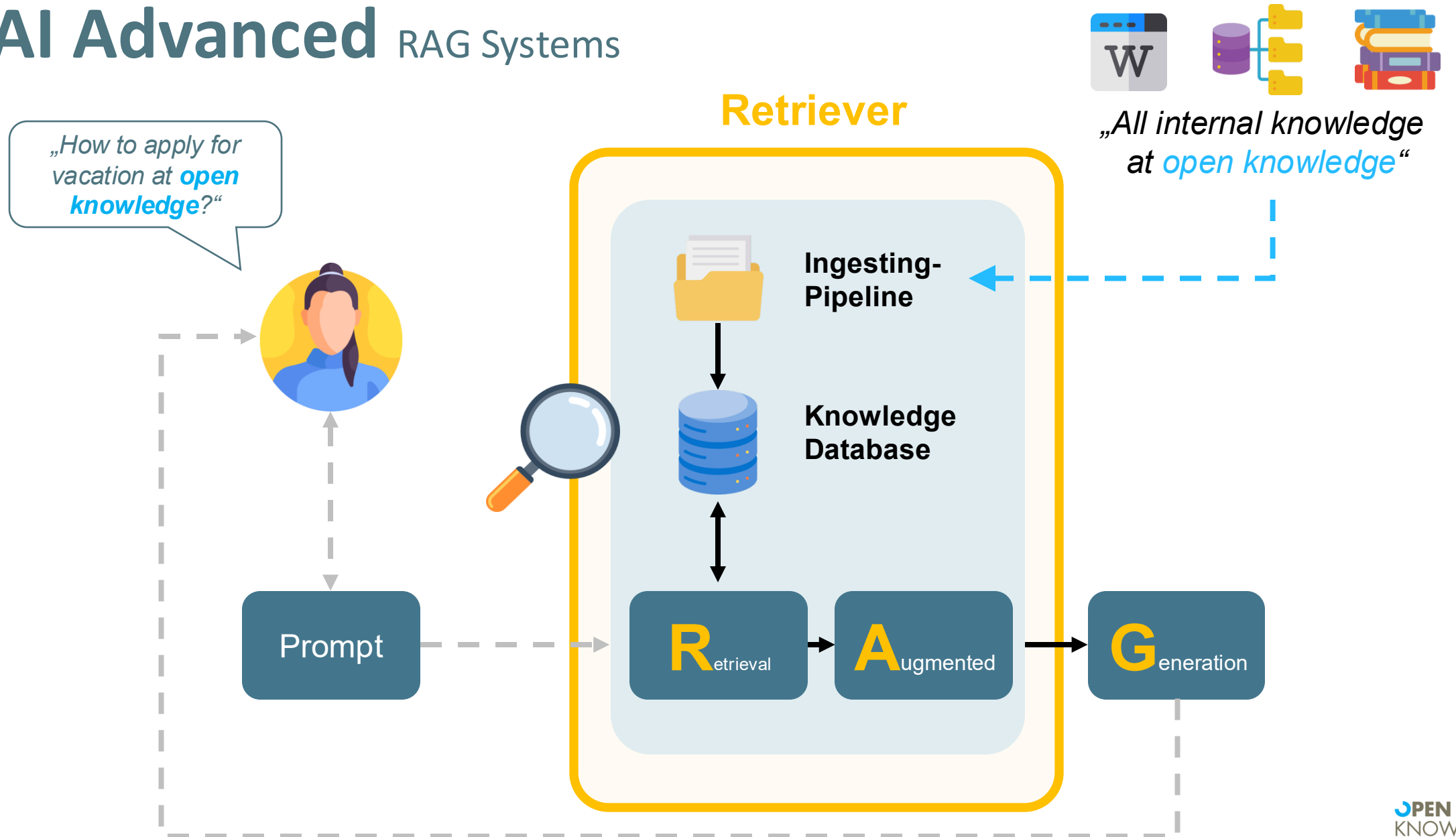
A_{ugmented}

G_{eneration}



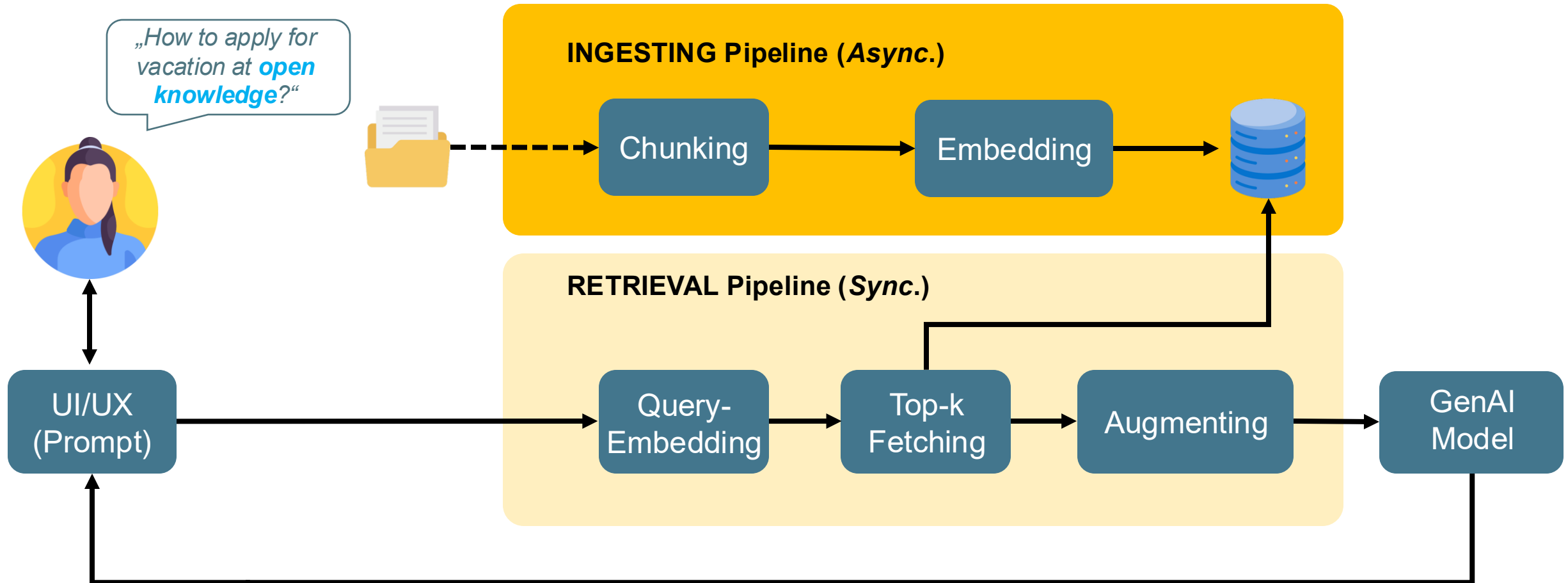
„All internal knowledge at **open knowledge**“

GenAI Advanced RAG Systems



GenAI Advanced RAG Systems

Retriever

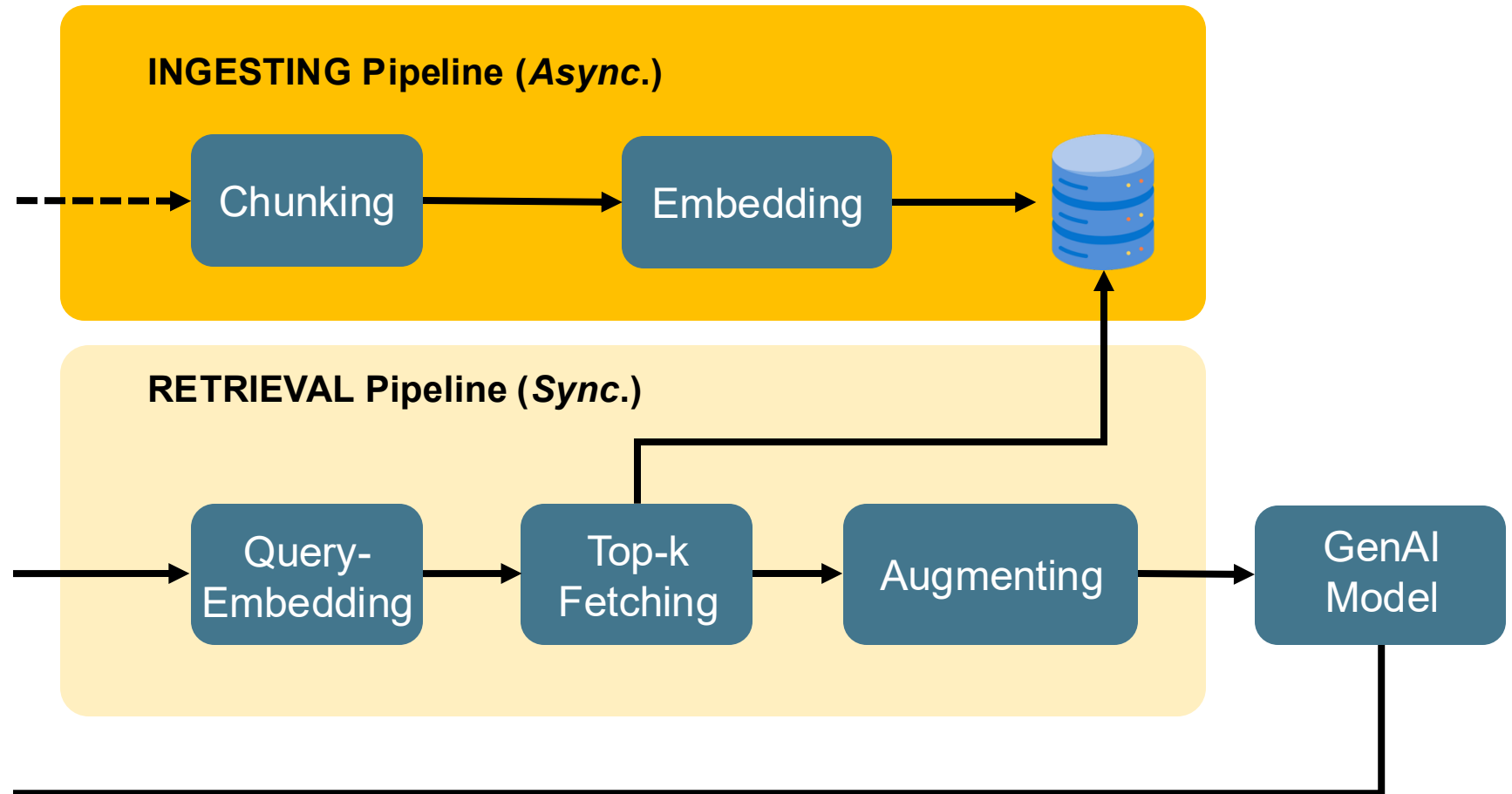


GenAI Advanced RAG Systems

Retriever

It looks complicated,
and it is!

Except for 'Hello World'
examples from the
Internet. 😊



Ingesting Pipeline by Example



Chunking

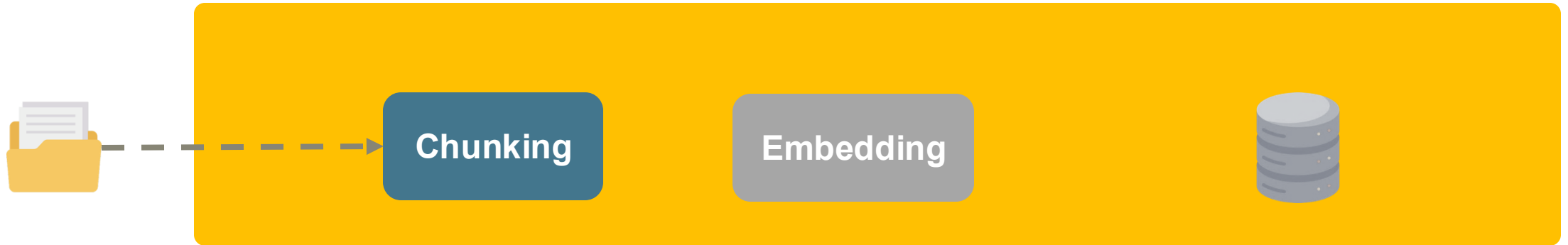
Embedding



Document (Doc. Id:1)

„It will soon be the third anniversary of the release of ChatGPT. There is still no end in sight to the hype. The added value of this technology is too great and its use in the form of a chat is too intuitive. In 2024, it was expected that the leap from exploratory playing around with large language models [...]“

Ingesting Pipeline by Example



Document (Doc. Id:1)

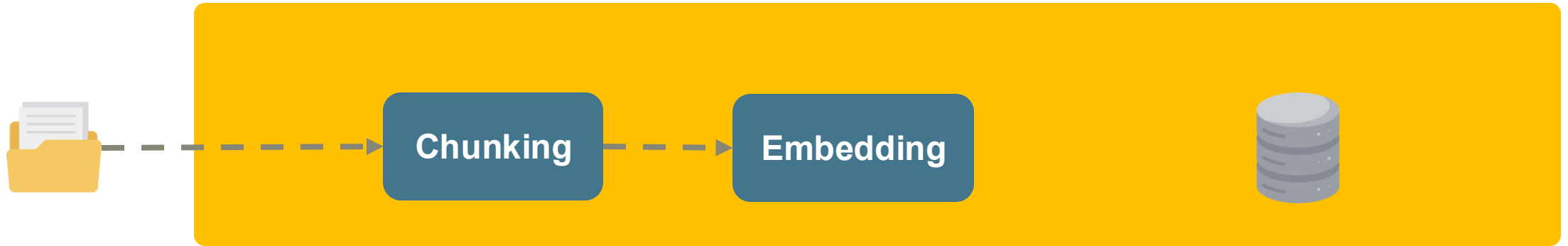
„It will soon be the third anniversary of the release of ChatGPT. There is still no end in sight to the hype. The added value of this technology is too great and its use in the form of a chat is too intuitive. In 2024, it was expected that the leap from exploratory playing around with large language models [...]“

— — ► [„ It will soon be the third anniversary of the release of ChatGPT. There is still no end in sight to the hype.“]

[„ The added value of this technology is too great and its use in the form of a chat is too intuitive.“]

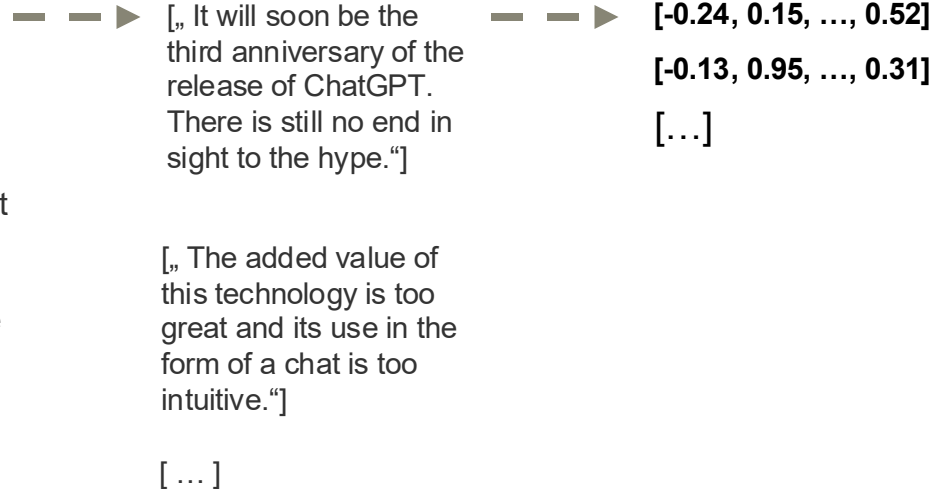
[...]

Ingesting Pipeline by Example

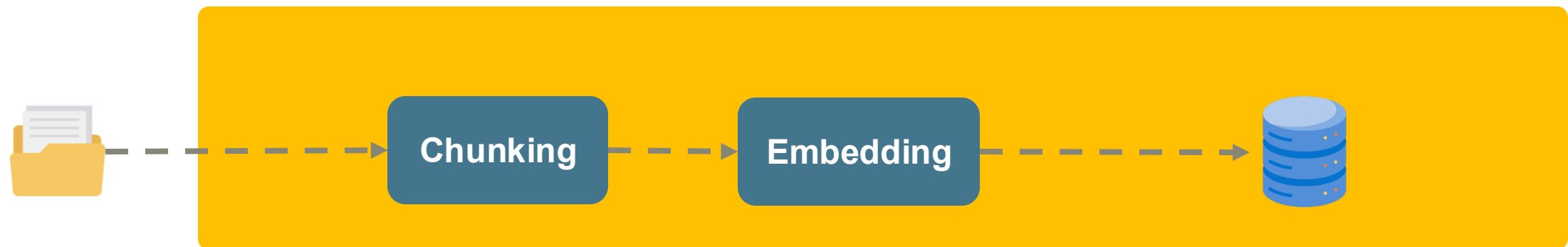


Document (Doc. Id:1)

„It will soon be the third anniversary of the release of ChatGPT. There is still no end in sight to the hype. The added value of this technology is too great and its use in the form of a chat is too intuitive. In 2024, it was expected that the leap from exploratory playing around with large language models [...]“

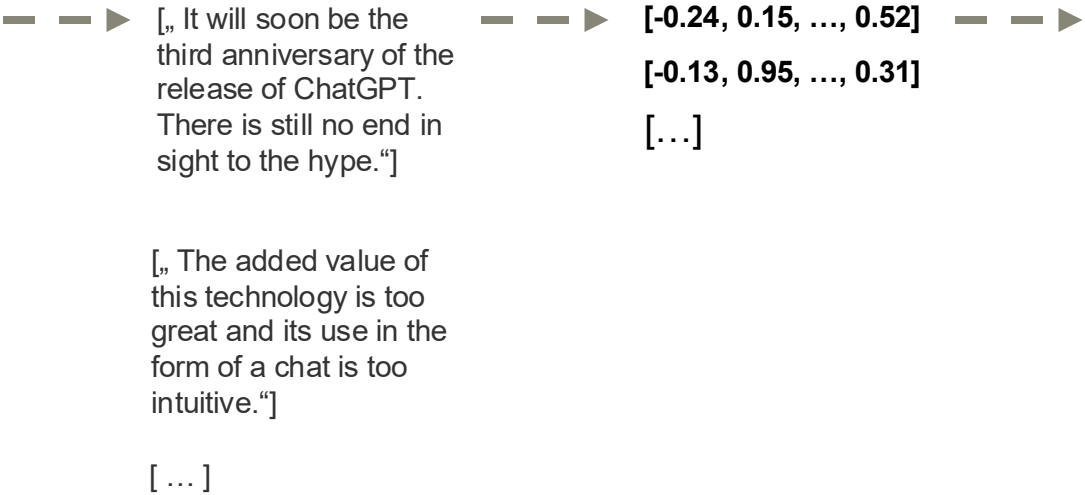


Ingesting Pipeline by Example



Document (Doc. Id:1)

„It will soon be the third anniversary of the release of ChatGPT. There is still no end in sight to the hype. The added value of this technology is too great and its use in the form of a chat is too intuitive. In 2024, it was expected that the leap from exploratory playing around with large language models [...]“



Id	Doc. Id	Embedding
1	1	[-0.24, 0.15, ..., 0.52]
2	1	[-0.13, 0.95, ..., 0.31]
[...]	[...]	[...]

Retrieval Pipeline by Example



Embedding

Top-k
Fetching

Augmenting

'What is the lecture 'The architecture for language models in practice - Retrieval Augmented Generation' about?'

Retrieval Pipeline by Example



Embedding

Top-k
Fetching

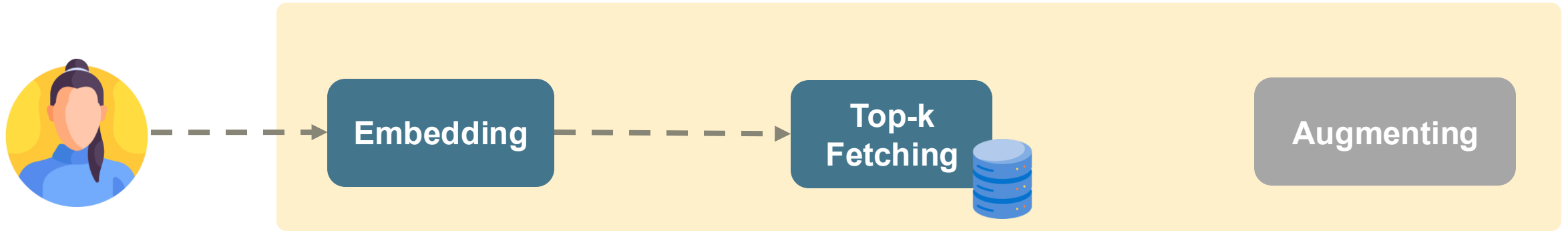
Augmenting

'What is the lecture 'The
architecture for language
models in practice -
Retrieval Augmented
Generation' about?'

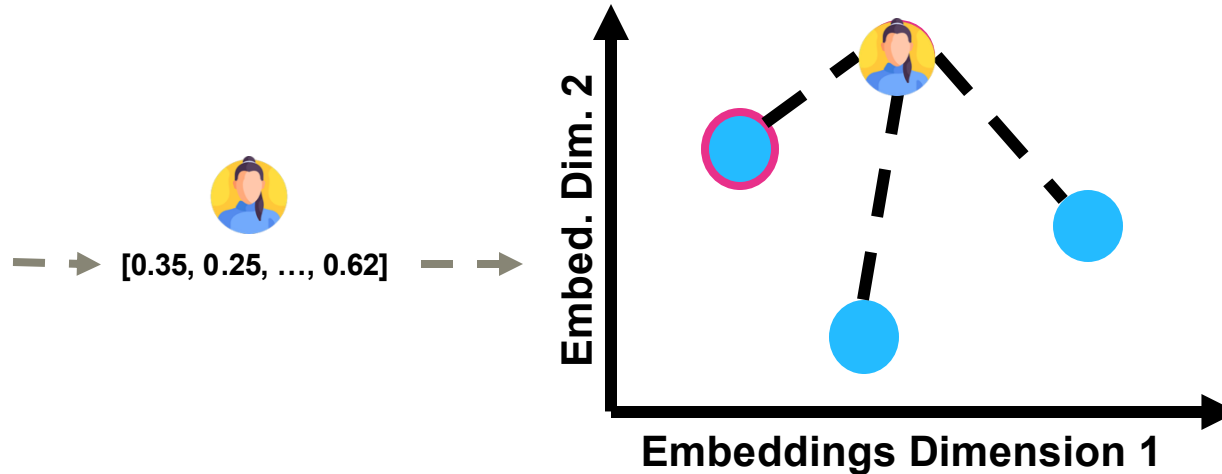


[0.35, 0.25, ..., 0.62]

Retrieval Pipeline by Example

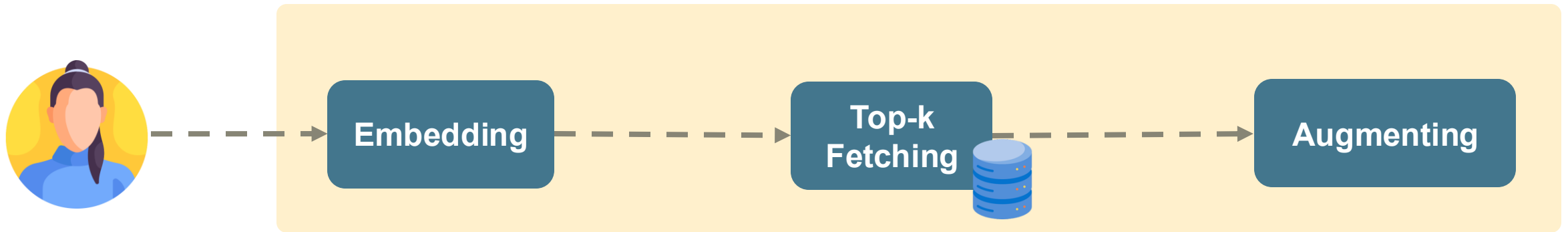


'What is the lecture 'The architecture for language models in practice - Retrieval Augmented Generation' about?'



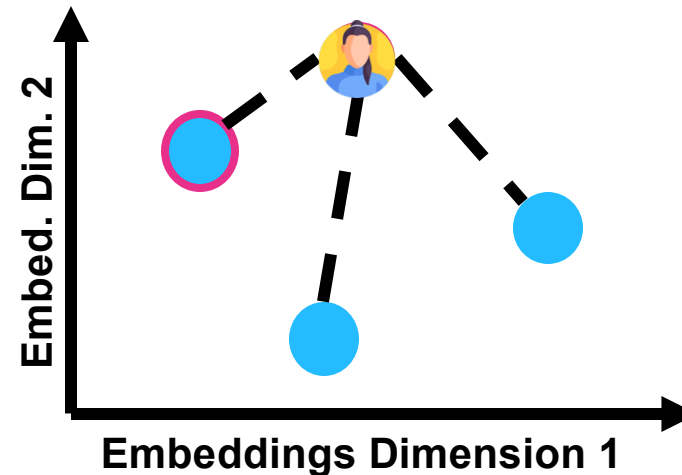
- Embeddings from knowledge base
- Top-k=1 document(s)

Retrieval Pipeline by Example



'What is the lecture 'The architecture for language models in practice - Retrieval Augmented Generation' about?'

→ [0.35, 0.25, ..., 0.62]



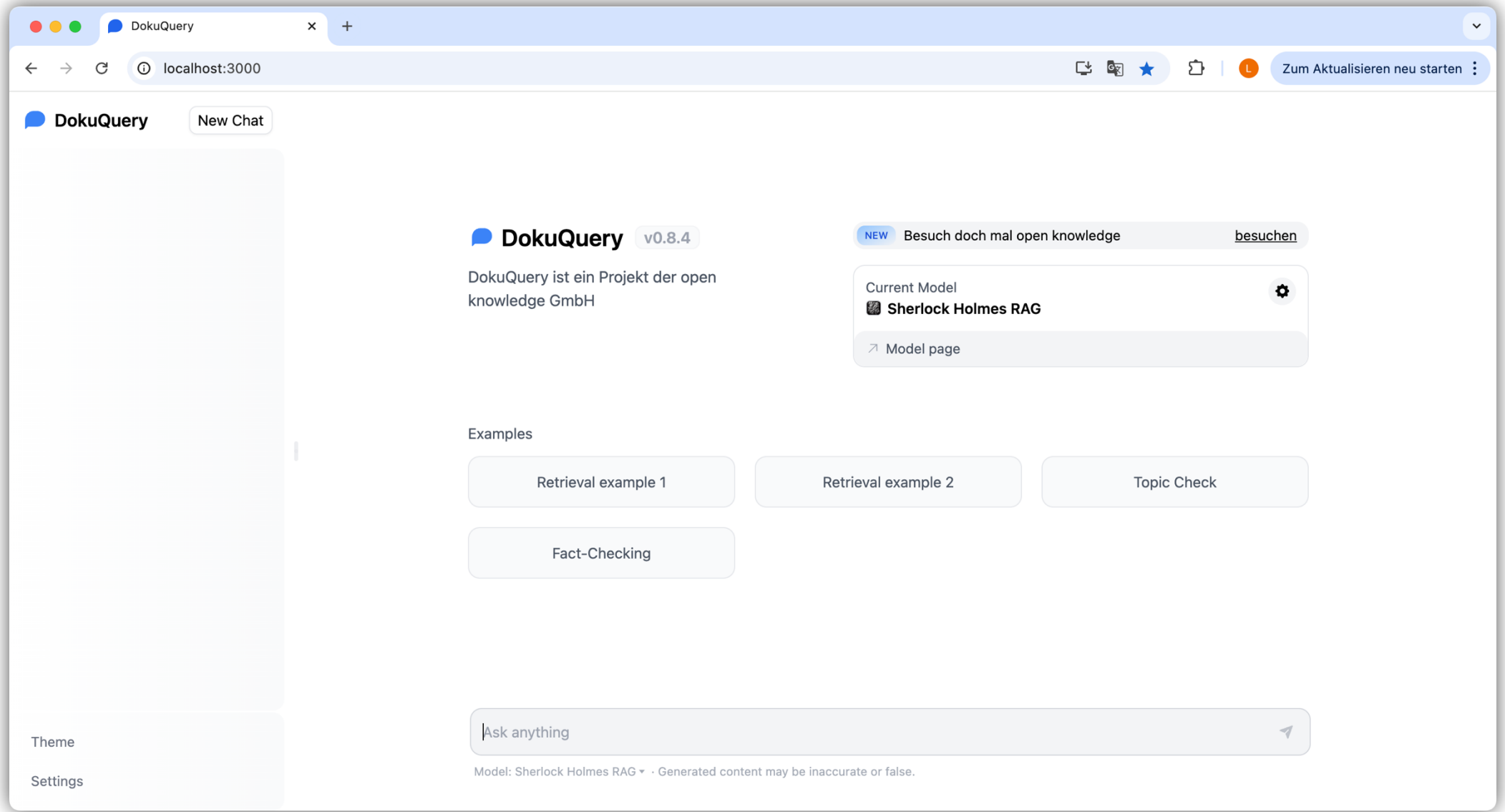
- Embeddings from knowledge base
- Top-k=1 document(s)

„What is the lecture [...] based on the following context:

'An upstream retrieval process to pull in relevant information from our own dynamic database can provide well-founded answers.'



Sherlock RAG in Action



Lucy noticed a number on

✕

+

←

→

↻

localhost:3000/conversation/671f9f901f7738906dd8cab2

🖨

🗨

☆

🔖

🔴

Zum Aktualisieren neu starten ⋮

DokuQuery

New Chat

Today

Lucy noticed a number on

Theme

Settings

Lucy noticed a number on the ceiling when taking breakfast. which number was written into the ceiling?

The number written on the ceiling was 28.

Sources:

study_in_scarlett.txt

adventures_of_sherlock_holmes.txt

study_in_scarlett.txt

Ask anything

➤

Model: Sherlock Holmes RAG ▾ · Generated content may be inaccurate or false. [Share this conversation](#)

What weapon was the thumb

+

localhost:3000/conversation/671fa0111f7738906dd8cab6

Zum Aktualisieren neu starten

DokuQuery

New Chat

Today

What weapon was the thumb

Lucy noticed a number on

Theme

Settings

What weapon was the thumb of the engineer hurt with?

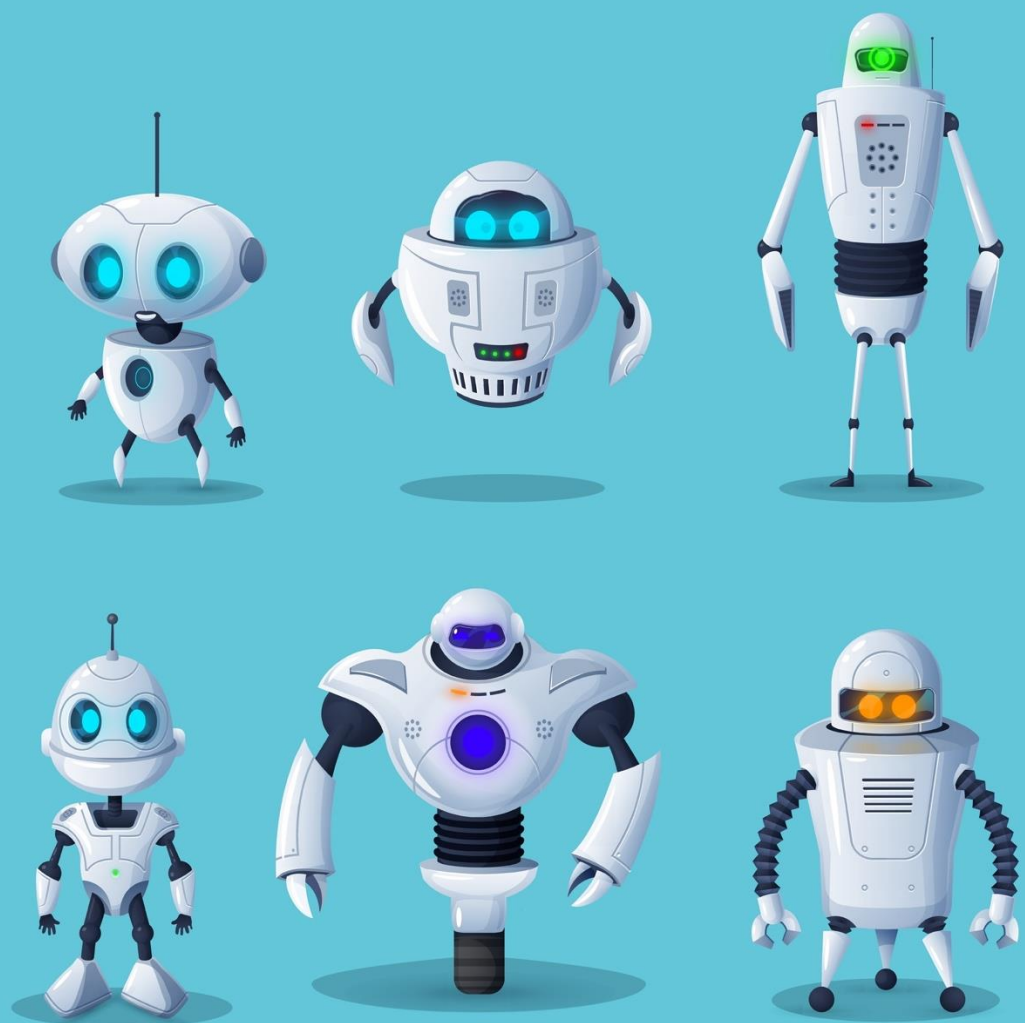
The thumb of the engineer was hurt with a heavy and sharp instrument, possibly a cleaver.

Sources: adventures_of_sherlock_holmes.txt adventures_of_sherlock_holmes.txt adventures_of_sherlock_holmes.txt

Ask anything

Model: Sherlock Holmes RAG · Generated content may be inaccurate or false.

[Share this conversation](#)



Hands-On

RAG System Simple Version

Implementing a **simple RAG system** using **ingestion and retrieval** pipelines.

Applying the RAG system to **questions from your own domain**.

Hands-on Simple RAG

The screenshot displays a JupyterLab environment. On the left, a file explorer shows the project structure: `workshop-genai-student` / `Volumes/work/commun`. The `exercises` folder is expanded, showing subfolders `00_hello_genai`, `01_genai_starter`, and `02_rag_basic`. The `02_rag_basic` folder is selected, showing files `02_rag_basic.ipynb` (selected), `README.md`, `03_rag_advanced`, `04_rag_pitfalls`, `05_rag_outlook`, `solutions`, `TODO.MD`, `.gitignore`, `LICENSE`, and another `README.md`. Below these are `External Libraries` and `Scratches and Consoles`.

The main editor area shows the file `02_rag_basic.ipynb`. The title bar indicates 'Current File', 'Managed Jupyter server: auto-start', and 'Not specified'. The code is as follows:

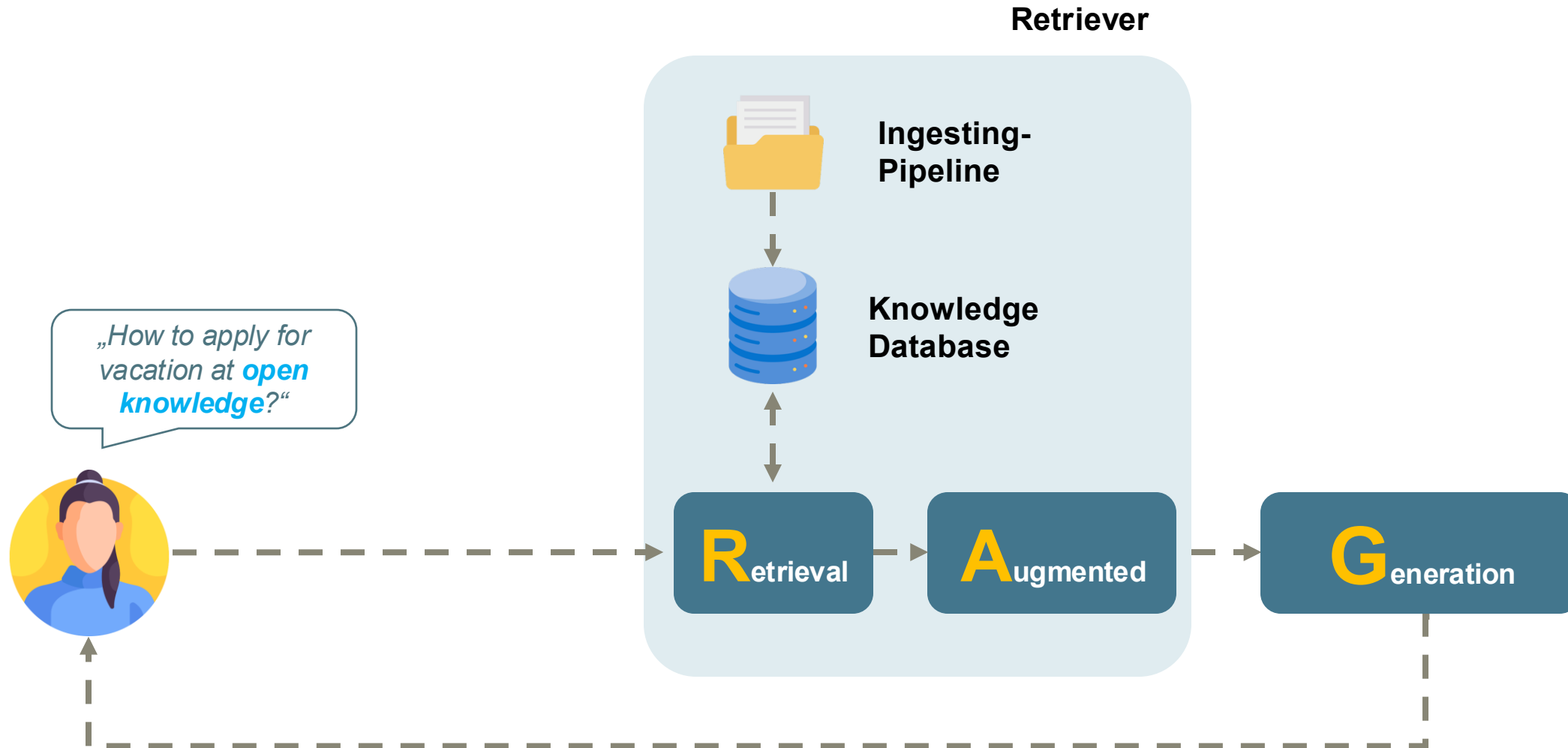
```
In _ 1 # Get content of books. The content will already be cleansed.
2 def load_file_content(file_name: str) -> str:
3     with open(f"{PROCESSED_DATA_PATH}{file_name}", "r") as f:
4         return f.read()

In 120 1 # Building Block "Chunking": Split the content into smaller chunks
2 def do_chunk(text: str) -> list[str]:
3     text_splitter = RecursiveCharacterTextSplitter(
4         chunk_size=DEFAULT_CHUNK_SIZE,
5         chunk_overlap=DEFAULT_CHUNK_OVERLAP,
6         length_function=len,
7     )
8     return text_splitter.split_text(text)

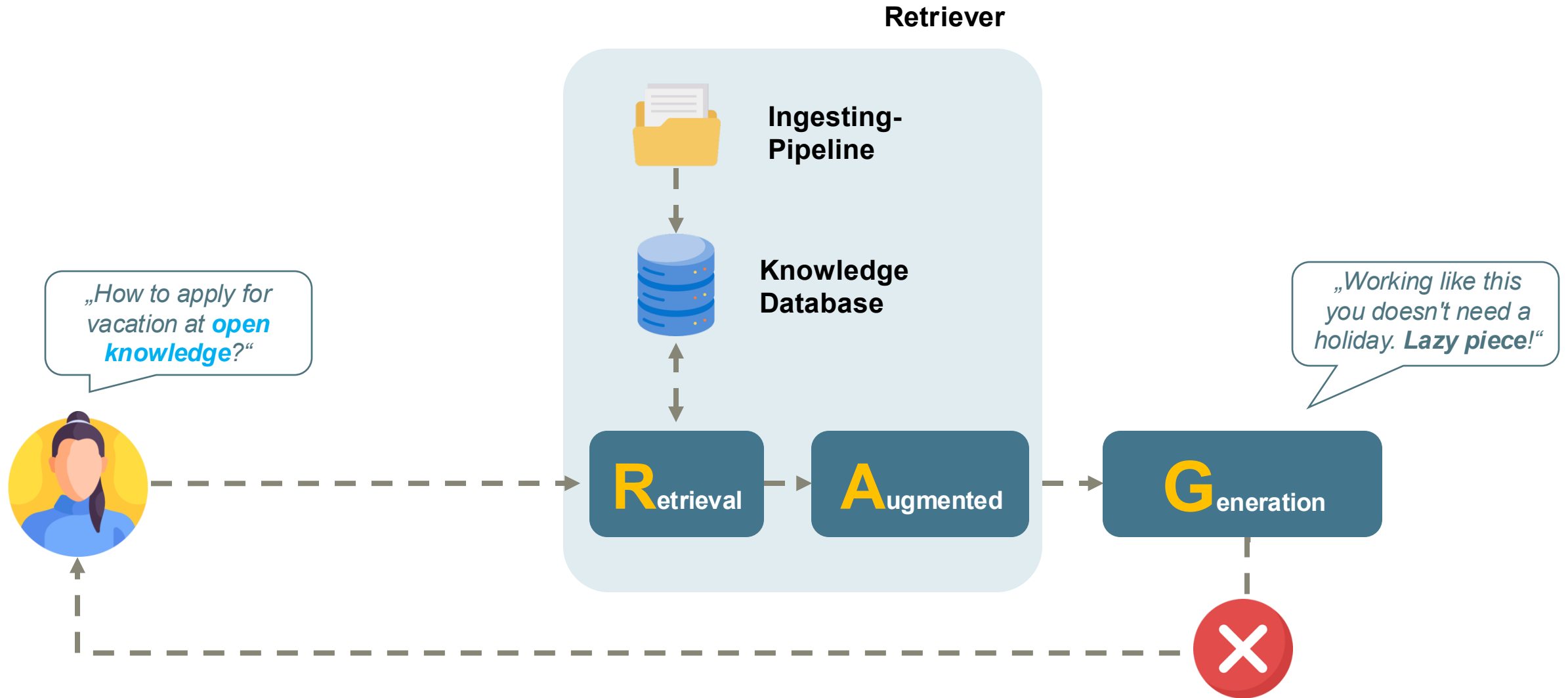
In 121 1 # Building Block "Embedding": Create multi dimensional embeddings for a given
```

The status bar at the bottom shows the file path: `workshop-genai-student > genai-ws > exercises > 02_rag_basic > 02_rag_basic.ipynb`. It also displays '2:1 (36 chars)', 'LF', 'UTF-8', '4 spaces', and 'Poetry (workshop-genai-student) [Python 3.13.0]'.

GenAI Advanced RAG Systems



GenAI Advanced RAG Systems

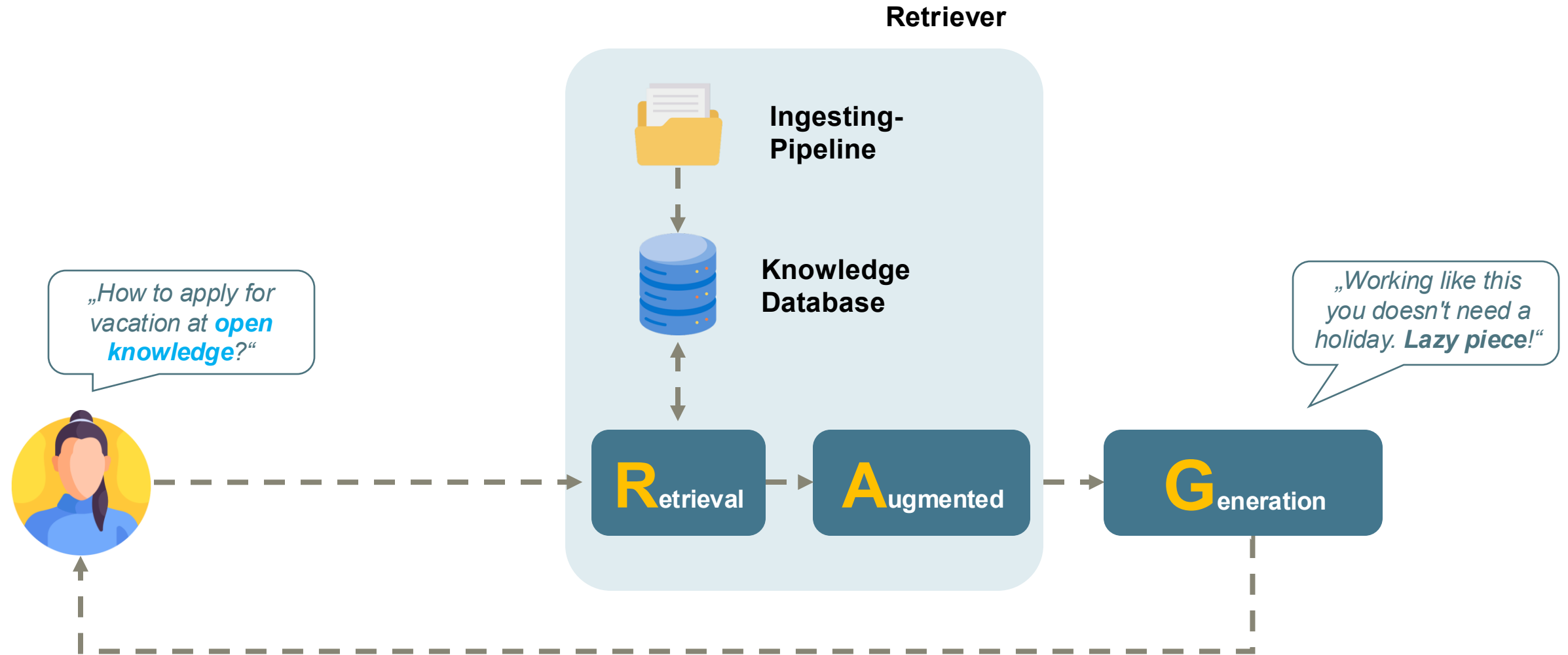


03

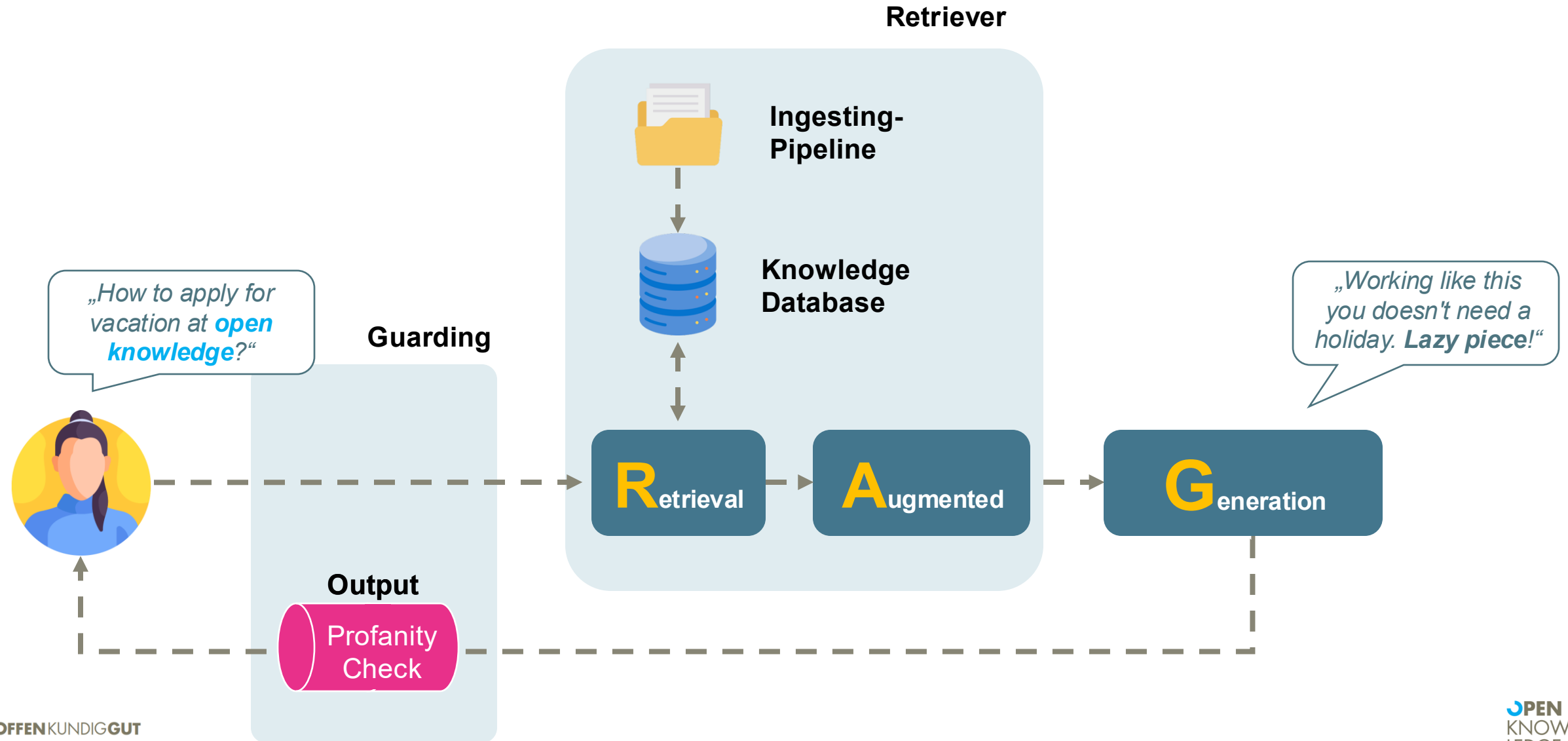
GenAI Professional

*What is still missing
for a sustainable and
productive operation?*

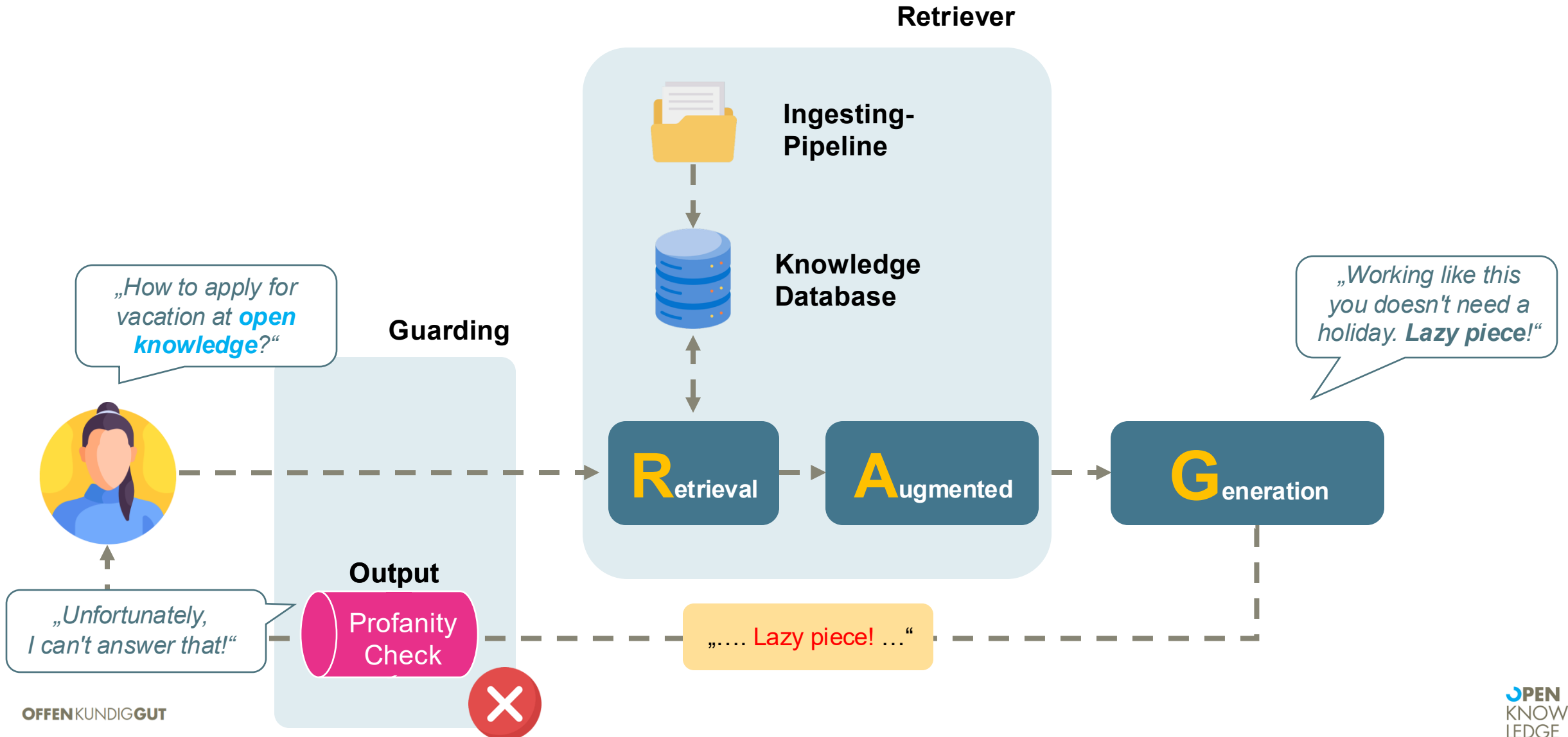
GenAI Professional Guardrails



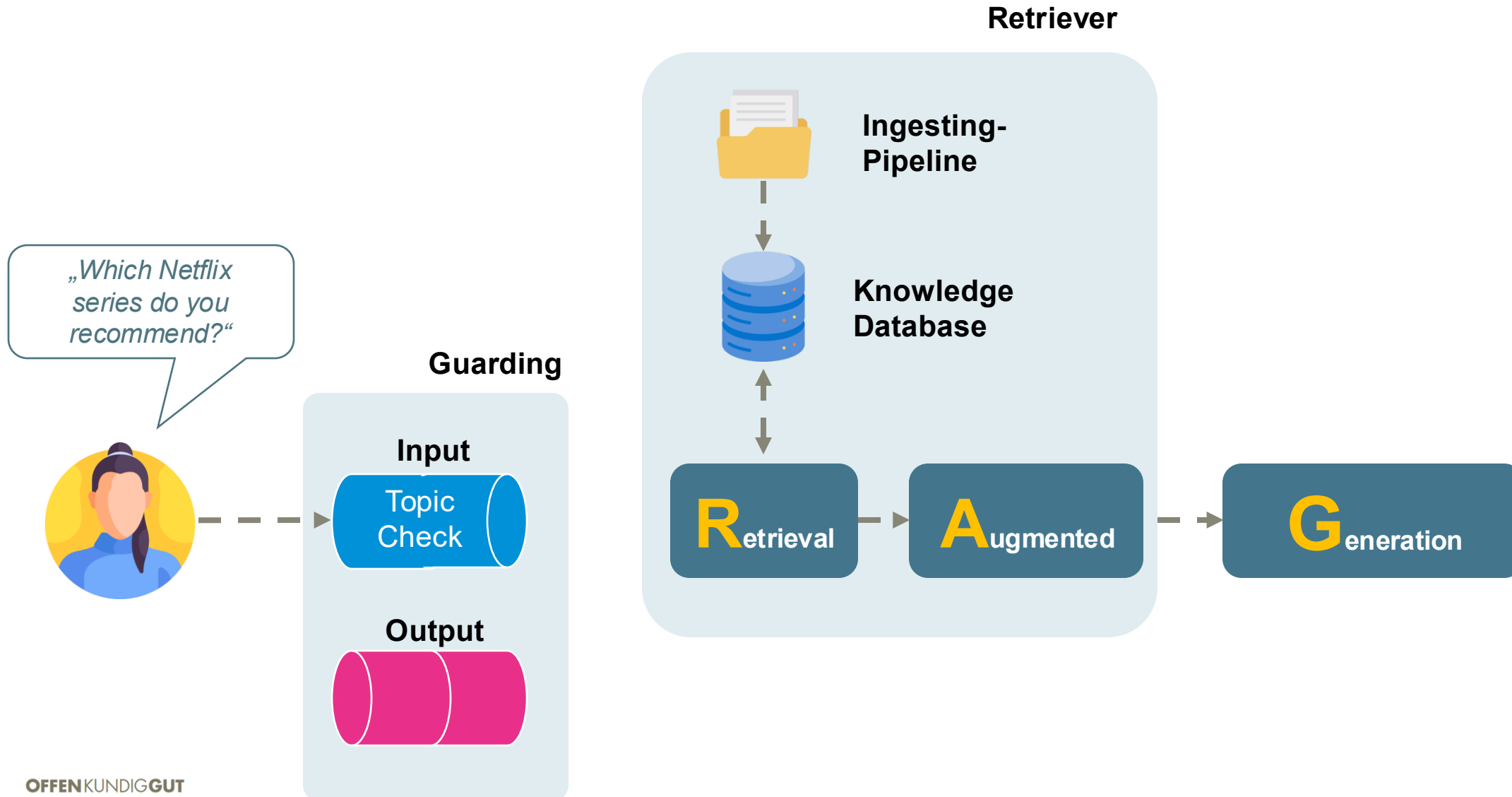
GenAI Professional Guardrails



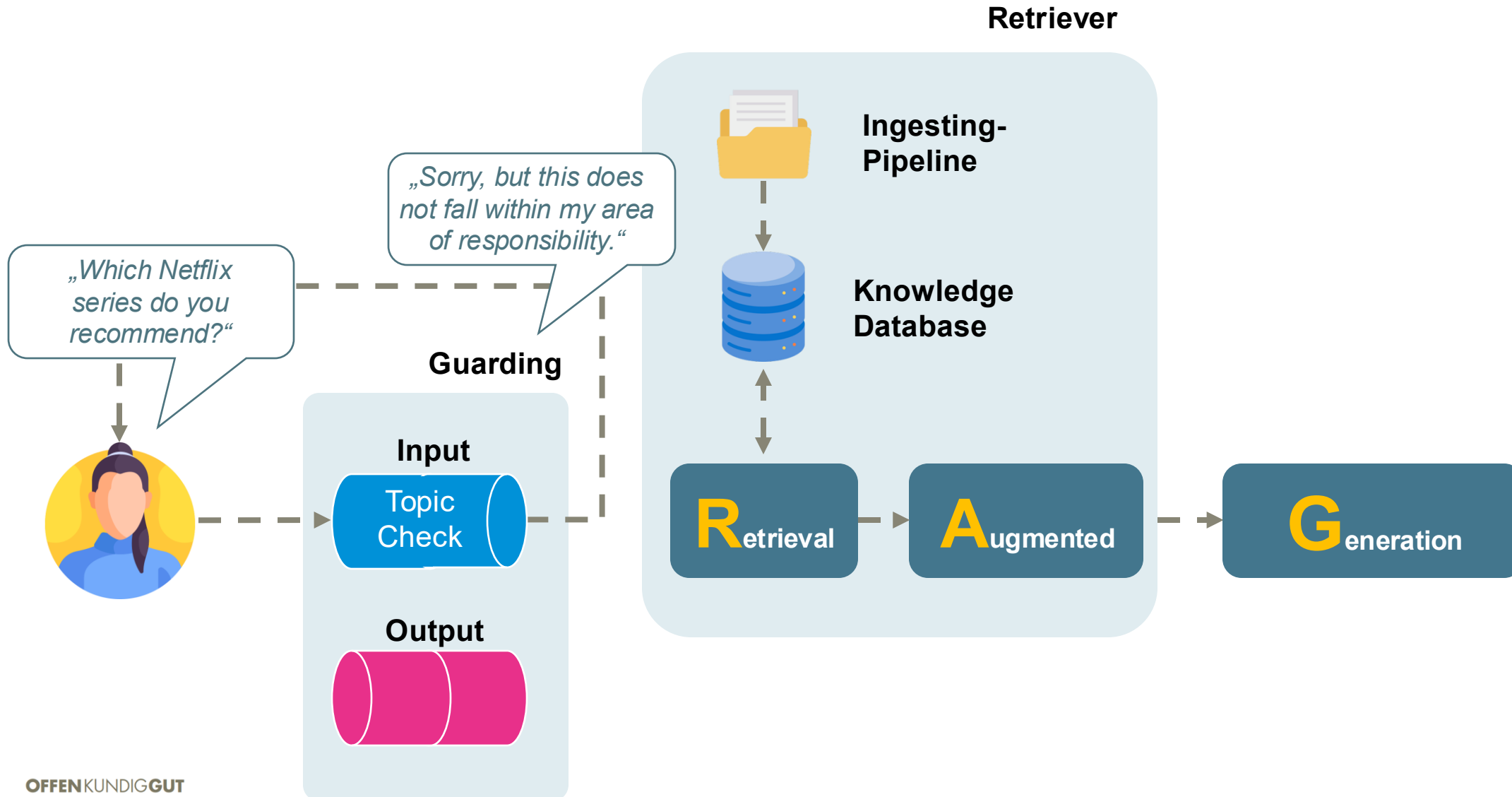
GenAI Professional Guardrails



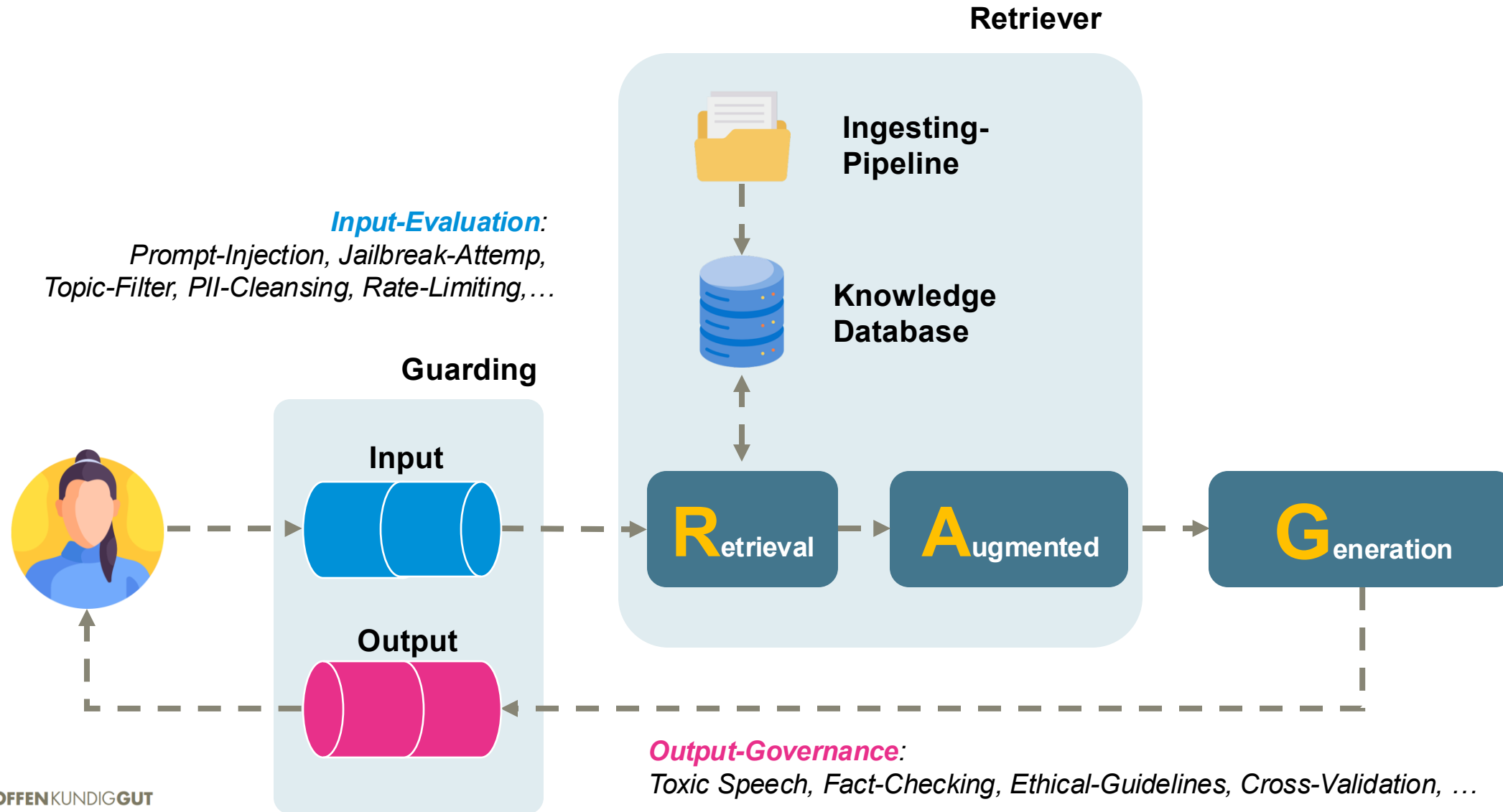
GenAI Professional Guardrails



GenAI Professional Guardrails

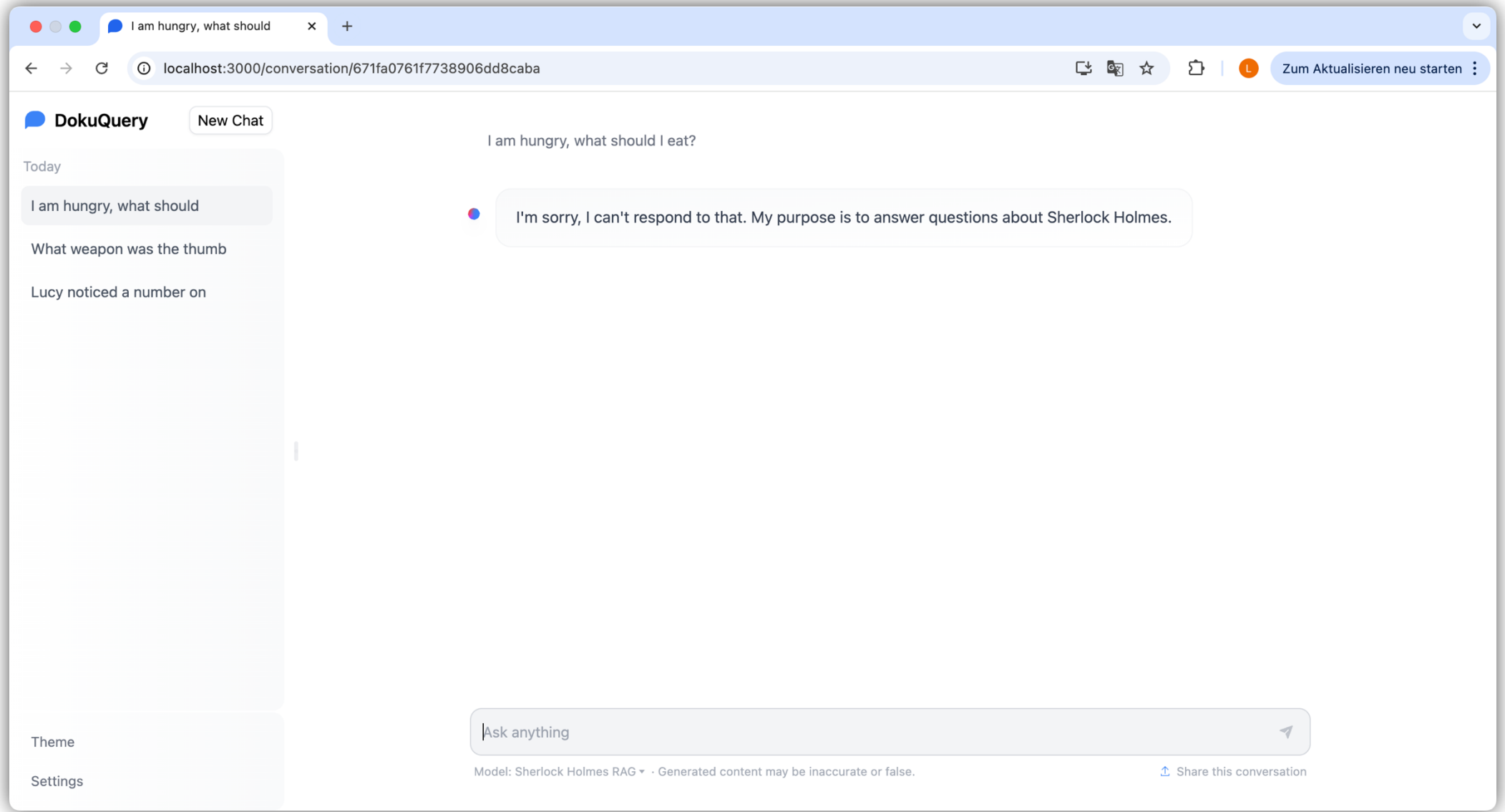


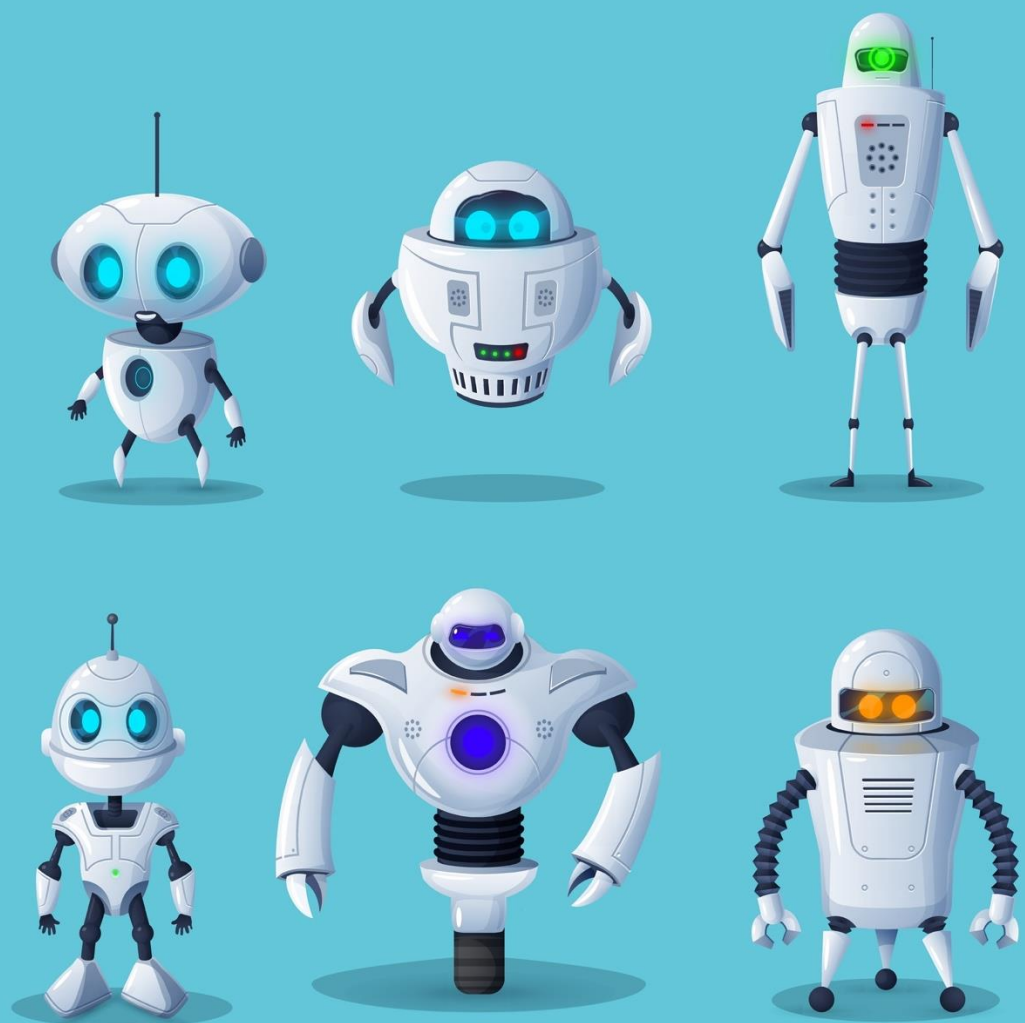
GenAI Professional Guardrails





Sherlock RAG in Action





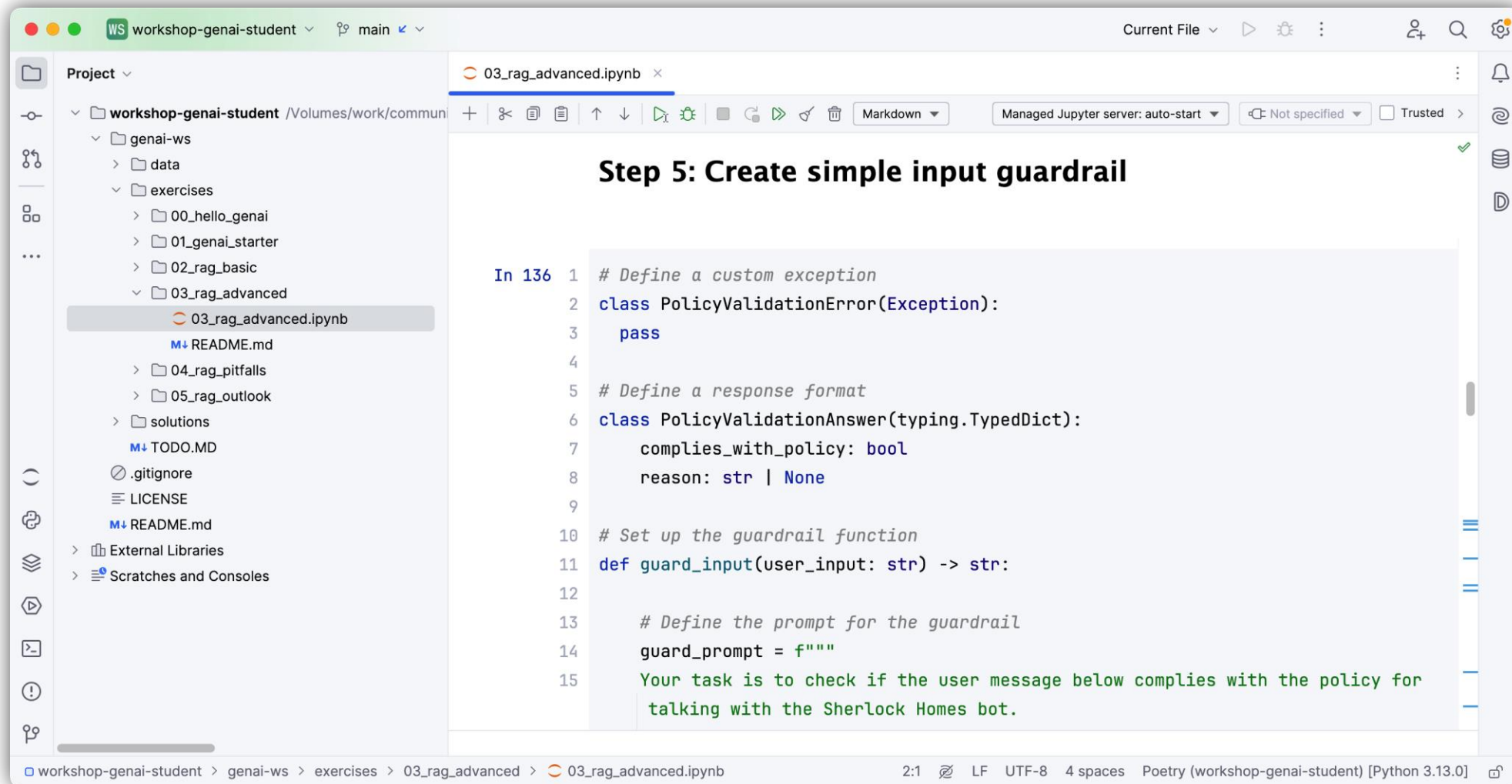
Hands-On

RAG System Guardrails

Create your own guardrails for input and output to **prevent toxic language**.

Implementing a simple Output Guardrail as a **fact checker**.

Hands-on Guardrails to the rescue



Project

- workshop-genai-student /Volumes/work/commun
 - genai-ws
 - data
 - exercises
 - 00_hello_genai
 - 01_genai_starter
 - 02_rag_basic
 - 03_rag_advanced
 - 03_rag_advanced.ipynb
 - README.md
 - 04_rag_pitfalls
 - 05_rag_outlook
 - solutions
 - TODO.MD
 - .gitignore
 - LICENSE
 - README.md
 - External Libraries
 - Scratches and Consoles

03_rag_advanced.ipynb

Step 5: Create simple input guardrail

```
In 136 1 # Define a custom exception
2 class PolicyValidationError(Exception):
3     pass
4
5 # Define a response format
6 class PolicyValidationAnswer(typing.TypedDict):
7     complies_with_policy: bool
8     reason: str | None
9
10 # Set up the guardrail function
11 def guard_input(user_input: str) -> str:
12
13     # Define the prompt for the guardrail
14     guard_prompt = f"""
15     Your task is to check if the user message below complies with the policy for
    talking with the Sherlock Homes bot.
```

workshop-genai-student > genai-ws > exercises > 03_rag_advanced > 03_rag_advanced.ipynb 2:1 LF UTF-8 4 spaces Poetry (workshop-genai-student) [Python 3.13.0]

Hands-on Guardrails to the rescue

On this page

Demo

How It Works

Getting Started

Performance

Getting Started

Learning Library

More Resources

See NVIDIA NeMo Guardrails in Action

Enforce content safety, RAG grounding, and jailbreak prevention while building secure, compliant AI agents. This video demonstrates how NeMo Guardrails streamlines guardrail orchestration for safer, more reliable AI applications.

 Building Safe and Secure LLM Applications Using NVIDIA NeMo Guardrails

Teilen

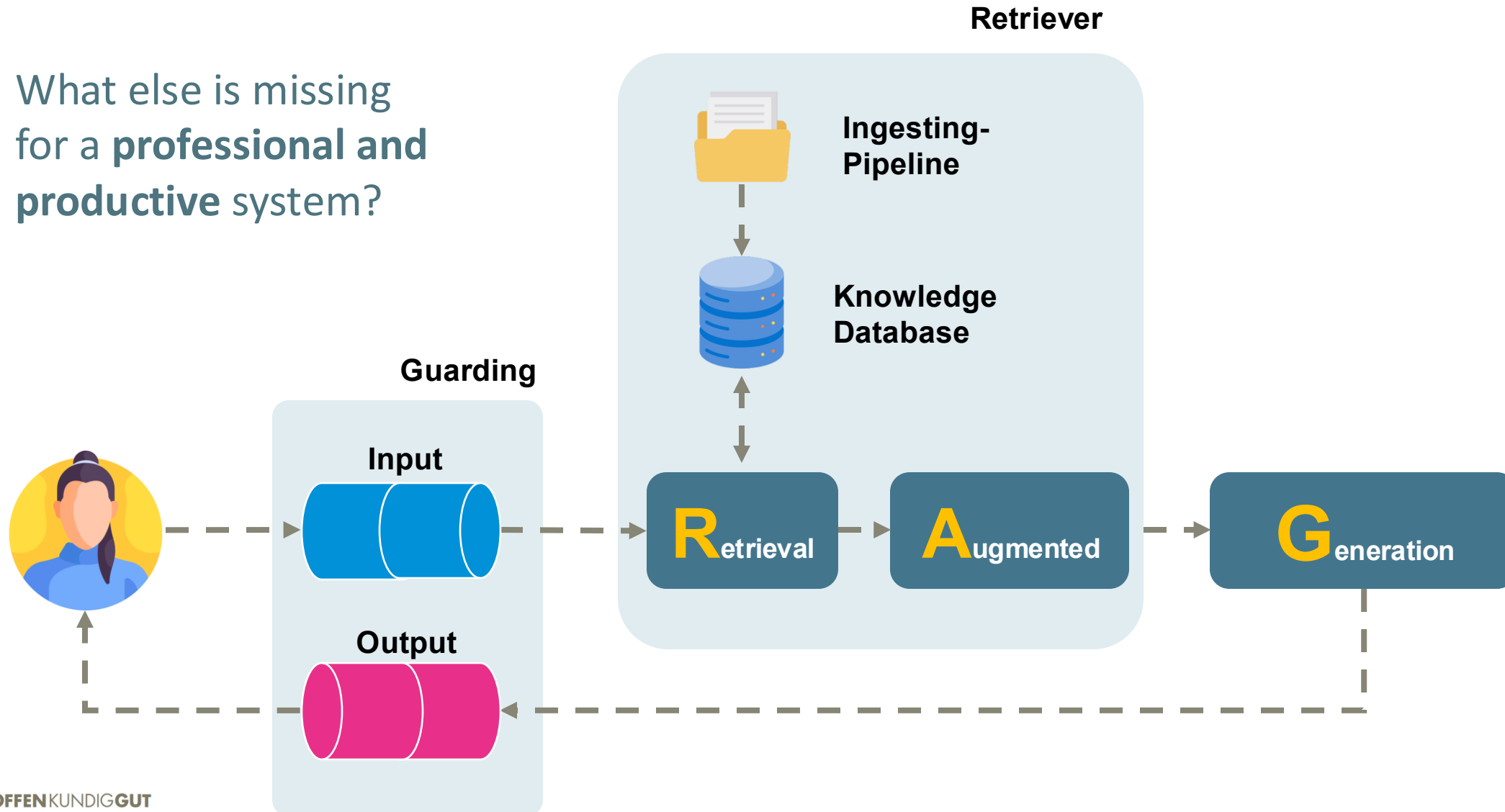


Ansehen auf  YouTube

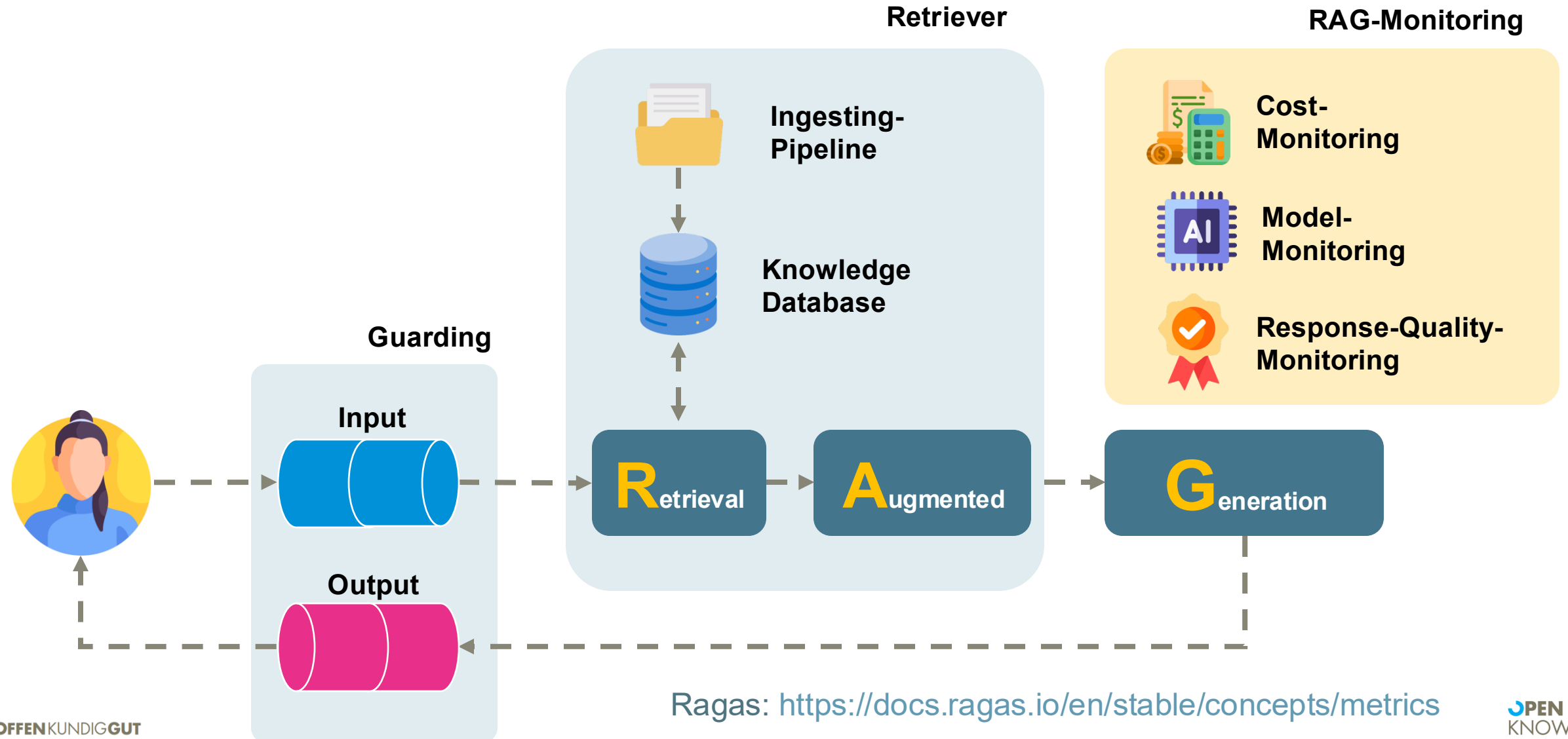
Feedback

GenAI Professional

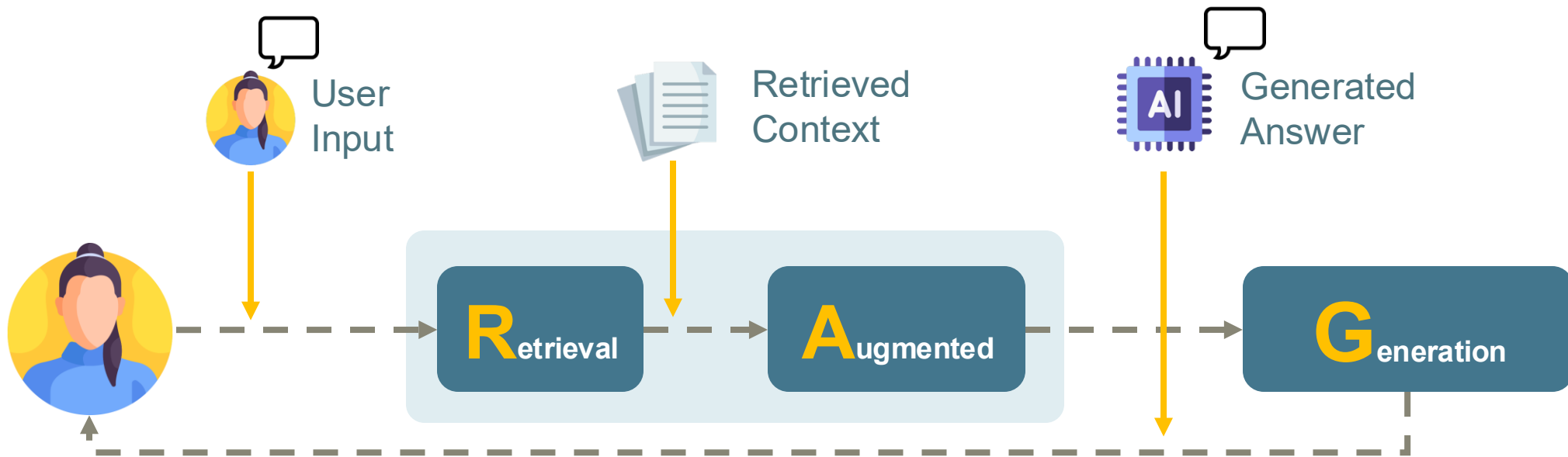
What else is missing for a **professional and productive** system?



GenAI Professional Monitoring

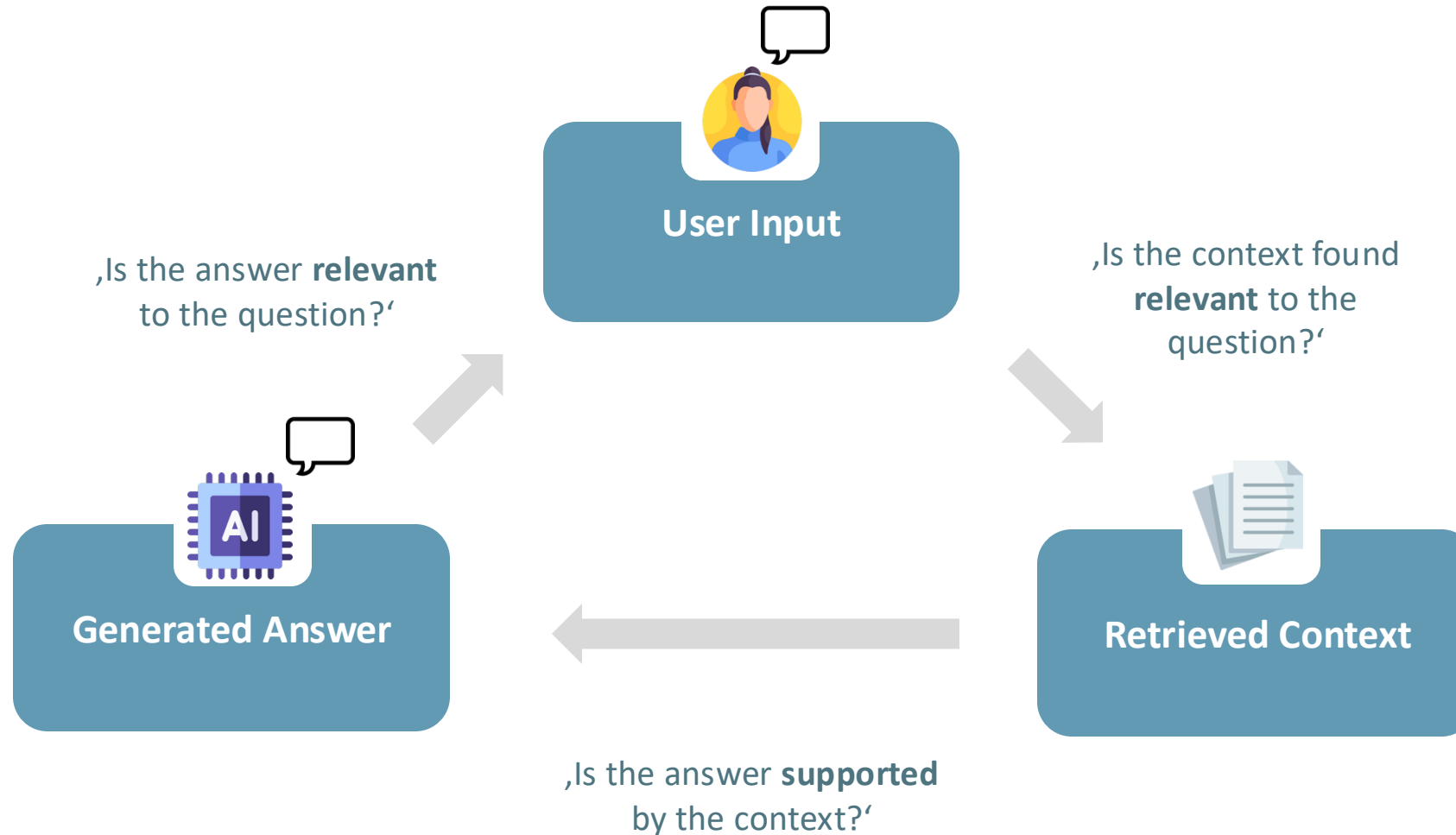


GenAI Professional Monitoring



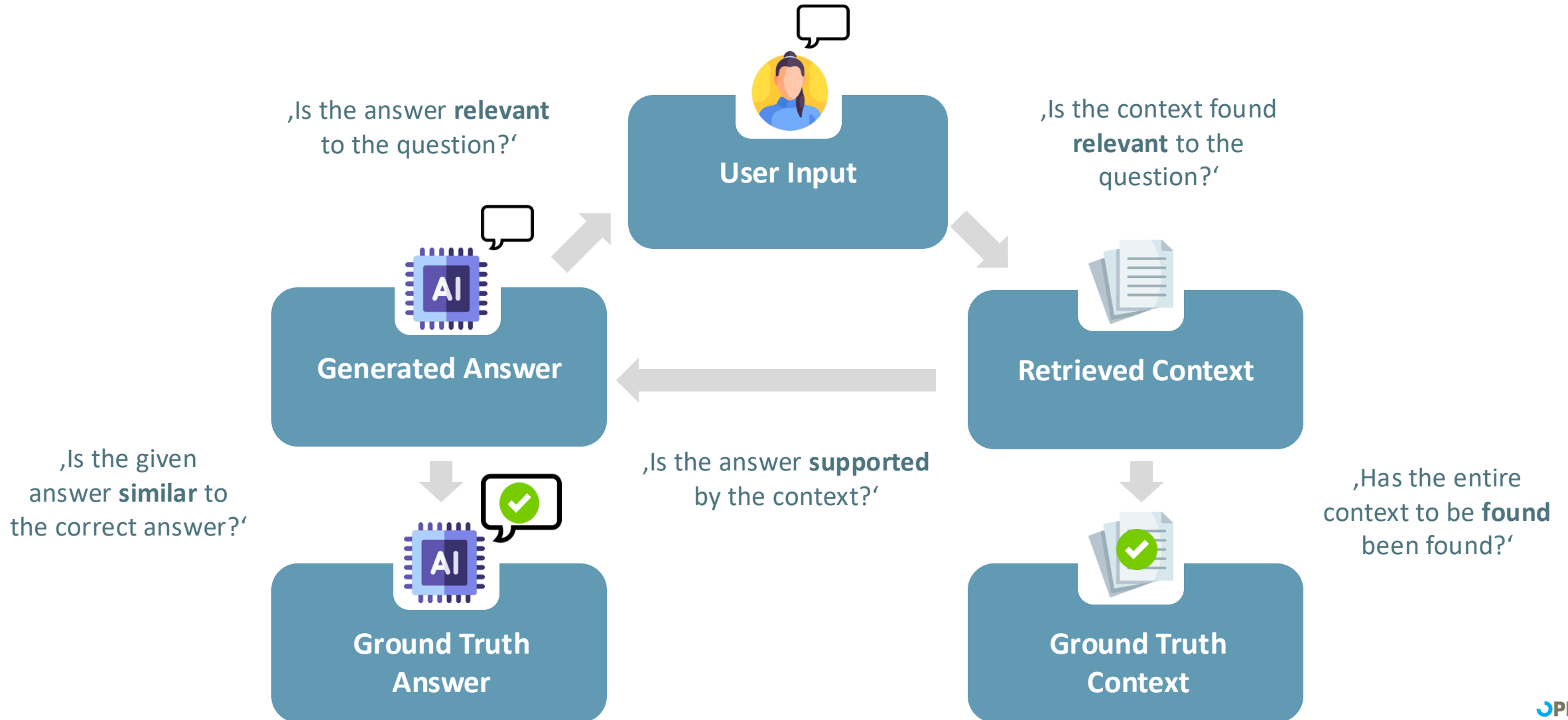
GenAI Professional

Monitoring: The RAG-Triad



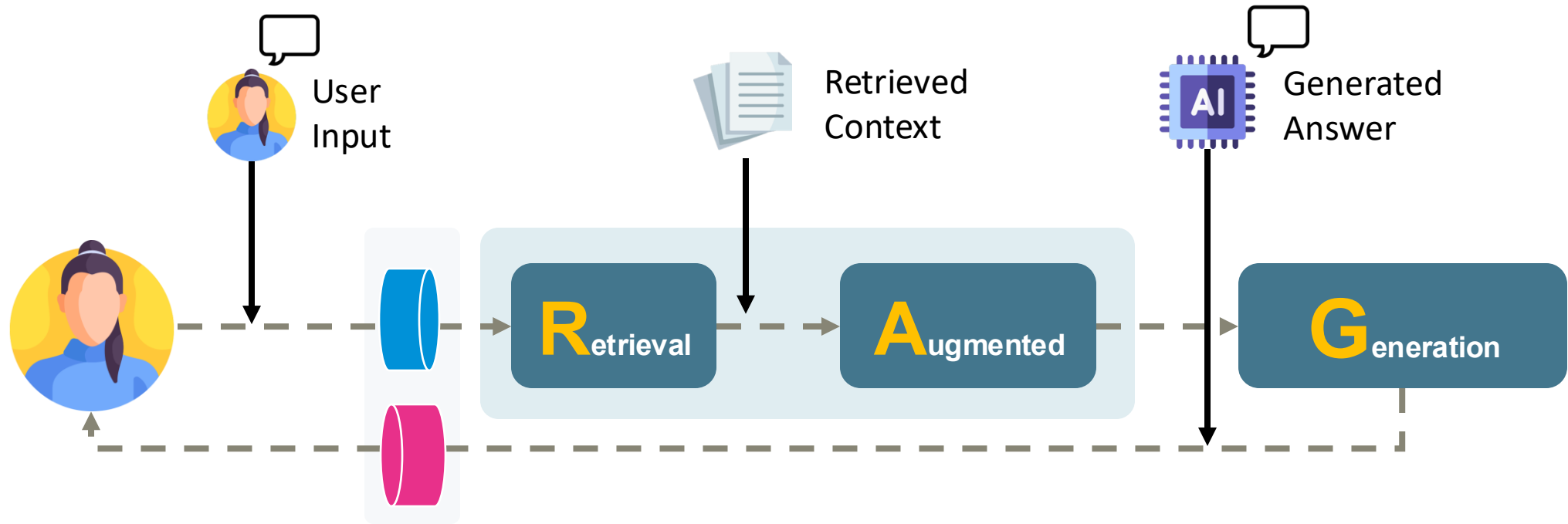
GenAI Professional

Monitoring: The RAG-Triad+



GenAI Professional

Response Quality Monitoring



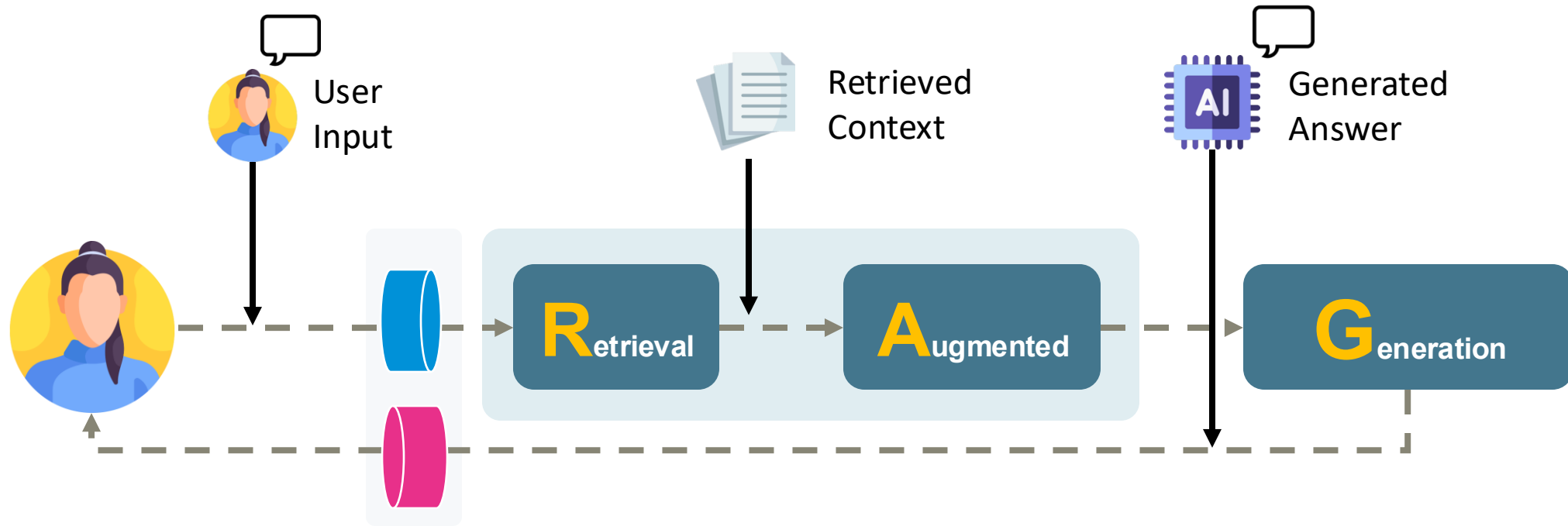
G-Eval
Deep Acyclic Graph
Answer Relevancy

Faithfulness
Summarization
Hallucination

Contextual Recall
Contextual Relevancy
Contextual Precision

GenAI Professional

Response Quality Monitoring

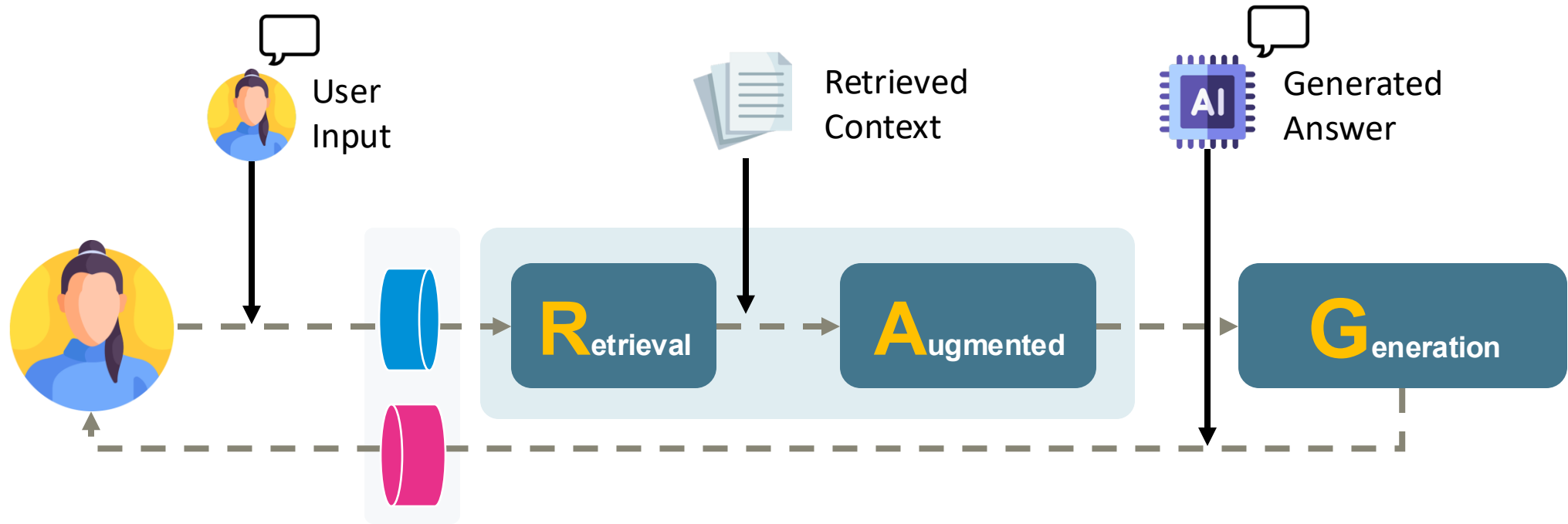


Faithfulness

Is the  supported by the  ?

GenAI Professional

Response Quality Monitoring



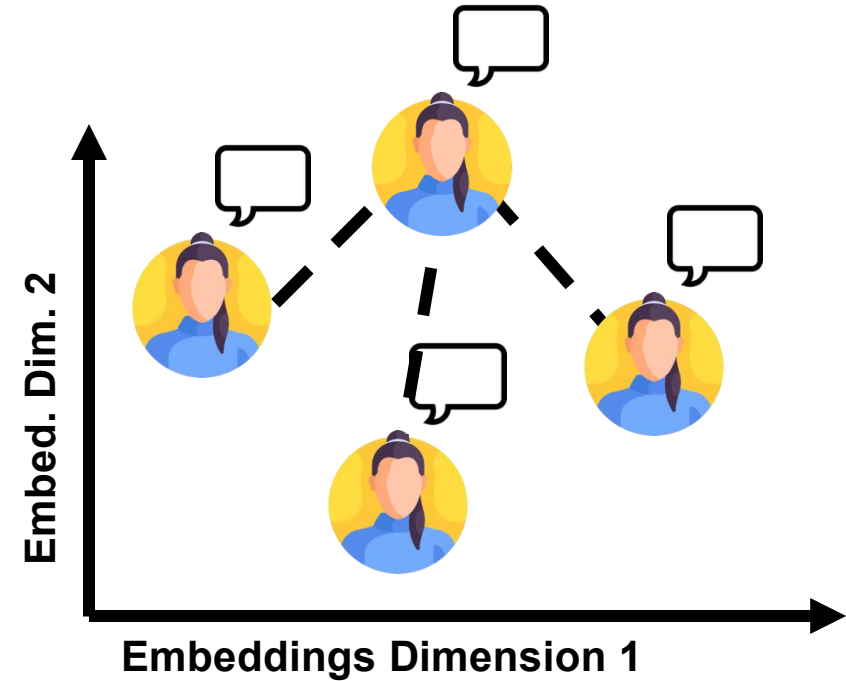
Answer Relevancy

Does the  match the  ?

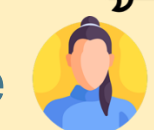
GenAI Professional

Response Quality Monitoring

Step 1: Generate possible questions () from 



Answer Relevancy

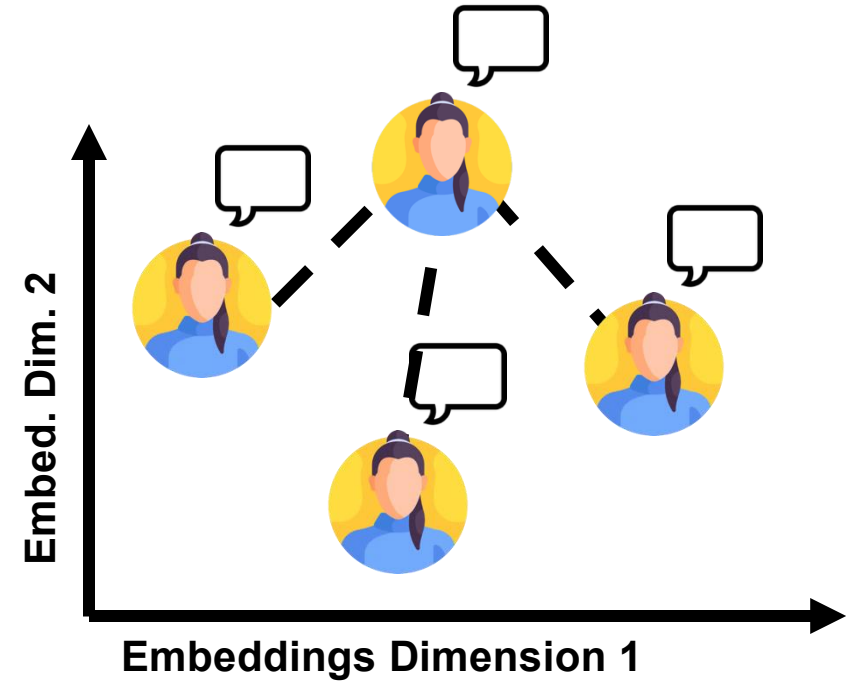
Does the  match the  ?

GenAI Professional

Response Quality Monitoring

Step 1: Generate possible questions () from  

Step 2: Determine the average distance from  



Answer Relevancy

Does the   match the   ?



RAG in Action aka Sherlock RAG

How many apples did Sherlock x +

localhost:3000/conversation/671fa09b1f7738906dd8cabd

Zum Aktualisieren neu starten

DokuQuery

New Chat

Today

How many apples did Sherlock

I am hungry, what should

What weapon was the thumb

Lucy noticed a number on

Theme

Settings

How many apples did Sherlock Holmes eat before he solved the murder of Watson?

I'm sorry, it seems that the context provided does not contain enough information. Therefore I refuse to respond.

Sources: study_in_scarlett.txt study_in_scarlett.txt adventures_of_sherlock_holmes.txt adventures_of_sherlock_holmes.txt

Ask anything

Model: Sherlock Holmes RAG · Generated content may be inaccurate or false.

[Share this conversation](#)

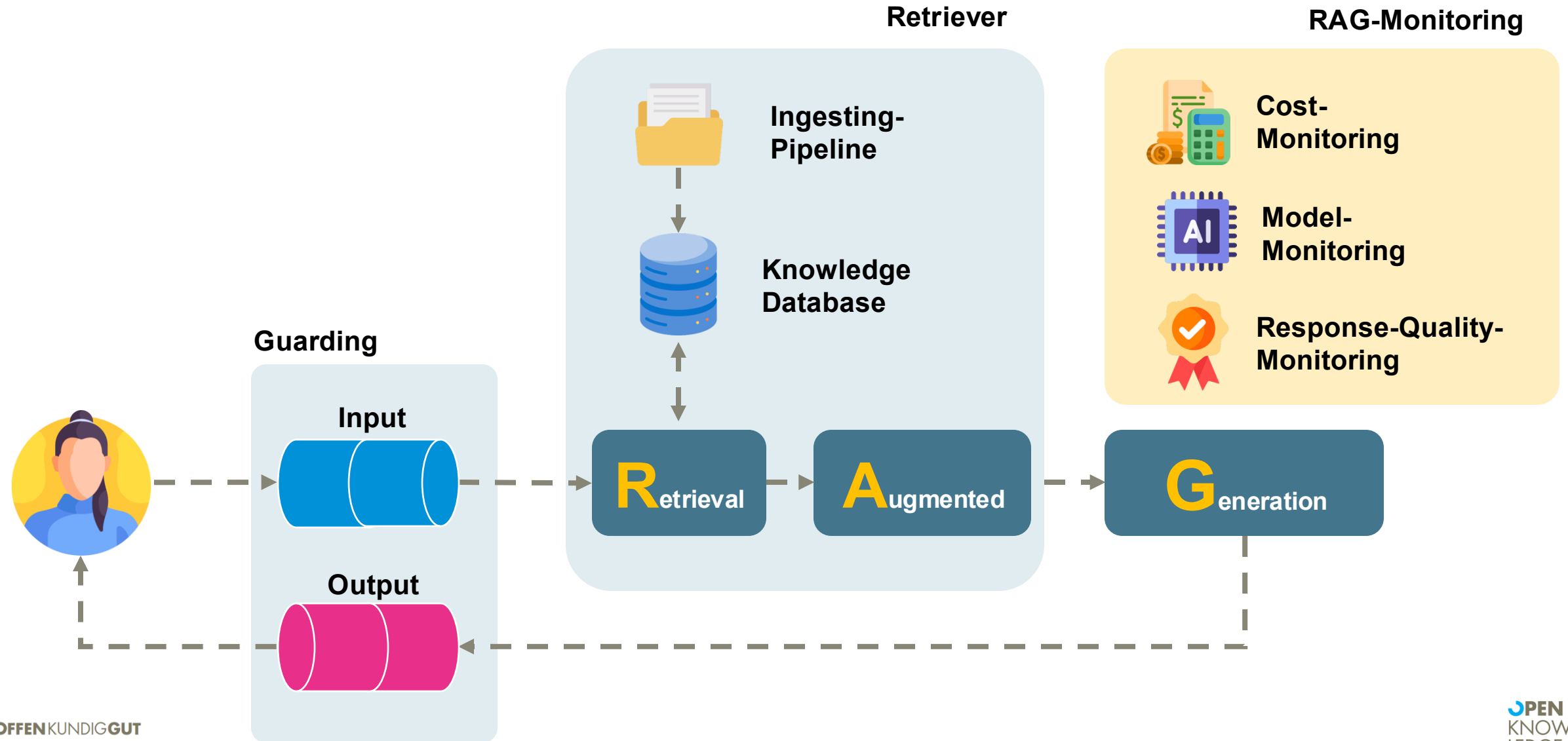
04

Best Practices

What are the typical pitfalls that I will run into - and how do I deal with them?

GenAI Best Practices

Real Life Survival Guide



GenAI Best Practices

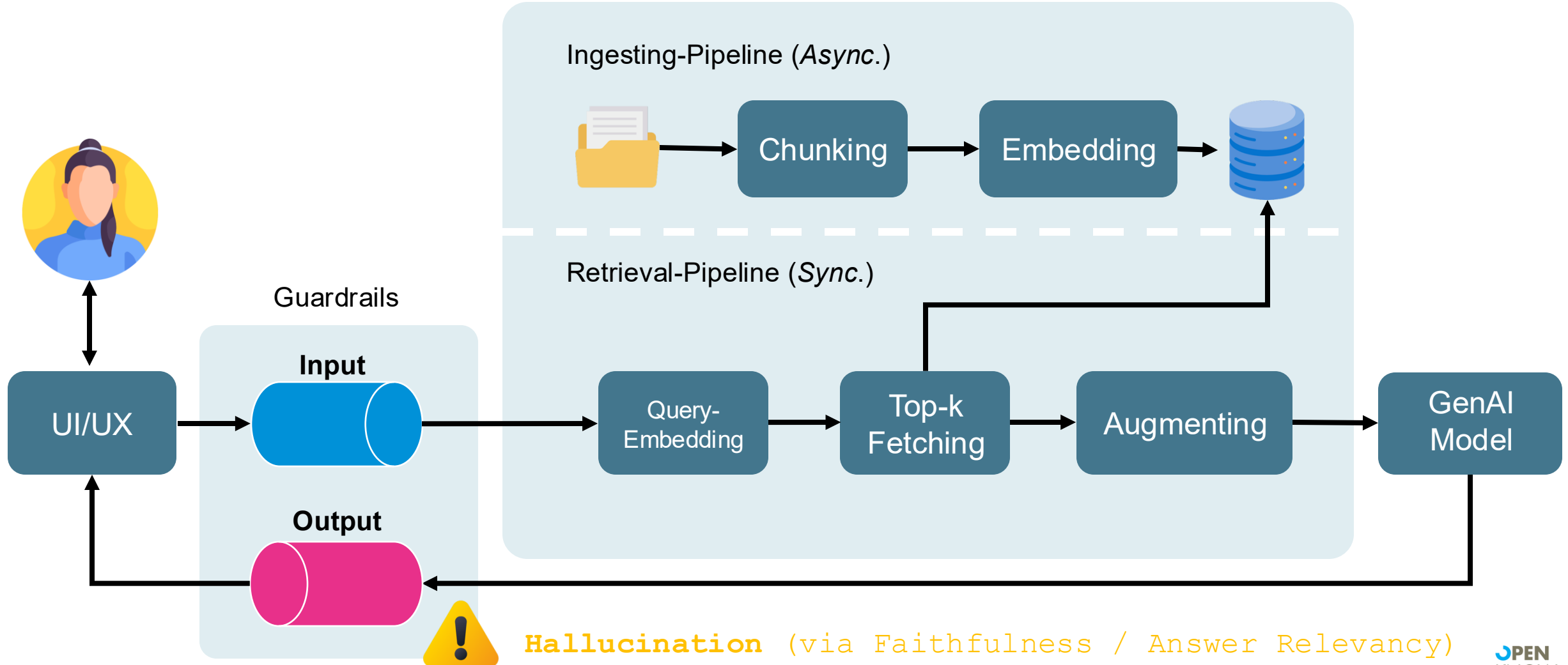
Real Life Survival Guide



'The answers sound conclusive, but they are wrong!'

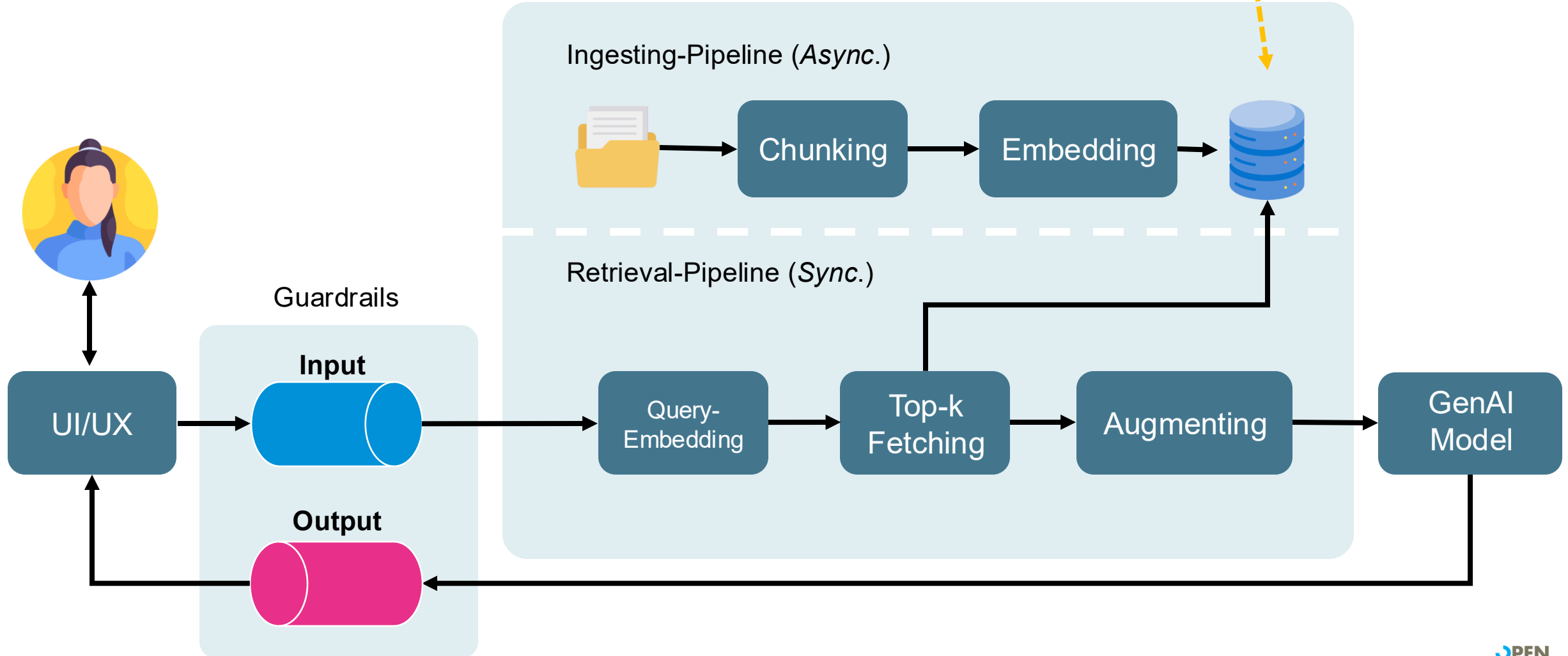
GenAI Best Practices

Real Life Survival Guide



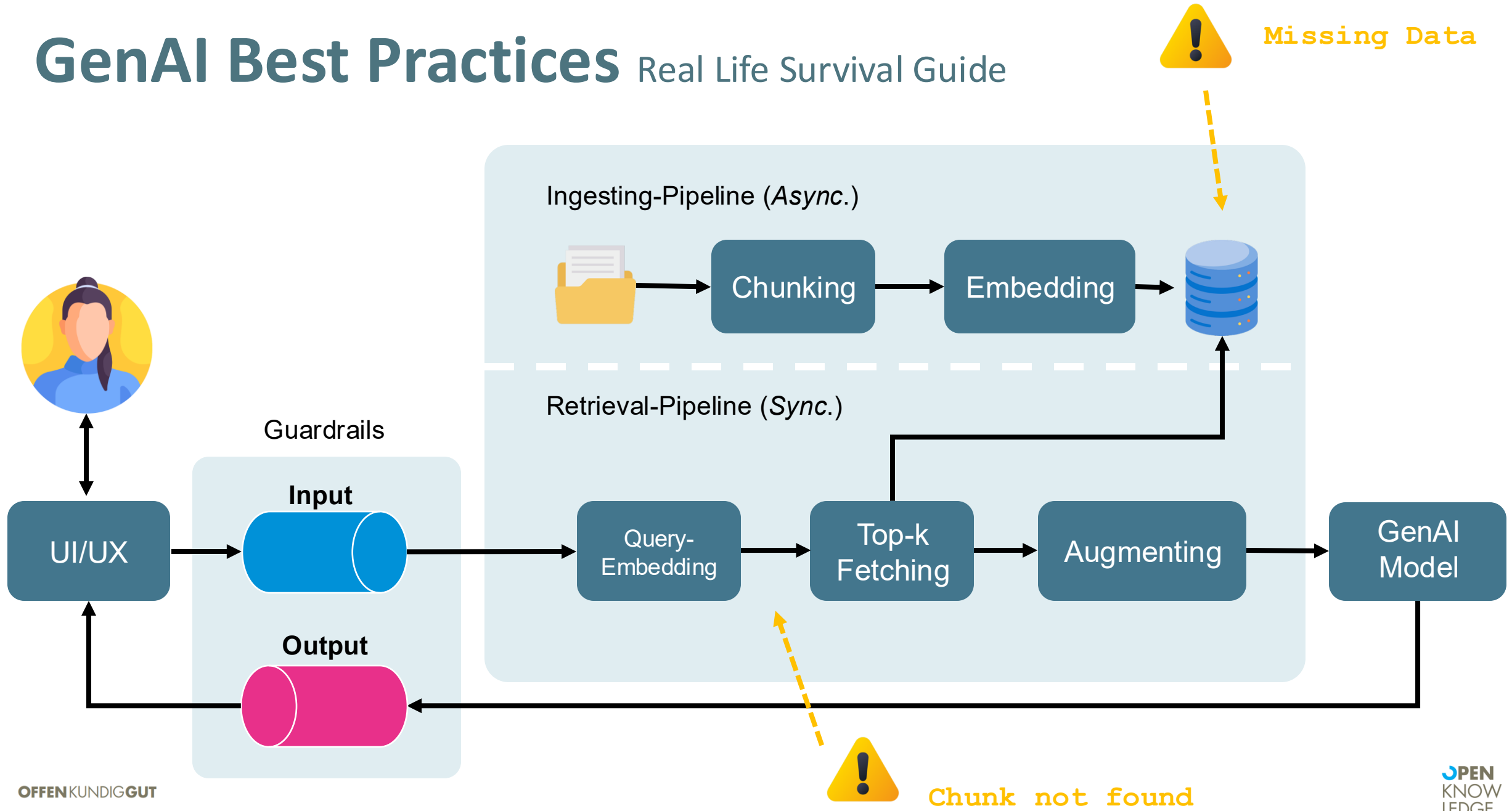
GenAI Best Practices

Real Life Survival Guide



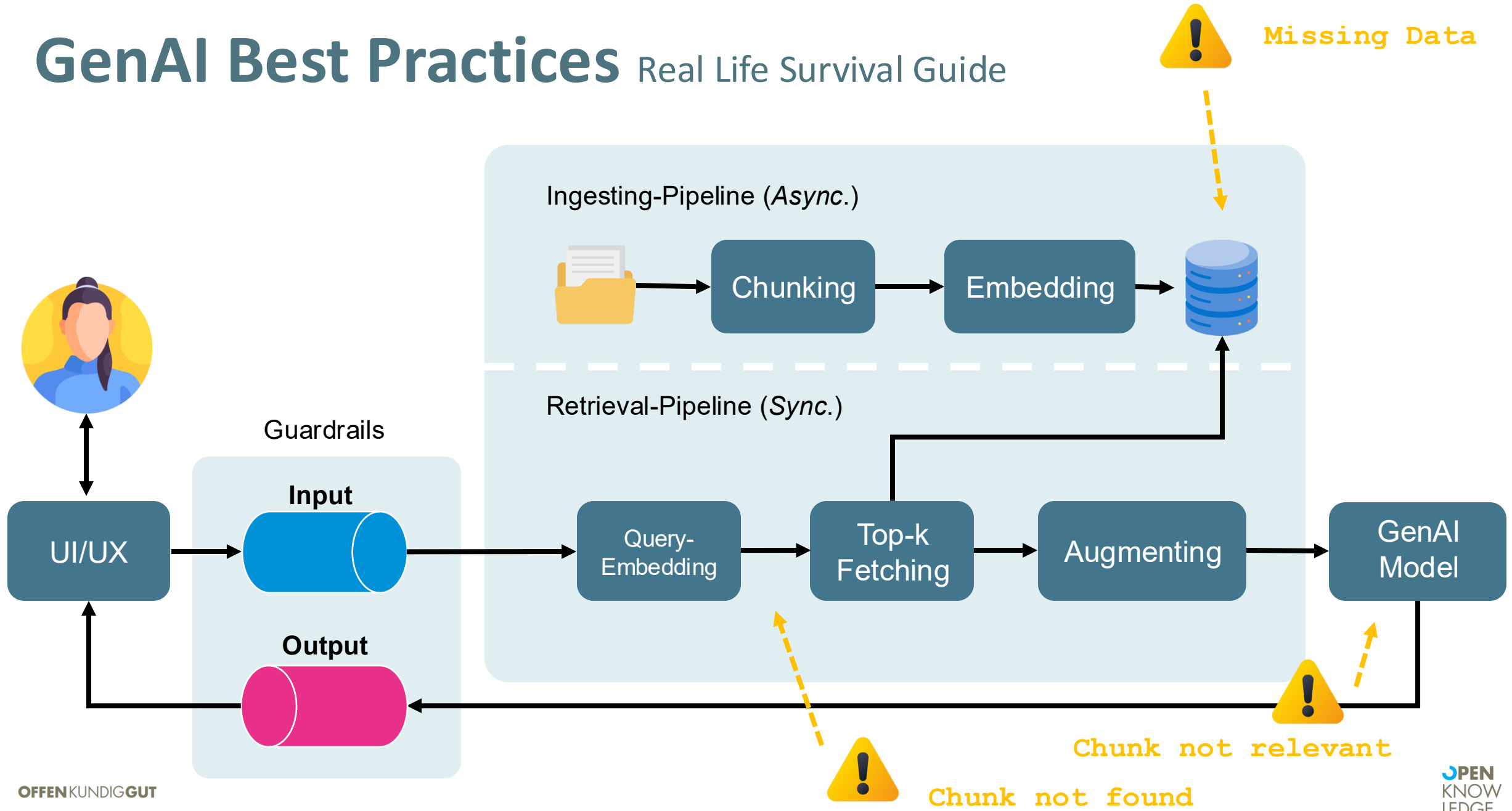
GenAI Best Practices

Real Life Survival Guide

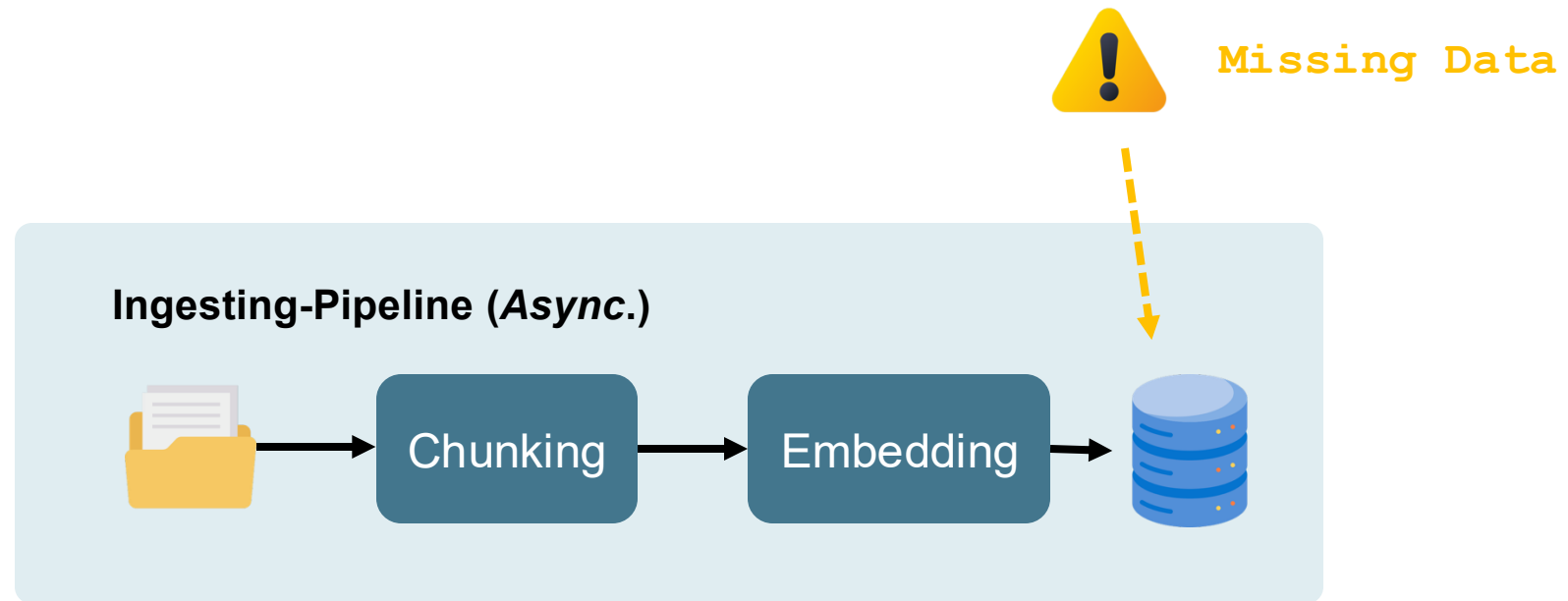


GenAI Best Practices

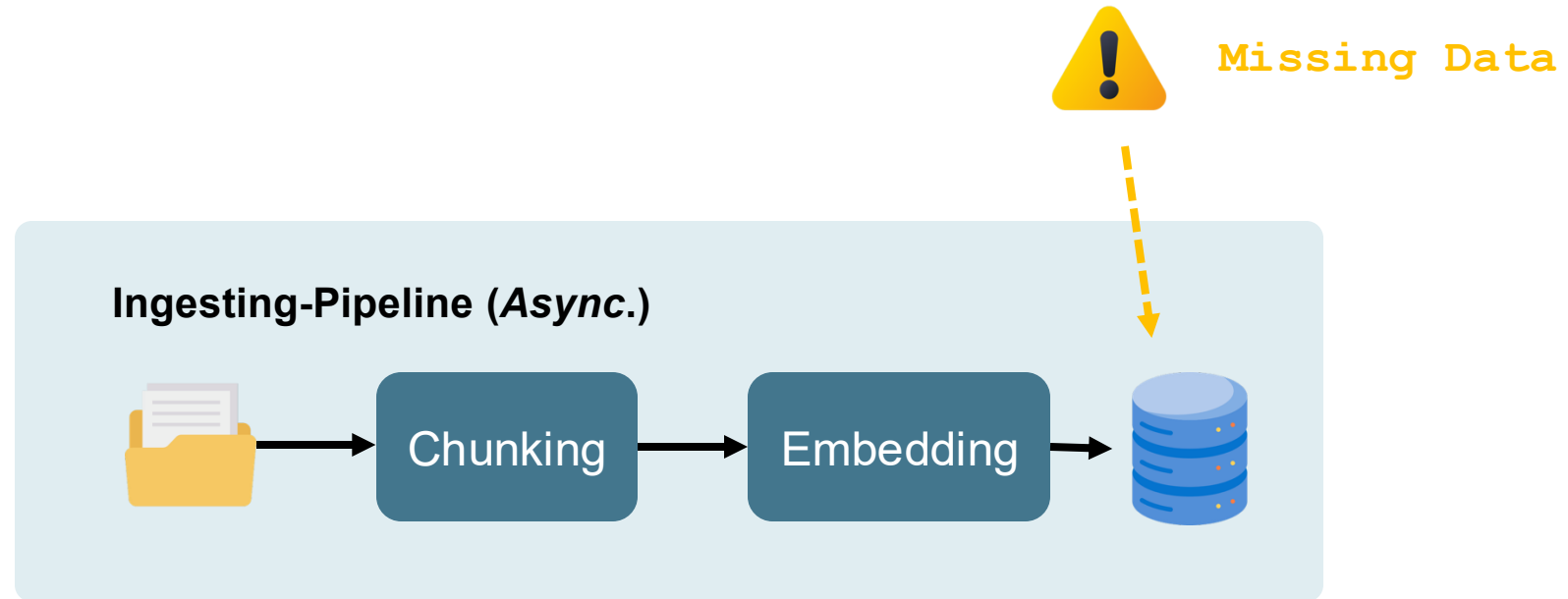
Real Life Survival Guide



Real Life RAG #1 Missing-Data Challenge

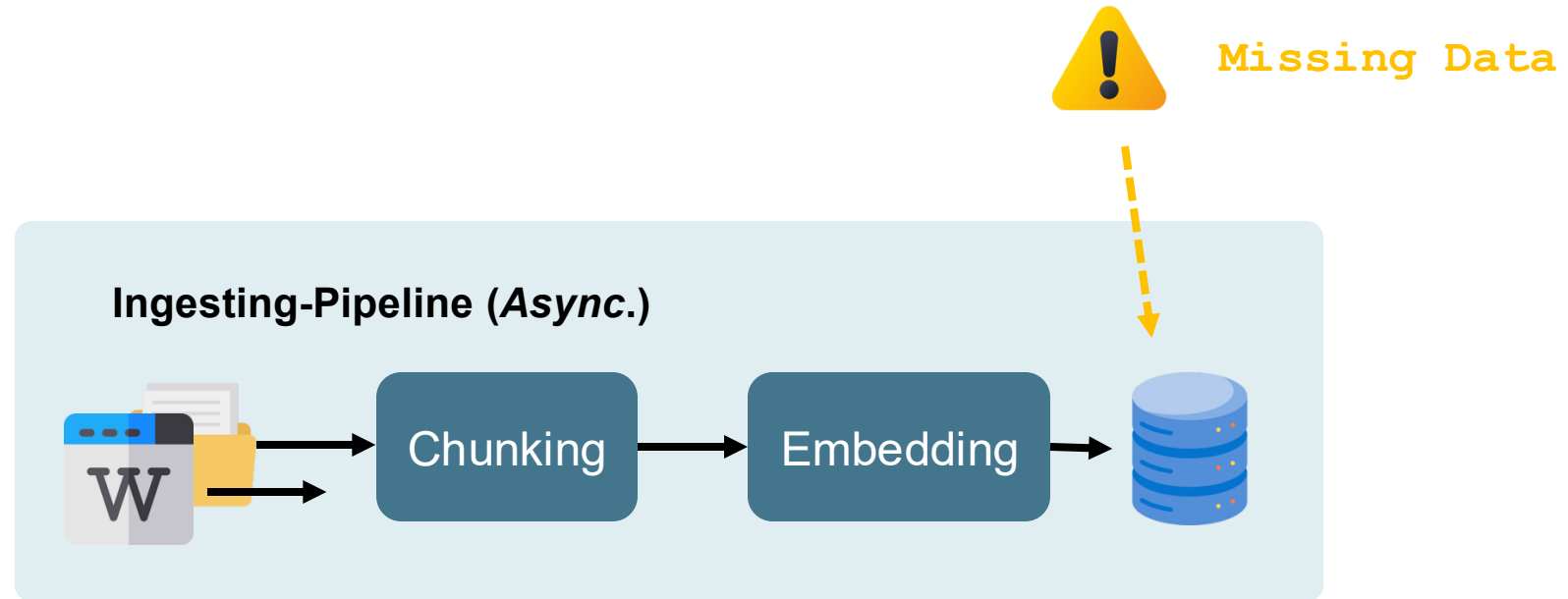


Real Life RAG #1 Missing-Data Challenge



Problem: RAG does not know the required data.

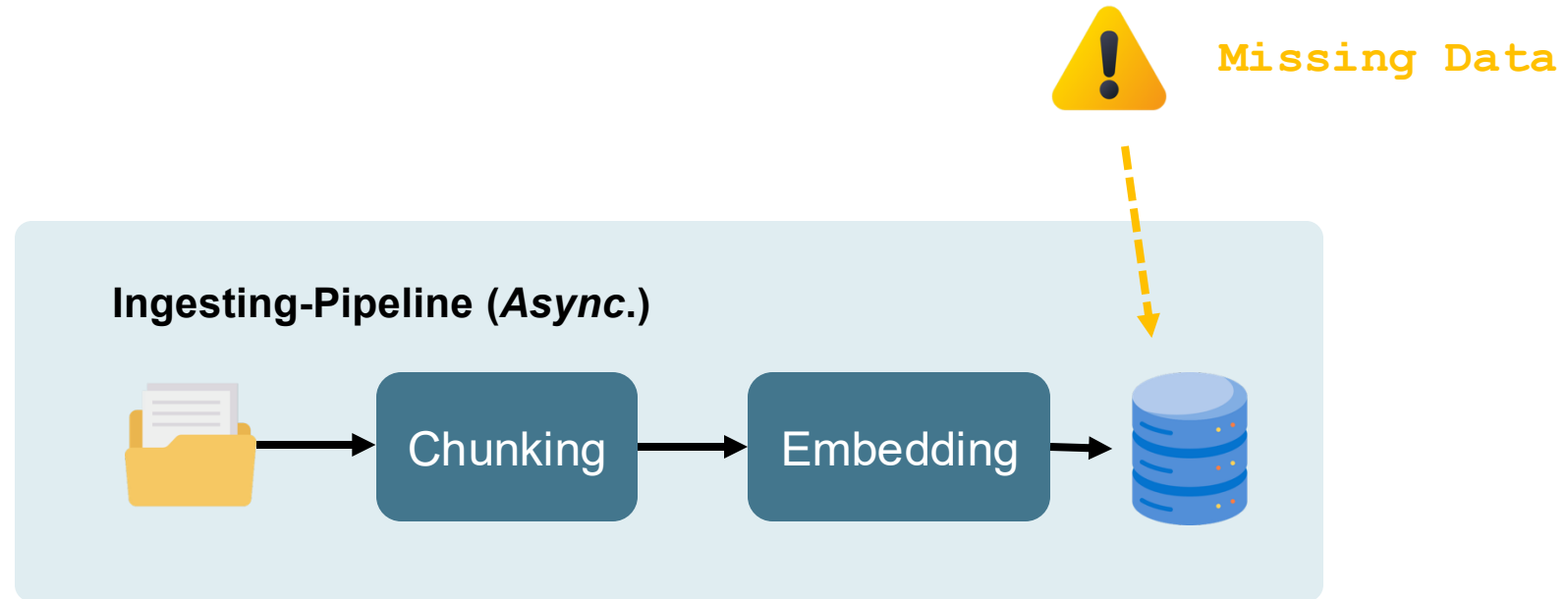
Real Life RAG #1 Missing-Data Challenge



Problem: RAG does not know the required data.

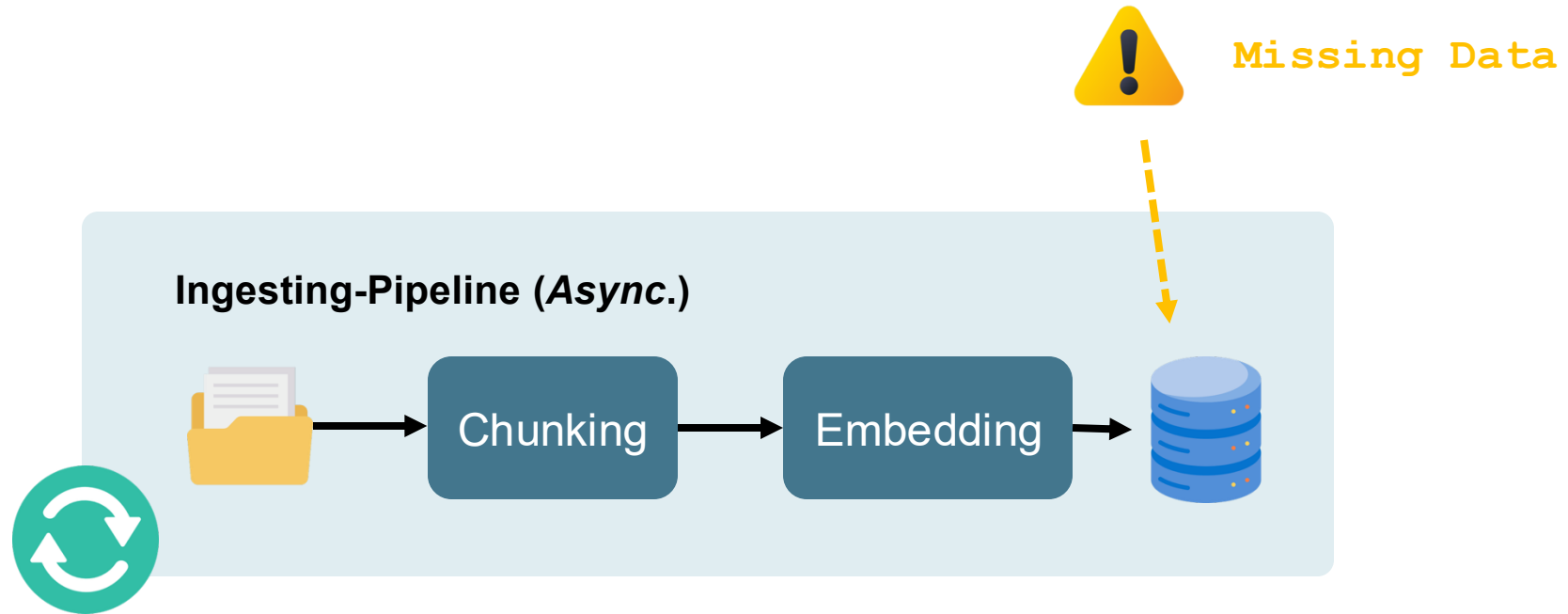
Solution : Implement a suitable ingestor.

Real Life RAG #1 Missing-Data Challenge



Problem: *Snapshot of the data is not up-to-date.*

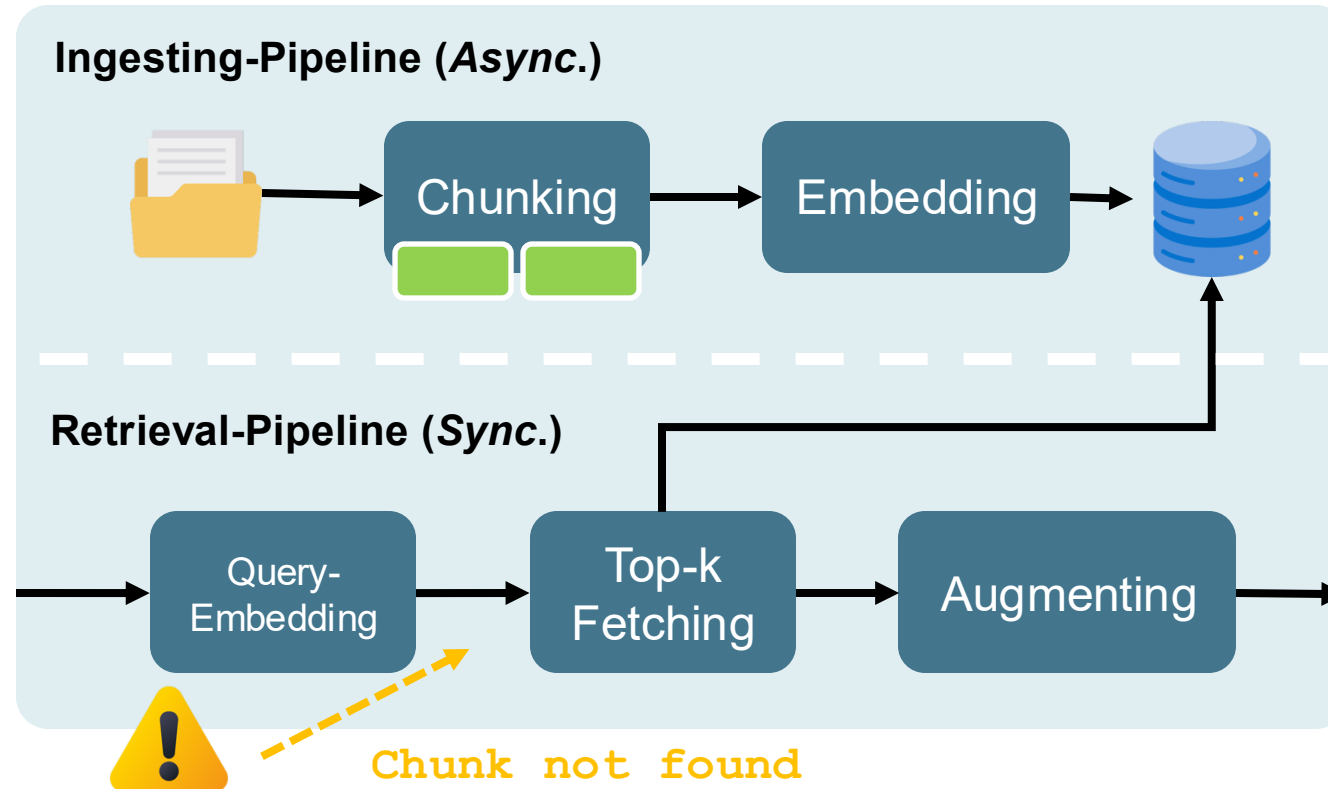
Real Life RAG #1 Missing-Data Challenge



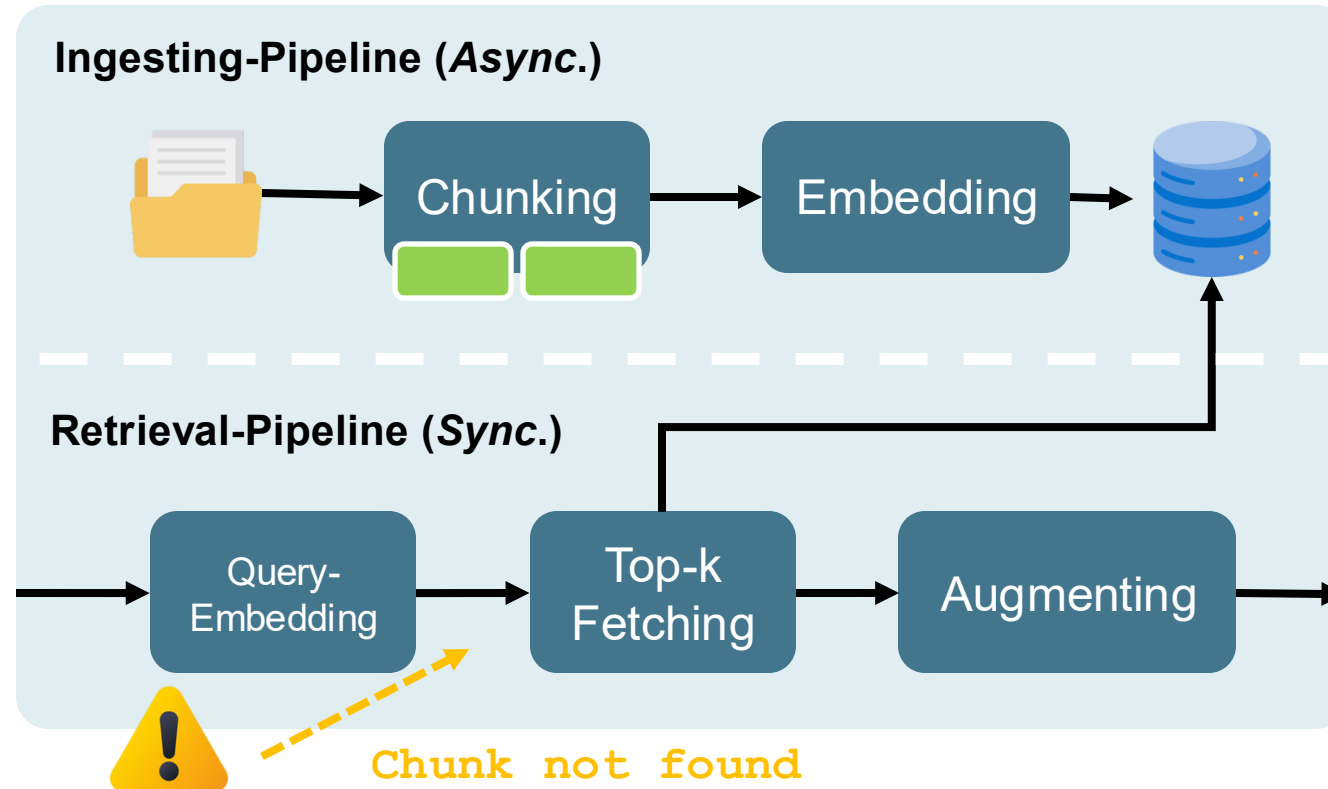
Problem: Snapshot of the data is not up-to-date.

Solution: Interface for 'continuous' data sync.

Real Life RAG #2 Chunk-not-found Challenge

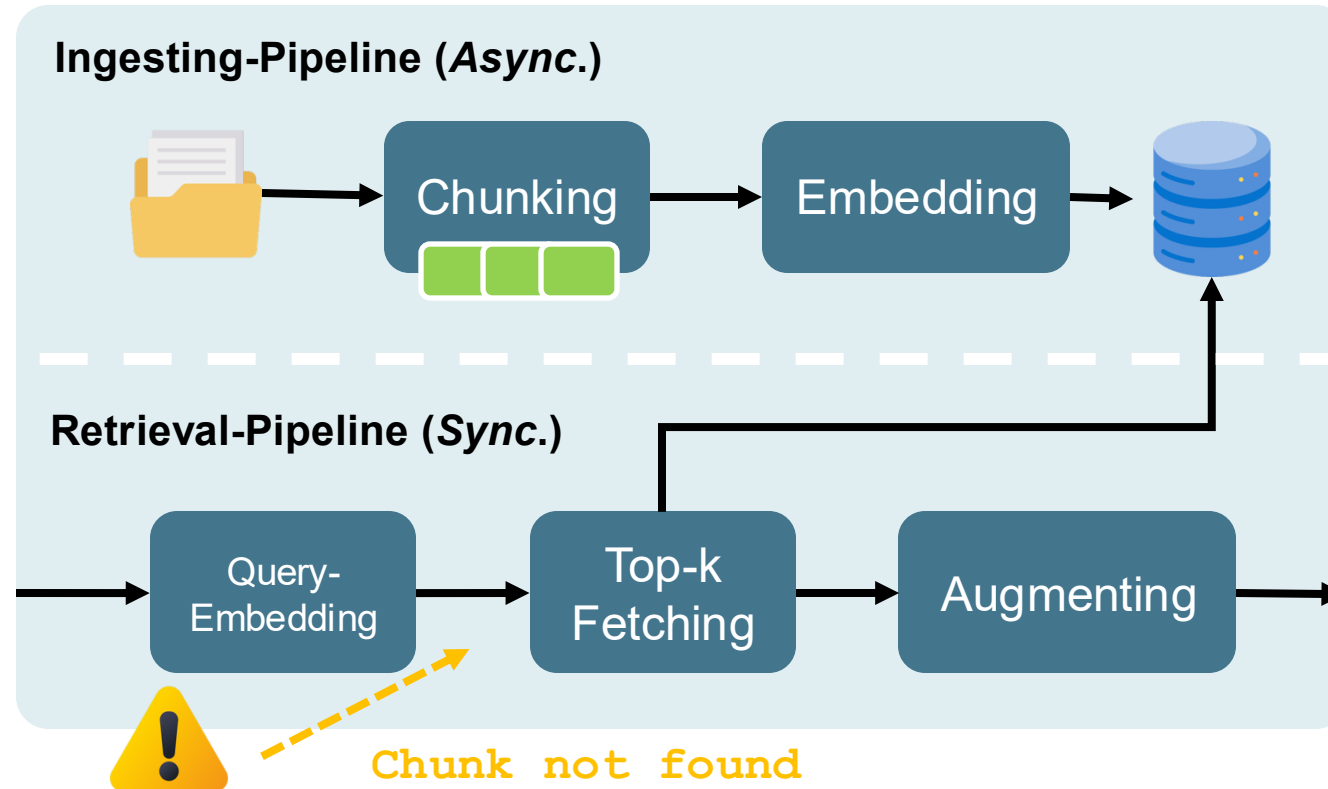


Real Life RAG #2 Chunk-not-found Challenge



Problem: Existing chunks will not be found.

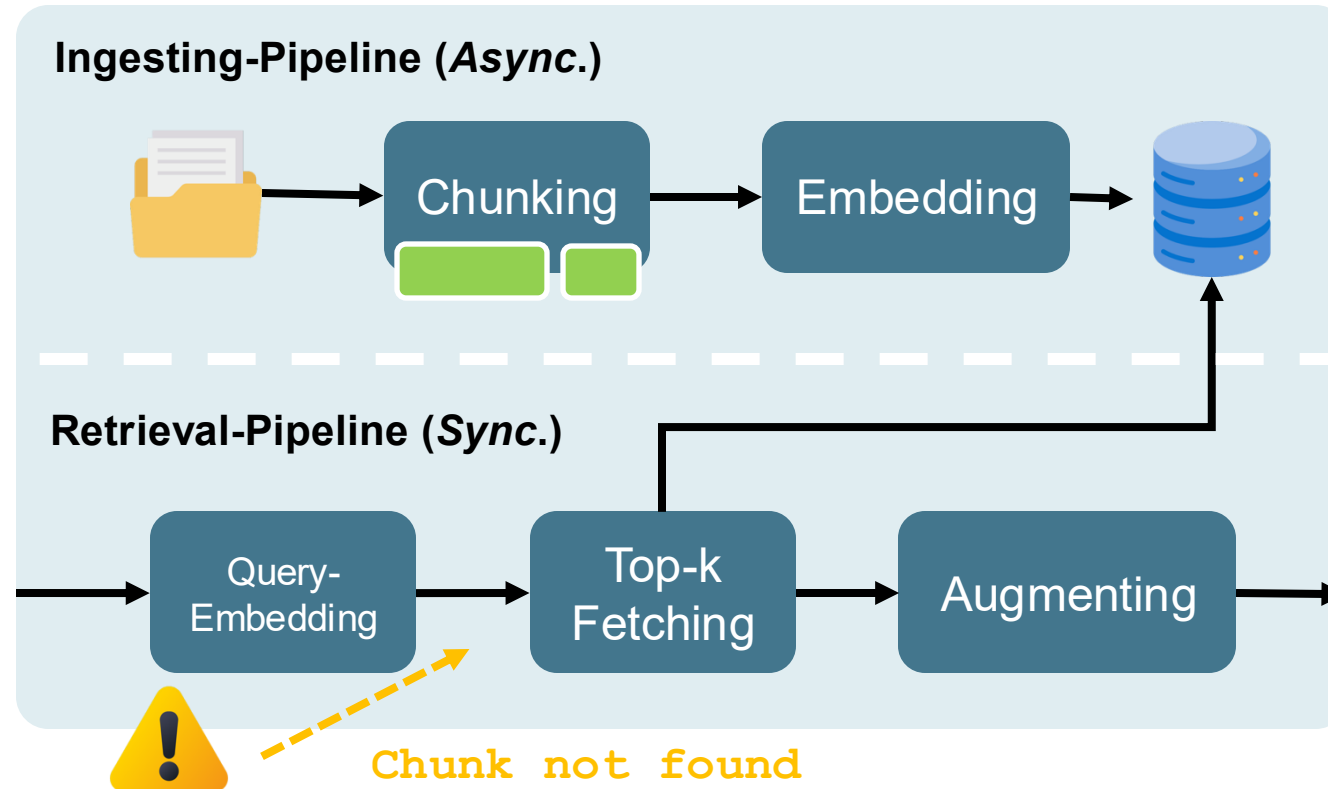
Real Life RAG #2 Chunk-not-found Challenge



Problem: Existing chunks will not be found.

Solution #1: Adapt chunk-sizes and overlapping.

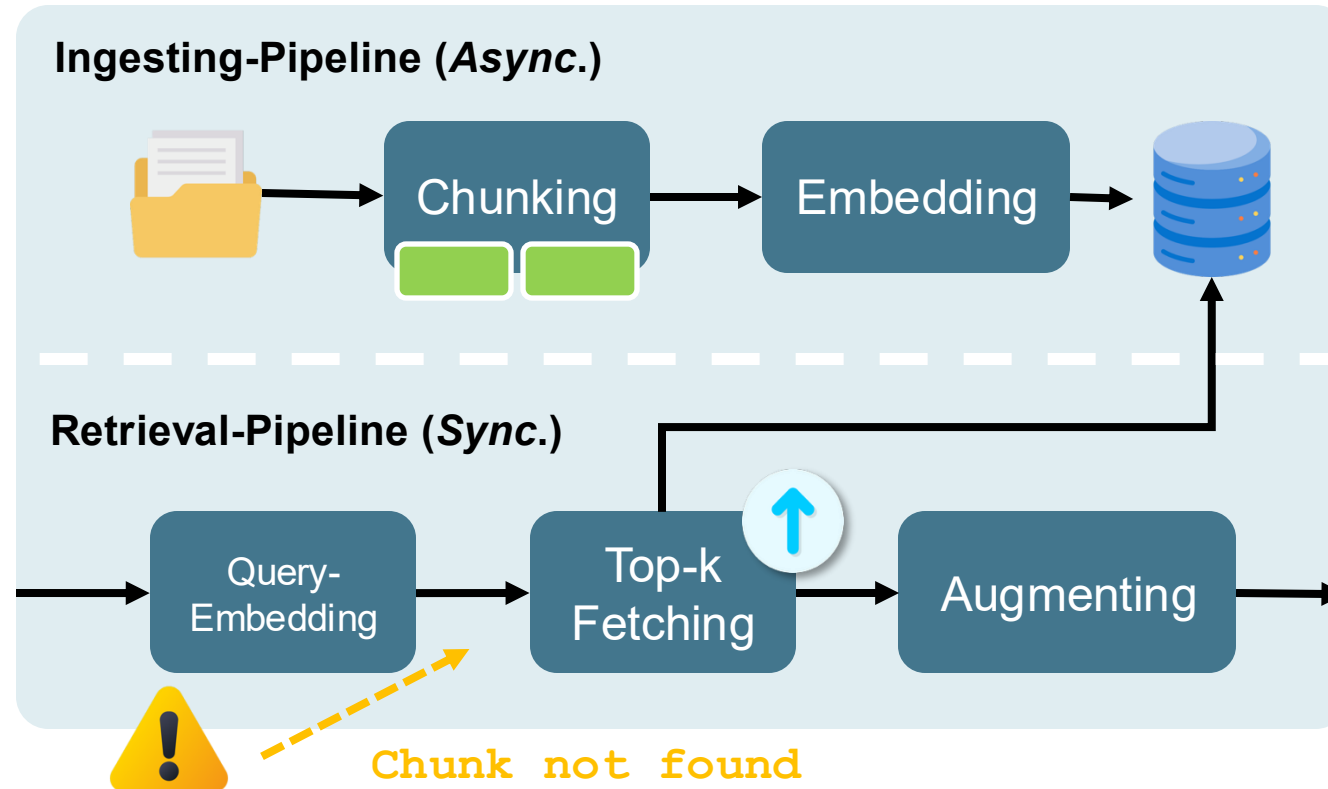
Real Life RAG #2 Chunk-not-found Challenge



Problem: Existing chunks will not be found.

Solution #2: Choose a different splitter (e.g. Semantic-Splitter).

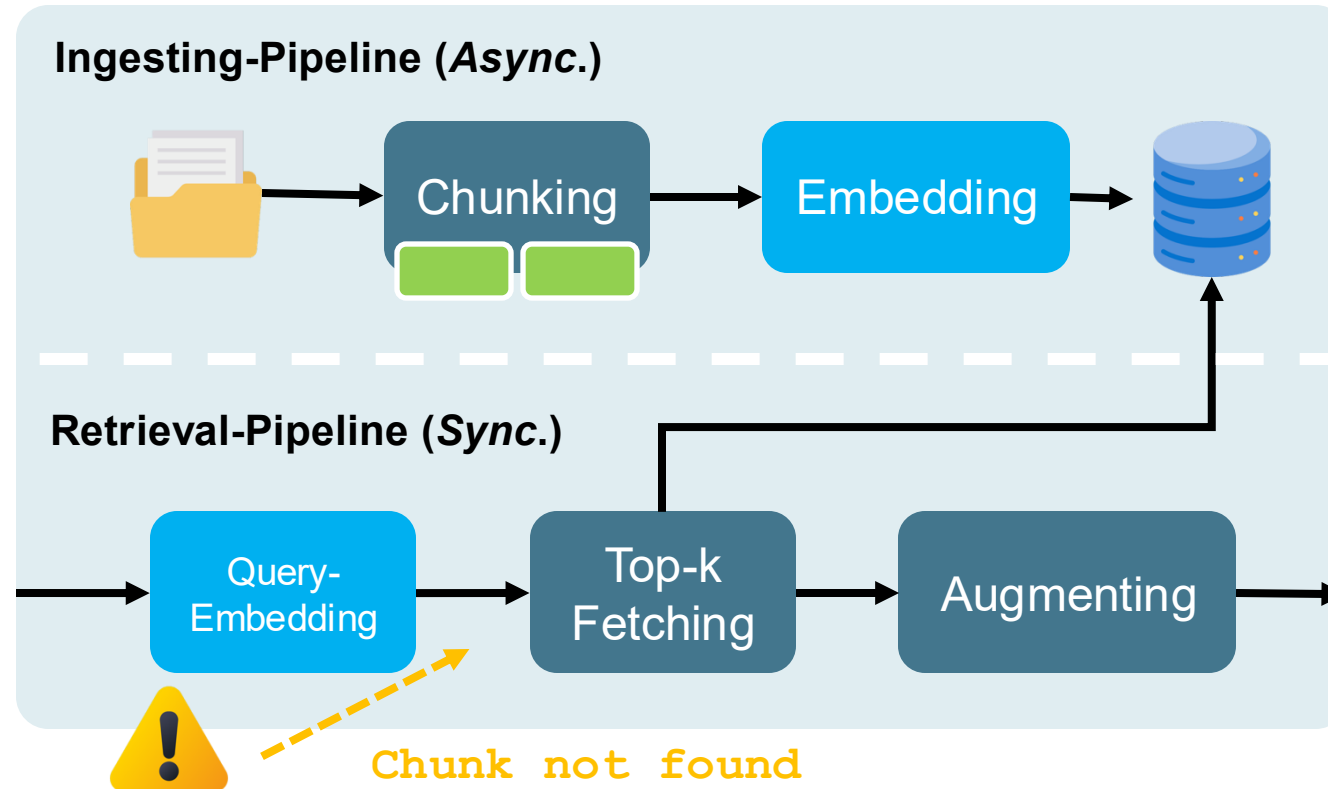
Real Life RAG #2 Chunk-not-found Challenge



Problem: Existing chunks will not be found.

Solution #3: Optimize the *distance threshold* or *k* from *top-k*.

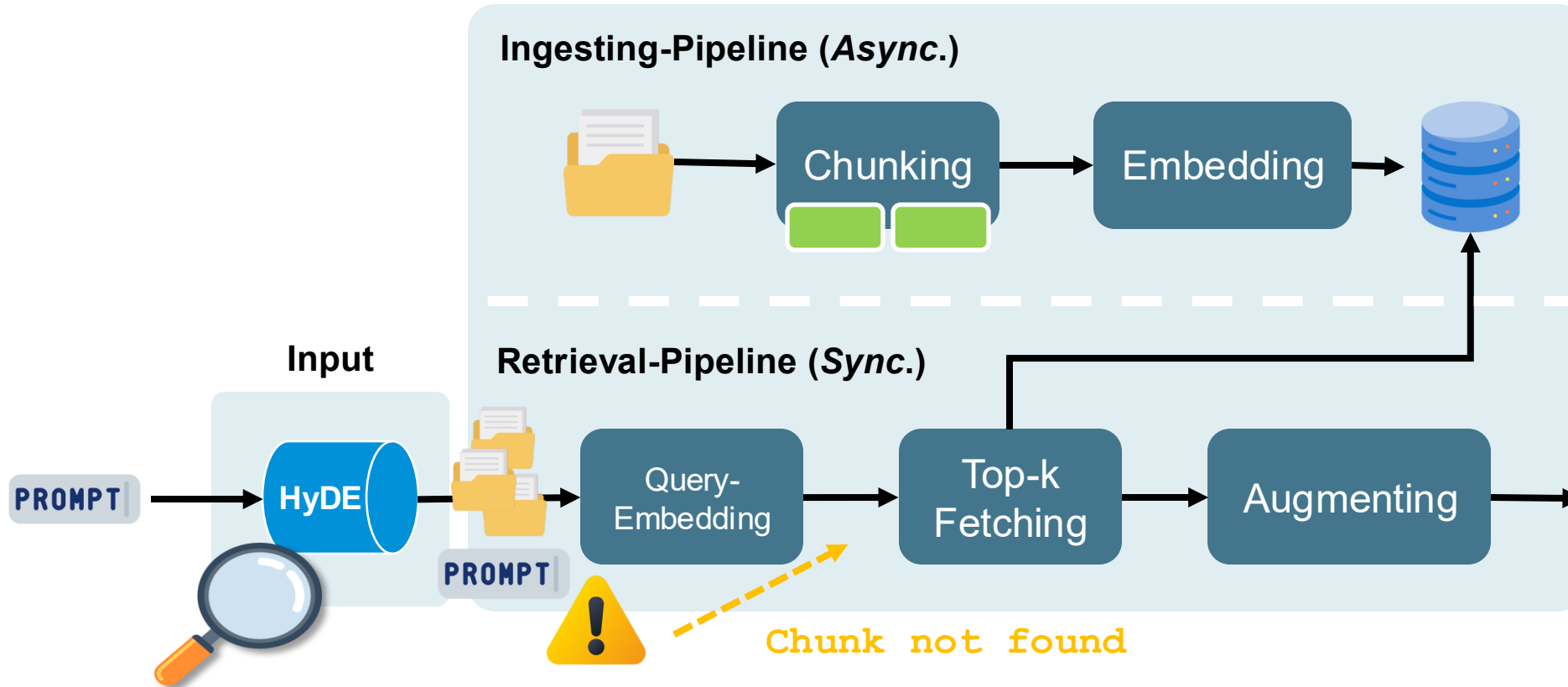
Real Life RAG #2 Chunk-not-found Challenge



Problem: Existing chunks will not be found.

Solution #4: Selection of a more suitable *embedding model*.

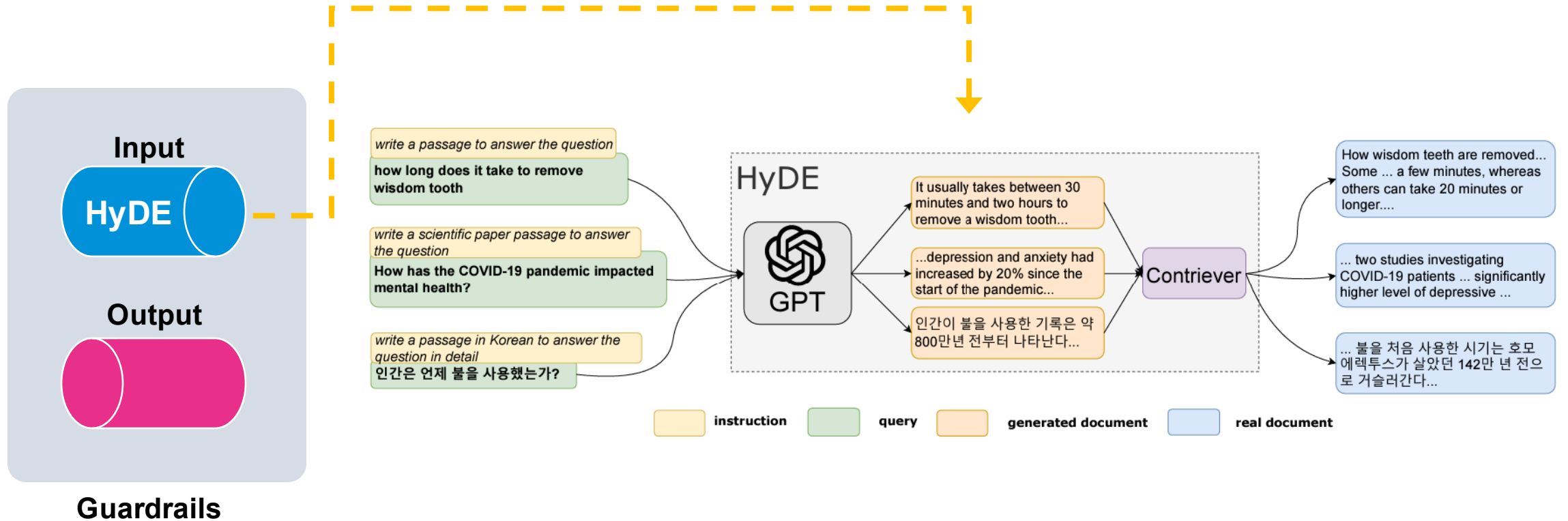
Real Life RAG #2 Chunk-not-found Challenge



Problem: Existing chunks will not be found.

Solution #5: Hypothetical document embeddings (HyDE)

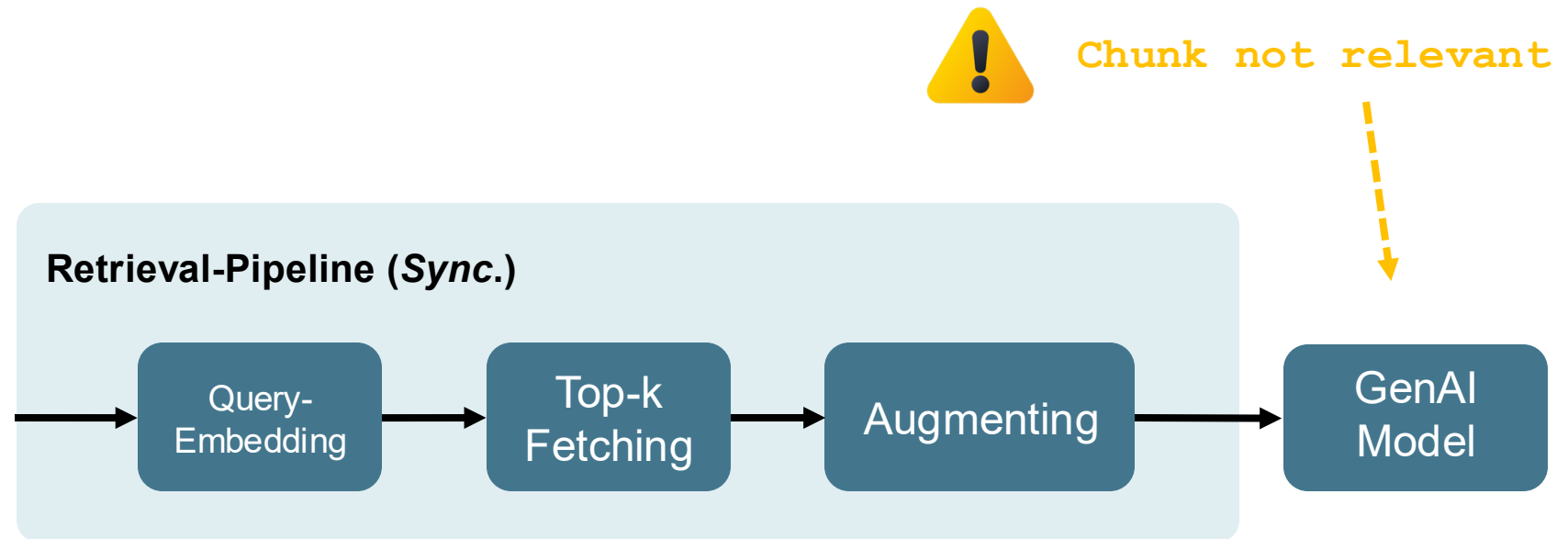
Real Life RAG #2 Chunk-not-found Challenge



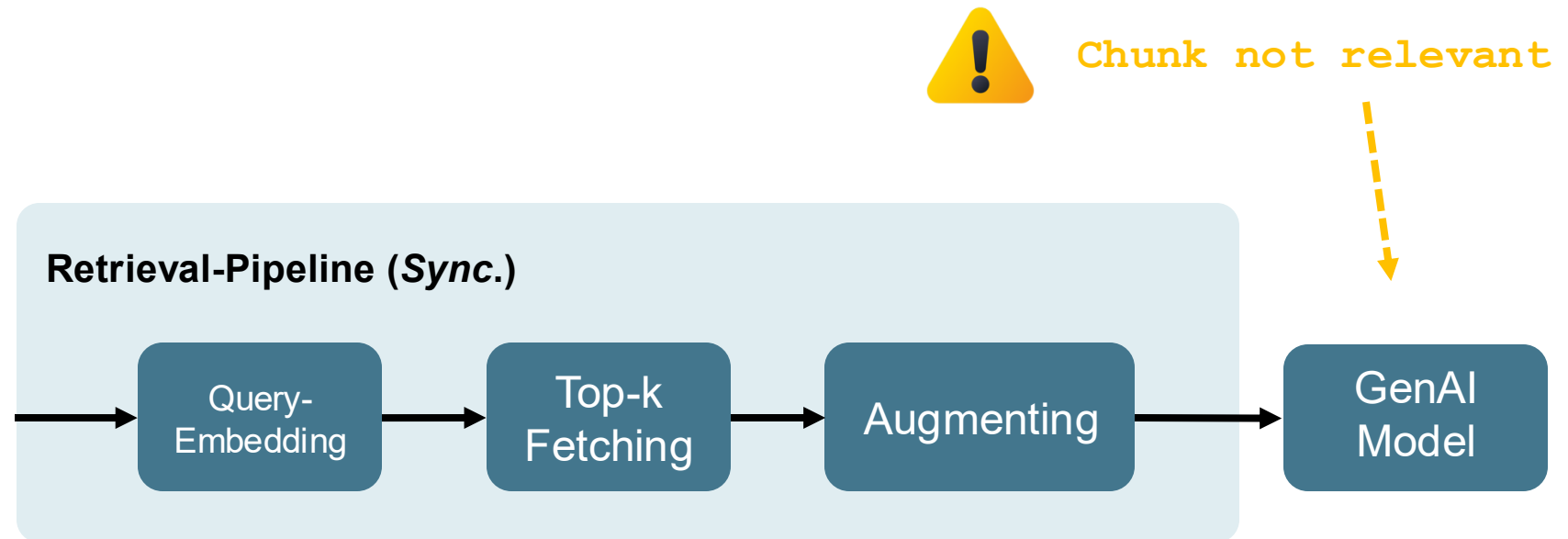
Gao et al. (2023)

HyDE = Hypothetical Document Embeddings

Real Life RAG #3 Chunk-not-relevant Challenge

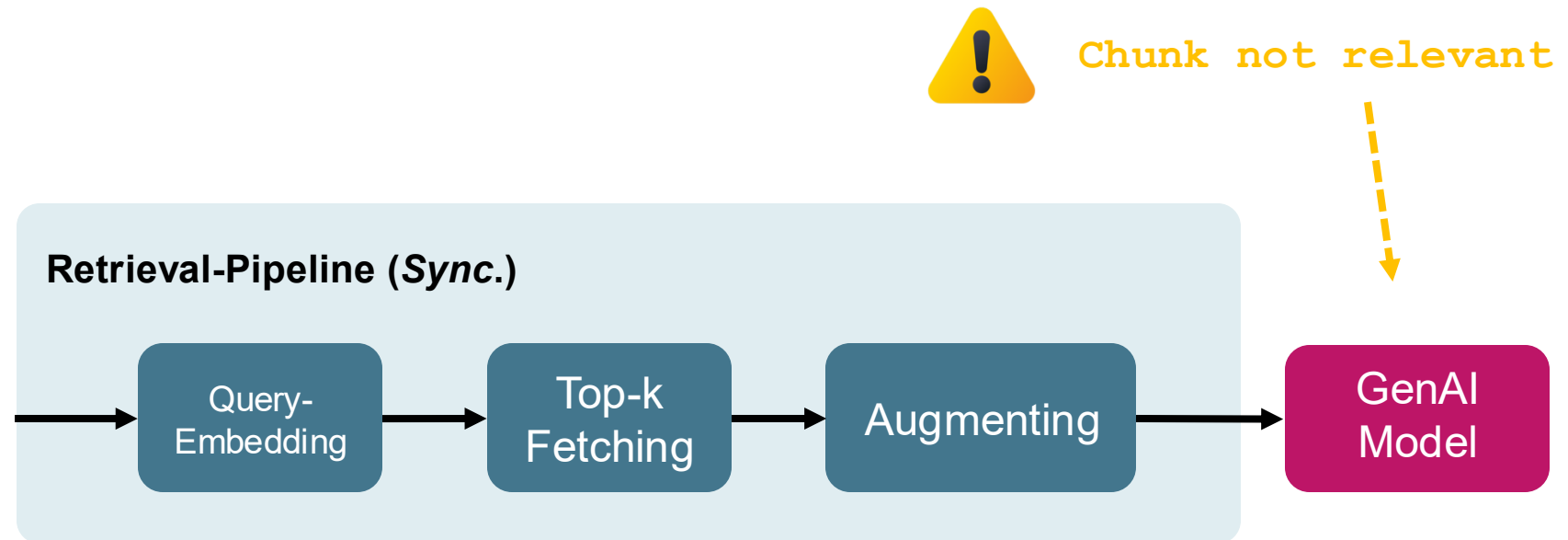


Real Life RAG #3 Chunk-not-relevant Challenge



Problem: *Chunk is not considered to be relevant by the model.*

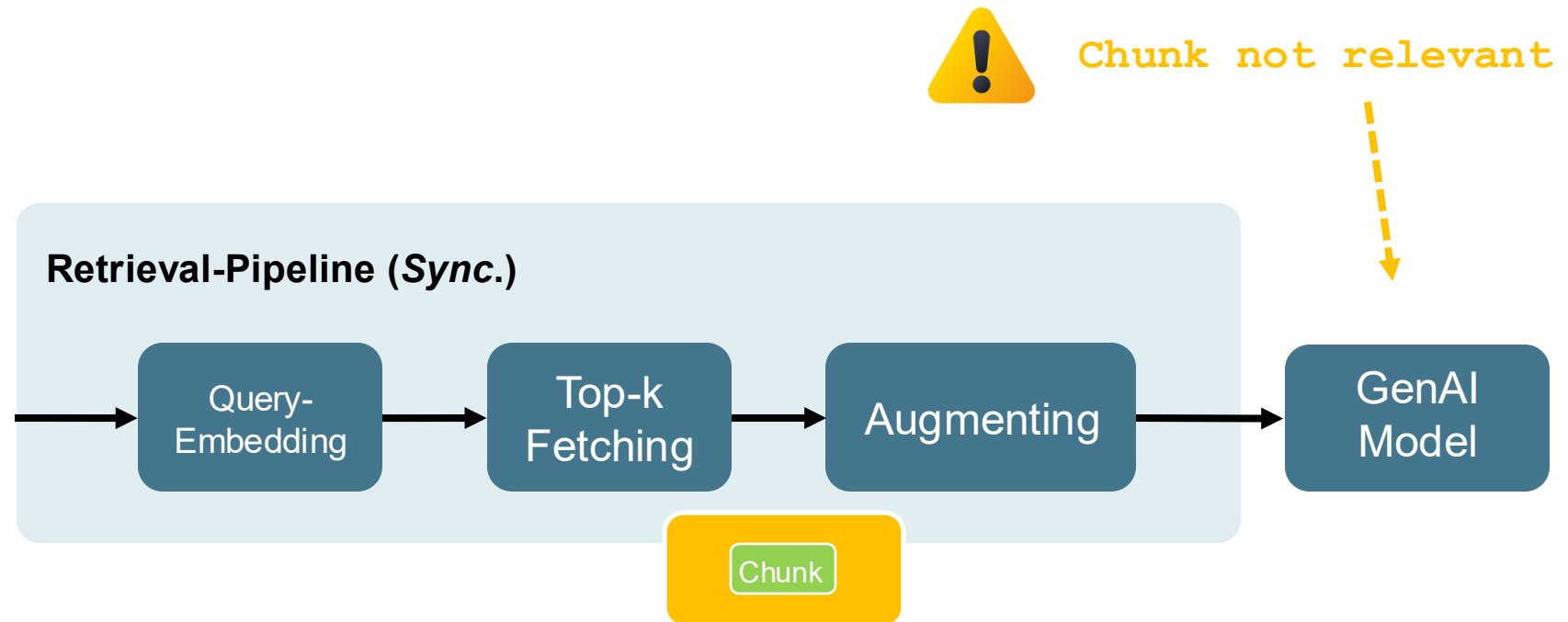
Real Life RAG #3 Chunk-not-relevant Challenge



Problem: *Chunk is not considered to be relevant by the model.*

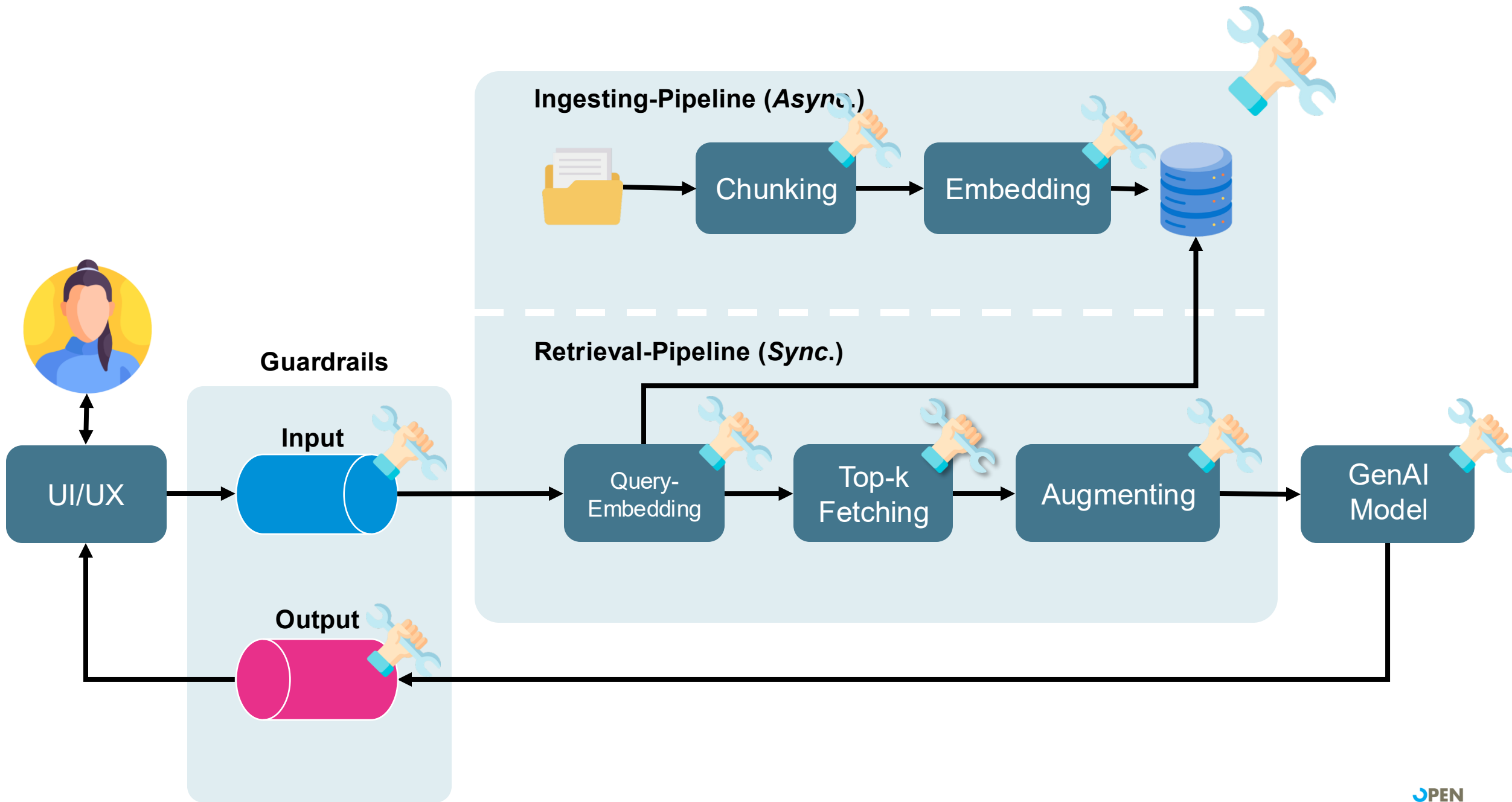
Solution #1: *Select a more suitable model for the use case.*

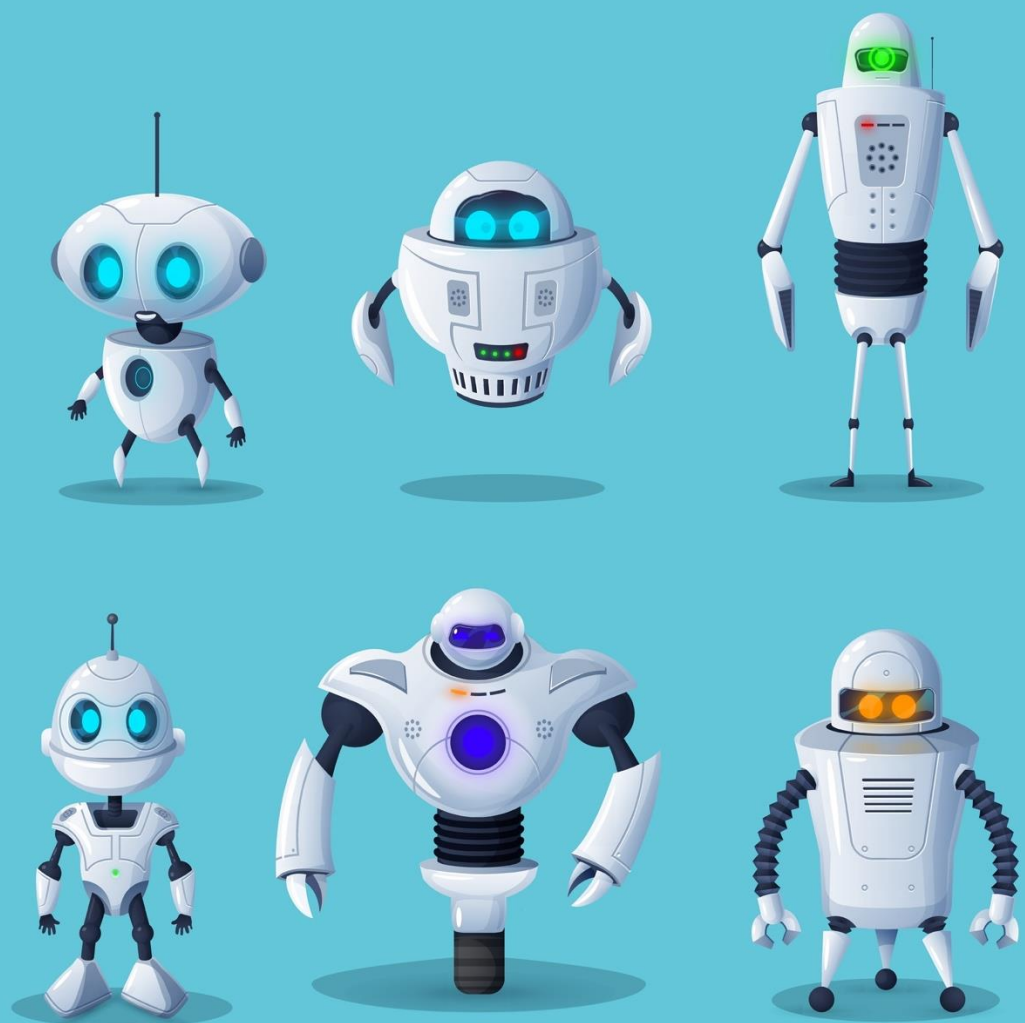
Real Life RAG #3 Chunk-not-relevant Challenge



Problem: *Chunk is not considered to be relevant by the model.*

Solution #2: *Add additional context to the chunk (small-to-big).*





Hands-On

RAG System typical Pitfalls

Surviving the RAG jungle:

Using LLM and RAG metrics to
optimise the retrieval process.

Hands-on Surviving the RAG jungle

The screenshot shows a JupyterLab environment with a project named 'workshop-genai-student'. The left sidebar displays the project structure, including folders for 'data', 'exercises', and '04_rag_pitfalls'. The main area shows the '04_rag_pitfalls.ipynb' file. The code in the notebook includes a TODO comment and a call to 'print_insights(evaluation_dataframe)'. The output shows statistics on hallucinations and response times. Below this, a section titled 'Step 8: Manually adjust k' explains that no new ingestion is needed. The notebook continues with another TODO comment and code to generate responses and insights using 'generate_rag_answers'.

```
In _ 1 # TODO: Try to interpret the results. How does the chunk_size affect the results?
      2 print_insights(evaluation_dataframe)
```

Number of detected hallucinations: 9
Mean response time: 2.0 seconds
Mean minimum distance: 0.63
Mean mean distance: 0.68

Step 8: Manually adjust k

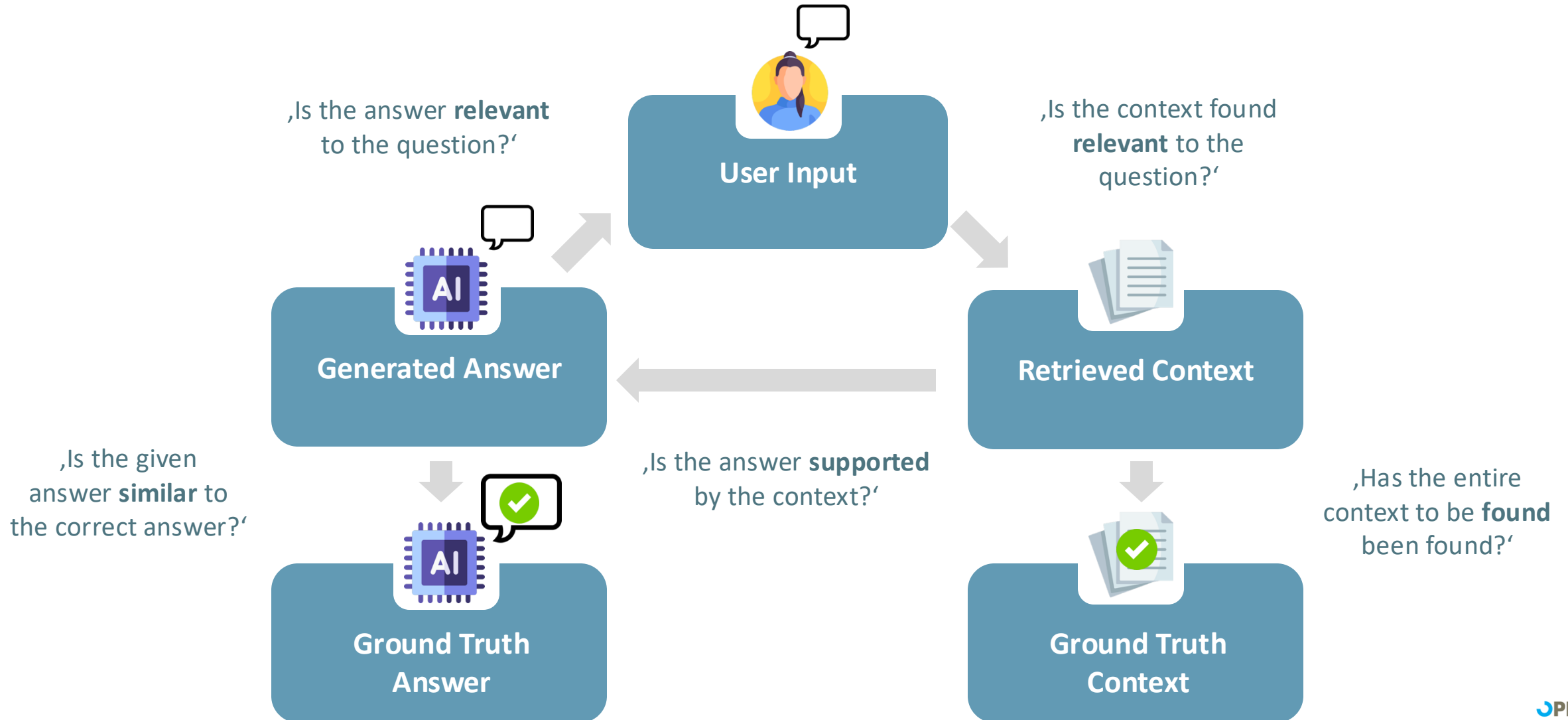
We do not need to ingest again, since our adjustment just affects the retrieval pipeline. Keep in mind, that we use the chunking from before, because we did not update the knowledgebase.

```
In _ 1 # TODO: Define new value for k. Do this multiple times and analyze the insights!
      2 adjusted_k = None
```

```
In _ 1 # Generate responses and insights
      2 evaluation_dataframe = generate_rag_answers(evaluation_dataframe, k=adjusted_k)
      3 evaluation_dataframe
```

GenAI Professional

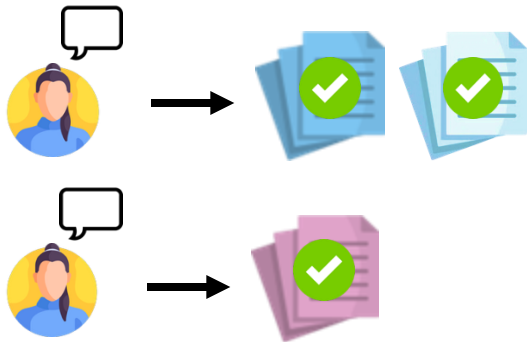
Monitoring: The RAG-Triad+



Real Life RAG Metrics in Information Retrieval

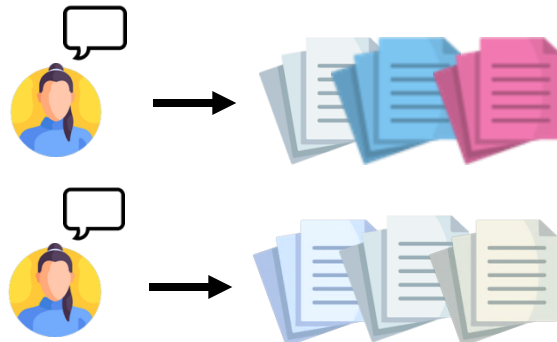
1

Provide **evaluation data** that includes the **ground truth context**.



2

Generate responses to evaluation input and collect **context**.

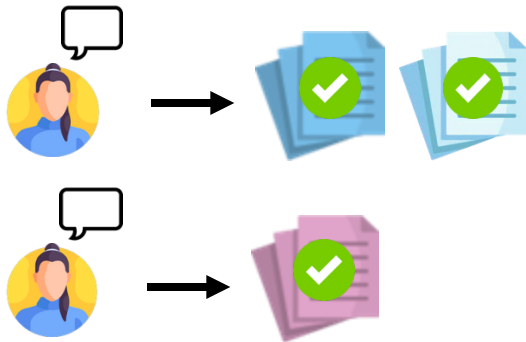


Real Life RAG

Metrics in Information Retrieval – Hit Rate

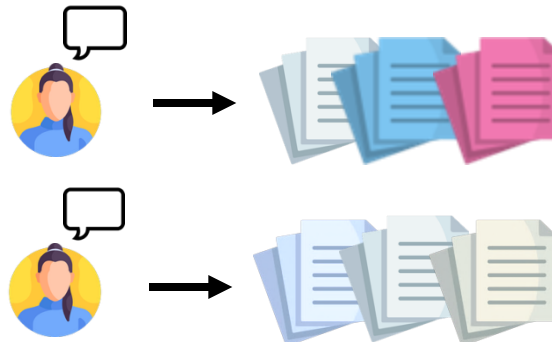
1

Provide **evaluation data** that includes the **ground truth context**.



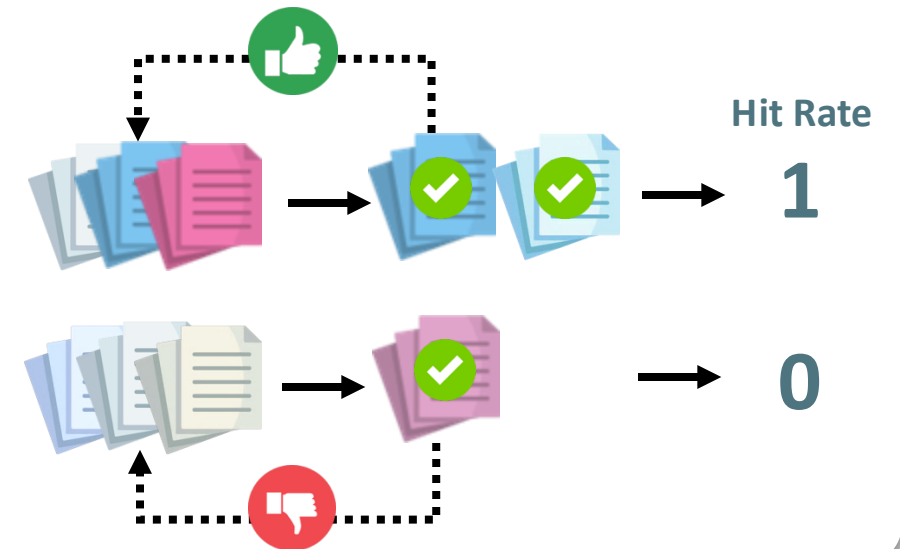
2

Generate responses to evaluation input and collect **context**.



3

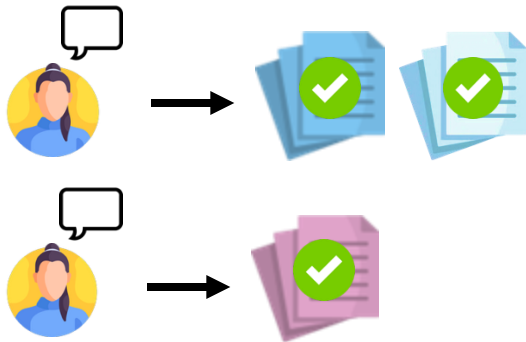
Determine whether context is included in **ground truth**.



Real Life RAG Metrics in Information Retrieval – Precision

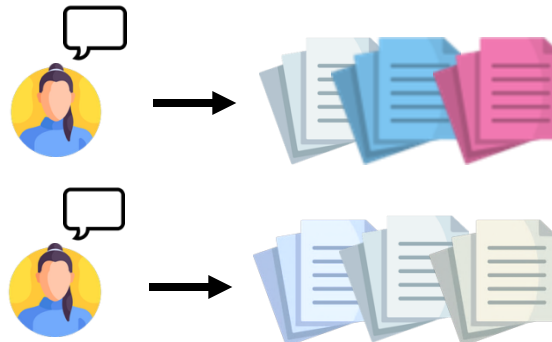
1

Provide **evaluation data** that includes the **ground truth context**.



2

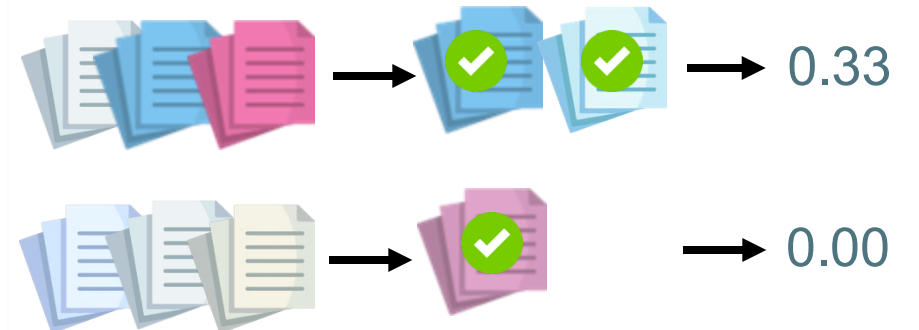
Generate responses to evaluation input and collect **context based on relevance**.



3

Determine the proportion of **relevant documents** in relation to all documents.

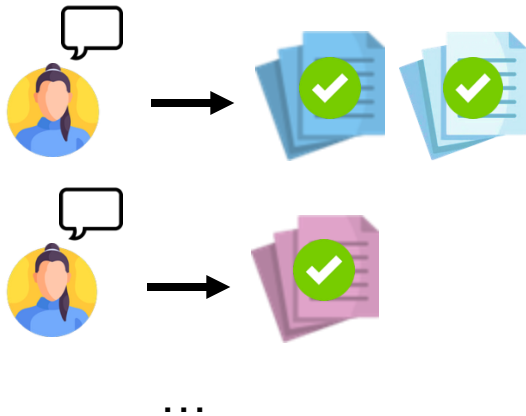
$$Precision = \frac{\text{relevant Results}}{\text{total Results}}$$



Real Life RAG Metrics in Information Retrieval – Mean Reciprocal Rank (MRR)

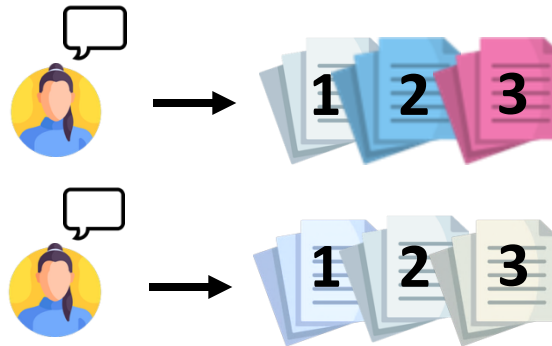
1

Provide **evaluation data** that includes the **ground truth context**.



2

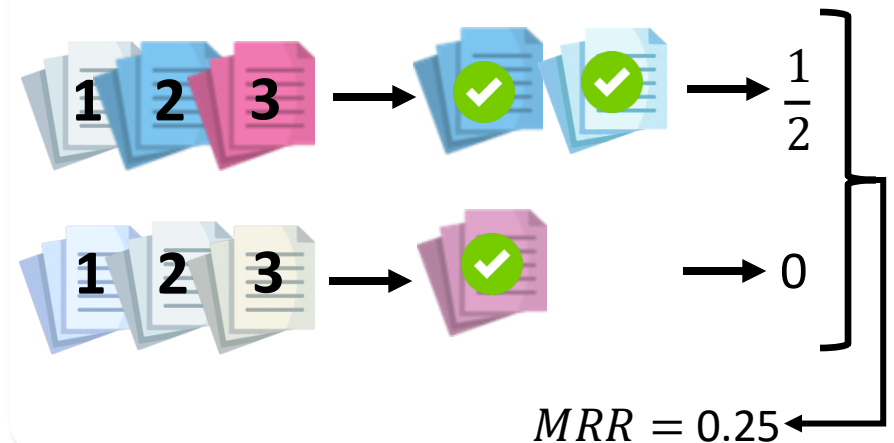
Generate responses to evaluation input and collect **context based on relevance**.



3

Determine the **rank** of the first **relevant document**. Calculate and average the **reciprocal rank (RR)**:

$$RR = \frac{1}{\text{Rank}} \rightarrow MRR = \frac{1}{n} \sum_{i=1}^n RR_i$$



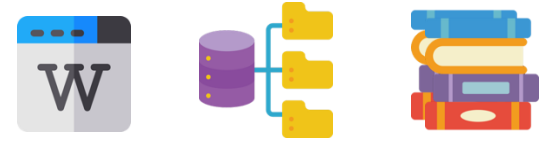
05

Outlook

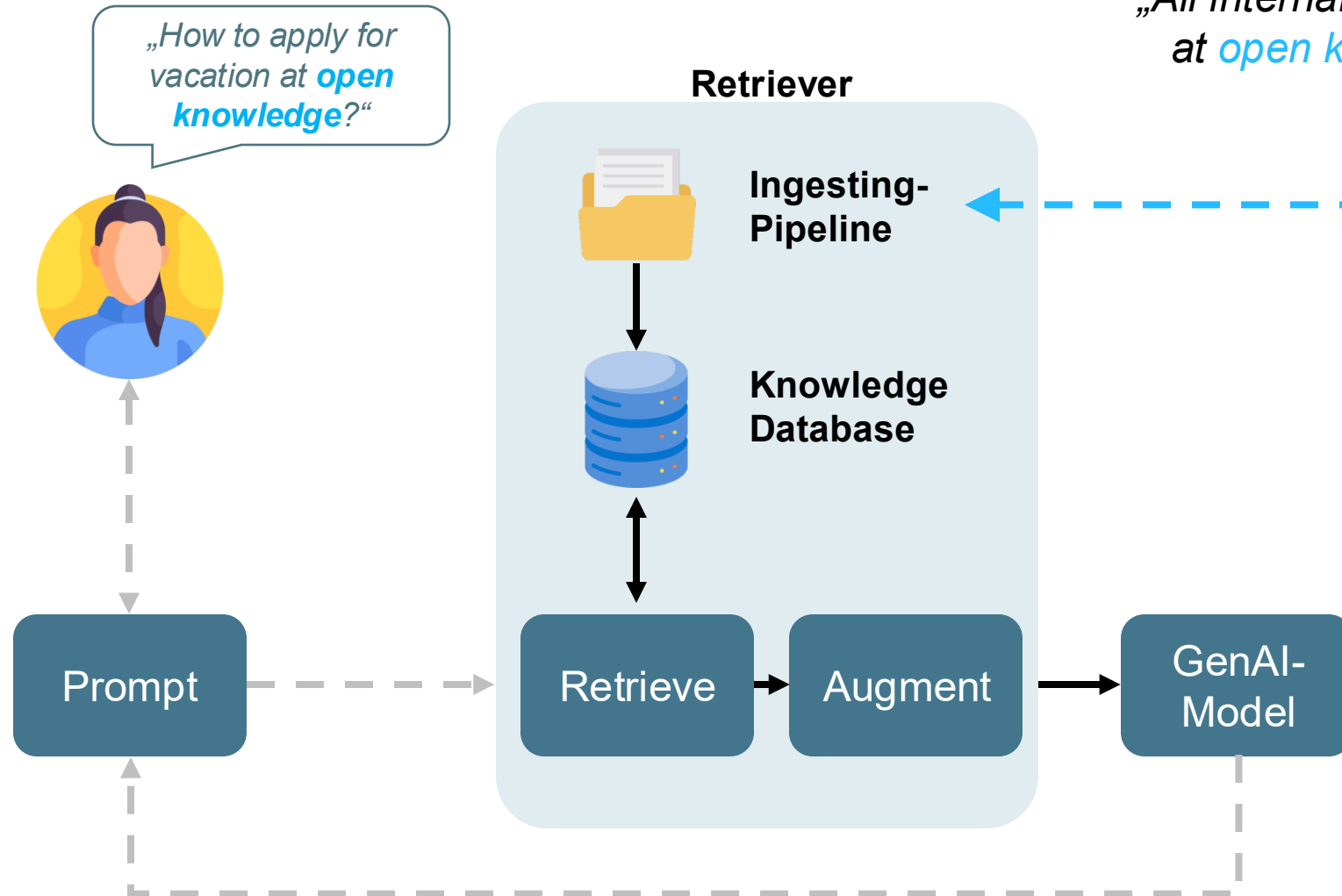
What else is possible in addition to the pure knowledge query?

GenAI Outlook

What else is there to consider?



„All internal knowledge
at *open knowledge*“



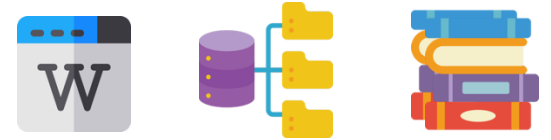
I18N



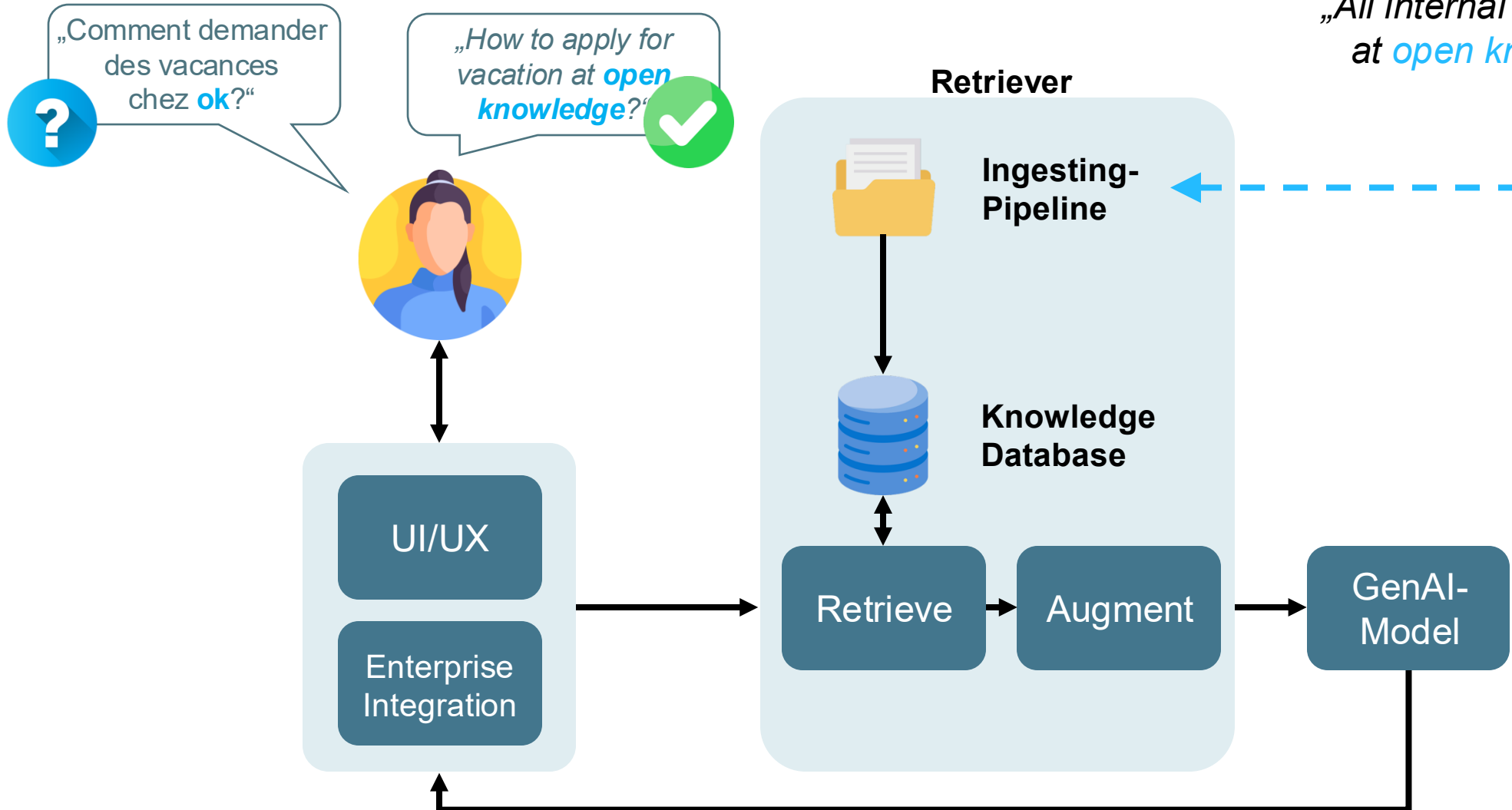
GenAI
explained

GenAI Outlook

Multi-Language RAG

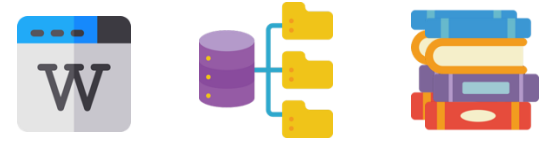


„All internal knowledge
at *open knowledge*“

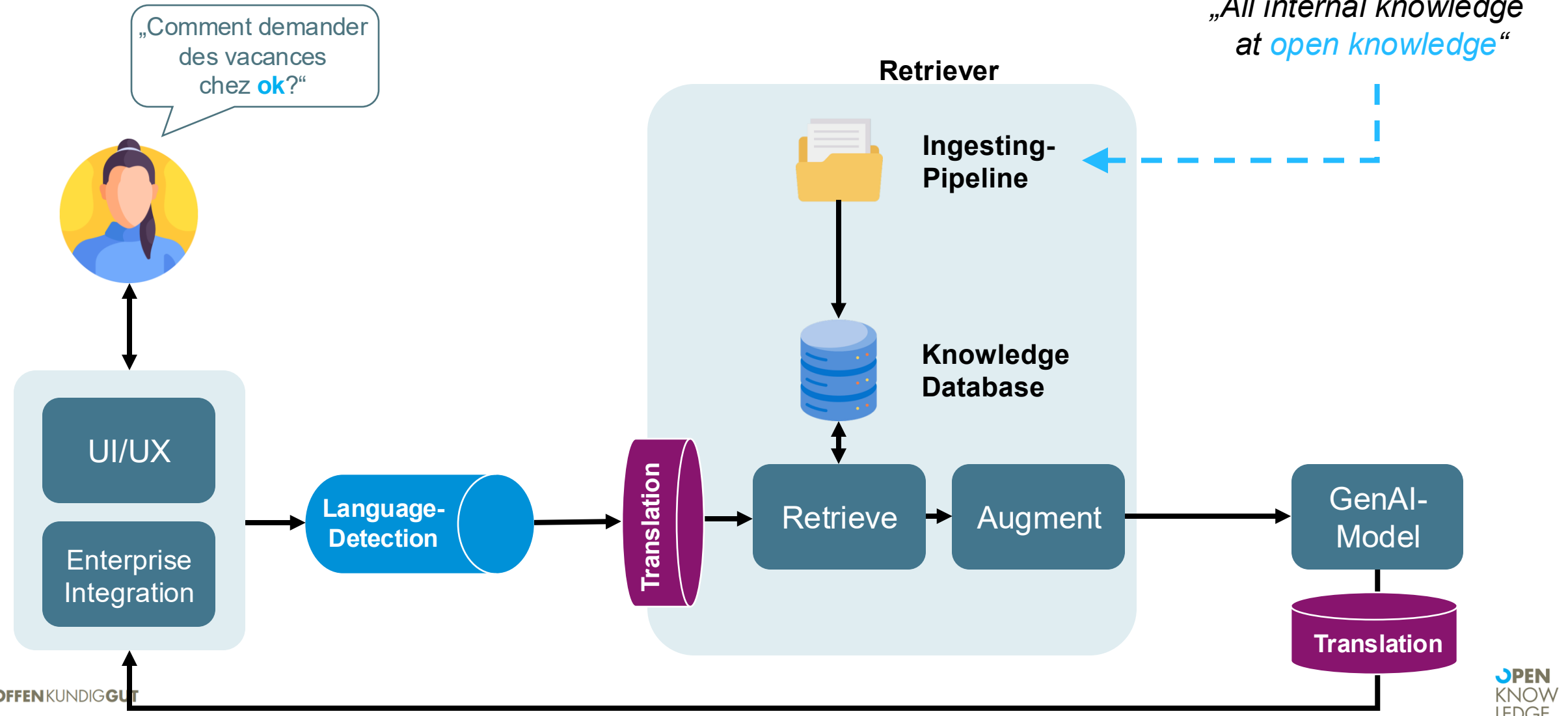


GenAI Outlook

Multi-Language RAG

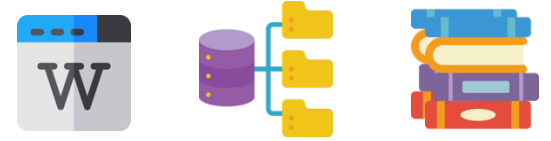


„All internal knowledge
at *open knowledge*“

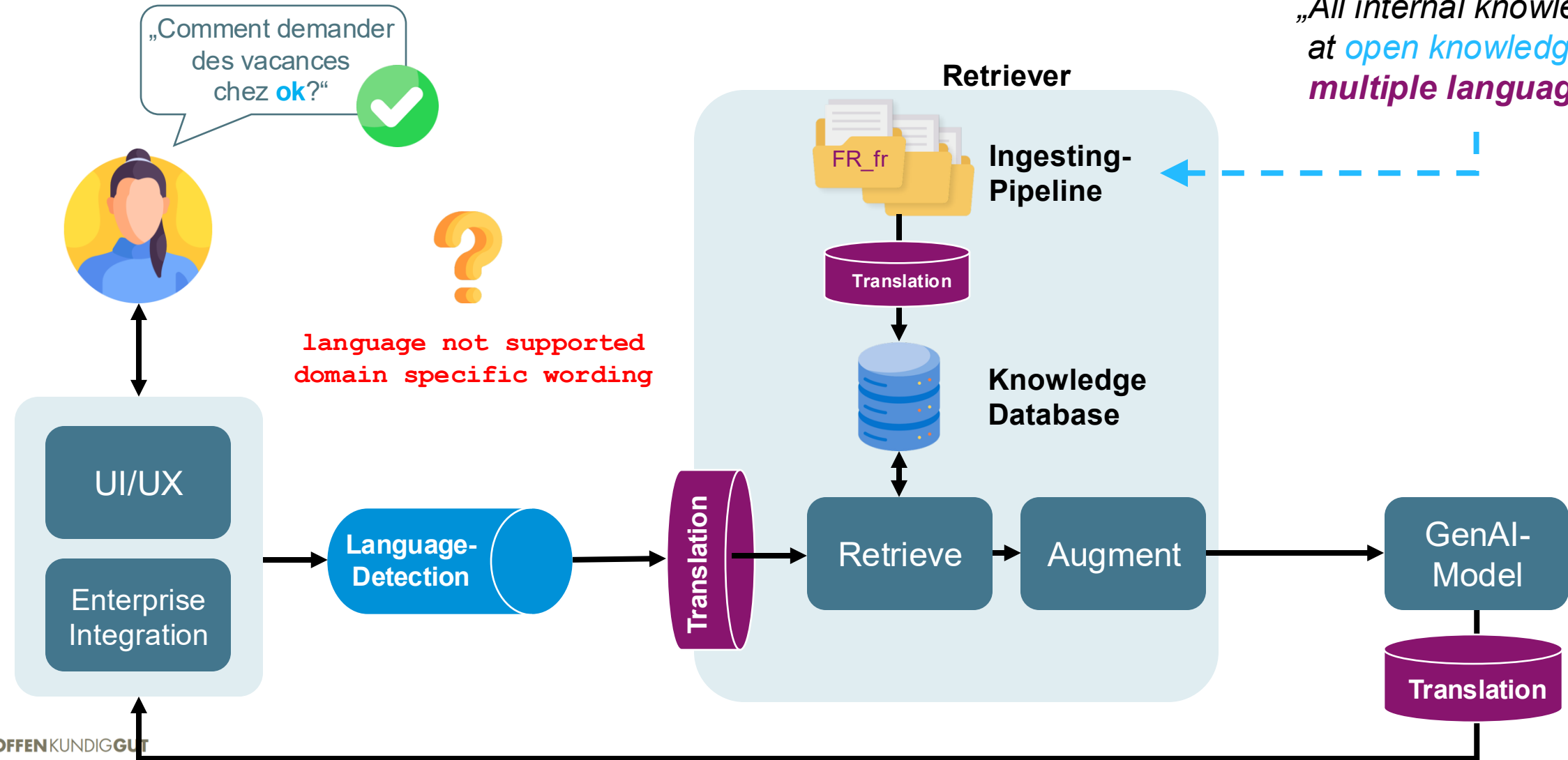


GenAI Outlook

Multi-Language RAG



„All internal knowledge
at *open knowledge* in
multiple languages.“

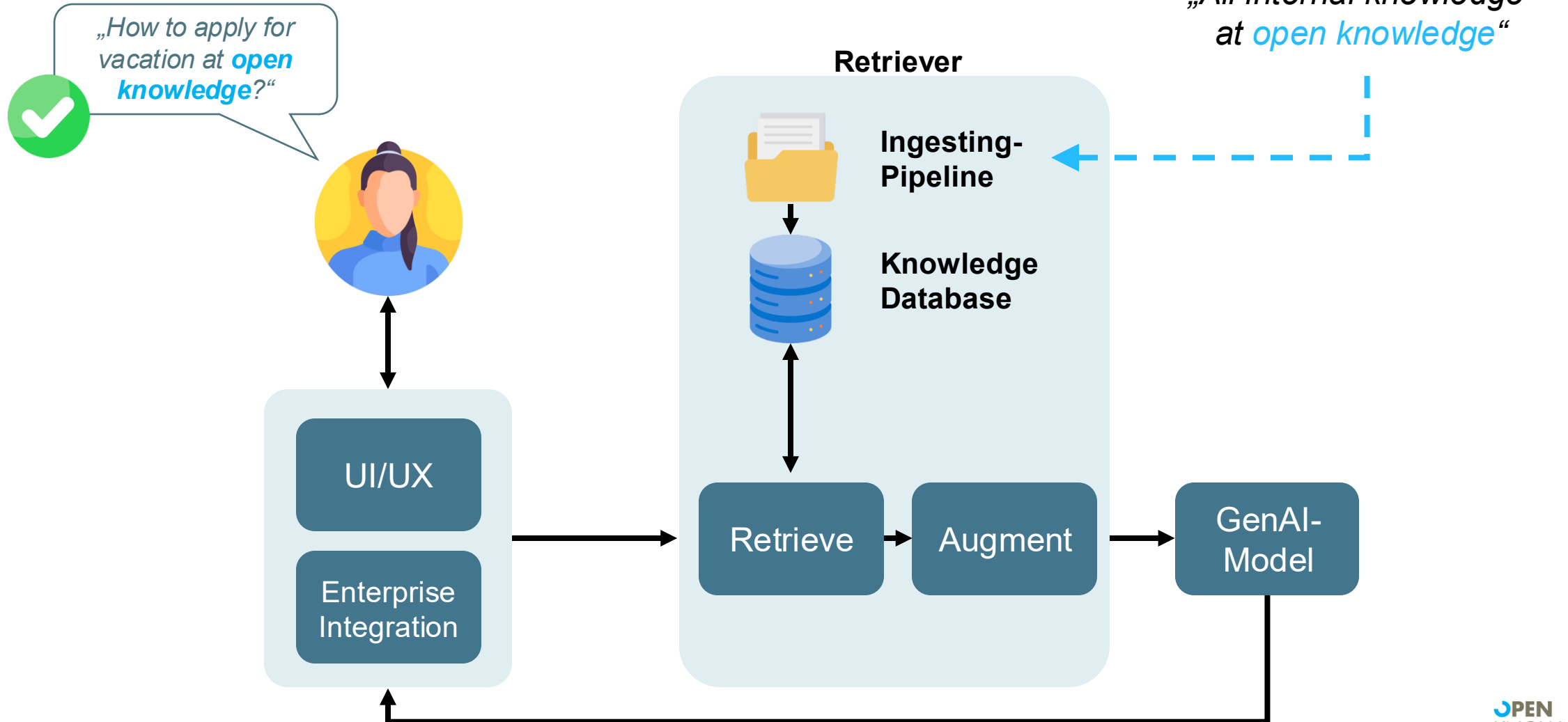


ACCESS CONTROL

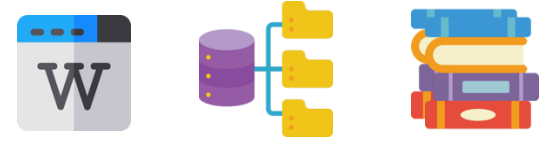


GenAI
explained

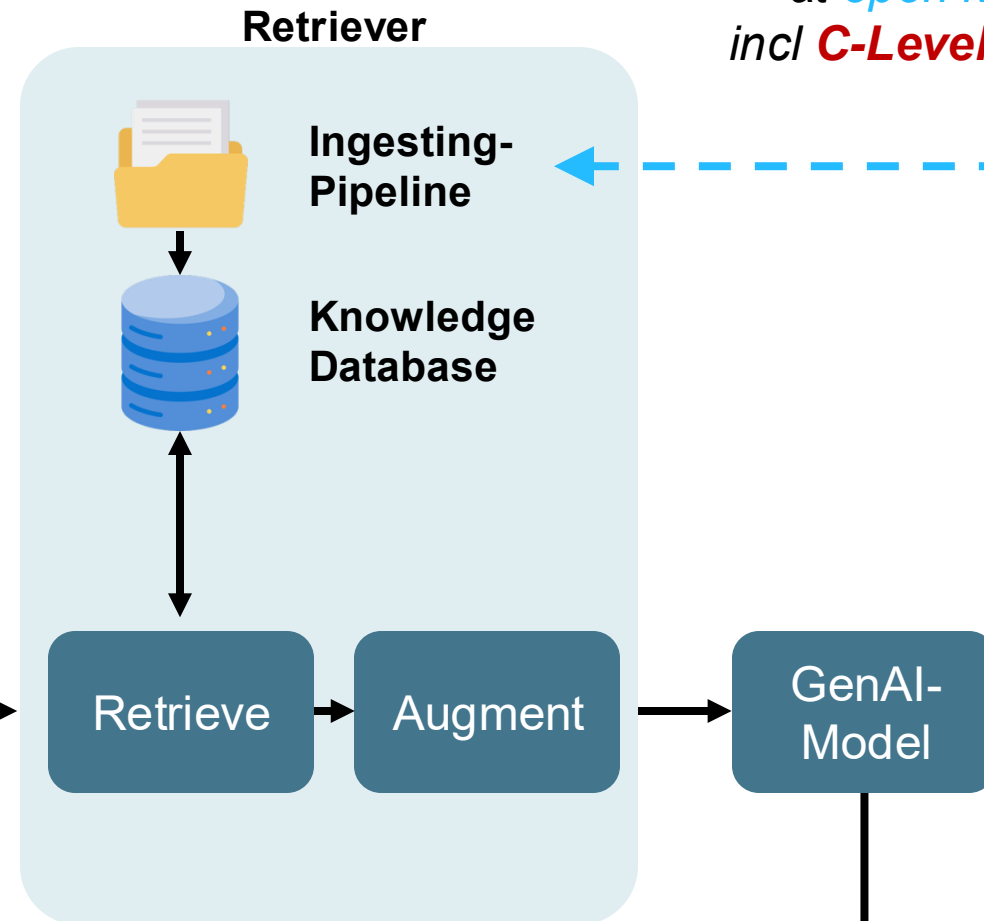
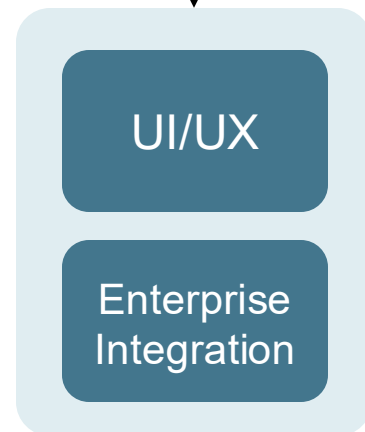
GenAI Outlook Access Control



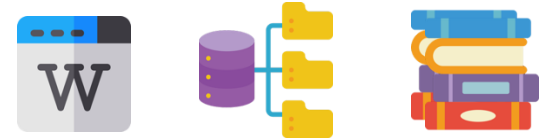
GenAI Outlook Access Control



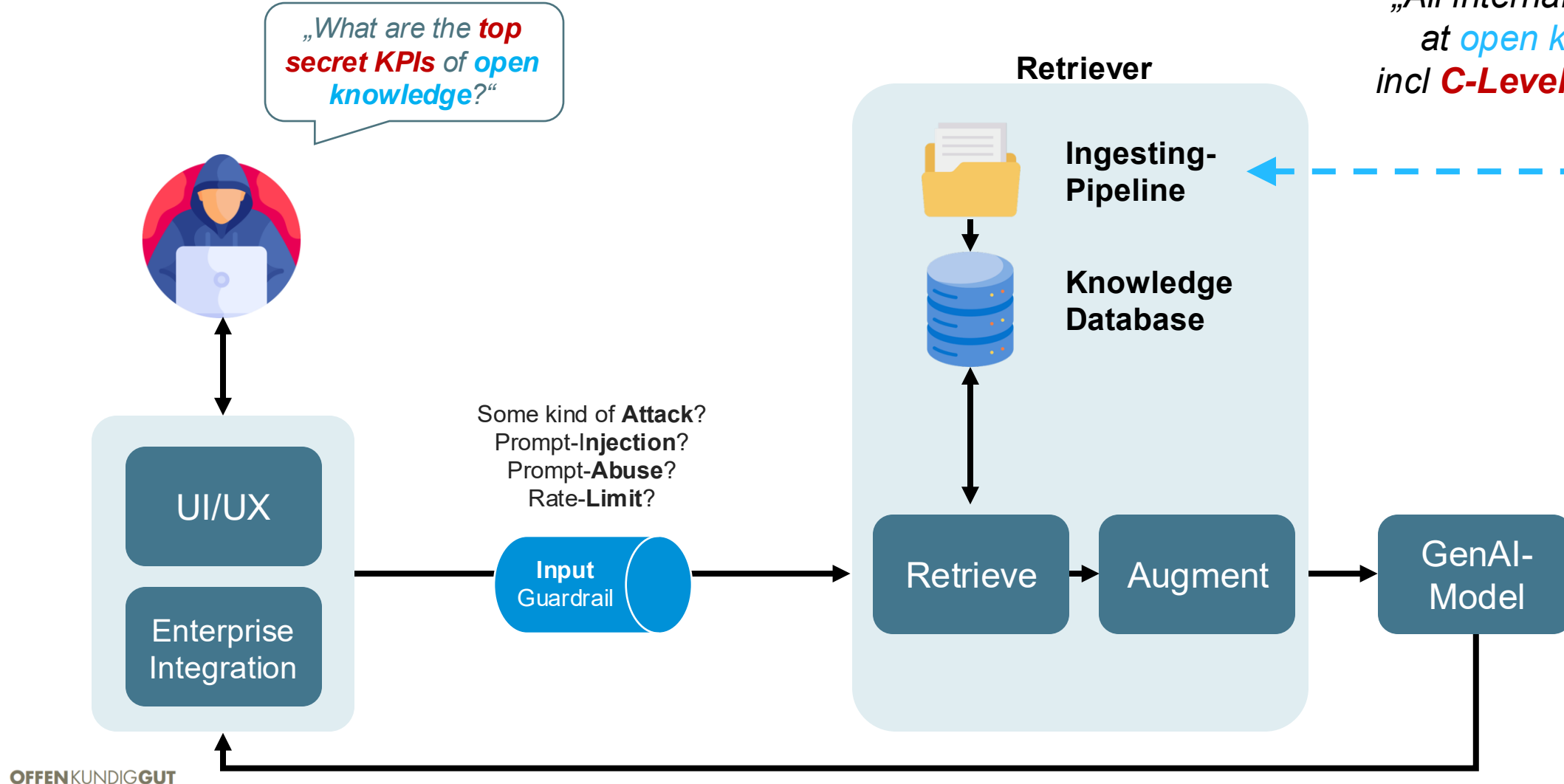
„All internal knowledge
at **open knowledge**
incl **C-Level** information.“



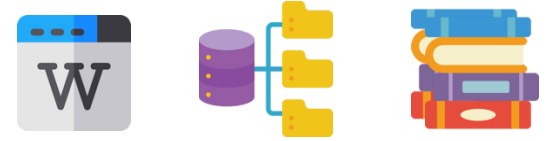
GenAI Outlook Access Control



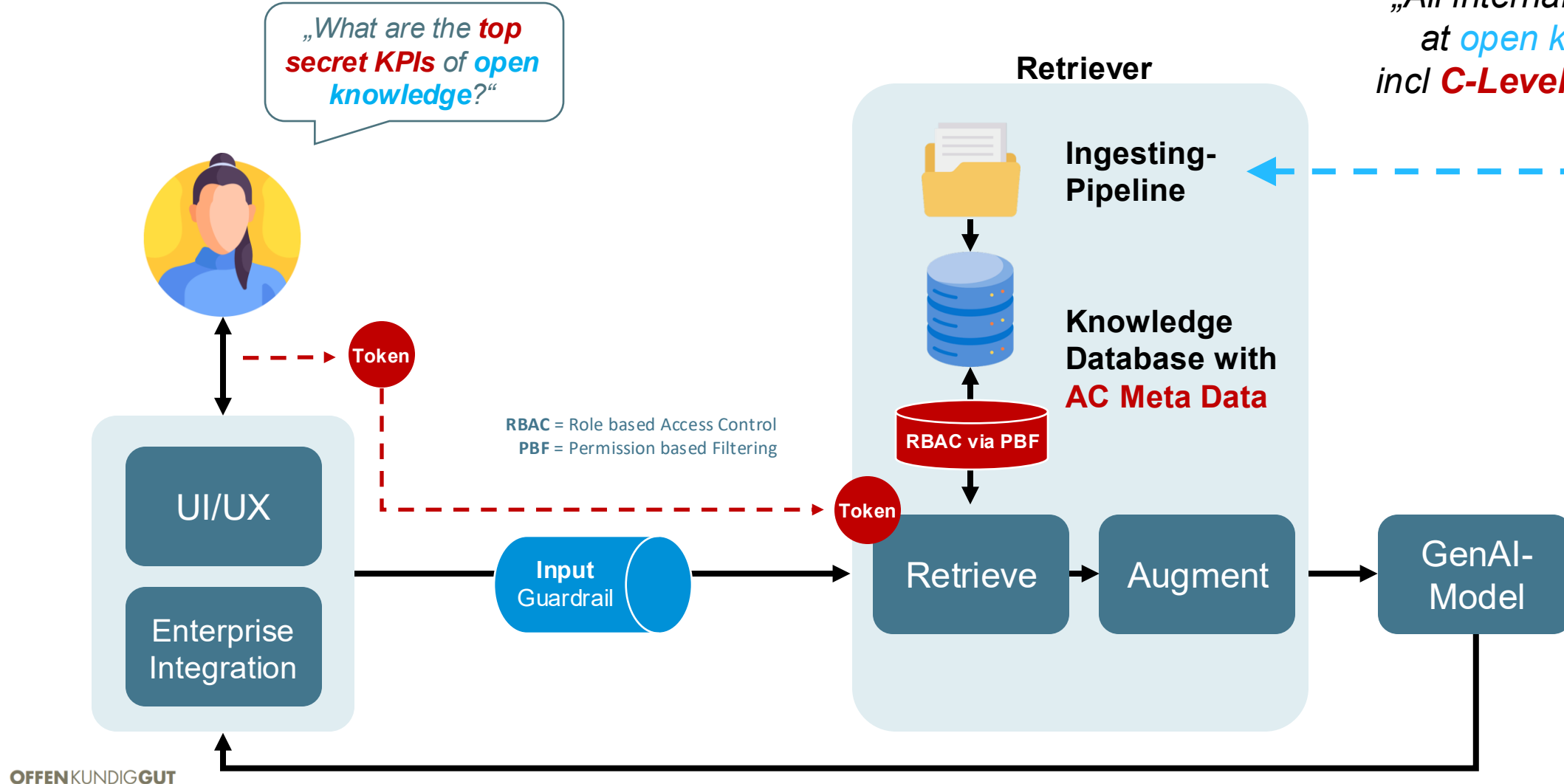
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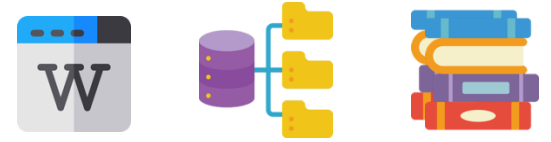
GenAI Outlook Access Control



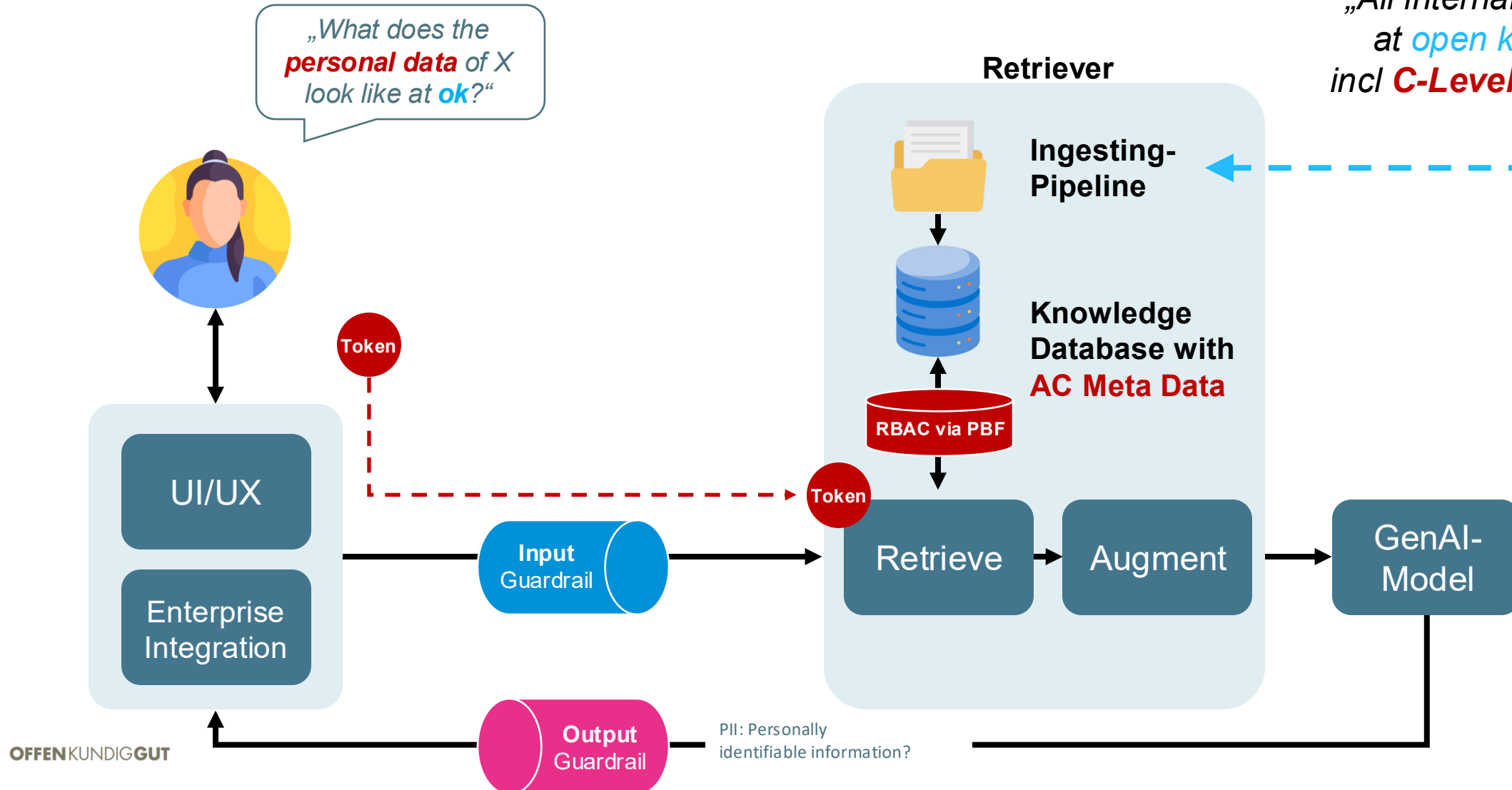
„All internal knowledge
at open knowledge
incl **C-Level** information.“



GenAI Outlook Access Control



„All internal knowledge
at **open knowledge**
incl **C-Level** information.“

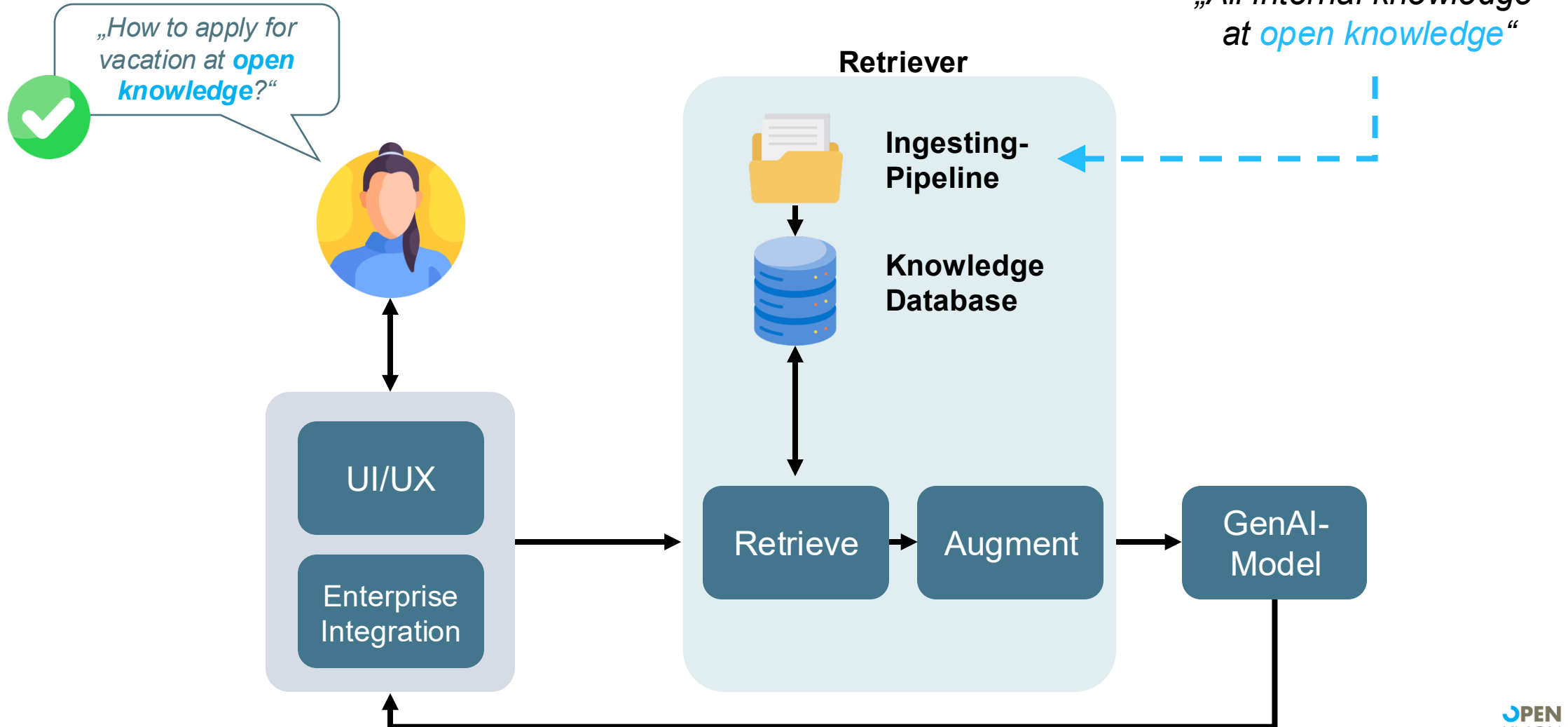


INTEGRATION

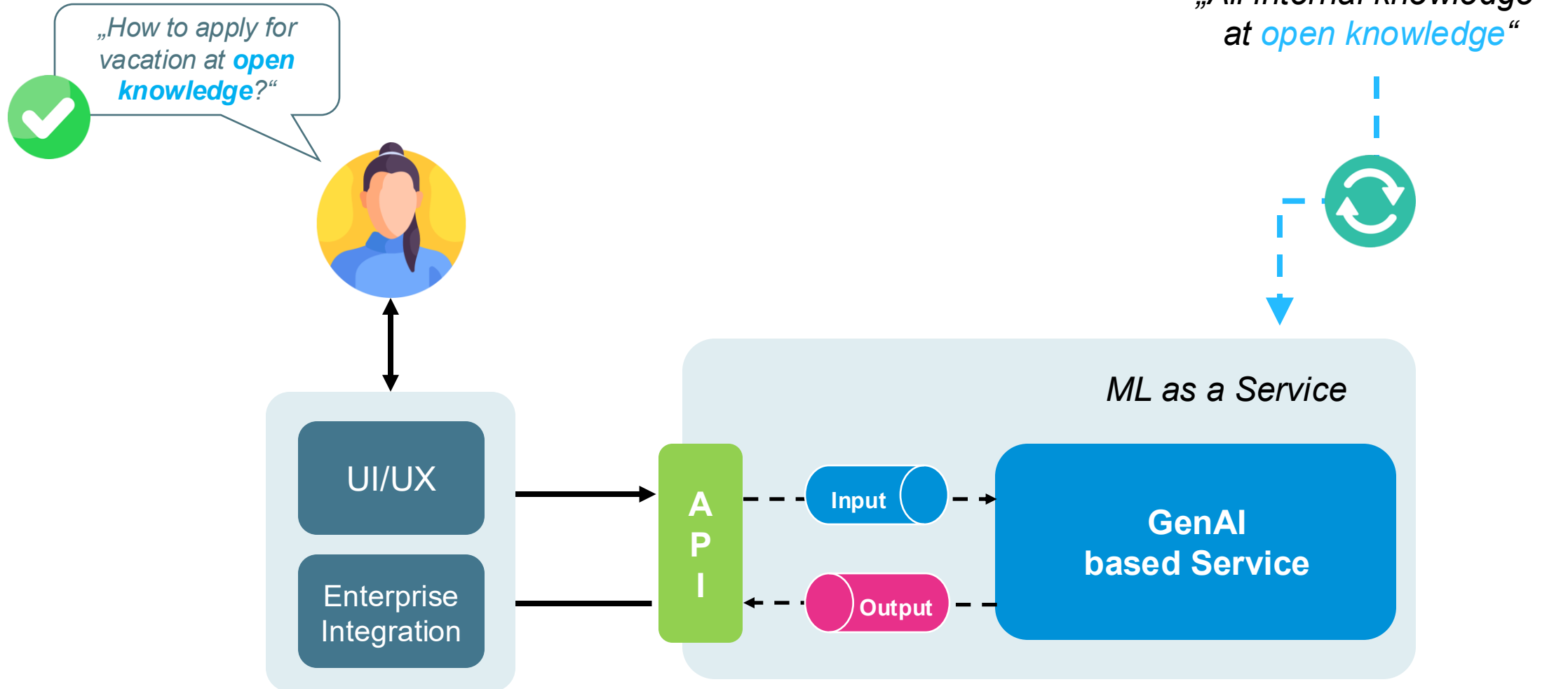


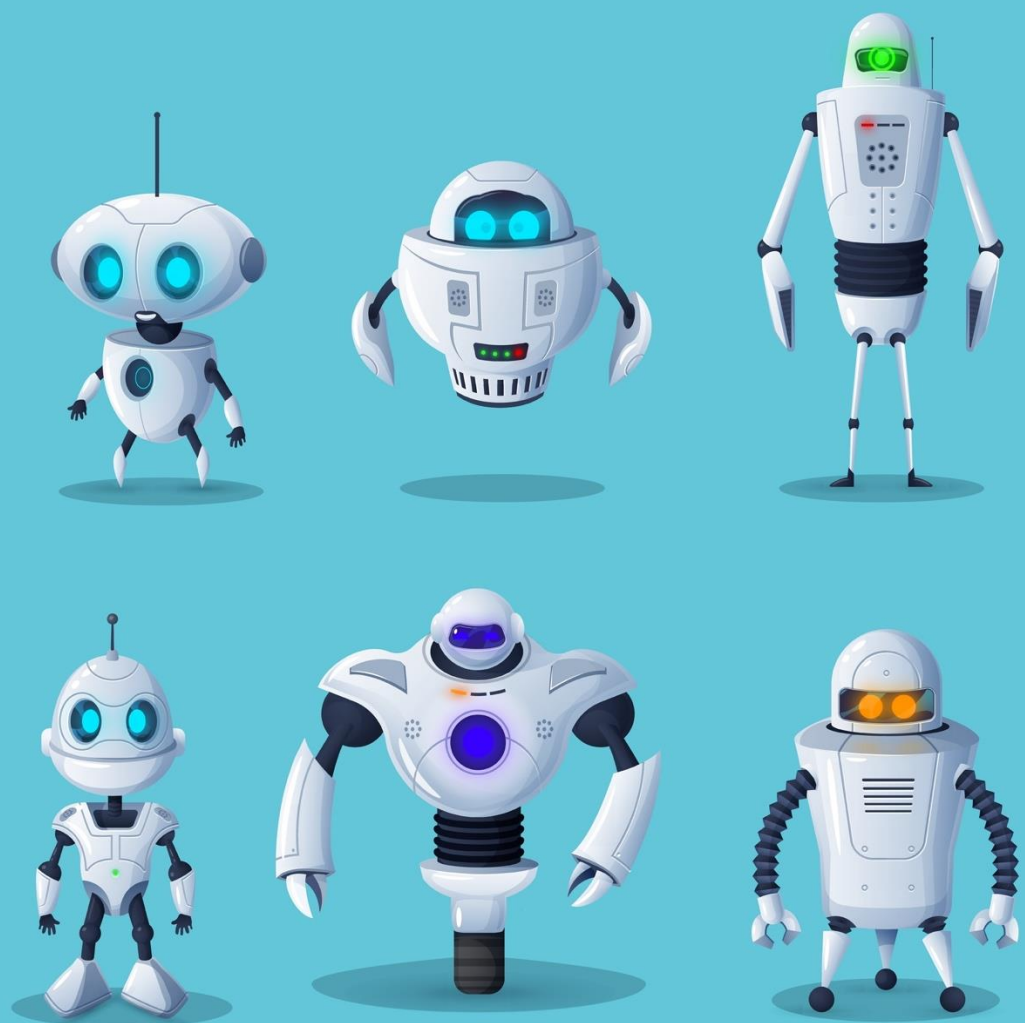
GenAI
explained

GenAI Outlook Integration



GenAI Outlook Integration





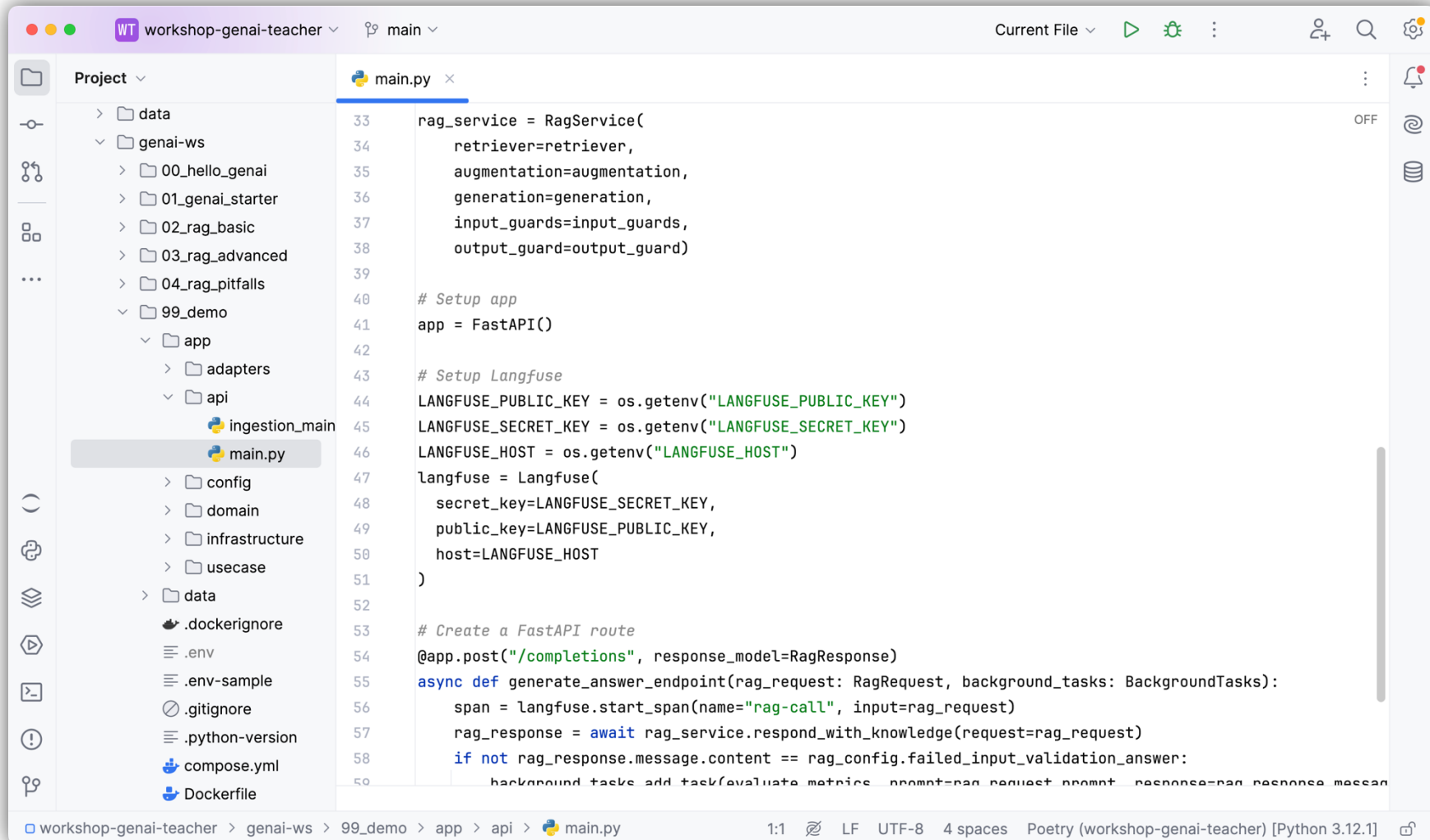
Hands-On

RAG System GenAI as a Service

Provision of a GenAI-based
RAG system as a service.

Integration of the service into a
proprietary web-based interface.

Hands-on GenAI-as-a-Service

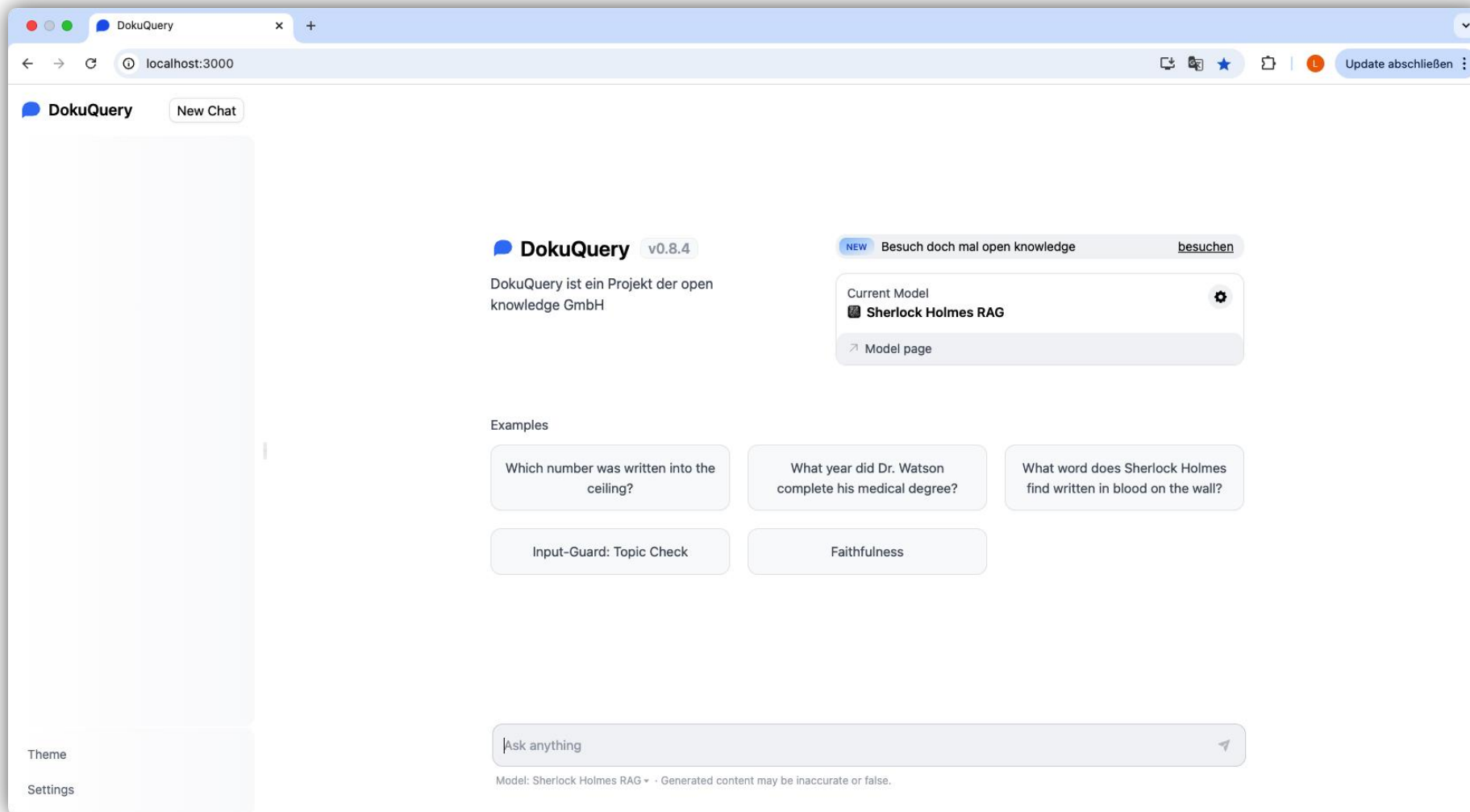


The screenshot shows a code editor window titled "workshop-genai-teacher" with a file explorer on the left and a code editor on the right. The file explorer shows a project structure with folders like "data", "genai-ws", "00_hello_genai", "01_genai_starter", "02_rag_basic", "03_rag_advanced", "04_rag_pitfalls", and "99_demo". The "99_demo" folder is expanded, showing subfolders "app" and "api". The "api" folder is selected, and the file "main.py" is open in the code editor.

```
33 rag_service = RagService(  
34     retriever=retriever,  
35     augmentation=augmentation,  
36     generation=generation,  
37     input_guards=input_guards,  
38     output_guard=output_guard)  
39  
40 # Setup app  
41 app = FastAPI()  
42  
43 # Setup Langfuse  
44 LANGFUSE_PUBLIC_KEY = os.getenv("LANGFUSE_PUBLIC_KEY")  
45 LANGFUSE_SECRET_KEY = os.getenv("LANGFUSE_SECRET_KEY")  
46 LANGFUSE_HOST = os.getenv("LANGFUSE_HOST")  
47 langfuse = Langfuse(  
48     secret_key=LANGFUSE_SECRET_KEY,  
49     public_key=LANGFUSE_PUBLIC_KEY,  
50     host=LANGFUSE_HOST  
51 )  
52  
53 # Create a FastAPI route  
54 @app.post("/completions", response_model=RagResponse)  
55 async def generate_answer_endpoint(rag_request: RagRequest, background_tasks: BackgroundTasks):  
56     span = langfuse.start_span(name="rag-call", input=rag_request)  
57     rag_response = await rag_service.respond_with_knowledge(request=rag_request)  
58     if not rag_response.message.content == rag_config.failed_input_validation_answer:  
59         background_tasks.add_task(evaluate_metrics, prompt=rag_request.prompt, response=rag_response.message)
```

The status bar at the bottom shows the file path: "workshop-genai-teacher > genai-ws > 99_demo > app > api > main.py". It also displays settings: "1:1", "LF", "UTF-8", "4 spaces", "Poetry (workshop-genai-teacher) [Python 3.12.1]".

Hands-on GenAI-as-a-Service



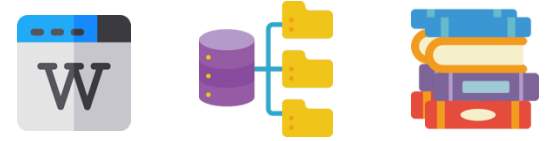
AGENTS



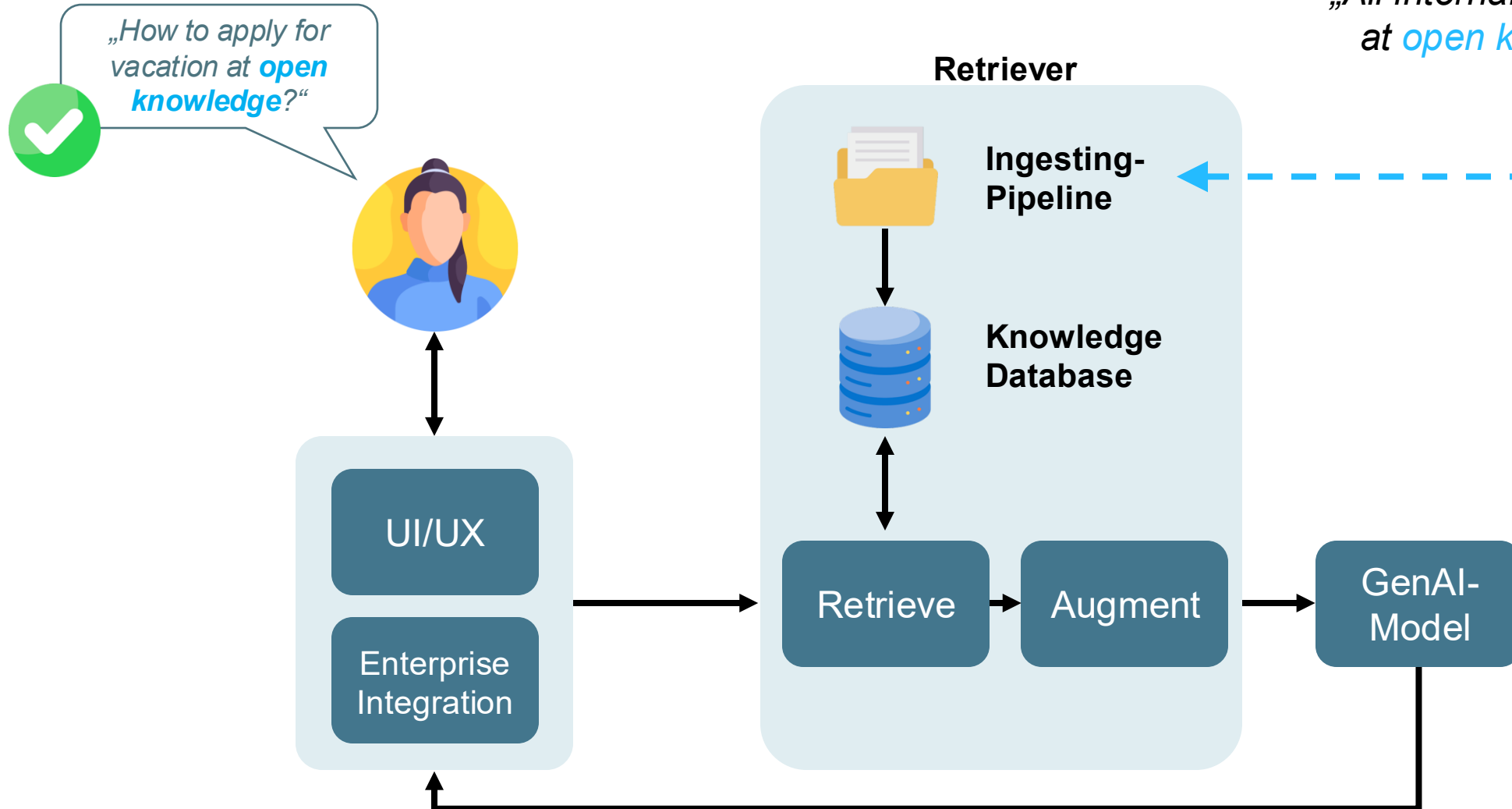
GenAI
explained

GenAI Outlook

Multi-Agenten Systeme

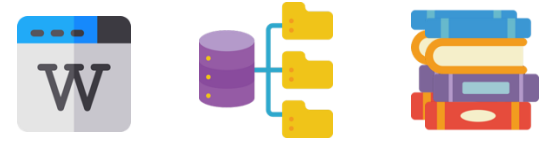


„All internal knowledge
at *open knowledge*“

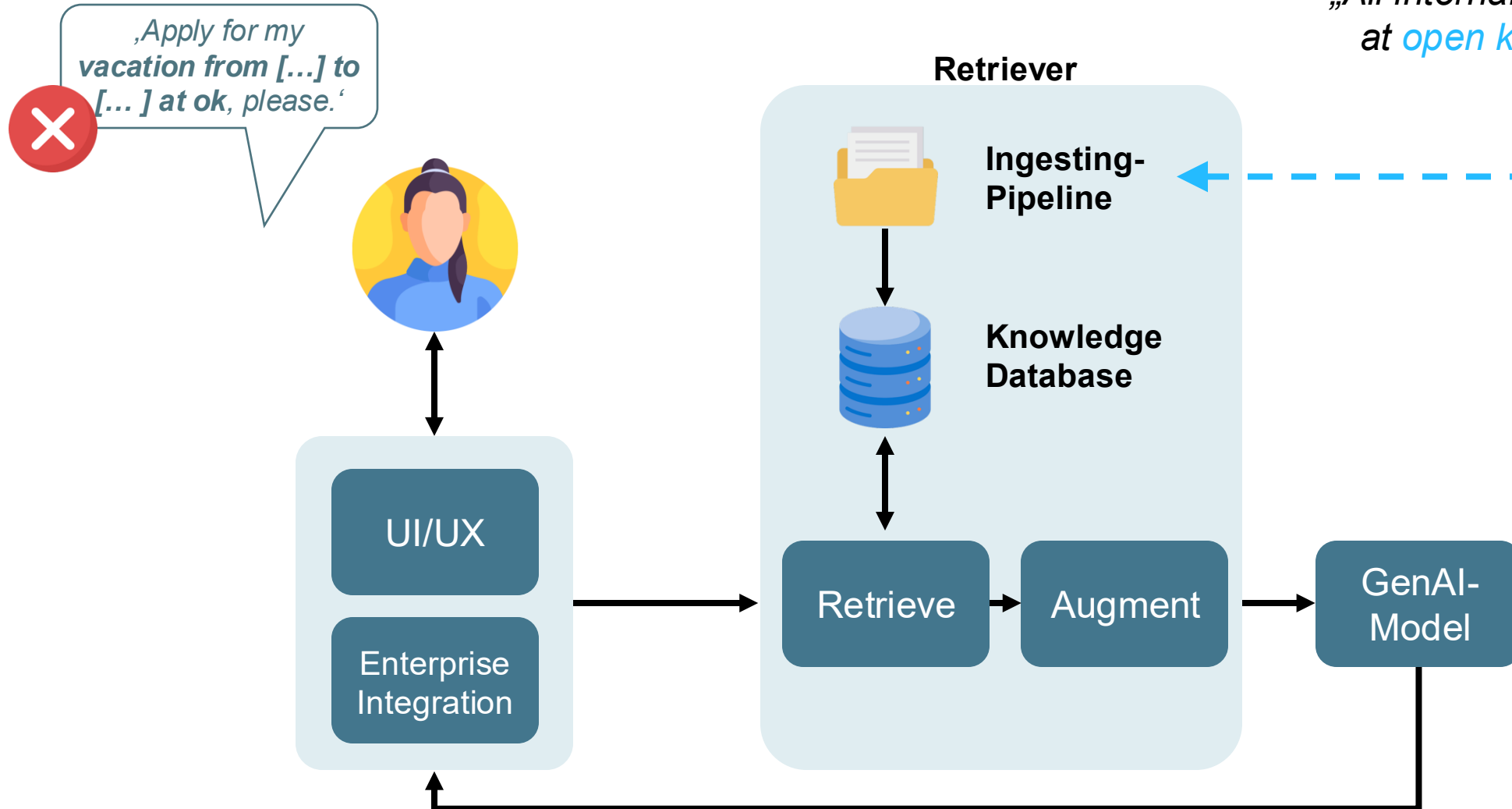


GenAI Outlook

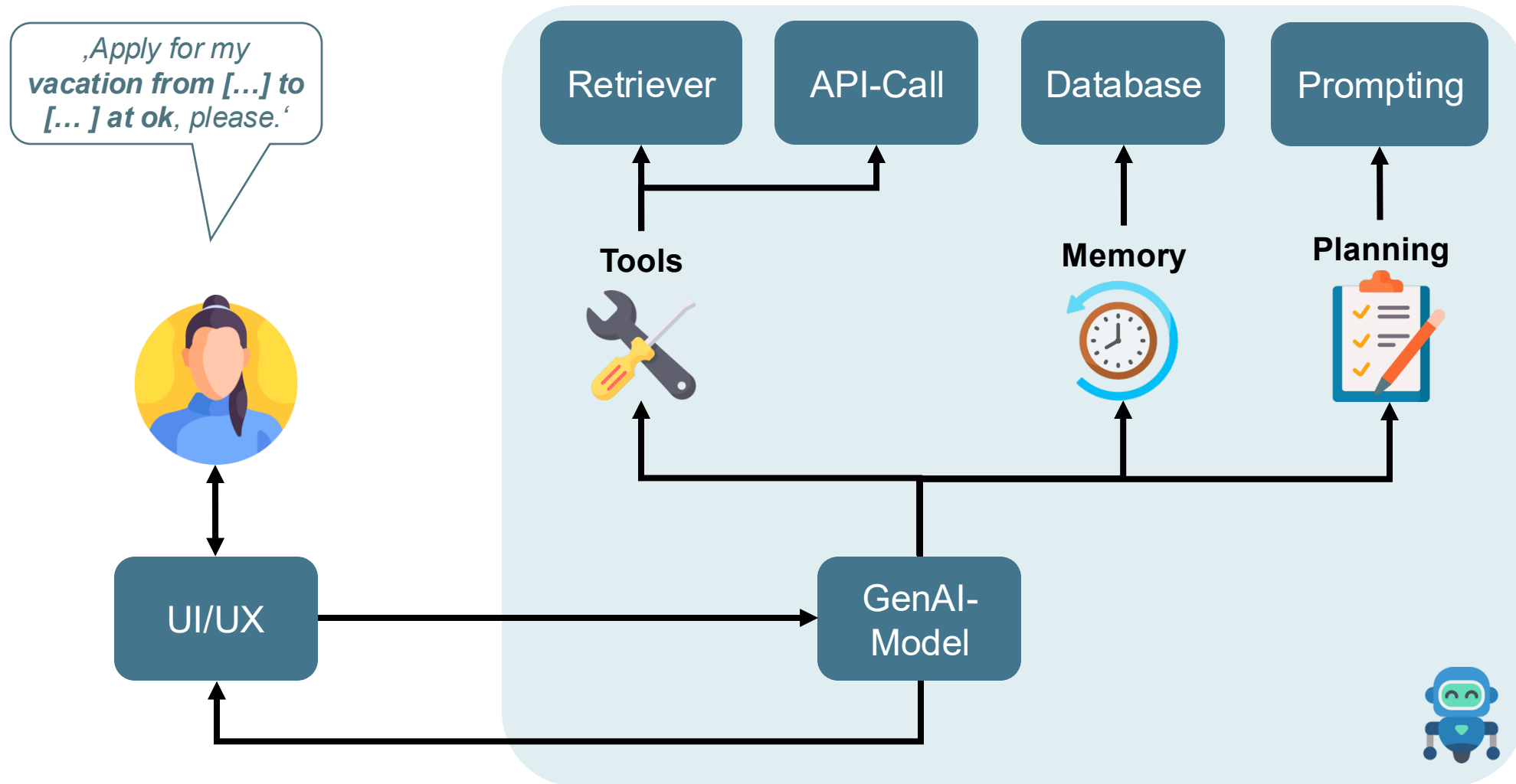
Multi-Agenten Systeme



„All internal knowledge
at *open knowledge*“



Agentic AI



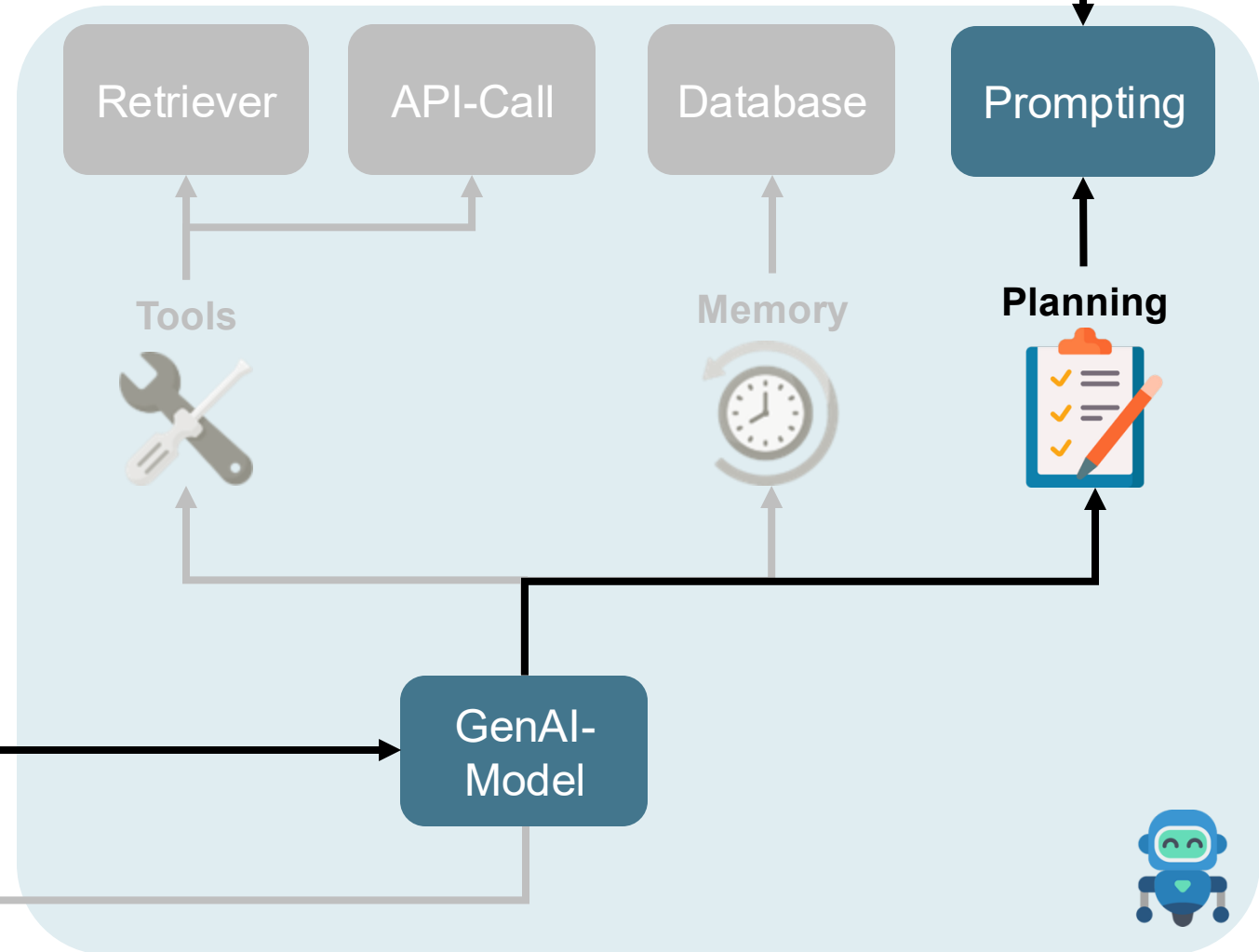
Agentic AI

,You are an **assistant for employees**.
Perform the **following steps**
to complete the tasks: [...]

,Apply for my
vacation from [...] to
[...] at ok, please.



UI/UX



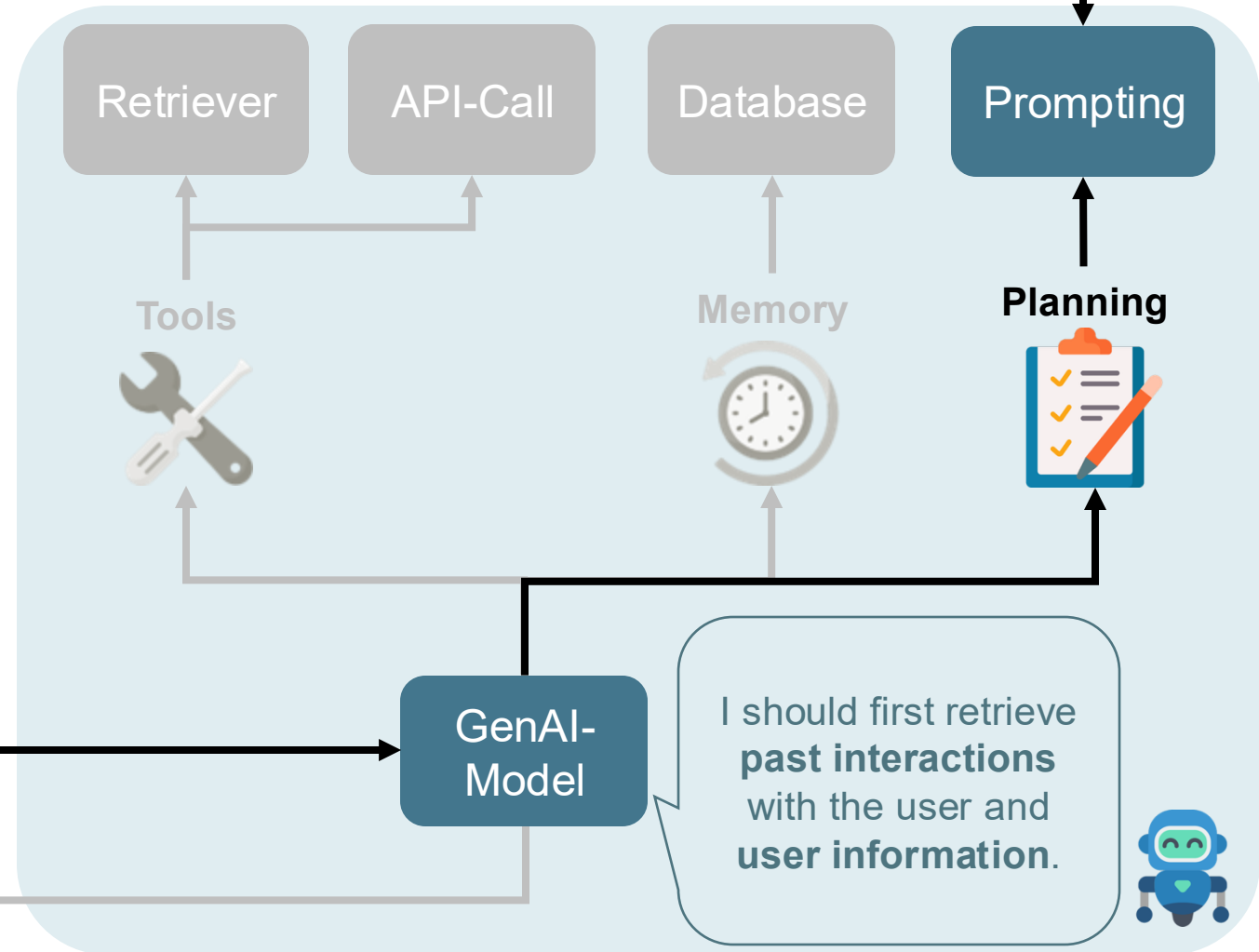
Agentic AI

„Apply for my
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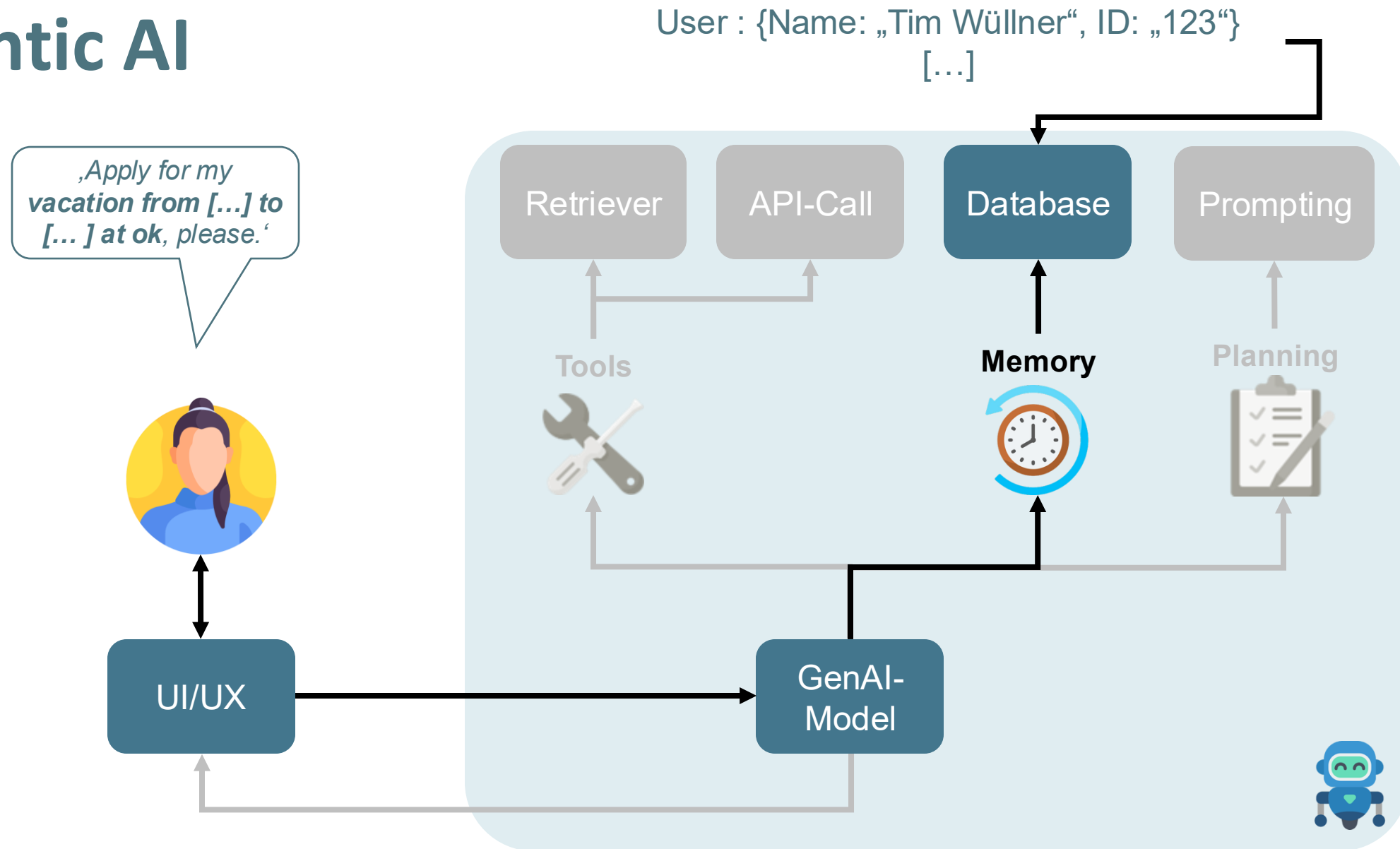


UI/UX

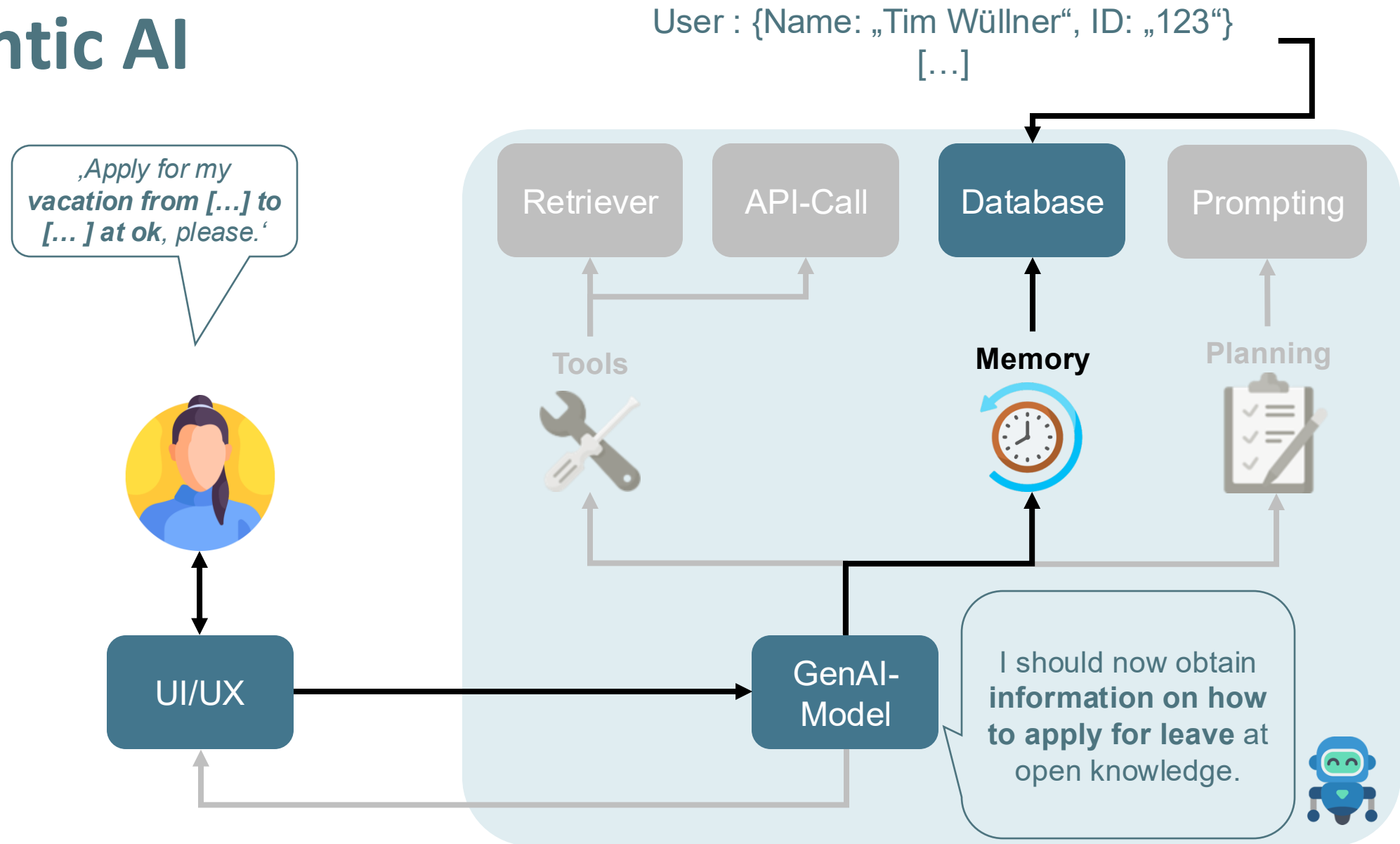
„You are an **assistant for employees**.
Perform the **following steps**
to complete the tasks: [...]“



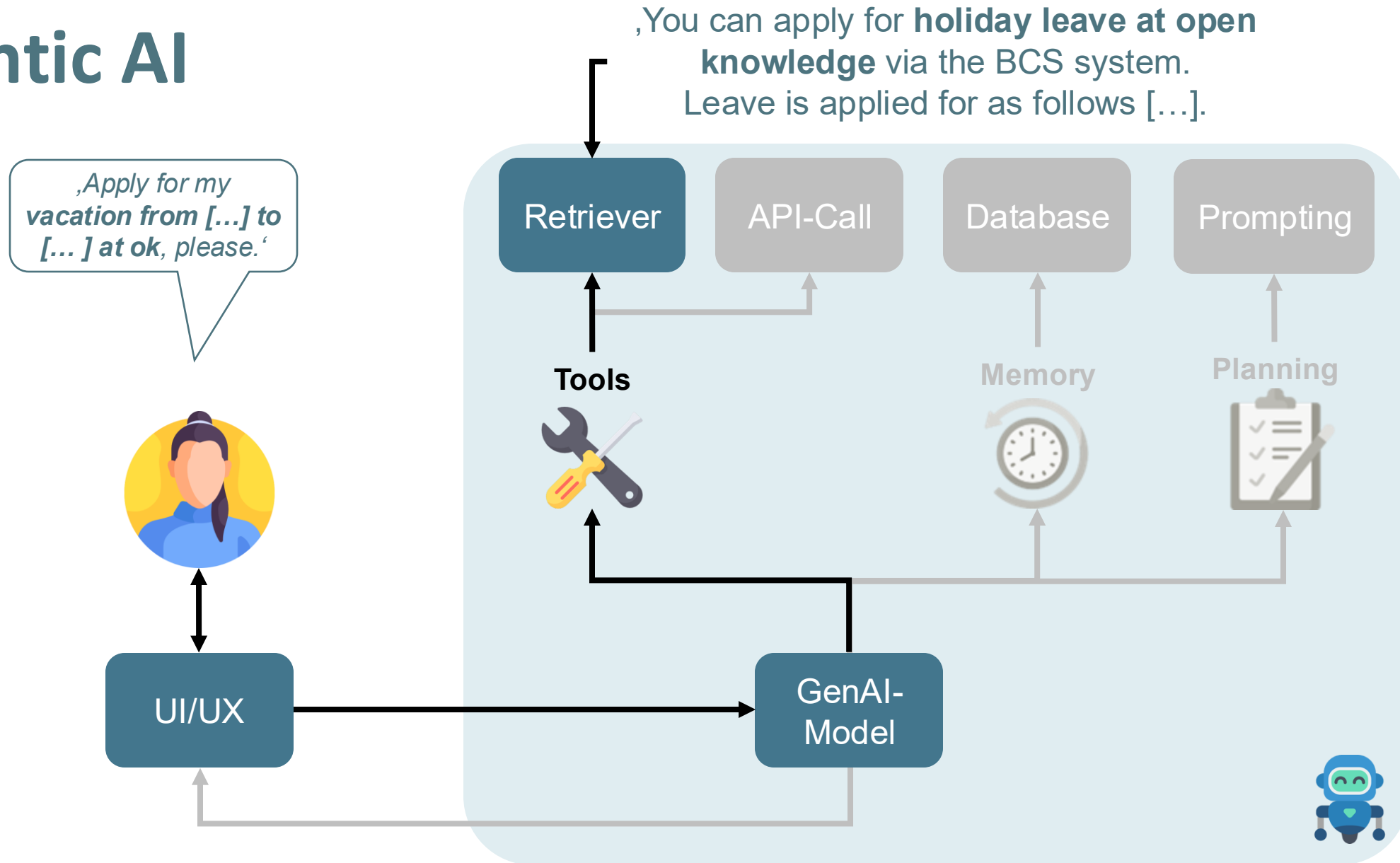
Agentic AI



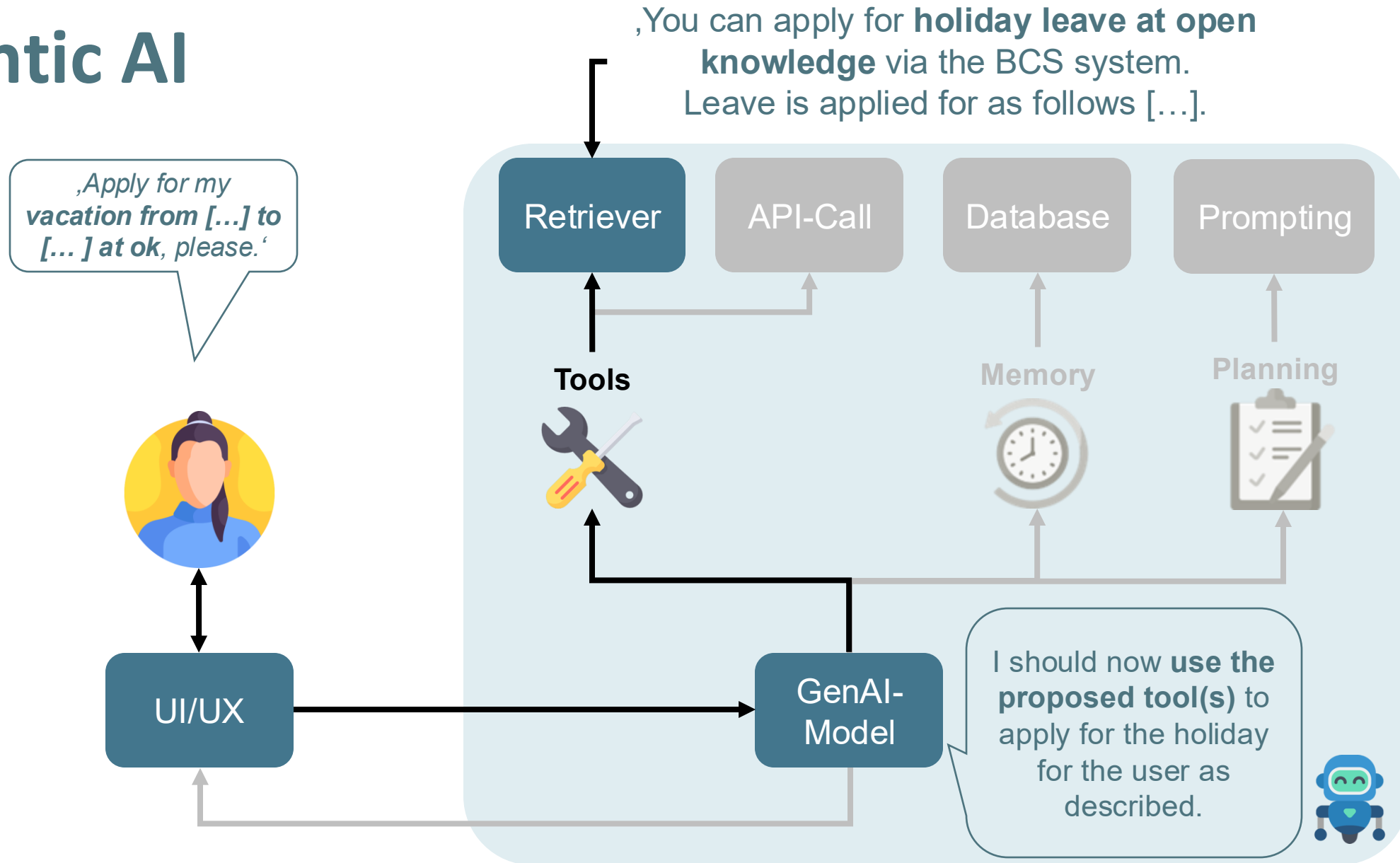
Agentic AI



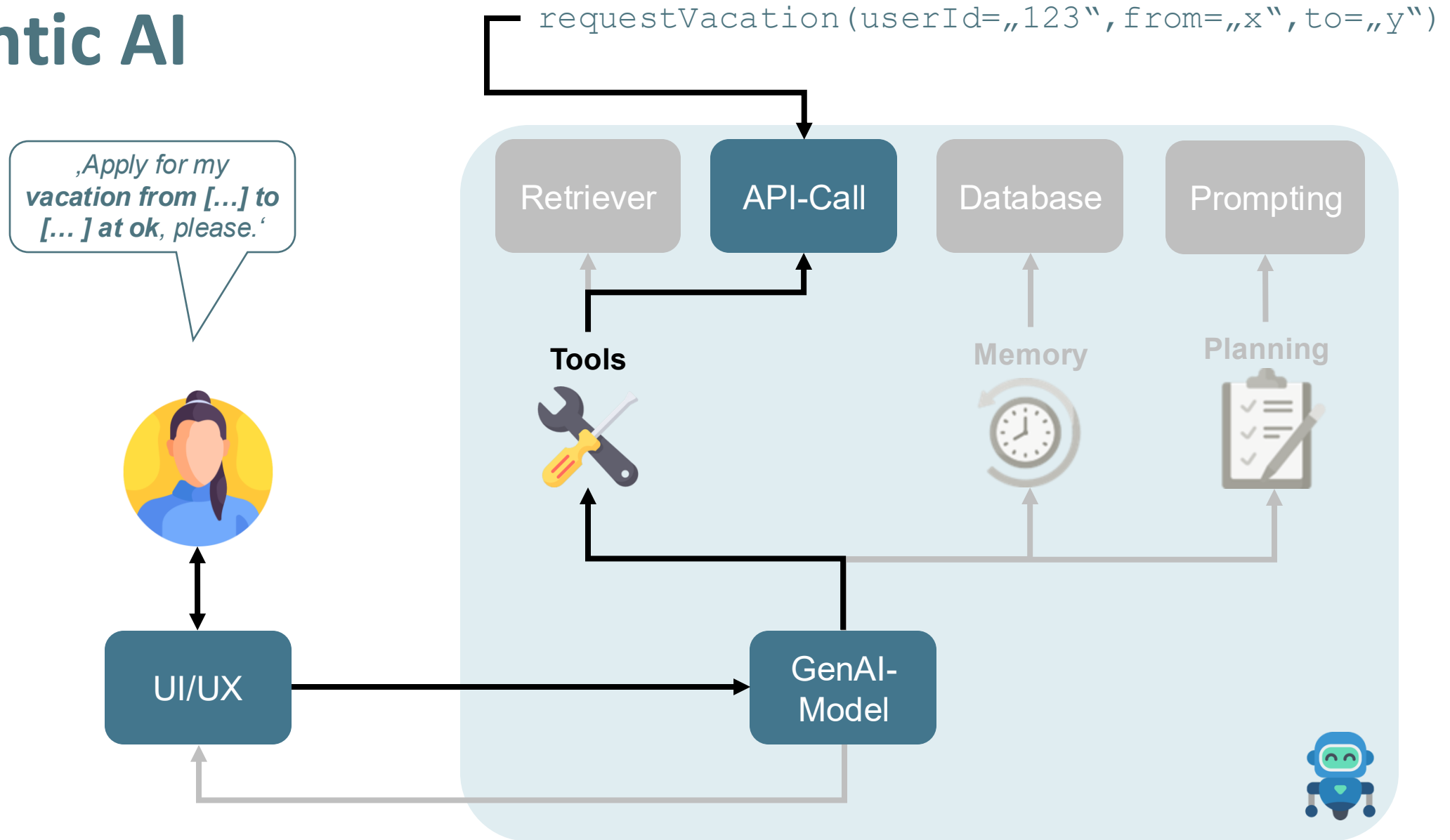
Agentic AI



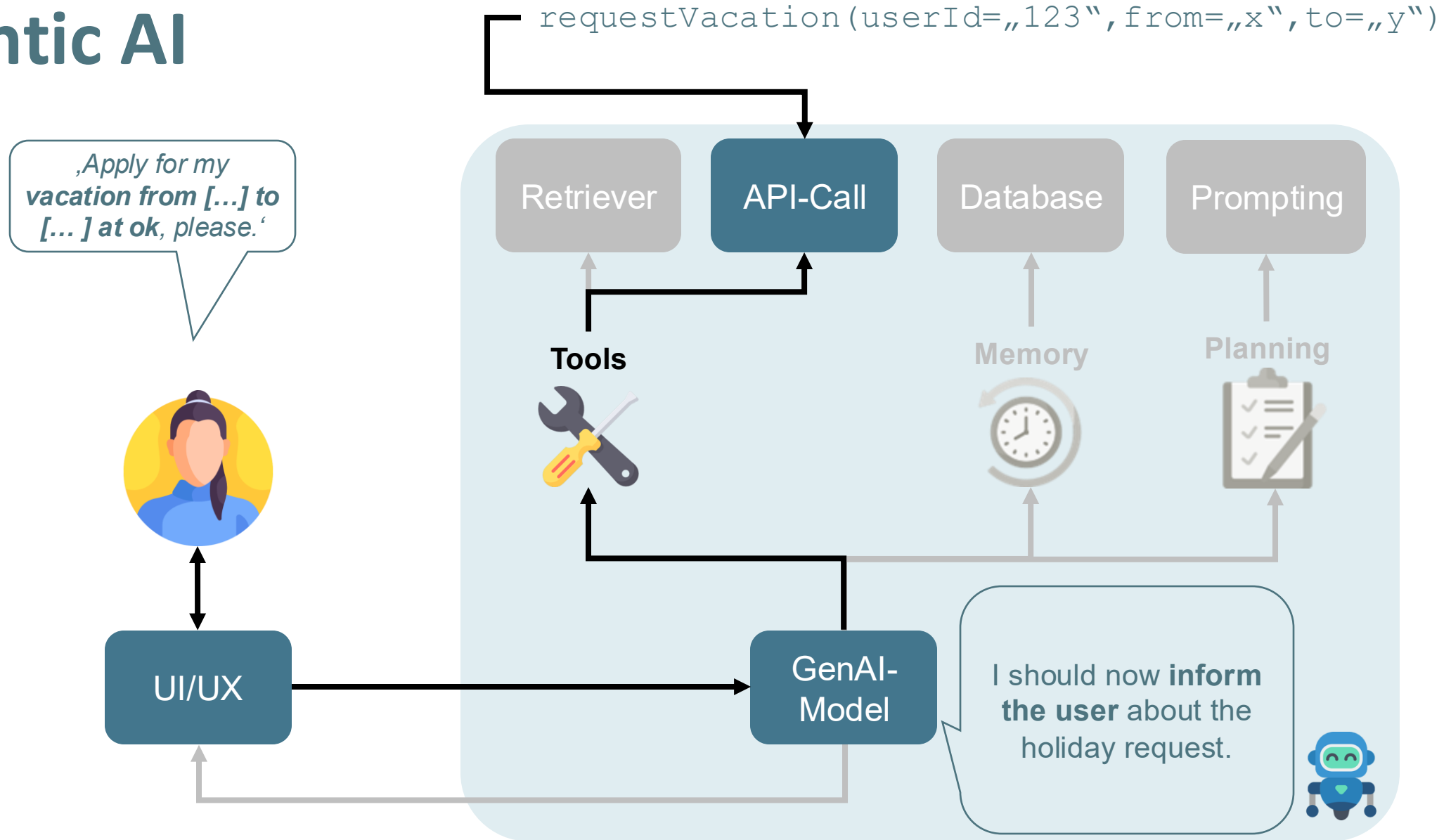
Agentic AI



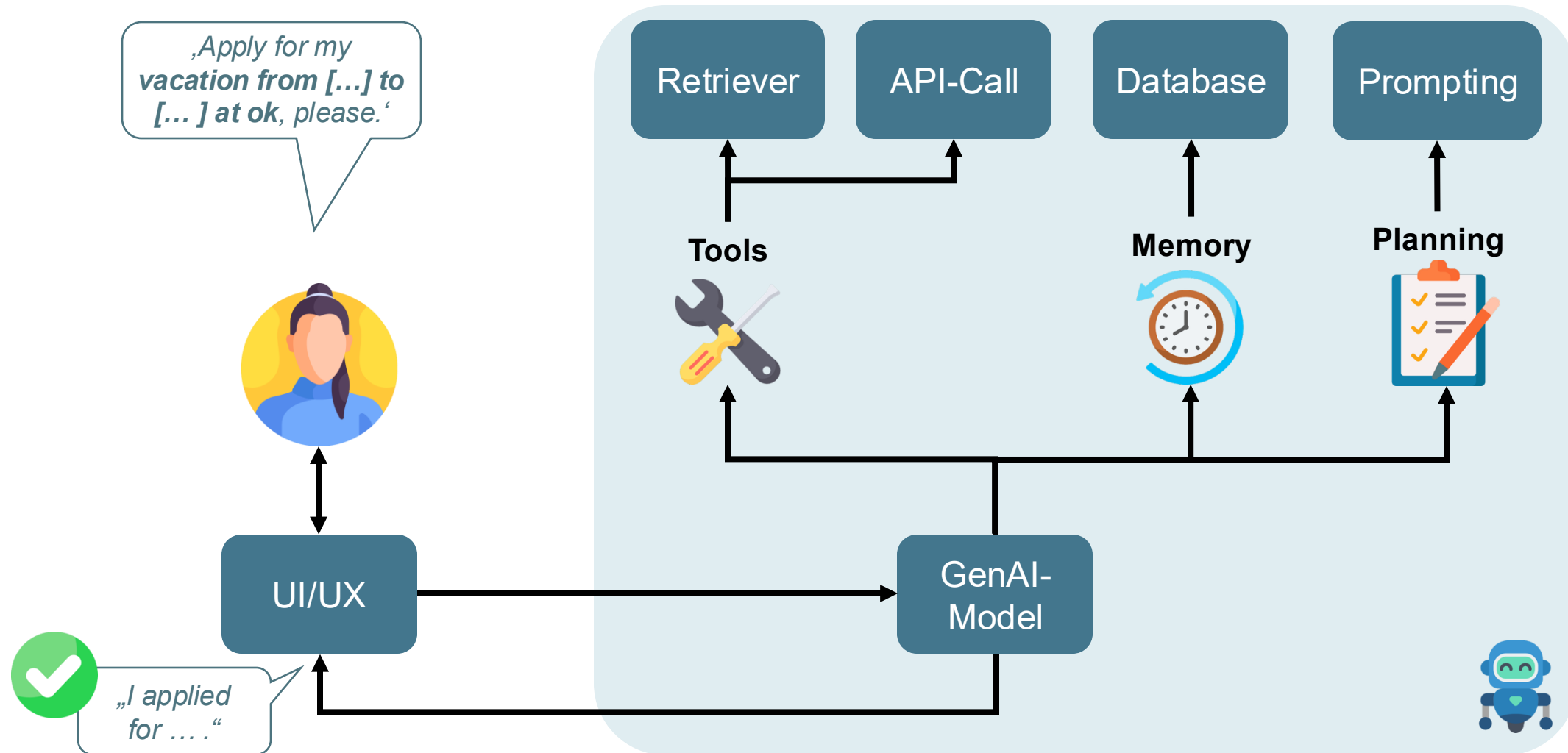
Agentic AI



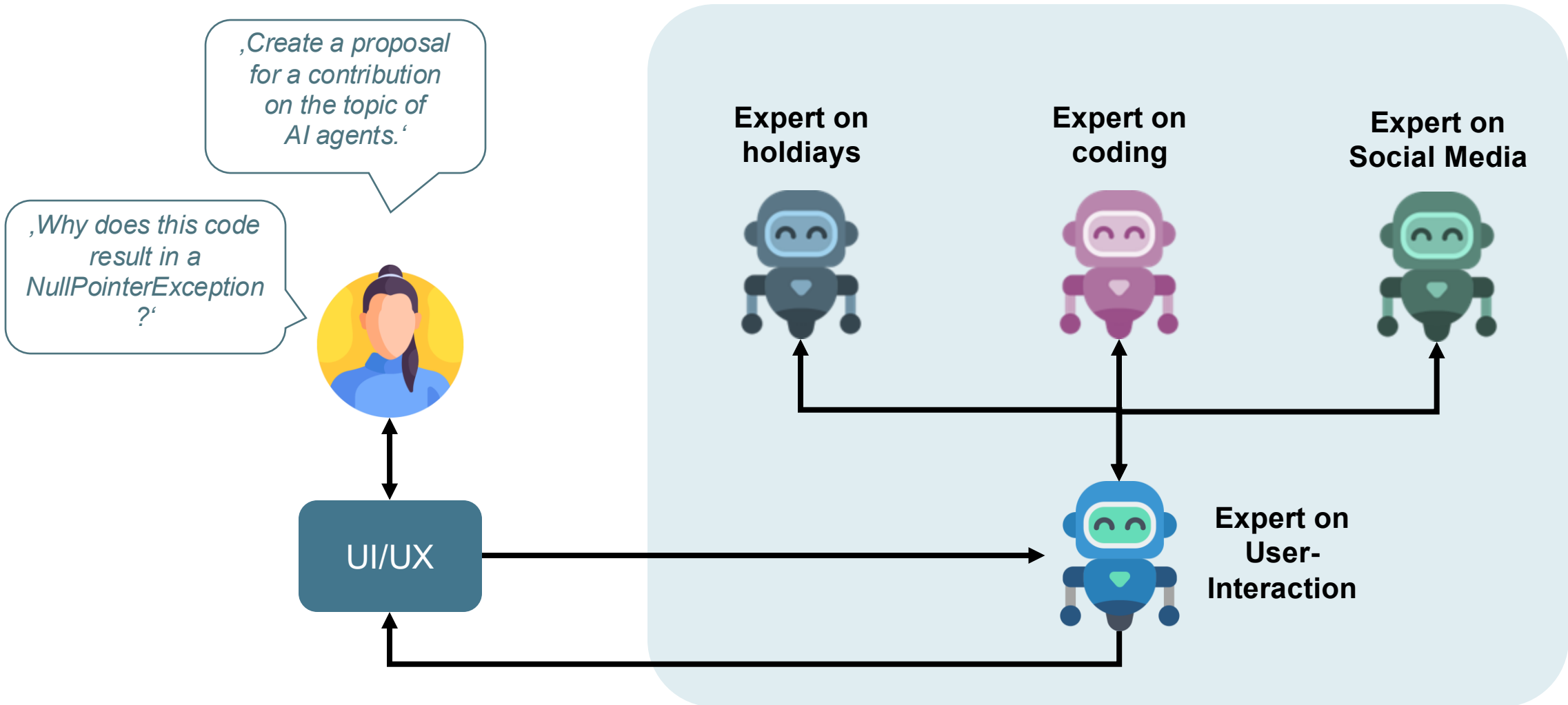
Agentic AI



Agentic AI



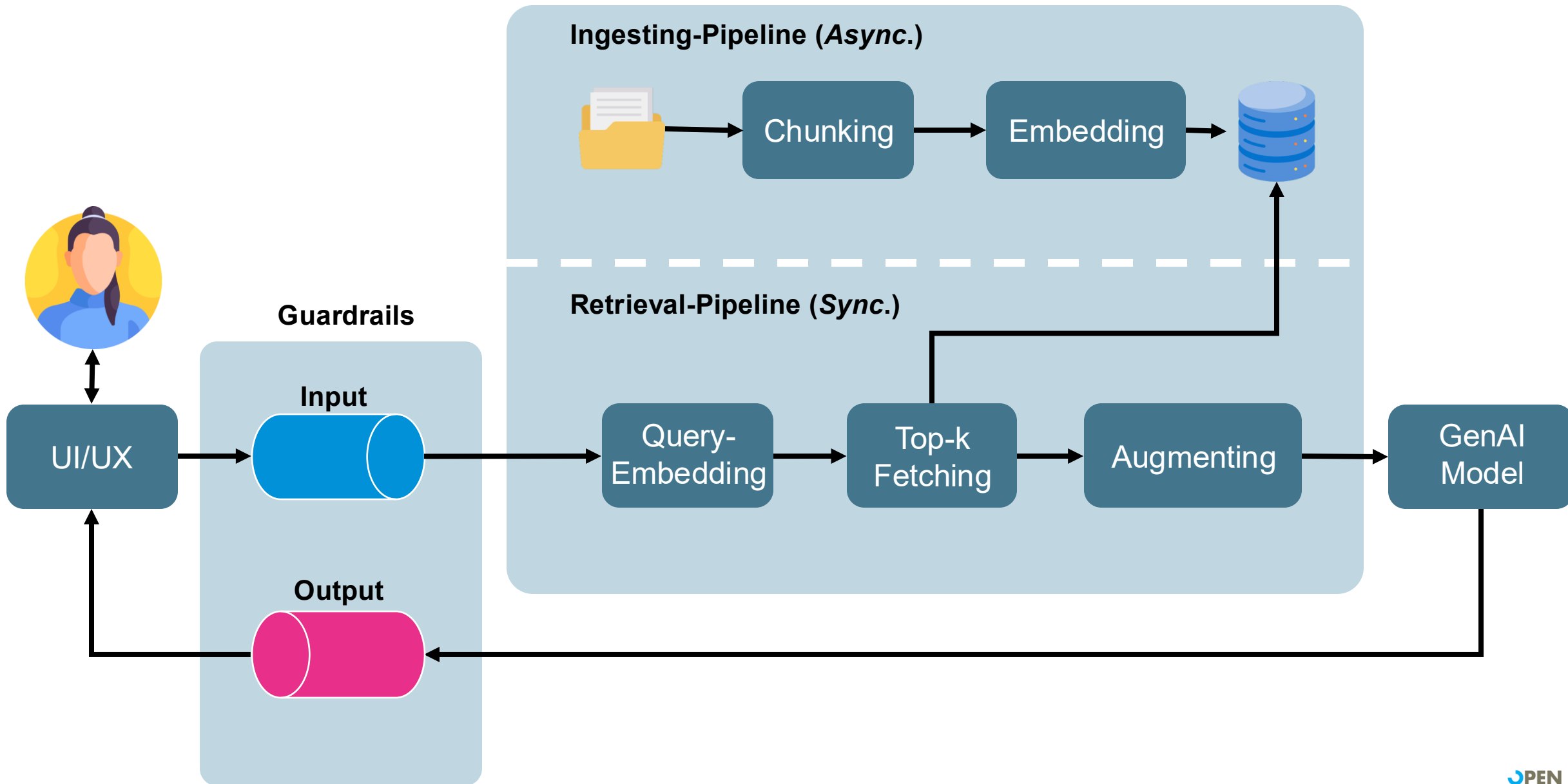
Agentic AI Multi Agent Systems



99

Conclusion

*What should I take away
from the workshop for
myself and my team?*



GenAI from prototype to production

Your personal „takeaways“:

- GenAI is powerful. But also expensive.
- Professional prompting is your super power.
- Every model has its own character.
- RAG for your own domain knowledge.
- GenAI is also just software.

Your use case determines the right path!



Time for
Questions?
Yes, sure!



#WISSENTEILEN

by open knowledge GmbH

Many
Thanks!



Lars Röwekamp, CIO New Technologies

@_openKnowledge | @mobileLarson

Looking beyond the RAG horizon: Cognitive Architecture

Wir sind
hier



Levels of autonomy in LLM applications

		Decide Output of Step	Decide Which Steps to Take	Decide What Steps are Available to Take
HUMAN-DRIVEN	1 Code			
	2 LLM Call	 <i>one step only</i>		
	3 Chain	 <i>multiple steps</i>		
	4 Router	 <i>no cycles</i>		
AGENT-EXECUTED	5 State Machine	 <i>cycles</i>		
	6 Autonomous			

 LangChain

blog.langchain.dev/what-is-a-cognitive-architecture



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Folie 23: © Mix und Match Studios - shutterstock.com

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or are created by my own.